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Theory

Sheaf Cohomology and Obstructions to Global Lifts

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Vorgelegt von:

Jeannine L. Döhrer

jeannine.doehrer@uni-ulm.de



Gutachter:

Prof. Dr. Jeroen Sijsling

Prof. Dr. Stefan Wewers

Betreuer:

Prof. Dr. Jeroen Sijsling

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To my dear cousins, Pauline, Fabi, and Nikki.

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Abstract

Sheaf cohomology provides a systematic way of measuring the failure of local data to glue to global objects. While Čech cohomology offers a concrete and computational approach to these obstructions, its dependence on open covers and their refinements can obscure the underlying categorical structure.

This thesis develops Čech cohomology from a categorical and homological perspective. After establishing the necessary foundations in category theory, sheaf theory, homological algebra, and derived functors, Čech cohomology is studied as a directed system over open covers. Using the exactness of filtered colimits in a suitable category such as the category of abelian groups, the first Čech cohomology group is identified with the first cohomology of the colimit of the Čech complexes. This separates artefacts arising from individual covers from genuine global obstructions and clarifies the role of filtered colimits in the construction of Čech cohomology.

The resulting framework is applied to the problem of gluing local square roots of holomorphic functions to global ones.

This yields a direct cohomological explanation for the existence of global square roots on the slit plane, whenever local square roots exist, and for the possible global obstruction on $\mathbb{C}^\times$. For the sheaf μ_2 , the spaces $\mathbb{C} \setminus (-\infty, 0]$ and $\mathbb{C}^\times$ are compared by purely Čech-theoretic methods. It is shown that every Čech 1-cocycle on the slit plane becomes a coboundary after refinement, whereas on the punctured plane a refinement-stable obstruction remains, corresponding to nontrivial monodromy around the puncture.

Overall, the thesis places Čech cohomology into a unified categorical framework and demonstrates how global geometric phenomena can be detected by the interaction of local data, filtered colimits, and the combinatorial structure of open covers.

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Introduction

The aim of this thesis is to present the categorical framework underlying Čech cohomology. To this end, the necessary concepts from category theory, sheaf theory, and homological algebra are developed, before Čech cohomology is constructed, and its relationship with sheaf cohomology is studied. The resulting theory is then applied to the existence of global square roots of holomorphic functions and to the interpretation of cohomological obstructions in terms of the topology of the underlying space.

A recurring problem in geometry is the passage from local to global data. Objects or constructions may exist locally on the members of an open cover without admitting a compatible global counterpart. Sheaf theory provides a natural language for describing local data and their compatibility, while cohomology measures obstructions to their globalisation. In the Čech approach, these obstructions are represented concretely by cocycles associated with open covers.

The dependence on a particular open cover, however, introduces information that need not reflect an intrinsic property of the underlying space or sheaf. Passing to refinements allows such cover-dependent phenomena to be distinguished from obstructions that persist independently of the chosen cover. From a categorical perspective, the collection of open covers and their refinements form a directed system, and Čech cohomology is obtained by passing to the corresponding colimit. The exactness of filtered colimits in suitable categories then makes it possible, in degree one, to identify this colimit of cohomology groups with the cohomology of a single colimit complex. In this way, the categorical structure of refinements becomes directly connected with the distinction between local artefacts and genuine global obstructions.

This perspective becomes particularly concrete in the problem of gluing local square roots of holomorphic functions. Local square roots determine transition data on

intersections, and the resulting Čech cocycle measures whether these local choices can be made compatible globally. On the slit plane, such an obstruction disappears after suitable refinement, whereas on the punctured plane nontrivial monodromy around the puncture can produce an obstruction that persists under refinement. Thus, the Čech–theoretic construction provides a direct link between the algebraic obstruction to gluing, and the topology of the underlying space.

The thesis is organised as follows. Chapters 1–3 develop the categorical, sheaf–theoretic, and homological foundations required for the construction of Čech cohomology. Chapter 4 studies singular cohomology, sheaf cohomology, and Čech cohomology, as well as their interrelations. Chapter 5 applies the developed theory to descent and cohomological obstructions. The appendices collect technical proofs and supplementary material supporting the main exposition.

Chapter 1. In this chapter, the language of category theory is introduced, and the categorical framework of this thesis is established. After defining categories, functors, and natural transformations, this chapter discusses universal constructions, the Yoneda perspective, limits and colimits, which are recurring universal constructions throughout this thesis, as well as additive and abelian categories, which provide the natural framework for sequences and exactness. These notions provide the common language for the subsequent development of sheaf theory and homological algebra.

Chapter 2. The foundations of sheaf theory are developed in this chapter. Beginning with presheaves and Grothendieck topologies, sheaves are introduced, and their categorical properties studied. Furthermore, this chapter discusses global sections and torsors. These constructions provide the setting in which sheaf cohomology is later defined.

Chapter 3. Homological algebra is presented in this chapter, which is essential to the construction of cohomology theories. After introducing chain complexes and abstract cohomology, injective resolutions, derived functors, and acyclic objects are presented. The chapter concludes with bicomplexes, filtrations, and spectral sequences, which provide important computational tools used later in this thesis.

Chapter 4. This chapter introduces the cohomology theories used throughout this thesis. After briefly discussing singular cohomology, sheaf cohomology is constructed by means of derived functors. Flabby sheaves are studied, before progressing with the development of Čech cohomology, and the nerve theorem, as well as the Leray theorem.

Chapter 5. The preceding theories come together in this chapter, where three examples of obstructions to global lifts are given, before the results are generalised. The examples study descent and the cohomological obstructions to lifting local data to global objects. The resulting perspective explains torsors and descent in terms of cohomological invariants and illustrates how cohomology naturally measures the failure of local constructions to globalise.

Appendix A. This chapter collects the set-theoretic, analytic, algebraic, and topological background used throughout the thesis. The appendix establishes several standard results and fixes notation in order to keep the main exposition largely self-contained.

Appendix B. A summary of algebraic categories and the required notions can be found in this chapter. The free-forgetful adjunction is explained, which allows certain constructions to be transferred from the category of sets $\mathbf{Set}$ into certain algebraic categories.

Appendix C. An intermediate result for cocomplete categories of the Freyd–Mitchell embedding theorem, as well as the necessary tools, are presented in this chapter.

Appendix D. This chapter elaborates on the connection between topological covers and sieves. Furthermore, several detailed proofs regarding sheaves on Grothendieck topologies and sheafification are given, and the necessary tools are introduced.

Appendix E. This chapter provides the necessary tools to prove the comparison theorem of sheaf cohomology and singular cohomology.

The following notions are assumed to be familiar:

1. The notions of injectivity, surjectivity, and bijectivity
2. The set of positive integers $\mathbb{N}$
3. The set of non-negative integers $\mathbb{N}_0$
4. The set of integers $\mathbb{Z}$
5. The set of positive real numbers $\mathbb{R}_{>0}$
6. The field with two elements $\mathbb{F}_2$
7. The field of complex numbers $\mathbb{C}$ with real part Re , imaginary part Im , the complex conjugate $\bar{z}$ of a complex number z , and the representation via polar coordinates
8. The notion of partially ordered sets (posets) and the notion of chains, which are totally ordered subsets of partially ordered sets
9. Throughout this thesis, the notation $a \ll b$ (or equivalently, $b \gg a$) is used informally to indicate that b is sufficiently large compared to a . The precise meaning depends on the surrounding argument.
10. The disjoint union of sets A and B is denoted by $A \dot{\cup} B$
11. Proper subsets are denoted by $\subsetneq$, while not necessarily proper subsets are denoted by $\subseteq$.
12. Convex combinations and convex hulls
13. The n -disc $D^n := \{x \in \mathbb{R}^n : \|x\| \leq 1\}$
14. The n -sphere $S^n := \{x \in \mathbb{R}^{n+1} : \|x\| = 1\}$

1 A brief Introduction to Category Theory

This chapter introduces the categorical background required throughout this thesis. Section 1.1 provides a brief introduction to the basic notions of category theory, including objects, morphisms, and diagrams. Functors are discussed in Section 1.2, followed by the notion of natural transformations and universal properties introduced in Section 1.3. The Yoneda perspective, presented in Section 1.4, provides a fundamental way of understanding representability and universal properties. Universal properties in turn underline the constructions of limits and colimits, introduced in Section 1.5. In Section 1.6, categories with distinguished properties, that is, additive and abelian categories, are developed. Finally, Section 1.7 discusses sequences and exactness, used throughout the remainder of the thesis.

1.1 Categories

This section provides a first introduction to the notion of categories, objects, diagrams, and morphisms.

Categories.

Definition 1.1.1 (Category). A category $\mathcal{C}$ consists of the following data:

- (C1) A *class* of **objects**, denoted by $\text{Ob}(\mathcal{C})$.
- (C2) Between every pair of objects $A, B \in \text{Ob}(\mathcal{C})$, there exists a class (which may be empty) of **morphisms** from A to B , denoted by $\text{Hom}_{\mathcal{C}}(A, B)$. The collection of all morphisms in a category $\mathcal{C}$ is denoted by $\text{Mor}(\mathcal{C})$.

For all $A, B, C, D \in \text{Ob}(\mathcal{C})$ and arbitrary morphisms $f : A \rightarrow B$, $g : B \rightarrow C$ and $h : C \rightarrow D$ in $\mathcal{C}$, the following axioms hold:

(C3) The composite $g \circ f : A \rightarrow C$ is a morphism in $\mathcal{C}$.

(C4) The composition is associative.

(C5) For all $A \in \text{Ob}(\mathcal{C})$ there exists a unique morphism $\text{id}_A \in \text{Hom}_{\mathcal{C}}(A, A)$, such that for every morphism $f \in \text{Hom}_{\mathcal{C}}(A, B)$

$$f \circ \text{id}_A = f = \text{id}_B \circ f.$$

The morphism id_A is unique and called the **identity** of the object A (see [8, Ch. 2, §2, Def. 2.2.2]).

Categories generalise mathematical objects such as abelian groups, commutative rings, and topological spaces, and the morphisms between these objects. An overview of categories used throughout this thesis can be found in Appendix E. A category is called **small** if both the collection of its objects as well as the collection of its morphisms are sets. A category where only the collection of morphisms between any two objects is a set is called **locally small**. Any other category is called **large** (see [3, Ch. 2, §1, Def. 2.1.15, Rem. 2.1.7]).

A central category in cohomology is the simplex category.

Example 1.1.2 (Simplex category). The **simplex category** Δ is the category whose objects are the finite non-empty ordered sets

$$[n] := \{0, \dots, n\}, \quad n \in \mathbb{N}_0$$

and whose morphisms are order-preserving maps. The category is generated by the coface maps

$$d^i : [n-1] \rightarrow [n], \quad d^i(j) = \begin{cases} j, & j < i \\ j+1, & j \geq i \end{cases}$$

and codegeneracy maps

$$s^i : [n+1] \rightarrow [n], \quad s^i(j) = \begin{cases} j, & j \leq i \\ j-1, & j > i \end{cases}$$

defined by repeating i (see [8, Ch. 9, §2, Ex. 9.2.21], [31, Ch. 7, §1, pp. 101 ff.]).

In some situations, it is useful to reverse the direction of the morphisms of a category.

Definition 1.1.3 (Dual category). Let $\mathcal{C}$ be a category. The **dual category** $\mathcal{C}^{\text{op}}$ consists of the following data:

1. The category $\mathcal{C}^{\text{op}}$ has the same objects as $\mathcal{C}$, that is $\text{Ob}(\mathcal{C}) = \text{Ob}(\mathcal{C}^{\text{op}})$.
2. $\text{Hom}_{\mathcal{C}^{\text{op}}}(A, B) := \text{Hom}_{\mathcal{C}}(B, A)$
3. For arbitrary morphisms $f \in \text{Hom}_{\mathcal{C}^{\text{op}}}(A, B)$, $g \in \text{Hom}_{\mathcal{C}^{\text{op}}}(B, C)$, the composite is defined as

$$\begin{aligned} \circ^{\text{op}} : \text{Hom}_{\mathcal{C}^{\text{op}}}(A, B) \times \text{Hom}_{\mathcal{C}^{\text{op}}}(B, C) &\rightarrow \text{Hom}_{\mathcal{C}^{\text{op}}}(A, C) \\ g \circ^{\text{op}} f &:= f \circ g \end{aligned} \tag{1.1.4}$$

(see [8, Ch. 2, §6, Def. 2.6.3]).

Thus, if $U \subseteq V$, the corresponding morphism in the category $\text{Open}(X)^{\text{op}}$ is directed from V to U . This direction is precisely the one required for a presheaf on X to assign to the inclusion $U \subseteq V$ a restriction morphism.

Definition 1.1.5 (Filtered category). A **filtered category** $\mathcal{I}$ is a non-empty category such that the following properties are satisfied:

1. For any pair $i, j \in \text{Ob}(\mathcal{I})$, there exists $k \in \text{Ob}(\mathcal{I})$ and morphisms $i \rightarrow k$ and $j \rightarrow k$.
2. For any pair of parallel morphisms $f, g : i \rightrightarrows j$ there exists a morphism $h : j \rightarrow k$ such that $h \circ f = h \circ g$

(see [34, p. 211]).

Filtered categories are important to the theory of colimits. In particular, stalks are defined as filtered colimits over neighbourhood systems, which is discussed in Chapter 2

Objects. Some objects play distinguished roles in a category.

Definition 1.1.6 (Initial-, terminal- and zero objects). Let $\mathcal{C}$ be a category.

1. An object $A \in \text{Ob}(\mathcal{C})$ is **initial** if for every object $B \in \text{Ob}(\mathcal{C})$ there exists a unique morphism $A \rightarrow B$ in $\mathcal{C}$.
2. An object $A \in \text{Ob}(\mathcal{C})$ is **terminal** if for every object $B \in \text{Ob}(\mathcal{C})$ there exists a unique morphism $B \rightarrow A$ in $\mathcal{C}$. Terminal objects are sometimes called **final**.
3. An object which is initial and terminal is a **zero object** (see [8, Ch. 2, §5, Def. 2.5.1]).

Example 1.1.7 (Zero object in Ab). One can show that the trivial group 0 is a zero object in Ab .

Claim: The trivial group is an initial object in Ab .

Let $A \in \text{Ob}(\text{Ab})$. There exists a unique morphism $0 \rightarrow A$, defined by $0 \mapsto 0$. Hence the trivial group is initial.

Claim: The trivial group is a terminal object in Ab .

For $A \in \text{Ob}(\text{Ab})$, a homomorphism $A \rightarrow 0$ must map every $a \in A$ to the unique element $0 \in \{0\}$. Hence there exists exactly one such morphism, making 0 a terminal object in Ab .

Since 0 is both initial and terminal, it is a zero object in Ab (see [8, Ch. 2, §5, Ex. 2.5.2]).

Initial, terminal, and zero objects are a first example of universal properties. They are unique up to a unique isomorphism. The defining properties of initial and terminal objects guarantee the uniqueness of morphisms from initial objects and to terminal objects, respectively. Not every category has initial (terminal) objects, like the category of all fields Fld , which has neither initial nor terminal objects. Moreover,

a category can have several initial (terminal) objects, for instance $\mathbf{Set}$, where any singleton is a terminal object, and $\emptyset$ is an initial object (see [3, Ch. 2, §7, Def. 2.7.11 ff.]). Diagrams are a central tool in category theory, encoding objects and the relation between them.

Diagrams.

Definition 1.1.8 (Diagram). Let $\mathcal{C}$ be a category and let $\Gamma = (V, E)$ be a directed multigraph. A **diagram** $\mathbf{D}$ of shape Γ in $\mathcal{C}$ consists of the following data:

1. For every vertex $v \in V$, $\mathbf{D}(v)$ is an object in $\mathcal{C}$.
2. For every edge $e := (u \rightarrow v) \in E$, $\mathbf{D}(e) : \mathbf{D}(u) \rightarrow \mathbf{D}(v)$ is a morphism in $\mathbf{Hom}_{\mathcal{C}}(\mathbf{D}(u), \mathbf{D}(v))$.

A diagram $\mathbf{D}$ is called **commutative** if for any pair of directed $u - v$ -edge-paths $e_1 \dots e_n$ and $f_1 \dots f_m$

$$\mathbf{D}(e_n) \circ \dots \circ \mathbf{D}(e_1) = \mathbf{D}(f_m) \circ \dots \circ \mathbf{D}(f_1)$$

(see [8, Ch. 2, §4, Def. 2.4.3]).

Not every assignment of morphisms to a graph gives rise to a diagram with the desired compatibility properties.

Definition 1.1.9 (Legal diagram). A diagram $\mathbf{D}$ of shape $\Gamma \otimes \Lambda$ is called **legal** if for every edge $v \rightarrow v'$ in Γ and for every edge $w \rightarrow w'$ in Λ , the induced diagram

$$\begin{array}{ccc} \mathbf{D}(v, w) & \longrightarrow & \mathbf{D}(v, w') \\ \downarrow & & \downarrow \\ \mathbf{D}(v', w) & \longrightarrow & \mathbf{D}(v', w') \end{array}$$

commutes. Thus, every edge $e : v \rightarrow v'$ in Γ induces a morphism between diagrams $\mathbf{D}(v, -) \rightarrow \mathbf{D}(v', -)$ of shape Λ (see [8, Ch. 6, §6, Def. 6.6.3]).

Legal diagrams are useful, for example, in proving that limits commute with limits.

Definition 1.1.10 (Directed diagram). A directed poset $(I, \leq)$ gives rise to a diagram with vertex set I and edge set of unique morphisms $e : i \rightarrow j$ whenever $i \leq j$ for $i, j \in I$. Such a diagram is called a **directed diagram**. Note that this diagram is commutative (see [8, Ch. 6, §5, Def. 6.5.15]).

The term "directed" refers to the underlying indexing category rather than to the visual appearance of the graph.

Diagrams also provide a convenient reformulation of classical notions such as deck transformations.

Definition 1.1.11 (Deck transformation). Let $\pi : Y \rightarrow X$ be a covering. A **deck transformation** is a homeomorphism $d : Y \rightarrow Y$ such that

$$\begin{array}{ccc} Y & \xrightarrow{d} & Y \\ & \searrow \pi & \swarrow \pi \\ & X & \end{array}$$

commutes. The collection of deck transformations together with the composition of maps is a group $\text{Deck}(\pi)$, which agrees with the automorphism group $\text{Aut}(\pi)$ (see [31, Ch. 9, §2, p. 147 ff.]).

Morphisms.

Definition 1.1.12 (Isomorphism, Epimorphism, Monomorphism). Let $\mathcal{C}$ be a category and let $f \in \text{Hom}_{\mathcal{C}}(A, B)$ and $C \in \text{Ob}(\mathcal{C})$.

1. If there exists a morphism $g \in \text{Hom}_{\mathcal{C}}(B, A)$, such that

- a) $f \circ g = \text{id}_B$

- b) $g \circ f = \text{id}_A$,

then f is called an **isomorphism**. If there exists an isomorphism between two objects A and B , the objects are isomorphic, denoted by $A \cong B$. The morphism g is unique. Furthermore, it is denoted by f^{-1} and called the **inverse** morphism of f (see [8, Ch. 2, §3, Def. 2.3.1]).

2. If for all morphisms $h_1, h_2 \in \text{Hom}_{\mathcal{C}}(C, A)$

$$f \circ h_1 = f \circ h_2 \Rightarrow h_1 = h_2$$

holds, then f is called a **monomorphism**, denoted by $f : A \hookrightarrow B$.

3. If for all morphisms $h_1, h_2 \in \text{Hom}_{\mathcal{C}}(B, C)$

$$h_1 \circ f = h_2 \circ f \Rightarrow h_1 = h_2$$

holds, then f is called an **epimorphism**, and denoted by $f : A \twoheadrightarrow B$

(see [8, Ch. 6, §7, Def. 6.7.1]).

In concrete categories, epimorphisms generalise surjectivity, and monomorphisms generalise injectivity. However, not every epimorphism is surjective, and not every monomorphism is injective. In general, the notions of surjectivity and epimorphism, as well as the notions of injectivity and monomorphisms, cannot be treated as synonymous or interchangeable.

Remark 1.1.13 (Composition of epimorphisms and their cancellation property). Consider epimorphisms a and b with

$$A \xrightarrow{b} B \xrightarrow{a} C.$$

Then $a \circ b$ is an epimorphism.

Indeed, suppose that

$$f \circ (a \circ b) = g \circ (a \circ b).$$

Then

$$(f \circ a) \circ b = (g \circ a) \circ b.$$

Since b is epimorphic, it follows that

$$f \circ a = g \circ a.$$

Since a is epimorphic, it follows that $f = g$, making $a \circ b$ epimorphic (see [1, Ch. 2, §7, Def. 7.39]).

If $f = g \circ e$, and f is epimorphic, then g is epimorphic. Suppose that

$$u \circ g = v \circ g,$$

then

$$u \circ f = u \circ g \circ e = v \circ g \circ e = v \circ f.$$

Since f is epimorphic, $u = v$. Hence g is epimorphic (compare [1, Ch. 2, §7, Def. 7.39]).

The preceding notions describe important morphisms. In practice, however, one is often interested in expressing a morphism as a composite of simpler morphisms.

Definition 1.1.14 (Factorisation of a morphism). A **factorisation of a morphism** $f : A \rightarrow B$ in a category $\mathcal{C}$ is a commutative diagram

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ & \searrow g & \nearrow h \\ & C & \end{array} \quad (1.1.15)$$

Given two morphisms f and g and assuming a morphism h exists, such that

$$f = h \circ g,$$

then f **factors through** g , or equivalently, f **continues along** g **to** h (see [8, Ch. 2, §4, Def. 2.4.8]).

The definition makes neither claims about the existence of a factorisation, nor about its uniqueness, and it is one of the central notions in category-theoretic approaches to mathematics. It appears in many different forms, for example, when one seeks a morphism that completes a commutative square.

Definition 1.1.16 (Lift). Let

$$\begin{array}{ccc} A & \xrightarrow{f} & X \\ \downarrow u & & \downarrow p \\ B & \xrightarrow{g} & Y \end{array}$$

be a commutative diagram in a category $\mathcal{C}$. A **lift** is a morphism $\ell : B \rightarrow X$ such that

$$\ell \circ u = f, \quad p \circ \ell = g$$

(see [63]).

Lifts are a special case of the factorisation of a morphism, and they are crucial in sheaf theory, for instance showing how sheaves formalise local-to-global principles.

More generally, one may ask for objects that are characterised by universal properties with respect to such diagrams. Many universal constructions are closely related to the behaviour of subobjects.

Definition 1.1.17 (Subobject). Let $\mathcal{C}$ be a category and $A \in \text{Ob}(\mathcal{C})$. Given two monomorphisms

$$\begin{aligned} f_1 : B_1 &\hookrightarrow A \\ f_2 : B_2 &\hookrightarrow A \end{aligned}$$

define an equivalence relation by $f_1 \equiv f_2$ if there exists an isomorphism

$$\phi : B_1 \rightarrow B_2$$

such that

$$f_1 = f_2 \circ \phi.$$

A **subobject** of A is an equivalence class of monomorphisms $f : B \hookrightarrow A$. The collection of subobjects of A is denoted $\text{Sub}_{\mathcal{C}}(A)$ (see [3, Ch. 2, §8, Rem. 2.8.5 ff.]).

The collection of subobjects of an object A may be a proper class. While the notion of subobjects is intended to generalise concepts such as subsets of sets, and subgroups of groups (see [3, Ch. 2, §8, Rem. 2.8.5 ff.]).

The study of subobjects naturally leads to the question of when a morphism that is both monomorphic and epimorphic is already invertible.

Definition 1.1.18 (Balanced category). A category $\mathcal{C}$ is called **balanced** if every morphism $f : A \rightarrow B$ that is both a monomorphism and an epimorphism is necessarily an isomorphism (see [1, Ch. 2, §7, Def. 7.49]).

Examples of balanced categories are $\mathbf{Set}$, and $\mathbf{Mod}$. A category that is not balanced is the category of commutative rings, where $\mathbb{Z} \hookrightarrow \mathbb{Q}$ provides an example for a morphism that is a monomorphism as well as an epimorphism, but not an isomorphism (see [58, Rem. 2.7]).

While balanced categories ensure that certain morphisms are isomorphisms, there are categories in which every morphism is an isomorphism.

Definition 1.1.19 (Groupoid). A category $\mathcal{C}$ in which every morphism is invertible is called a **groupoid** (see [8, Ch. 2, §3, Rem. 2.3.16]).

Let G be a groupoid. A **subgroupoid** $H \subseteq G$ is a subcategory that itself is a groupoid.

The **fundamental groupoid** $\Pi_1(X)$ is a category consisting of the following data:

1. The objects are the points in the topological space X
2. The morphisms $x \rightarrow y$ are homotopy classes of paths

$$\gamma : [0, 1] \rightarrow X, \quad \gamma(0) = x, \quad \gamma(1) = y$$

3. The composition is the concatenation of paths
4. The identities are constant paths.

As every homotopy class of paths is invertible, as it can be represented by inverted paths

$$\gamma^{-1}(t) = \gamma(1 - t),$$

this category is indeed a groupoid (see [8, Ch. 2, §7, Ex. 2.24]).

Rather than imposing conditions on all morphisms of a category, one may study whether a single object is sufficiently rich to detect differences between morphisms.

Definition 1.1.20 (Generator). Let $\mathcal{C}$ be a category. A generator G is an object $G \in \mathbf{Ob}(\mathcal{C})$ if for every pair of distinct morphisms $f, g \in \mathbf{Hom}_{\mathcal{C}}(A, B)$ there exists a morphism $h \in \mathbf{Hom}_{\mathcal{C}}(G, A)$ such that

$$f \circ h \neq g \circ h$$

(compare [34, Ch. 5, §7, p. 127]).

Example 1.1.21. The category $\mathbf{Ab}$ has a generator, namely $\mathbb{Z}$. Indeed, let $f, g : A \rightarrow B$ be distinct morphisms in $\mathbf{Ab}$. Then there exists $a \in A$ such that

$$f(a) \neq g(a).$$

Let $h : \mathbb{Z} \rightarrow A$ be the unique group homomorphism determined by $h(1) = a$. Then

$$(f \circ h)(1) = f(a) \neq g(a) = (g \circ h)(1).$$

Hence

$$f \circ h \neq g \circ h,$$

so $\mathbb{Z}$ is a generator of $\mathbf{Ab}$ (see [34, Ch. 5, §7, p. 127]).

Morphisms compare objects with objects. One can also consider mappings that compare categories with categories.

1.2 Functors

Definition 1.2.1 (Functor). Let $\mathcal{C}$ and $\mathcal{D}$ be categories. A **covariant functor** $F : \mathcal{C} \rightarrow \mathcal{D}$ consists of the following data:

- (F1) For every $A \in \mathbf{Ob}(\mathcal{C})$ an object $F(A) \in \mathbf{Ob}(\mathcal{D})$ is assigned.
- (F2) For any $f \in \mathbf{Hom}_{\mathcal{C}}(A, B)$, a morphism $F(f) \in \mathbf{Hom}_{\mathcal{D}}(F(A), F(B))$ is assigned.
- (F3) For every object $A \in \mathbf{Ob}(\mathcal{C})$, it holds that $F(\mathrm{id}_A) = \mathrm{id}_{F(A)}$.
- (F4) For any two morphisms $f \in \mathbf{Hom}_{\mathcal{C}}(A, B)$, $g \in \mathbf{Hom}_{\mathcal{C}}(B, C)$, the equality

$$F(g \circ f) = F(g) \circ F(f)$$

holds

(see [8, Ch. 3, §2, Def. 3.2.1, Rem. 3.2.3]).

Functors are mappings between categories that preserve categorical structure, but not necessarily of objects, that is, they map objects to objects, and morphisms to morphisms while preserving identities and composition.

Remark 1.2.2. A **contravariant** functor from $\mathcal{C}$ to $\mathcal{D}$ is the dual notion to a covariant functor, that is, the same as a covariant functor $F : \mathcal{C}^{\text{op}} \rightarrow \mathcal{D}$, such that the composite and the identities are preserved, but the morphisms are reversed in direction (see [8, Ch. 3, §2, Rem. 3.2.3]).

An essential property of functors is that they preserve the commutativity of diagrams (see [8, Ch. 3, §2, Rem. 3.2.2]).

Example 1.2.3. 1. Let T be an arbitrary object in $\text{Ob}(\mathcal{D})$. For every category $\mathcal{C}$ the **constant functor** $T^{\text{cst}} : \mathcal{C} \rightarrow \mathcal{D}$ is defined by

- $T^{\text{cst}}(A) := T$ for all $A \in \text{Ob}(\mathcal{C})$.
- $T^{\text{cst}}(f) := \text{id}_T$ for all $f \in \text{Mor}(\mathcal{C})$

(see [8, Ch. 3, §2, Ex. 3.2.21, 2]).

2. The **identity functor** $\text{id}_{\mathcal{C}} : \mathcal{C} \rightarrow \mathcal{C}$ is given by the following data:

- $\text{id}_{\mathcal{C}}(A) := A$ for all $A \in \text{Ob}(\mathcal{C})$
- $\text{id}_{\mathcal{C}}(f) := f$ for all $f \in \text{Mor}(\mathcal{C})$

(see [8, Ch. 3, §2, Ex. 3.2.21, 1]).

3. The *fundamental group* is functorial in the sense that

$$\pi_1 : \text{Top}_* \rightarrow \text{Grp}, \quad (X, x_0) \mapsto \pi_1(X, x_0).$$

For a based continuous map

$$f : (X, x_0) \rightarrow (Y, y_0)$$

one has

$$f_* : \pi_1(X, x_0) \rightarrow \pi_1(Y, y_0), \quad [\gamma] \mapsto [f \circ \gamma]$$

(see [8, Ch. 3, §4, Ex. 3.4.9]).

4. Functoriality of diagrams: A diagram of shape J in a category $\mathcal{C}$ is a functor $\mathbf{D} : J \rightarrow \mathcal{C}$ (see [3, Ch. 4, §4.1, Def. 4.1.1]).
5. A **forgetful functor** $\mathfrak{F} : \mathcal{C} \rightarrow \mathbf{Set}$ is a functor from an algebraic category $\mathcal{C}$ into $\mathbf{Set}$. Intuitively, one forgets the structure imposed on the objects in $\mathcal{C}$ and considers only the underlying sets. Details can be found in Appendix B.
6. A **free functor** $\mathfrak{U} : \mathbf{Set} \rightarrow \mathcal{C}$ equips sets with a free algebraic structure (details can be found in Appendix B).
7. Let $\mathcal{C}$ be a locally small category and let $A \in \mathbf{Ob}(\mathcal{C})$ be arbitrary but fixed. The **covariant Hom-functor**

$$\mathrm{Hom}(A, -) : \mathcal{C} \rightarrow \mathbf{Set}$$

assigns to every object $B \in \mathbf{Ob}(\mathcal{C})$ the *set* of morphisms

$$B \mapsto \mathrm{Hom}_{\mathcal{C}}(A, B),$$

and to every morphism $f \in \mathrm{Hom}_{\mathcal{C}}(B, C)$, assigns the map

$$\begin{aligned} f_* : \mathrm{Hom}_{\mathcal{C}}(A, B) &\rightarrow \mathrm{Hom}_{\mathcal{C}}(A, C) \\ g &\mapsto f \circ g \end{aligned} \tag{1.2.4}$$

for any $g \in \mathrm{Hom}_{\mathcal{C}}(A, B)$. The axioms (C1)-(C5) of a category (see Definition 1.1.1) ensure that the properties of functors (F1)-(F4) (see Definition 1.2.1) hold:

- a) $(\mathrm{id}_B)_* = \mathrm{id}_{\mathrm{Hom}_{\mathcal{C}}(A, B)}$, since $\mathrm{id}_B \circ h = h$ for any $h \in \mathrm{Hom}_{\mathcal{C}}(A, B)$.
- b) $(f \circ g)_* = f_* \circ g_*$.

Applying the very same construction in the opposite category $\mathcal{C}^{\mathrm{op}}$ yields the **contravariant Hom-functor**

$$\mathrm{Hom}(-, A) : \mathcal{C}^{\mathrm{op}} \rightarrow \mathbf{Set},$$

which assigns to every object $B \in \mathbf{Ob}(\mathcal{C})$ the set of morphisms

$$B \mapsto \mathrm{Hom}_{\mathcal{C}}(B, A),$$

and to every morphism $f \in \text{Hom}_{\mathcal{C}}(B, C)$, the contravariant Hom-functor assigns the morphism

$$\begin{aligned} f^* : \text{Hom}_{\mathcal{C}}(C, A) &\rightarrow \text{Hom}_{\mathcal{C}}(B, A) \\ h &\mapsto h \circ f \end{aligned}$$

for any $h \in \text{Hom}_{\mathcal{C}}(C, A)$ (see [8, Ch. 5, §2, Def. 5.2.1]).

8. The Hom-functor in "two variables" is defined as

$$\text{Hom}(-, -) : \mathcal{C}^{\text{op}} \times \mathcal{C} \rightarrow \text{Set}$$

(see [8, Ch. 5, §2, Def. 5.2.1]).

These examples illustrate that functors occur both as basic categorical constructions, and as naturally arising mathematical assignments.

Definition 1.2.5 (Simplicial object). A **simplicial object** in a category $\mathcal{C}$ is a functor

$$S_{\bullet} : \Delta^{\text{op}} \rightarrow \mathcal{C}.$$

Equivalently, a simplicial object consists of objects $S_0, S_1, \dots$ in $\mathcal{C}$ together with **face morphisms**

$$d_i : S_n \rightarrow S_{n-1}, \quad n \in \mathbb{N}, \quad 0 \leq i \leq n,$$

and **degeneracy morphisms**

$$s_i : S_n \rightarrow S_{n+1}, \quad 0 \leq i \leq n$$

satisfying the simplicial identities

$$\begin{aligned} d_i \circ d_j &= d_{j-1} \circ d_i \text{ for } i < j \\ s_i \circ s_j &= s_{j+1} \circ s_i \text{ for } i \leq j, \end{aligned}$$

and

$$d_i \circ s_j = \begin{cases} s_{j-1} \circ d_i & i < j \\ \text{id} & i = j \text{ or } i = j + 1 \\ s_j \circ d_{i-1} & i > j + 1 \end{cases}$$

(see [34, Ch. 7, §5, pp. 175 ff.]).

Unlike simplicial complexes, simplices in a simplicial set need not be determined by vertices, different simplices may have the same boundary, and degeneracies are explicitly included.

Definition 1.2.6 (Cosimplicial object). A **cosimplicial object** in a category $\mathcal{C}$ is a functor

$$S^\bullet : \Delta \rightarrow \mathcal{C}.$$

Equivalently, a cosimplicial object consists of objects $S^0, S^1, \dots$ in $\mathcal{C}$ together with **coface morphisms**

$$d^i : S^n \rightarrow S^{n+1}, \quad 0 \leq i \leq n+1,$$

and **codegeneracy morphisms**

$$s^i : S^n \rightarrow S^{n-1}, \quad 0 \leq i \leq n-1$$

satisfying the dual simplicial identities

$$\begin{aligned} d^j \circ d^i &= d^i \circ d^{j-1} \text{ for } i < j \\ s^j \circ s^i &= s^i \circ s^{j+1} \text{ for } i \leq j, \end{aligned}$$

and

$$s^j \circ d^i = \begin{cases} d^i \circ s^{j-1} & i < j \\ \text{id} & i = j \text{ or } i = j+1 \\ d^{i-1} \circ s^j & i > j+1 \end{cases}$$

(see [34, Ch. 7, §5, pp. 175 ff.]).

Having introduced several examples of functors, one can study how functors can be composed.

Definition 1.2.7 (Composition of functors). For arbitrary functors $F : \mathcal{C} \rightarrow \mathcal{D}$ and $G : \mathcal{D} \rightarrow \mathcal{E}$, the **composition of functors** $G \circ F : \mathcal{C} \rightarrow \mathcal{E}$ is defined by

1. $(G \circ F)(A) := G(F(A))$ for all $A \in \text{Ob}(\mathcal{C})$.
2. $(G \circ F)(f) := G(F(f))$ for all morphisms f in $\mathcal{C}$

(see [8, Ch. 3, §3, Def. 3.3.1]).

One can examine how much information of morphisms a functor retains, leading to the notion of full and faithful functors.

Definition 1.2.8 (Full and faithful functors). A functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is **faithful** if

$$F(f) = F(g) \Rightarrow f = g$$

for all objects $A, B \in \text{Ob}(\mathcal{C})$, and for all $f, g \in \text{Hom}_{\mathcal{C}}(A, B)$.

A functor is **full** if for all $A, B \in \text{Ob}(\mathcal{C})$, and all $g : F(A) \rightarrow F(B)$ there exists a morphism $f : A \rightarrow B$ in $\mathcal{C}$ such that $F(f) = g$.

If a functor is faithful and full, it is **fully faithful** (see [3, Ch. 3, §3, Rem. 3.3.1 ff.]).

The functor axioms imply that functors preserve isomorphisms. However, the reverse statement needs stronger assumptions.

Lemma 1.2.9 (Functors preserve isomorphisms). Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a functor and let $f \in \text{Hom}_{\mathcal{C}}(A, B)$ be an isomorphism, then $F(f) : F(A) \rightarrow F(B)$ is an isomorphism in $\mathcal{D}$. Moreover

$$A \cong B \Rightarrow F(A) \cong F(B)$$

(see [8, Ch. 3, §2, Lemma 3.2.4]). If F is fully faithful, then

$$A \cong B \Leftrightarrow F(A) \cong F(B)$$

(see [8, Ch. 3, §6, Rem. 3.6.17]).

Proof. For any isomorphism it holds that

$$\begin{aligned} F(f^{-1}) \circ F(f) &= F(f^{-1} \circ f) \\ &= F(\text{id}_A) \\ &= \text{id}_{F(A)} \end{aligned}$$

and

$$\begin{aligned} F(f) \circ F(f^{-1}) &= F(f \circ f^{-1}) \\ &= F(\text{id}_B) \\ &= \text{id}_{F(B)}. \end{aligned}$$

Hence $F(f^{-1})$ is inverse to $F(f)$, and

$$A \cong B \Rightarrow F(A) \cong F(B)$$

(see [8, Ch. 3, §2, Lemma 3.2.4]).

Let

$$\alpha : F(A) \rightarrow F(B)$$

be an isomorphism. By fullness, there exist

$$f : A \rightarrow B \quad g : B \rightarrow A$$

such that

$$F(f) = \alpha, \quad F(g) = \alpha^{-1}.$$

Then

$$\begin{aligned} F(g \circ f) &= F(g) \circ F(f) \\ &= \alpha^{-1} \circ \alpha \\ &= \text{id}_{F(A)} \\ &= F(\text{id}_A). \end{aligned}$$

Faithfulness gives

$$g \circ f = \text{id}_A,$$

and an analogous computation yields

$$f \circ g = \text{id}_B.$$

Thus f is an isomorphism. It follows directly that

$$A \cong B \Leftarrow F(A) \cong F(B).$$

□

The first implication does not rely on F being fully faithful.

1.3 Natural Transformations and Universality

Having introduced functors between categories, one can consider transformations between functors.

Definition 1.3.1 (Natural transformation). Let $F, G : \mathcal{C} \rightarrow \mathcal{D}$ be functors. A **natural transformation** is a mapping $\alpha : F \rightarrow G$, such that the following properties are satisfied:

1. Every $A \in \text{Ob}(\mathcal{C})$ has a **component** $\alpha_A : F(A) \rightarrow G(A)$.
2. For every morphism $f \in \text{Hom}_{\mathcal{C}}(A, B)$, the diagram

$$\begin{array}{ccc} F(A) & \xrightarrow{\alpha_A} & G(A) \\ F(f) \downarrow & & \downarrow G(f) \\ F(B) & \xrightarrow{\alpha_B} & G(B) \end{array} \quad (1.3.2)$$

commutes. The commutativity of this diagram is called **naturality**.

The **identity natural transformation** $\text{id}_F : F \rightarrow F$ is defined via

$$\text{id}_F(A) = \text{id}_{F(A)}$$

for all $A \in \text{Ob}(\mathcal{C})$ (see [8, Ch. 3, §4, Def. 3.4.1]).

Natural transformations provide a notion of morphisms between functors that is compatible with the action of morphisms in the source category.

Definition 1.3.3 (Subfunctor). Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a functor. A **subfunctor** of F is a pair (G, α) where $G : \mathcal{C} \rightarrow \mathcal{D}$ is a functor and $\alpha : G \hookrightarrow F$ is a natural transformation such that every component

$$\alpha_A : G(A) \rightarrow F(A)$$

is a monomorphism in $\mathcal{D}$ (see [3, Ch. 4, §3, Ex. 4.3.13, 4]).

The notion of a subfunctor differs categorically from that of a restriction. A restriction of a functor modifies its domain, whereas a subfunctor modifies its values while keeping the domain fixed.

Definition 1.3.4 (Vertical composition). Let $F, G, H : \mathcal{C} \rightarrow \mathcal{D}$ be functors. Let $\alpha : F \rightarrow G$ and $\beta : G \rightarrow H$ be natural transformations. The composition of natural transformations $\beta \circ \alpha : F \rightarrow H$ is defined as

$$(\beta \circ \alpha)(A) = \beta(A) \circ \alpha_A : F(A) \rightarrow H(A)$$

for all $A \in \text{Ob}(\mathcal{C})$. This composition is called **vertical composition** (see [8, Ch. 3, §5, Def. 3.5.2]).

The composite of natural transformations is a natural transformation.

Definition 1.3.5 (Horizontal composition). Let $F, G : \mathcal{C} \rightarrow \mathcal{D}$ be functors and let $\alpha : F \rightarrow G$ be a natural transformation.

1. Let $H : \mathcal{D} \rightarrow \mathcal{E}$ be another functor. Then the **horizontal composition from the left** is defined as

$$\begin{aligned} (H \bullet \alpha) &: (H \circ F) \rightarrow (H \circ G) \\ (H \bullet \alpha)(A) &:= H(\alpha_A) \end{aligned}$$

for all $A \in \text{Ob}(\mathcal{C})$.

2. Let $K : \mathcal{E} \rightarrow \mathcal{C}$ be another functor. Then the **horizontal composition from the right** is defined as

$$\begin{aligned} (\alpha \bullet K) &: (F \circ K) \rightarrow (G \circ K) \\ (\alpha \bullet K)(A) &:= \alpha(K(A)) \end{aligned}$$

for all $A \in \text{Ob}(\mathcal{E})$.

Combining the horizontal composition from the left and the horizontal composition from the right motivates

$$H \bullet \alpha \bullet K : (H \circ F \circ K) \rightarrow (H \circ G \circ K).$$

Furthermore, for functors $\tilde{F}, \tilde{G} : \mathcal{D} \rightarrow \mathcal{E}$ and a natural transformation $\beta : \tilde{F} \rightarrow \tilde{G}$ the diagram

$$\begin{array}{ccc} \tilde{F} \circ F & \xrightarrow{\tilde{F} \bullet \alpha} & \tilde{F} \circ G \\ \beta \bullet F \downarrow & & \downarrow \beta \bullet G \\ \tilde{G} \circ F & \xrightarrow{\tilde{G} \bullet \alpha} & \tilde{G} \circ G \end{array} \quad (1.3.6)$$

commutes. Thus, the horizontal composition of natural transformations is defined as

$$\alpha \bullet \beta : (\tilde{F} \circ F) \rightarrow (\tilde{G} \circ G).$$

This is exactly the **diagonal** of the diagram 1.3.6, that is

$$\alpha \bullet \beta := (\beta \bullet G) \circ (\tilde{F} \bullet \alpha) = (\tilde{G} \bullet \alpha) \circ (\beta \bullet F)$$

(see [8, Ch. 3, §5, Def. 3.5.10 - Def. 3.5.12]).

Vertical and horizontal composition express the compatibility of natural transformations with composition in the relevant categories.

Remark 1.3.7 (Functor category). Functors from a category $\mathcal{C}$ to a category $\mathcal{D}$ form a category where the functors are objects, and the morphisms are natural transformations. The category of functors from $\mathcal{C}$ to $\mathcal{D}$ is denoted by $[\mathcal{C}, \mathcal{D}]$ (see [8, Ch. 3, §5, Def. 3.5.5]).

Example 1.3.8. Let $\mathcal{C}$ and $\mathcal{D}$ be categories, and let $A \in \text{Ob}(\mathcal{C})$. The **evaluation functor**

$$\text{ev}_A : [\mathcal{C}, \mathcal{D}] \rightarrow \mathcal{D}$$

at A is defined by

$$F \mapsto F(A)$$

on objects, and

$$\alpha : (F \rightarrow G) \mapsto (\alpha_A : F(A) \rightarrow G(A))$$

on morphisms (see [1, Ch. 7, §27, p. 437]).

Since evaluation extracts the components of a natural transformation, it provides a convenient criterion for recognising isomorphisms between functors.

Definition 1.3.9 (Natural isomorphism). A natural transformation $\alpha : F \rightarrow G$ is a **natural isomorphism** between functors if all components are isomorphisms (see [8, Ch. 3, §5, Def. 3.5.3]). Equivalently, α is an isomorphism in $[\mathcal{C}, \mathcal{D}]$

Natural isomorphisms provide the appropriate notion of equivalence between functors. Configurations consisting of objects equipped with morphisms between their images under given functors are not of this form.

Definition 1.3.10 (Comma category). Let $F : \mathcal{C} \rightarrow \mathcal{E}$ and $G : \mathcal{D} \rightarrow \mathcal{E}$ be functors. Their **comma category** $(F \downarrow G)$ consists of the following data:

1. Objects are triples (C, D, φ) , where $C \in \text{Ob}(\mathcal{C})$, $D \in \text{Ob}(\mathcal{D})$, and $\varphi : F(C) \rightarrow G(D)$ is a morphism in $\mathcal{E}$.
2. Morphisms from (C_1, D_1, φ_1) to (C_2, D_2, φ_2) are pairs (f, g) , where $f \in \text{Hom}_{\mathcal{C}}(C_1, C_2)$ and $g \in \text{Hom}_{\mathcal{D}}(D_1, D_2)$ such that

$$\varphi_2 \circ F(f) = G(g) \circ \varphi_1.$$

3. The composition of morphisms is the component-wise composition of morphisms in $\mathcal{C}$ and $\mathcal{D}$, that is,

$$(f_2, g_2) \circ (f_1, g_1) = (f_2 \circ f_1, g_2 \circ g_1)$$

(see [4, Ch. 1, §2, Def. 2.2.2]).

Remark 1.3.11 (Category of elements). The category $\int F$ is the same as the comma category $(* \downarrow F)$, where $*$ is a singleton, that is, $* : 1 \rightarrow \text{Set}$ where 1 denotes the terminal category, and $F : \mathcal{C} \rightarrow \text{Set}$ a covariant functor. Its objects are (A, x) with $A \in \text{Ob}(\mathcal{C})$ and $x \in F(A)$, and its morphisms are

$$f : (A_1, x_1) \rightarrow (A_2, x_2)$$

such that $f : A_1 \rightarrow A_2$ satisfies $F(f)(x_1) = x_2$. Furthermore, there is a canonical

projection

$$\begin{aligned} \text{pr} : \int F &\rightarrow \mathcal{C} \\ (A, x) &\mapsto A \end{aligned}$$

(see [34, Ch. 3, §7, p. 76]).

Remark 1.3.12 (Slice category). The comma category $(\text{id}_{\mathcal{C}} \downarrow A^{\text{cnst}})$ for $A \in \text{Ob}(\mathcal{C})$ is the **slice category of objects over** A , often denoted by $(\mathcal{C} \downarrow A)$ (see [3, Ch. 2, §6, Def. 2.6.9]).

Comma categories organise morphism-shaped data relative to functors into a category. Although they are not independent of any size considerations, their definition does not require local smallness.

Definition 1.3.13 (Universal property). Let $F : \mathcal{D} \rightarrow \mathcal{C}$ be a functor. An object $A \in \text{Ob}(\mathcal{D})$ together with a morphism $u \in \text{Hom}_{\mathcal{C}}(B, F(A))$ for $B \in \text{Ob}(\mathcal{C})$ satisfies a **universal property**, or equivalently, a **universal morphism from** B **to** F if the pair (A, u) is an initial object in the comma category $(B^{\text{cnst}} \downarrow F)$ (see [34, Ch. 2, §. 7, p. 50]).

In contrast to characterising an object set-theoretically by its elements, universal properties define objects by the morphisms to and from them. While the classical definition of universal properties via Hom -sets relies on locally small categories, this definition makes no assumptions on the size of the category under consideration. However, it assumes the existence of initial objects in the comma category $(B^{\text{cnst}} \downarrow F)$.

Definition 1.3.14 (Adjunction). Let $F : \mathcal{C} \rightarrow \mathcal{D}$ and $G : \mathcal{D} \rightarrow \mathcal{C}$ be functors. An **adjunction between** F **and** G consists of two natural transformations

$$\begin{aligned} \eta : \text{id}_{\mathcal{C}} &\rightarrow G \circ F \\ \varepsilon : F \circ G &\rightarrow \text{id}_{\mathcal{D}} \end{aligned}$$

such that the triangle identities

$$\begin{array}{ccc}
 F \circ \text{id}_{\mathcal{C}} & \xrightarrow{F \bullet \eta} & F \circ G \circ F \\
 & \searrow & \downarrow \varepsilon \bullet F \\
 & & \text{id}_{\mathcal{D}} \circ F
 \end{array}
 \qquad
 \begin{array}{ccc}
 \text{id}_{\mathcal{C}} \circ G & \xrightarrow{\eta \bullet G} & G \circ F \circ G \\
 & \searrow & \downarrow G \bullet \varepsilon \\
 & & G \circ \text{id}_{\mathcal{D}}
 \end{array}$$

hold.

Considering an adjunction $(F, G, \eta, \varepsilon) : \mathcal{C} \rightarrow \mathcal{D}$ between F and G , F is called **left-adjoint to G** , in signs $F \vdash G$, or respectively, G is **right-adjoint to F** . The natural transformation η is called **unit** and ε is called **counit** of the adjunction (see [8, Ch. 7, §2, Def. 7.2.2]).

Remark 1.3.15. For locally small categories, an adjunction $F \dashv G$ may be characterised by a family of natural bijections

$$\text{Hom}_{\mathcal{D}}(F(C), D) \cong \text{Hom}_{\mathcal{C}}(C, G(D))$$

natural in objects $C \in \text{Ob}(\mathcal{C})$ and $D \in \text{Ob}(\mathcal{D})$ (see [8, Ch. 7, §2, Def. 7.2.2]).

Adjunctions encode universal behaviour intrinsic to the ambient categories via unit and counit, independent of any size assumptions of the categories. Therefore, in locally small settings, they produce representability.

1.4 The Yoneda Perspective

In this section, locally small categories are considered.

Lemma 1.4.1 (Yoneda lemma). Let $\mathcal{C}$ be a locally small category, $F : \mathcal{C} \rightarrow \text{Set}$ a functor, and $A \in \text{Ob}(\mathcal{C})$. Then there is a natural bijection

$$\text{Hom}(\text{Hom}(A, -), F) \cong F(A)$$

(see [8, Ch. 5, §2, Lemma 5.2.5]).

Proof. Let $F : \mathcal{C} \rightarrow \mathbf{Set}$ be a functor on a locally small category, and

$$\alpha : \mathrm{Hom}(A, -) \rightarrow F$$

a natural transformation. Any natural transformation gives rise to an element

$$u_\alpha := (\alpha_A)(\mathrm{id}_A) \in F(A). \quad (1.4.2)$$

On the other hand, any element $u \in F(A)$ gives rise to a natural transformation

$$\begin{aligned} \alpha_u : \mathrm{Hom}(A, -) &\rightarrow F \\ (\alpha_u(B))(f) &:= (F(f))(u) \end{aligned} \quad (1.4.3)$$

with $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$. This construction is sometimes referred to as the **Yoneda construction** (see [8, Ch. 5, §2, Def. 5.2.4]).

To show that α_u is a natural transformation, it suffices to prove that the naturality diagram (see Definition 1.3.1, Diagram 1.3.2)

$$\begin{array}{ccc} \mathrm{Hom}(A, B) & \xrightarrow{\alpha_u(B)} & F(B) \\ g_* \downarrow & & \downarrow F(g) \\ \mathrm{Hom}(A, C) & \xrightarrow{\alpha_u(C)} & F(C) \end{array}$$

commutes for every morphism $g : B \rightarrow C$:

Consider a morphism $g \in \mathrm{Hom}_{\mathcal{C}}(B, C)$. For every morphism $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$ it holds that

$$\begin{aligned} (F(g) \circ \alpha_u(B))(f) &\stackrel{1.4.3}{=} F(g \circ f)(u) \\ &\stackrel{1.4.3}{=} \alpha_u(C)(g \circ f) \\ &\stackrel{7}{=} (\alpha_u(C) \circ g_*)(f). \end{aligned}$$

Thus, the diagram commutes and α_u is a natural transformation (see [8, Ch. 5, §2, Def. 5.2.4]).

Evaluating the diagram at id_A gives

$$\begin{aligned}
 \alpha_B(f) &= \alpha_B(f_*(\text{id}_A)) \\
 &\stackrel{\substack{\text{commutativity} \\ \text{of diagram}}}{=} F(f)(\alpha_A(\text{id}_A)) \\
 &= F(f)(u_\alpha) \\
 &= \alpha_{u_\alpha}(B)(f).
 \end{aligned}$$

Thus, α and α_{u_α} have the same components for every morphism $f \in \text{Hom}_{\mathcal{C}}(A, B)$, which makes them equal as natural transformations by definition. Hence $\alpha = \alpha_{u_\alpha}$.

Furthermore,

$$u_{\alpha_u} = (\alpha_u)_A(\text{id}_A) = F(\text{id}_A)(u) = u$$

(see [8, Ch. 5, §2, Lemma 5.2.5]). □

By the Yoneda lemma, objects can be studied through their relations to other objects. Moreover, it allows one to define universal properties without the notion of comma categories. Unlike the Set-valued representability formulation, the definition in terms of initial objects in comma categories does not require local smallness.

Definition 1.4.4 (Representable functor). A covariant functor $F : \mathcal{C} \rightarrow \text{Set}$ is called **representable** if there exists an object $A \in \text{Ob}(\mathcal{C})$ together with a natural isomorphism

$$F \cong \text{Hom}_{\mathcal{C}}(A, -).$$

In this case, A is called a **representing object** of F (see [8, Ch. 5, §2, Def. 5.2.12]).

A contravariant functor $F : \mathcal{C}^{\text{op}} \rightarrow \text{Set}$ is representable if

$$F \cong \text{Hom}_{\mathcal{C}}(-, A).$$

Representability allows one to describe a functor entirely by a single object. By the Yoneda lemma, a representable functor is determined, up to isomorphism, by its representing object. Consequently, many universal constructions can be formulated as statements about representing objects rather than individual elements.

Remark 1.4.5 (*U*-points). Let $\mathcal{C}$ be a category and let $X \in \text{Ob}(\mathcal{C})$. By the contravariant Yoneda embedding, X determines the representable functor

$$h_X := \text{Hom}_{\mathcal{C}}(-, X).$$

For any $U \in \text{Ob}(\mathcal{C})$, the set

$$h_X(U) = \text{Hom}_{\mathcal{C}}(U, X)$$

is called the set of ***U*-points** of X . Thus a *U*-point of X is simply a morphism $U \rightarrow X$ (see [8, Ch. 5, §5, Def. 5.5.1]).

The Yoneda lemma does more than identify natural transformations from representable functors with elements: applied to another representable functor, it identifies morphisms in $\mathcal{C}$ with natural transformations between the corresponding Hom-functors.

Corollary 1.4.6. Let $A, B \in \text{Ob } \mathcal{C}$, then

$$\begin{aligned} \text{Hom}(A, B) &\rightarrow \text{Hom}(\text{Hom}(B, -), \text{Hom}(A, -)) \\ f &\mapsto f^*, \end{aligned}$$

is bijective, and motivates a fully faithful functor

$$\begin{aligned} \mathcal{C}^{\text{op}} &\rightarrow [\mathcal{C}, \text{Set}] \\ A &\mapsto \text{Hom}(A, -) \\ f &\mapsto f^* \end{aligned}$$

(see [8, Ch. 5, §2, Corollary 5.2.7]).

Proof. This is a direct consequence of the Yoneda lemma. □

Since this embedding is fully faithful, it reflects isomorphisms. Consequently, an object is determined up to isomorphism by its representable Hom-functor.

Corollary 1.4.7. If $\text{Hom}(A, -) \cong \text{Hom}(B, -)$ for $A, B \in \text{Ob}(\mathcal{C})$, then $A \cong B$ (see [8, Ch. 5, §2, Corollary 5.2.8]).

Proof. This follows directly from Corollary 1.4.6 and Lemma 1.2.9 (see [8, Ch. 5, §2, Corollary 5.2.8]). $\square$

This corollary allows one to determine whether objects are isomorphic only by studying the isomorphism between their Hom -functors.

1.5 Limits and Colimits

In this section, $\mathcal{C}$ shall be an arbitrary category, and I shall be a small category. One of the central insights of the Yoneda lemma is that objects may be characterised by their relationships with other objects. Limits and colimits exploit this principle by defining objects through universal mapping properties rather than explicit Set-theoretic constructions.

1.5.1 Limits

Definition 1.5.1 (Cone). A **cone** to the diagram $\mathbf{D} : I \rightarrow \mathcal{C}$ is a pair $(T, (\text{pr}_i)_{i \in \text{Ob}(I)})$ consisting of a tip $T \in \text{Ob}(\mathcal{C})$ and morphisms $(\text{pr}_i : T \rightarrow \mathbf{D}(i))_{i \in I}$ such that for every morphism $e_{ij} : i \rightarrow j$ in I , the diagram

$$\begin{array}{ccc} & T & \\ \text{pr}_i \swarrow & & \searrow \text{pr}_j \\ \mathbf{D}(i) & \xrightarrow{\mathbf{D}(e_{ij})} & \mathbf{D}(j) \end{array} \quad (1.5.2)$$

commutes.

The condition

$$\mathbf{D}(e_{ij}) \circ \text{pr}_i = \text{pr}_j$$

is called the **cone condition**. The collection of cones from T to the diagram $\mathbf{D}$ is denoted by $\text{Cone}(T, \mathbf{D})$ (see [8, Ch. 6, §2, Def. 6.2.1], [3, Ch. 8, §2, Def. 8.2.1]).

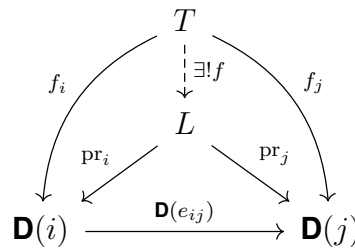
The diagram $\mathbf{D} : I \rightarrow \mathcal{C}$ is not necessarily commutative.

Definition 1.5.3 (Limit). The **limit** of a diagram $\mathbf{D} : I \rightarrow \mathcal{C}$ is a cone $(L, (\text{pr}_i)_{i \in \text{Ob}(I)})$ to $\mathbf{D}$ such that for every cone $(T, (f_i)_{i \in \text{Ob}(I)})$ to $\mathbf{D}$ the **universal limit property** holds:

There exists a unique morphism $f : T \rightarrow L$ such that

$$\text{pr}_i \circ f = f_i$$

for all $i \in \text{Ob}(I)$, that is,



commutes (see [3, Ch. 8, §2, Def. 8.2.3]).

A category that admits all small limits is called **complete** (see [8, Ch. 6, §4, Def. 6.4.3]).

The existence of the limit is not guaranteed. If the limit exists, it is unique up to a unique isomorphism. In literature, limits, projective limits, and inverse limits are often treated as synonyms.

The terms *inverse limit* and *projective limit* are generally used for limits of diagrams indexed by cofiltered categories. In particular, an inverse system indexed by a directed poset I is a functor $I^{\text{op}} \rightarrow \mathcal{C}$.

Definition 1.5.4 (Product). The **product** $(P, (\text{pr}_i)_{i \in I})$ of a family of objects $(\mathbf{D}(i))_{i \in I}$ in $\mathcal{C}$ consists of an object

$$P := \prod_{i \in I} \mathbf{D}(i) \in \text{Ob}(\mathcal{C})$$

and morphisms

$$(\text{pr}_i : P \rightarrow \mathbf{D}(i))_{i \in I}$$

such that the **universal product property** holds:

For any $T \in \text{Ob}(\mathcal{C})$ and any family of morphisms $(f_i : T \rightarrow \mathbf{D}(i))_{i \in I}$ there exists a unique morphism $f : T \rightarrow P$ such that

$$\text{pr}_i \circ f = f_i$$

for all $i \in I$ (see [3, Ch. 5, §3, Def. 5.3.], [8, Ch. 6, §2, Def. 6.2.4]).

Products are precisely limits of discrete diagrams; a product is **small** if the corresponding discrete indexing category is small.

Example 1.5.5 ($\mathbf{Ab}$ has all small products). For any small family $(A_i)_{i \in I}$ of abelian groups, define

$$\prod_{i \in I} A_i := \{(a_i)_{i \in I} \mid a_i \in A_i\}$$

with componentwise addition:

$$(a_i)_{i \in I} + (b_i)_{i \in I} = (a_i + b_i)_{i \in I}, \quad -(a_i)_{i \in I} = (-a_i)_{i \in I}.$$

This is an abelian group. For each $j \in I$ define a projection

$$\text{pr}_j : \prod_{i \in I} A_i \rightarrow A_j, \quad (a_i)_{i \in I} \mapsto a_j.$$

Let T be any abelian group with maps $f_j : T \rightarrow A_j$ for all $j \in I$. Define

$$f : T \rightarrow \prod_{i \in I} A_i, \quad t \mapsto (f_i(t))_{i \in I}.$$

Then

$$\text{pr}_j \circ f = f_j$$

for every $j \in I$. Furthermore, f is the unique morphism with this property, since any map into the product is determined by all its components. Therefore $\prod_i A_i$ satisfies the universal property of the product (compare [8, Ch. 6, §2, Ex. 6.2.5]).

Having established the existence of products, one can consider a more general construction. Given morphisms

$$A \rightarrow C \leftarrow B$$

one may ask for an object that behaves like a product of A and B while respecting the maps into C .

Definition 1.5.6 (Pullback). Consider a category $\mathcal{C}$, and two morphisms

$$f : A \rightarrow C, \quad g : B \rightarrow C.$$

A **pullback** of f and g is an object, commonly denoted by $A \times_C B$, together with morphisms

$$p_A : A \times_C B \rightarrow A, \quad p_B : A \times_C B \rightarrow B$$

such that

$$f \circ p_A = g \circ p_B,$$

satisfying the **universal pullback property**:

For every $T \in \text{Ob}(\mathcal{C})$ with morphisms

$$h_A : T \rightarrow A, \quad h_B : T \rightarrow B$$

satisfying

$$f \circ h_A = g \circ h_B$$

there exists a unique morphism

$$h : T \rightarrow A \times_C B$$

such that

$$p_A \circ h = h_A,$$

and

$$p_B \circ h = h_B$$

(see [8, Ch. 6, §2, Def. 6.2.16]).

The pullback $A \times_C B$ is also called the **fibre product** of A and B over C .

In contrast to the category-theoretic product, the fibre product is defined with respect to an object C , and it is not a special case of the product in the considered category $\mathcal{C}$. However, the fibre product is a product in the slice category $\mathcal{C}/C$.

In the poset category $\text{Open}(X)$, the pullback of inclusions $U_1 \hookrightarrow U$, $U_2 \hookrightarrow U$ is the intersection $U_1 \cap U_2$, equipped with its inclusions into U_1 and U_2 .

Products admit a distinguished canonical morphism obtained from their universal property. This morphism plays a central role in the construction of equalisers and pullbacks.

Definition 1.5.7 (Diagonal morphisms). Consider a category $\mathcal{C}$ with products, and let $A \in \text{Ob}(\mathcal{C})$. The **diagonal morphism**

$$\blacktriangle_A : A \rightarrow A \times A$$

is the unique morphism such that

$$\begin{array}{ccc} A & \xrightarrow{\text{id}_A} & A \\ \text{id}_A \downarrow & \searrow \blacktriangle_A & \uparrow \text{pr}_2 \\ A & \xleftarrow{\text{pr}_1} & A \times A \end{array}$$

commutes (compare [1, Ch. 3, §10, 10V]).

The diagonal morphism provides a canonical way of expressing equality within a category. Indeed, two parallel morphisms agree precisely when the induced morphism

$$(f, g) : A \rightarrow B \times B$$

factors through the diagonal. This observation leads to the notion of equalisers.

Definition 1.5.8 (Equaliser). A pair $(E, \text{eq}(f, g))$ with $E \in \text{Ob}(\mathcal{C})$ and morphism $\text{eq}(f, g) : E \rightarrow A$ satisfying

$$f \circ \text{eq}(f, g) = g \circ \text{eq}(f, g)$$

is the **equaliser** of parallel morphisms $f, g \in \text{Hom}_{\mathcal{C}}(A, B)$, satisfying the **universal equaliser property**:

Given any pair (T, h) with $T \in \text{Ob}(\mathcal{C})$ and $h \in \text{Hom}_{\mathcal{C}}(T, A)$ such that

$$f \circ h = g \circ h,$$

there exists a unique morphism $\eta : T \rightarrow E$ such that

$$\text{eq}(f, g) \circ \eta = h$$

(see [3, Ch. 8, §1, Def. 8.1.2]).

Although the morphism $E \rightarrow A$ is essential to the notion of equalisers, it is often the object E that is referred to as the equaliser. The equaliser identifies the greatest subobject of A on which the two parallel morphisms f and g agree.

The equaliser is the limit of a diagram of the shape $\bullet \rightrightarrows \bullet$. A category might have equalisers, but not products, and therefore need not admit all finite limits.

Remark 1.5.9. For an arbitrary family of morphisms

$$(f_i : A \rightarrow B)_{i \in I},$$

the multiple equaliser $\text{eq}((f_i)_{i \in I})$ is defined analogously (see [8, Ch. 6, §2, Def. 6.2.13]).

Example 1.5.10 (Equalisers in Ab). Let $f, g \in \text{Hom}_{\text{Ab}}(A, B)$, and define

$$E := \{a \in A \mid f(a) = g(a)\}.$$

Since

$$f(a_1 + a_2) = f(a_1) + f(a_2), \quad g(a_1 + a_2) = g(a_1) + g(a_2),$$

E is a subgroup of A . Let $\iota : E \hookrightarrow A$ denote the inclusion. Then

$$f \circ \iota = g \circ \iota.$$

Suppose that $h : T \rightarrow A$ satisfies

$$f \circ h = g \circ h.$$

Then for every $t \in T$, $h(t) \in E$. Hence there exists a unique morphism $\varphi : T \rightarrow E$ with

$$\iota \circ \varphi = h.$$

Therefore E satisfies the universal property of the equaliser (compare [8, Ch. 6, §2, Ex. 6.2.14]).

Example 1.5.11. Let $A^\bullet : \Delta \rightarrow \mathcal{C}$ be a cosimplicial object in a category $\mathcal{C}$ admitting equalisers. Then

$$\lim_{\Delta} A^\bullet \cong \text{eq} \left(A^0 \begin{smallmatrix} \xrightarrow{d^0} \\ \xleftarrow{d^1} \end{smallmatrix} A^1 \right).$$

The equaliser $A^0 \rightrightarrows A^1$ represents exactly the cones over the truncated diagram. The cosimplicial identities imply that every cone over $A^0 \rightrightarrows A^1$ extends uniquely to a cone over $A^\bullet$. Hence the two universal properties coincide (see [15, p. 2]).

In detail, consider

$$\text{eq}(d^0, d^1) : E \rightarrow A^0.$$

For each n , one can choose a map

$$\iota : [0] \rightarrow [n]$$

in Δ and define

$$e_n := A^\bullet(\iota) \circ \text{eq}(d^0, d^1) : E \rightarrow A^n.$$

The equaliser relation

$$d^0 \circ \text{eq}(d^0, d^1) = d^1 \circ \text{eq}(d^0, d^1),$$

together with the cosimplicial identities, makes e_n independent of the choice of ι , and the family of morphisms $(e_n)_n$ is a cone:

Any two morphisms $[0] \rightarrow [n]$ can be connected by a sequence of adjacent vertex maps. Repeated application of the coface identities, together with the equaliser relation, shows that their images under $A^\bullet$ agree after precomposition with $\text{eq}(d^0, d^1)$. Since any cone

$$(t_n : T \rightarrow A^n)_n$$

satisfies

$$d^0 \circ t_0 = t_1 = d^1 \circ t_0,$$

the morphism t_0 factors uniquely through $\text{eq}(d^0, d^1)$. The remaining components t_n are forced by the cone condition.

Remark 1.5.12 (Functoriality of limits). Let I be a small category, and let $\mathbf{D}_1, \mathbf{D}_2 : I \rightarrow \mathcal{C}$ be two diagrams admitting limits, and let $\alpha : \mathbf{D}_1 \rightarrow \mathbf{D}_2$ be a natural transformation. Then there exists a unique morphism

$$\lim \mathbf{D}_1 \rightarrow \lim \mathbf{D}_2$$

making the induced diagram

$$\begin{array}{ccc} \mathbf{D}^{\text{cnst}}(\lim \mathbf{D}_1) & \longrightarrow & \mathbf{D}^{\text{cnst}}(\lim \mathbf{D}_2) \\ \downarrow & & \downarrow \\ \mathbf{D}_1 & \longrightarrow & \mathbf{D}_2 \end{array}$$

commute, where $\mathbf{D}^{\text{cnst}} : \mathcal{C} \rightarrow [I, \mathcal{C}]$ denotes the constant diagram functor. The vertical morphisms are the universal cones. By the universal property of $\lim \mathbf{D}_2$ on the cone from $\lim \mathbf{D}_1$ to $\mathbf{D}_2$,

$$\mathbf{D}^{\text{cnst}} \lim \mathbf{D}_1 \rightarrow \mathbf{D}_1 \rightarrow \mathbf{D}_2,$$

the following holds:

Let f, g be two morphisms between diagrams. By uniqueness in the universal property,

$$\text{id} = \lim(\text{id})$$

and

$$\lim(g \circ f) = \lim(g) \circ \lim(f)$$

hold. Thus, assuming that $\mathcal{C}$ has limits of shape I , this gives a functor

$$\lim : \text{Diag}_I(\mathcal{C}) \rightarrow \mathcal{C},$$

or equivalently,

$$\lim : [I, \mathcal{C}] \rightarrow \mathcal{C}.$$

Defining the limit as a functor requires choosing one limit for each diagram; such global choices generally invoke the axiom of choice (see [8, Ch. 6, §2, Rem. 6.2.24]).

A core principle of category theory is duality. Consequently, every construction and result concerning limits admits a dual construction, and a dual result.

1.5.2 Colimits

The notions dual to limits are colimits. Coproducts generalise disjoint unions, whereas coequalisers formalise identifications and gluing.

Definition 1.5.13 (Cocone). A **cocone** under the diagram $\mathbf{D} : I \rightarrow \mathcal{C}$ is a pair $(T^{\text{co}}, (\iota_i)_{i \in \text{Ob}(I)})$ consisting of $T^{\text{co}} \in \text{Ob}(\mathcal{C})$ and morphisms $(\iota_i : \mathbf{D}(i) \rightarrow T^{\text{co}})_{i \in \text{Ob}(I)}$, such that for every morphism $e_{ij} : i \rightarrow j$ in I

$$\begin{array}{ccc} \mathbf{D}(i) & \xrightarrow{\mathbf{D}(e_{ij})} & \mathbf{D}(j) \\ & \searrow \iota_i \quad \swarrow \iota_j & \\ & T^{\text{co}} & \end{array}$$

commutes.

The condition

$$\iota_i = \iota_j \circ \mathbf{D}(e_{ij})$$

is called the **cocone condition**. The collection of cocones from a diagram $\mathbf{D}$ to an object T is denoted by $\text{Cocone}(\mathbf{D}, T)$ (see [3, Ch. 8, §5, Def. 8.5.1]).

Cocones in $\mathcal{C}$ are cones in $\mathcal{C}^{\text{op}}$.

Definition 1.5.14 (Colimit). The **colimit** of a diagram $\mathbf{D} : I \rightarrow \mathcal{C}$ is a cocone $(L^{\text{co}}, (\iota_i)_{i \in \text{Ob}(I)})$ to $\mathbf{D}$ such that for every cocone $(T^{\text{co}}, (f_i)_{i \in \text{Ob}(I)})$ to $\mathbf{D}$ the **universal colimit property** holds:

There exists a unique morphism

$$f : L^{\text{co}} \rightarrow T^{\text{co}}$$

such that

$$f \circ \iota_i = f_i$$

for all $i \in \text{Ob}(I)$ (see [3, Ch. 8, §5, Def. 8.5.1]).

A category that admits all small colimits is called **cocomplete** (see [8, Ch. 6, §5, Def. 6.5.2]).

Equivalently, the colimit of $\mathbf{D} : I \rightarrow \mathcal{C}$ is the limit of the opposite diagram $\mathbf{D}^{\text{op}} : I^{\text{op}} \rightarrow \mathcal{C}^{\text{op}}$. There is no implication between the existence of limits and colimits over the same category.

Definition 1.5.15 (Coproduct). The **coproduct** $(P^{\text{co}}, (\iota_i)_{i \in I})$ of a family of objects $(\mathbf{D}(i))_{i \in I}$ in $\mathcal{C}$ consists of

$$P^{\text{co}} := \coprod_{i \in I} \mathbf{D}(i) \in \text{Ob}(\mathcal{C})$$

and a family of morphisms

$$(\iota_i : \mathbf{D}(i) \rightarrow P^{\text{co}})_{i \in I}$$

such that the **universal coproduct property** holds:

For any $T^{\text{co}} \in \text{Ob}(\mathcal{C})$ and any family of morphisms

$$(f_i : \mathbf{D}(i) \rightarrow T^{\text{co}})_{i \in I}$$

there exists a unique morphism $f : P^{\text{co}} \rightarrow T^{\text{co}}$ such that

$$f \circ \iota_i = f_i$$

for all $i \in I$ (see [8, Ch. 6, §3, Def. 6.3.5]).

Coproducts are precisely colimits of discrete diagrams.

Example 1.5.16 (Coproducts in Top). Let $(\mathbf{D}(v))_{v \in V}$ be a family of objects in Top . The coproduct is the disjoint union of topological spaces

$$\coprod_{v \in V} \mathbf{D}(v) := \dot{\bigcup}_{v \in V} (\mathbf{D}(v) \times \{v\}).$$

The injections are defined as

$$\begin{aligned}\iota_w : \mathbf{D}(w) &\rightarrow \coprod_{v \in V} \mathbf{D}(v) \\ x &\mapsto (x, w)\end{aligned}$$

for an element $x \in \mathbf{D}(w)$. The topology on $\coprod_{v \in V} \mathbf{D}(v)$ is the finest topology, such that all injections are continuous.

Let T be an arbitrary object with a family of morphisms $(f_v : \mathbf{D}(v) \rightarrow T)_{v \in V}$. Define the unique morphism

$$f : \coprod_{v \in V} \mathbf{D}(v) \rightarrow T$$

such that $f_v = f \circ \iota_v$ for all $v \in V$, as

$$f((x, v)) := f_v(x)$$

(see [8, Ch. 6, §5, Proof 6.5.6]).

Note that every f_v is continuous by the definition of the category $\mathbf{Top}$. By the coproduct topology,

$$f : \coprod_{v \in V} \mathbf{D}(v) \rightarrow T$$

is continuous precisely when each composite

$$f \circ \iota_v = f_v$$

is continuous. Hence f is continuous.

Moreover, the canonical images $\iota_v(\mathbf{D}(v))$ are open and closed.

Example 1.5.17 ($\mathbf{Ab}$ has all small coproducts). Let $(A_i)_{i \in I}$ be a small family of objects in $\mathbf{Ab}$. Define

$$\coprod_{i \in I} A_i := \left\{ (a_i)_{i \in I} \in \prod_{i \in I} A_i \mid a_i \neq 0 \text{ for finitely many } i \right\}.$$

This is an abelian group under componentwise addition. For each $j \in I$, define the

inclusion

$$\iota_j : A_j \rightarrow \coprod_{i \in I} A_i$$

by

$$(\iota_j(a))_i = \begin{cases} a, & i = j \\ 0, & i \neq j. \end{cases}$$

Let T be an abelian group with morphisms $f_j : A_j \rightarrow T$, and define

$$f : \coprod_{i \in I} A_i \rightarrow T, \quad f((a_i)_{i \in I}) = \sum_{i \in I} f_i(a_i).$$

This sum is finite because $(a_i)_{i \in I}$ has finite support. Then

$$f \circ \iota_j = f_j$$

for every j , and f is unique with this property. Therefore $\coprod_{i \in I} A_i$ is the coproduct in $\mathbf{Ab}$ (see [8, Ch. 6, §5, Ex. 6.5.21]).

Definition 1.5.18 (Pushout). Let $\mathcal{C}$ be a category, and let

$$f : C \rightarrow A, \quad g : C \rightarrow B$$

be morphisms in $\mathcal{C}$. The **pushout** is an object $A \sqcup_C B$ with morphisms

$$q_A : A \rightarrow A \sqcup_C B, \quad q_B : B \rightarrow A \sqcup_C B$$

such that

$$q_A \circ f = q_B \circ g,$$

satisfying the **universal pushout property**:

For every $T \in \mathbf{Ob}(\mathcal{C})$ with morphisms

$$h_A : A \rightarrow T, \quad h_B : B \rightarrow T$$

satisfying

$$h_A \circ f = h_B \circ g,$$

there exists a unique morphism

$$h : A \sqcup_C B \rightarrow T$$

such that

$$h \circ q_A = h_A$$

and

$$h \circ q_B = h_B$$

(see [8, Ch. 6, §3, Def. 6.3.16]).

Pushouts are the dual notion of pullbacks. They are also referred to as *fibred coproduct*, *fibred sum*, or *amalgamated sum*. The universal property gives the most general way to complete the commutative square.

This is the coproduct in the undercategory $(C \downarrow \mathcal{C})$, dual to the description of a pullback as a product in $\mathcal{C}/C$. The existence of coproducts does not imply the existence of colimits. The universal property of coproducts provides canonical morphisms from each object into the coproduct. In the special case where all objects coincide, the universal property yields a distinguished morphism in the opposite direction.

Definition 1.5.19 (Codiagonal morphisms). Consider $B \in \text{Ob}(\mathcal{D})$, where $\mathcal{D}$ is a category with coproducts. Then the **codiagonal** is the unique morphism

$$\nabla_B : B \sqcup B \rightarrow B$$

such that

$$\begin{array}{ccc} B & \xleftarrow{\text{id}_B} & B \\ \text{id}_B \uparrow & \swarrow \nabla_B & \downarrow \iota_2 \\ B & \xrightarrow{\iota_1} & B \sqcup B \end{array}$$

commutes (see [43, Ch. 1, §1, p. 17]).

More generally, one is often interested in forcing two arbitrary morphisms to become equal. This leads to the formalisation of gluing.

Definition 1.5.20 (Coequaliser). A pair $(E^{\text{co}}, \text{coeq}(f, g))$ with $E^{\text{co}} \in \text{Ob}(\mathcal{C})$ and morphism

$$\text{coeq}(f, g) : B \rightarrow E^{\text{co}}$$

satisfying

$$\text{coeq}(f, g) \circ f = \text{coeq}(f, g) \circ g$$

is the **coequaliser** of parallel morphisms $f, g \in \text{Hom}_{\mathcal{C}}(A, B)$, satisfying the **universal coequaliser property**:

Given any pair (T^{co}, h) with $T^{\text{co}} \in \text{Ob}(\mathcal{C})$ and $h \in \text{Hom}_{\mathcal{C}}(B, T^{\text{co}})$ such that

$$h \circ g = h \circ f,$$

there exists a unique morphism $\eta : E^{\text{co}} \rightarrow T^{\text{co}}$ such that

$$\eta \circ \text{coeq}(f, g) = h$$

(see [3, Ch. 8, §5, Def. 8.5.1]).

The coequaliser is the colimit of a diagram of the shape $\bullet \rightrightarrows \bullet$. The existence of all coequalisers alone does not imply the existence of all colimits, and the existence of all coproducts alone does not imply the existence of all colimits.

Example 1.5.21 (Coequalisers in Ab). Consider $f, g \in \text{Hom}_{\text{Ab}}(A, B)$, and define the subgroup N of B , generated by

$$N := \langle f(a) - g(a) \mid a \in A \rangle.$$

Set $Q := B/N$, and let $q : B \rightarrow B/N$ be the quotient map. Since

$$f(a) - g(a) \in N,$$

it follows that

$$q(f(a)) = q(g(a)).$$

Thus

$$q \circ f = q \circ g.$$

Suppose that $h : B \rightarrow C$ satisfies

$$h \circ f = h \circ g.$$

Then for every $a \in A$,

$$h(f(a) - g(a)) = h(f(a)) - h(g(a)) = 0.$$

Thus h vanishes on every generator of N , hence $N \subseteq \ker(h)$. Therefore there exists a unique morphism $\varphi : B/N \rightarrow C$ such that

$$\varphi \circ q = h.$$

Thus B/N with the quotient map q is the coequaliser of f and g (compare [8, Ch. 6, §3, Ex. 6.3.11]).

Example 1.5.22 (Coequalisers in $\mathbf{Top}$). Let $f, g \in \mathbf{Hom}_{\mathbf{Top}}(X, Y)$ be parallel morphisms. Define an equivalence relation $\sim$ on Y as the smallest equivalence relation satisfying

$$f(x) \sim g(x)$$

for all $x \in X$. Let

$$q : Y \rightarrow Y / \sim$$

be the canonical quotient map, where $Y / \sim$ is equipped with the quotient topology. Then

$$q \circ f = q \circ g.$$

Let $h \in \mathbf{Hom}_{\mathbf{Top}}(Y, Z)$ be a continuous map satisfying

$$h \circ f = h \circ g.$$

Then h is constant on equivalence classes of $\sim$, and therefore induces a unique map

$$\varphi : Y / \sim \rightarrow Z$$

such that

$$\begin{array}{ccccc} X & \xrightleftharpoons[g]{f} & Y & \xrightarrow{q} & Y/\sim \\ & & \searrow h & & \downarrow \varphi \\ & & & & Z \end{array}$$

commutes.

Since $Y/\sim$ carries the quotient topology, the morphism φ is continuous if and only if

$$\varphi \circ q = h$$

is continuous. Hence φ is continuous. Therefore

$$q : Y \rightarrow Y/\sim$$

satisfies the universal property of a coequaliser in $\mathbf{Top}$ (see [8, Ch. 6, §5, Proof 6.5.6]).

While coequalisers and coproducts are colimits, coequalisers or coproducts alone are not sufficient for arbitrary colimits to exist. Combined, they provide a concrete criterion for cocompleteness.

Theorem 1.5.23 (Construction of Colimits). Let $\mathcal{C}$ be a category that has all coproducts and all coequalisers, and let $\mathbf{D} : I \rightarrow \mathcal{C}$ be a diagram. Then $\mathcal{C}$ admits all small colimits, which can be constructed as the coequaliser of suitable morphisms

$$a, b : \coprod_{e_{ij} \in \mathbf{Mor}(I)} \mathbf{D}(i) \rightrightarrows \coprod_{j \in \mathbf{Ob}(I)} \mathbf{D}(j)$$

(see [8, Ch. 6, §5, Thm. 6.5.1]).

Proof. First construct the coproduct:

For a family of objects $(\mathbf{D}(i))_{i \in I}$ construct the coproduct

$$P^{\text{co}} := \coprod_{i \in I} \mathbf{D}(i)$$

with morphisms

$$\iota_i : \mathbf{D}(i) \rightarrow P^{\text{co}}$$

for all $i \in I$. Since I is not necessarily a discrete category, and since the canonical morphisms $(\iota_i)_i$ do not in general come from a cocone over $\mathbf{D}$, the diagram might not commute.

Consider the coproducts

$$Q^{\text{co}} := \coprod_{e_{ij} \in \text{Mor}(I)} \mathbf{D}(i)$$

with canonical coprojections

$$\kappa_{e_{ij}} : \mathbf{D}(i) \rightarrow Q^{\text{co}}$$

and $(P^{\text{co}}, (\iota_i)_{i \in I})$ as constructed. Define morphisms $a, b : Q^{\text{co}} \rightarrow P^{\text{co}}$ such that

$$\begin{aligned} a \circ \kappa_{e_{ij}} &:= \iota_i \\ b \circ \kappa_{e_{ij}} &:= \iota_j \circ \mathbf{D}(e_{ij}) \end{aligned}$$

for each $e_{ij} : i \rightarrow j$ and take the coequaliser

$$q := \text{coeq}(a, b) : P^{\text{co}} \rightarrow E^{\text{co}}.$$

One verifies that $(q \circ \iota_i)_{i \in I}$ is a cocone:

For every morphism $e : i \rightarrow j$,

$$\begin{aligned} q \circ \iota_i &= q \circ a \circ \kappa_e \\ &= q \circ b \circ \kappa_e \\ &= q \circ \iota_j \circ \mathbf{D}(e). \end{aligned}$$

Thus the morphisms $q \circ \iota_i$ satisfy the cocone condition.

Since a cocone $(T^{\text{co}}, (f_i)_{i \in I})$ induces a unique morphism

$$\tilde{f} : P^{\text{co}} \rightarrow T^{\text{co}}$$

by the universal property of the coproduct such that

$$\tilde{f} \circ \iota_i = f_i.$$

The cocone condition implies

$$\tilde{f} \circ a = \tilde{f} \circ b,$$

so $\tilde{f}$ factors uniquely through q as

$$\tilde{f} = f \circ q.$$

Finally, the uniqueness of f proves the universal property of the colimit (compare [8, Ch. 6, §4, Proof Thm. 6.4.1]). $\square$

This reveals not only a criterion for the existence of colimits, but also how colimits can be constructed. A dual statement holds for limits (see [8, Ch. 6, §4, Thm. 6.4.1]).

Example 1.5.24 ($\mathbf{Ab}$ is cocomplete). By Example 1.5.17, $\mathbf{Ab}$ has all small coproducts, and by Example 1.5.21, $\mathbf{Ab}$ has all coequalisers. By Theorem 1.5.23, $\mathbf{Ab}$ has all colimits. Hence $\mathbf{Ab}$ is cocomplete (see [8, Ch. 6, §5, Ex. 6.5.3]).

Example 1.5.25 ($\mathbf{Top}$ is cocomplete). As established in Example 1.5.16, $\mathbf{Top}$ has all coproducts, and all coequalisers by Example 1.5.22. Hence $\mathbf{Top}$ is cocomplete (see [8, Ch. 6, §5, Thm. 6.5.6]).

A special colimit that arises naturally in sheaf theory, for example in the construction of stalks and in the definition of Čech cohomology, is the directed colimit.

Definition 1.5.26 (Directed colimit). A colimit on a directed diagram is called a **directed colimit**, and shall be denoted by $\varinjlim$ (see [8, Ch. 6, §5, Def. 6.5.15]).

Directed colimits are also called **direct limit** or **inductive limit**. In essence, directed colimits provide a mechanism to "complete" structures by taking their universal "union" in a directed manner, fundamental for understanding how properties defined by finite operations manifest in larger, potentially infinite structures.

Definition 1.5.27 (Filtered colimit). A **filtered colimit** is a colimit taken over a filtered category (see [34, Ch. 9, §1]).

Notice that although filtered colimits and directed colimits are closely related, they are not the same. While every directed colimit is a filtered colimit, not every filtered colimit is a directed colimit.

1.5.3 Navigating Limits and Colimits

This subsection provides a basic guide on how limits interact with limits, and in particular, how limits and colimits interact.

Theorem 1.5.28 (Nested Limits). Let $\Gamma = (V, E)$ and $\Lambda = (W, F)$ be directed graphs and let $\Gamma \otimes \Lambda$ be their Cartesian product. Consider a legal diagram $\mathbf{D}$ of shape $\Gamma \otimes \Lambda$ in an arbitrary category.

Assume that the limit of the diagram $\mathbf{D}(v, -)$ of shape Λ exists for every $v \in V$. This limit is denoted by $\lim_{w \in W} \mathbf{D}(v, w)$. By Remark 1.5.12, every edge $v \rightarrow v'$ in E induces a morphism

$$\lim_{w \in W} \mathbf{D}(v, w) \rightarrow \lim_{w \in W} \mathbf{D}(v', w).$$

This gives rise to a diagram $\lim_{w \in W} \mathbf{D}(-, w)$ of shape Γ .

The cones over the diagram $\lim_{w \in W} \mathbf{D}(-, w)$ correspond naturally to the cones over $\mathbf{D}$. In particular,

$$\lim_{v \in V} \lim_{w \in W} \mathbf{D}(v, w)$$

exists if and only if

$$\lim_{(v,w) \in V \times W} \mathbf{D}(v, w)$$

exists. In that case, there exists a natural isomorphism

$$\lim_{v \in V} \lim_{w \in W} \mathbf{D}(v, w) \cong \lim_{(v,w) \in V \times W} \mathbf{D}(v, w)$$

(see [8, Ch. 6, §6, Thm. 6.6.4]).

Proof. By the functoriality of limits, every edge $e : v \rightarrow v'$ in Γ induces a morphism

$$\lim_{w \in W} \mathbf{D}(v, w) \rightarrow \lim_{w \in W} \mathbf{D}(v', w),$$

making

$$\lim_{w \in W} \mathbf{D}(-, w)$$

a diagram of shape Γ . Let $T \in \text{Ob}(\mathcal{C})$.

Claim: Cones from T to $\lim_{w \in W} \mathbf{D}(-, w)$ correspond naturally to cones from T to $\mathbf{D}$.

A cone from T to $\lim_{w \in W} \mathbf{D}(-, w)$ consists of morphisms

$$\alpha_v : T \rightarrow \lim_{w \in W} \mathbf{D}(v, w)$$

for $v \in V$, compatible with the edges of Γ . By the universal property of

$$\lim_{w \in W} \mathbf{D}(v, w),$$

each morphism α_v corresponds uniquely to a cone with components

$$(\alpha_{v,w} : T \rightarrow \mathbf{D}(v, w))_{w \in W}$$

over the diagram $\mathbf{D}(v, -)$.

The compatibility of the morphisms α_v with the edges of Γ is equivalent to the compatibility of the family $(\alpha_{v,w})_{v,w}$ with the edges in the first variable. Since $\mathbf{D}$ is legal, the family is therefore a cone from T to the diagram $\mathbf{D}$.

Conversely, every cone from T to $\mathbf{D}$ restricts, for each $v \in V$, to a cone over $\mathbf{D}(v, -)$, and hence induces a unique morphism

$$\alpha_v : T \rightarrow \lim_{w \in W} \mathbf{D}(v, w).$$

The compatibility of the original cone in the first variable implies that the morphisms α_v form a cone over $\lim_{w \in W} \mathbf{D}(-, w)$.

These constructions are inverse to each other and natural in T . Hence cones over $\lim_{w \in W} \mathbf{D}(-, w)$ correspond naturally to cones over $\mathbf{D}$. Consequently, a universal cone over $\lim_{w \in W} \mathbf{D}(-, w)$ exists if and only if a universal limit over $\mathbf{D}$ exists. In that case,

$$\lim_{v \in V} \lim_{w \in W} \mathbf{D}(v, w) \cong \lim_{(v,w) \in V \times W} \mathbf{D}(v, w)$$

(see [8, Ch. 6, §6, Proof 6.6.4]). □

A consequence of this theorem is the commutativity of limits.

Theorem 1.5.29 (Commutativity of Limits). Let $\Gamma := (V, E)$ and $\Lambda := (W, F)$ be two directed graphs and let $\mathbf{D}$ be a legal diagram of shape $\Gamma \otimes \Lambda$. If both iterated limits

exist,

$$\lim_{v \in V} \lim_{w \in W} \mathbf{D}(v, w) \cong \lim_{w \in W} \lim_{v \in V} \mathbf{D}(v, w)$$

(see [8, Ch. 6, §6, Thm. 6.6.5]).

Proof. By Theorem 1.5.28, both iterated limits are canonically isomorphic to the limit over $\Gamma \otimes \Lambda$, and hence

$$\lim_{v \in V} \lim_{w \in W} \mathbf{D}(v, w) \cong \lim_{(v, w) \in V \times W} \mathbf{D}(v, w) \cong \lim_{w \in W} \lim_{v \in V} \mathbf{D}(v, w)$$

(see [8, Ch. 6, §6, Proof 6.6.5]). □

The dual statement holds for colimits.

Initial and terminal objects arise as colimits and limits over the empty diagram.

Corollary 1.5.30 (Terminal and initial objects as (co-)limits). Let $\emptyset$ denote the empty category, and let $\emptyset \rightarrow \mathcal{C}$ be the unique functor. Terminal objects in $\mathcal{C}$ are limits of the empty diagram and initial objects are colimits of the empty diagram (see [1, Ch. 3, §10, p. 174], [37, pp. 1-2]).

Proof. Since a cone over the empty diagram consists only of its tip, there are no objects and therefore no structure morphisms to specify. Consequently, every object of $\mathcal{C}$ determines a cone over the empty diagram.

A limit of the empty diagram is therefore an object T such that, for every object $A \in \text{Ob}(\mathcal{C})$, there exists a unique morphism $A \rightarrow T$. Since cones over the empty diagram consist only of their tips, this means that for every object $A \in \text{Ob}(\mathcal{C})$ there exists a unique morphism

$$A \rightarrow T.$$

This is precisely the universal property of a terminal object.

Dually, a cocone over the empty diagram consists only of its tip. Hence a colimit of the empty diagram is an object T' such that for every object $A' \in \text{Ob}(\mathcal{C})$ there exists a unique morphism

$$T' \rightarrow A',$$

which is exactly the definition of an initial object (see [1, Ch. 3, §10, p. 174], [37, pp. 1-2]). □

As a consequence, since terminal objects are limits, and limits commute with limits, the limit of terminal objects is a terminal object.

A standard example of a directed colimit is the directed colimit in $\mathbf{Set}$. The category $\mathbf{Set}$ provides a concrete model for directed colimits.

Example 1.5.31 (Directed colimit in $\mathbf{Set}$). Let $\mathbf{D} = (\mathbf{D}(v))_{v \in V}$ be a directed diagram of sets. Define an equivalence relation $\sim$ on $\coprod_{v \in V} \mathbf{D}(v)$ by declaring $x \sim y$ (for $x \in \mathbf{D}(v)$ and $y \in \mathbf{D}(w)$) if there exists $k \in V$ with $v \leq k$ and $w \leq k$ such that

$$\mathbf{D}(v \rightarrow k)(x) = \mathbf{D}(w \rightarrow k)(y).$$

The quotient set $\coprod_{v \in V} \mathbf{D}(v) / \sim$ is the colimit of $\mathbf{D}$ (see [8, Ch. 6, §5, Thm. 6.5.16]).

Proof. Claim: $\sim$ is an equivalence relation:

It is reflexive, since every object $\mathbf{D}(v)$ is equipped with the identity, and thus, $x \sim x$ holds for every element $x \in \mathbf{D}(v)$ for all $v \in V$.

Assume that $x \sim y$ holds (for $x \in \mathbf{D}(v)$ and $y \in \mathbf{D}(w)$). By definition of $\sim$ and since $\mathbf{D}$ is directed, there exists $k \in V$ such that $v \leq k$ and $w \leq k$ such that

$$\mathbf{D}(v \rightarrow k)(x) = \mathbf{D}(w \rightarrow k)(y).$$

The definition of $\sim$ is symmetric in x and y , hence $y \sim x$.

Let $v_1, v_2, v_3 \in V$ be vertices and let $x \in \mathbf{D}(v_1)$, $y \in \mathbf{D}(v_2)$ and $z \in \mathbf{D}(v_3)$ be elements such that $x \sim y$ and $y \sim z$. By definition of $\sim$ and since V is directed, there exist elements $w_1, w_2 \in V$ such that $v_1, v_2 \leq w_1$ and $v_2, v_3 \leq w_2$ such that

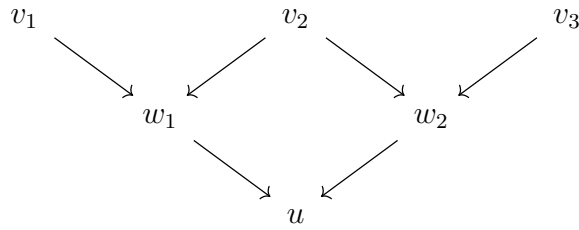
$$\mathbf{D}(v_1 \rightarrow w_1)(x) = \mathbf{D}(v_2 \rightarrow w_1)(y) =: x'$$

$$\mathbf{D}(v_2 \rightarrow w_2)(y) = \mathbf{D}(v_3 \rightarrow w_2)(z) =: y'.$$

Since V is directed, there exists $u \in V$ with $w_1, w_2 \leq u$ such that

$$\begin{aligned} \mathbf{D}(w_1 \rightarrow u)(x') &= \mathbf{D}(w_1 \rightarrow u)(\mathbf{D}(v_2 \rightarrow w_1)(y)) \\ &= \mathbf{D}(v_2 \rightarrow u)(y) \\ &= \mathbf{D}(w_2 \rightarrow u)(\mathbf{D}(v_2 \rightarrow w_2)(y)) \\ &= \mathbf{D}(w_2 \rightarrow u)(y'). \end{aligned}$$

Thus, $\sim$ is transitive and hence an equivalence relation. The diagram



visualises the strategy. Assume $v \leq w$, then for every $x \in \mathbf{D}(v)$, it holds that

$$x \sim \mathbf{D}(v \rightarrow w)(x)$$

with $\mathbf{D}(v \rightarrow w)(x) \in \mathbf{D}(w)$. Hence the quotient map determines a cocone with tip

$$\coprod_{v \in V} \mathbf{D}(v) / \sim$$

and morphisms

$$\iota_v : \mathbf{D}(v) \rightarrow \left(\coprod_{w \in V} \mathbf{D}(w) \right) / \sim, \quad x \mapsto [x].$$

Then

$$\begin{aligned} \iota_w(\mathbf{D}(v \rightarrow w)(x)) &= [\mathbf{D}(v \rightarrow w)(x)] \\ &= [x] \\ &= \iota_v(x). \end{aligned}$$

Claim: The cocone is universal.

Consider a family of morphisms

$$(f_v : \mathbf{D}(v) \rightarrow T)_{v \in V}$$

to some object T , such that for every $v \leq w$ the diagram

$$\begin{array}{ccc} \mathbf{D}(v) & \xrightarrow{\mathbf{D}(v \rightarrow w)} & \mathbf{D}(w) \\ & \searrow f_v \quad \swarrow f_w & \\ & T & \end{array}$$

commutes.

Indeed, if $x \sim y$, then by the cocone condition, there exists k with

$$\begin{aligned} f_v(x) &= f_k(\mathbf{D}(v \rightarrow k)(x)) \\ &= f_k(\mathbf{D}(w \rightarrow k)(y)) \\ &= f_w(y). \end{aligned}$$

Hence the value depends only on the equivalence classes of x regarding $\sim$. Thus, the morphisms f_v extend uniquely on

$$\coprod_{v \in V} \mathbf{D}(v) / \sim,$$

that is, there exists a unique morphism

$$\bar{f} : \left(\coprod_{v \in V} \mathbf{D}(v) \right) / \sim \rightarrow T, \quad \bar{f}([x]) = f_v(x),$$

which satisfies

$$\bar{f} \circ \iota_v = f_v$$

(see [8, Ch. 6, §5, Thm. 6.5.16]). □

Directed colimits arise not only in the definition of Čech cohomology, they also provide enough structure to behave well enough to interact predictably with finite limits.

Theorem 1.5.32 (Finite limits commute with directed colimits). Consider the category $\mathbf{Set}$. Let $\Gamma = (V, E)$ be a finite graph and let J be a directed poset. Then the limit of shape Γ commutes with the directed colimit of shape J (see [8, Ch. 6, §6, Thm. 6.6.16]).

Remark 1.5.33. Naturality in the second variable means that

$$\begin{array}{ccc} \mathbf{D}(v, i) & \xrightarrow{\mathbf{D}(e, \text{id}_i)} & \mathbf{D}(w, i) \\ \mathbf{D}(\text{id}_v, g) \downarrow & & \downarrow \mathbf{D}(\text{id}_w, g) \\ \mathbf{D}(v, j) & \xrightarrow{\mathbf{D}(e, \text{id}_j)} & \mathbf{D}(w, j) \end{array} \quad (1.5.34)$$

commutes, where $e : v \rightarrow w \in E$ and $g : i \rightarrow j$ is a morphism in J (see [8, Ch. 6, §6, Proof 6.6.16]).

Proof. Let $\mathbf{D}$ be a legal diagram of shape $(V, E) \otimes (J, \leq)$ which is commutative in the second variable. One shows that the canonical morphism

$$\alpha : \text{colim}_{j \in J} \lim_{v \in V} \mathbf{D}(v, j) \rightarrow \lim_{v \in V} \text{colim}_{j \in J} \mathbf{D}(v, j) \quad (1.5.35)$$

is an isomorphism:

Claim: The morphism α is injective.

Consider two elements $[x], [y] \in \text{colim}_{j \in J} \lim_{v \in V} \mathbf{D}(v, j)$ such that

$$\alpha([x]) = \alpha([y]).$$

Recall that elements of the colimit are equivalence classes $[x], [y]$ represented by compatible families $(x_v)_{v \in V}$ at some indexing element $i \in J$, and $(y_v)_{v \in V}$ at some indexing element $j \in J$ respectively.

Since by assumption, $\alpha([x]) = \alpha([y])$, it holds for all $v \in V$ that

$$[x_v] = [y_v]$$

in $\text{colim}_{j \in J} \mathbf{D}(v, j)$. Since J is directed, there exists an indexing element $k_v \in J$ for all $v \in V$, and morphisms

$$\phi_v : i \rightarrow k_v \quad (1.5.36)$$

$$\psi_v : j \rightarrow k_v \quad (1.5.37)$$

such that

$$\mathbf{D}(\mathrm{id}_v, \phi_v)(x_v) = \mathbf{D}(\mathrm{id}_v, \psi_v)(y_v) \quad (1.5.38)$$

in $\mathbf{D}(v, k_v)$.

Since V is finite, the set

$$K := \{k_v \mid v \in V\}$$

is finite, and since J is filtered, K has an upper bound k in J with morphisms

$$\lambda_v : k_v \rightarrow k. \quad (1.5.39)$$

This gives rise to morphisms

$$\beta_v := \lambda_v \circ \phi_v : i \rightarrow k \quad (1.5.40)$$

$$\gamma_v := \lambda_v \circ \psi_v : j \rightarrow k. \quad (1.5.41)$$

It follows that

$$\mathbf{D}(\mathrm{id}_v, \phi_v)(x_v) \stackrel{(1.5.36)}{=} \mathbf{D}(\mathrm{id}_v, \psi_v)(y_v) \stackrel{(1.5.37)}{=} \quad (1.5.42)$$

$$\mathbf{D}(\mathrm{id}_v, \beta_v)(x_v) \stackrel{(1.5.40)}{=} \mathbf{D}(\mathrm{id}_v, \lambda_v)(\mathbf{D}(\mathrm{id}_v, \phi_v)(x_v)) \quad (1.5.43)$$

$$\mathbf{D}(\mathrm{id}_v, \gamma_v)(y_v) \stackrel{(1.5.41)}{=} \mathbf{D}(\mathrm{id}_v, \lambda_v)(\mathbf{D}(\mathrm{id}_v, \psi_v)(y_v)) \quad (1.5.44)$$

by the definitions of the respective morphisms. Combined, (1.5.42)-(1.5.44) give rise to

$$\mathbf{D}(\mathrm{id}_v, \beta_v)(x_v) = \mathbf{D}(\mathrm{id}_v, \gamma_v)(y_v). \quad (1.5.45)$$

Define two cones in $\mathbf{D}(-, k)$ with

$$x' := (\mathbf{D}(\mathrm{id}_v, \beta_v)(x_v))_{v \in V}$$

and

$$y' := (\mathbf{D}(\mathrm{id}_v, \gamma_v)(y_v))_{v \in V}.$$

Equation (1.5.45) shows that the two families coincide componentwise. Hence they define the same element of $\lim_{v \in V} \mathbf{D}(v, k)$.

Thus, their equivalence classes in the colimit over $j \in J$ coincide and $[x] = [y]$.

This shows that α is injective.

Claim: The morphism α is surjective.

The central argument in showing surjectivity of α is that due to the finiteness of V , there are only finitely many indices to "amalgamate", and due to the directedness of J , there exists a single index j' , "dominating" any other index, which makes all equalities become strict equalities in the component j' .

Consider an element $([x_v])_{v \in V} \in \lim_{v \in V} \operatorname{colim}_{j \in J} \mathbf{D}(v, j)$. Choose for each equivalence class $[x_v] \in \operatorname{colim}_{j \in J} \mathbf{D}(v, j)$ a representative

$$x'_v \in \mathbf{D}(v, j_v)$$

for some $j_v \in J$. By the compatibility in the limit, for any edge $f : u \rightarrow v$ in V , the images of $[x'_u]$ and $[x'_v]$ in $\operatorname{colim}_{j \in J} \mathbf{D}(v, j)$ are equal.

Let $\mathbf{D}(w \rightarrow k)$ denote the structure morphism $\mathbf{D}(w, j_w) \rightarrow \mathbf{D}(w, k)$, which is induced by the morphism $j_w \rightarrow k$ in J . By the definition of the colimit in $\mathbf{Set}$, for each morphism $f : u \rightarrow v$ in V , there exists an index $k_f \in J$ and morphisms

$$\begin{aligned} j_u &\rightarrow k_f \\ j_v &\rightarrow k_f \end{aligned}$$

in J , such that

$$\mathbf{D}(f, k_f)(\mathbf{D}(u \rightarrow k_f)(x'_u)) = \mathbf{D}(v \rightarrow k_f)(x'_v)$$

is satisfied by the images of x'_u and x'_v in $\mathbf{D}(u, k_f)$ and $\mathbf{D}(v, k_f)$. Since V is finite, the set

$$S := \{j_v \mid v \in V\} \cup \{k_f \mid f \in E\}$$

is finite. There exists an index $j' \in J$ and morphisms $t_s : s \rightarrow j'$ for every $s \in S$, because J is filtered. Define

$$y_v := \mathbf{D}(v, t_{j_v})(x'_v) \in \mathbf{D}(v, j')$$

for each $v \in V$. It holds that

$$\mathbf{D}(f, k_f)(\mathbf{D}(u \rightarrow k_f)(x'_u)) = \mathbf{D}(v \rightarrow k_f)(x'_v)$$

for any morphism $f : u \rightarrow v$ and index $k_f \in S$, and the functoriality of the structure maps, that is the composition with morphisms $k_f \rightarrow j'$, gives

$$\mathbf{D}(f, j')(\mathbf{D}(u \rightarrow j')(x'_u)) = \mathbf{D}(v \rightarrow j')(x'_v),$$

which is in essence

$$\mathbf{D}(f, j')(y_u) = y_v.$$

Thus the family $(y_v)_{v \in V}$ is a cone in $\lim_{v \in V} \mathbf{D}(v, j')$, and its image under the canonical map

$$\lim_{v \in V} \mathbf{D}(v, j') \rightarrow \lim_{v \in V} \operatorname{colim}_{j \in J} \mathbf{D}(v, j)$$

is precisely the original element $([x'_v])_v$.

This shows that the preimage of any element in $\lim_{v \in V} \operatorname{colim}_{j \in J} \mathbf{D}(v, j)$ exists, and α is surjective.

Hence, α is bijective and therefore an isomorphism of sets (see [8, Ch. 6, §6, Proof 6.6.16]). $\square$

Remark 1.5.46. The preceding theorem is specific to categories with sufficiently well-behaved filtered colimits. In particular, it holds for algebraic categories (see B):

The **forgetful functor** $F : \mathcal{C} \rightarrow \mathbf{Set}$ creates limits and filtered colimits whenever the defining monad is finitary. Since filtered colimits commute with finite limits in $\mathbf{Set}$, the same is true in every algebraic category.

While limits and colimits are generally independent constructions, products and coproducts often exhibit a remarkable compatibility in certain categories. These categories require additional structure, that is, the existence of zero morphisms, and an additive structure on $\mathbf{Hom}$ -sets.

1.6 Additive and Abelian Categories

Biproducts constitute a first indication of the additional algebraic structure present in many categories arising in algebra. The following notions make this structure explicit and culminate in the concept of an abelian category.

1.6.1 Preadditive Categories

The categories encountered in algebra are often equipped with more structure than is visible from their objects and morphisms alone. For example, morphisms between abelian groups can be added pointwise, and composition distributes over this addition. The following definition abstracts this phenomenon and provides the first step towards additive and abelian categories.

Definition 1.6.1 (Preadditive category). Let $\mathcal{C}$ be a category. For any two objects $A, B \in \text{Ob}(\mathcal{C})$ let $+_{A,B}$ be a binary operation on $\text{Hom}_{\mathcal{C}}(A, B)$ such that

$$(\text{Hom}_{\mathcal{C}}(A, B), +_{A,B})$$

is an abelian group. The category $\mathcal{C}$ is **preadditive** if the following property holds:

The composition of morphisms is **biadditive**, that is, for $f, f_1, f_2 \in \text{Hom}_{\mathcal{C}}(A, B)$ and $g, g_1, g_2 \in \text{Hom}_{\mathcal{C}}(B, C)$ it holds that

$$g \circ (f_1 + f_2) = (g \circ f_1) + (g \circ f_2)$$

and

$$(g_1 + g_2) \circ f = (g_1 \circ f) + (g_2 \circ f)$$

(see [34, Ch. 1, §8, p. 28]).

In a preadditive category $\mathcal{C}$, each Hom-set $\text{Hom}_{\mathcal{C}}(A, B)$ carries the structure of an abelian group. In particular, for every pair $A, B \in \text{Ob}(\mathcal{C})$ there exists a distinguished zero morphism $0_{A,B} \in \text{Hom}_{\mathcal{C}}(A, B)$, independently of the existence of zero objects.

Construction 1.6.2. Let $\mathcal{C}$ be a preadditive category. Then applying the Hom-functor to any object $A \in \text{Ob}(\mathcal{C})$ defines a functor with values in Ab , that is

$$\text{Hom}(-, A) : \mathcal{C}^{\text{op}} \rightarrow \text{Ab}$$

$$\text{Hom}(A, -) : \mathcal{C} \rightarrow \text{Ab}.$$

Fix an arbitrary $A \in \text{Ob}(\mathcal{C})$. For each object $X \in \text{Ob}(\mathcal{C})$, the set $\text{Hom}_{\mathcal{C}}(X, A)$ is an abelian group by preadditivity, hence an object in Ab . For a morphism $f \in \text{Hom}_{\mathcal{C}}(X, Y)$,

define

$$\begin{aligned} f^* : \text{Hom}_{\mathcal{C}}(Y, A) &\rightarrow \text{Hom}_{\mathcal{C}}(X, A) \\ g &\mapsto g \circ f \end{aligned} \tag{1.6.3}$$

for any $g \in \text{Hom}_{\mathcal{C}}(Y, A)$.

Claim: (1.6.3) is a group homomorphism.

For $\varphi, \psi \in \text{Hom}_{\mathcal{C}}(Y, A)$, bilinearity of composition gives

$$\begin{aligned} f^*(\varphi + \psi) &= (\varphi + \psi) \circ f \\ &= \varphi \circ f + \psi \circ f \\ &= f^*(\varphi) + f^*(\psi), \end{aligned}$$

and similarly $f^*(0_{Y,A}) = 0_{X,A}$. Thus f^* is a morphism in Ab .

The functoriality is immediate: For $f : X \rightarrow Y$, $g : Y \rightarrow Z$, and $\varphi : Z \rightarrow A$,

$$\begin{aligned} (\text{id}_X)^*(\varphi) &= \varphi \circ \text{id}_X = \varphi \\ (g \circ f)^*(\varphi) &= \varphi \circ (g \circ f) = (\varphi \circ g) \circ f = f^*(g^*(\varphi)). \end{aligned}$$

Hence $\text{Hom}_{\mathcal{C}}(-, A)$ is a well-defined functor $\mathcal{C}^{\text{op}} \rightarrow \text{Ab}$.

The proof for $\text{Hom}_{\mathcal{C}}(A, -) : \mathcal{C} \rightarrow \text{Ab}$ is analogous (see [42, Ch. 3, §1, Prop. 3.7]).

Example 1.6.4 (Ab is preadditive). For all abelian groups A, B , the set $\text{Hom}_{\text{Ab}}(A, B)$ is itself an abelian group. For $f, g \in \text{Hom}_{\text{Ab}}(A, B)$, define

$$(f + g)(a) := f(a) + g(a), \quad (-f)(a) := -f(a), \quad 0(a) := 0.$$

Then $\text{Hom}_{\text{Ab}}(A, B)$ is an abelian group under pointwise addition. One verifies bilinearity. For $f, g \in \text{Hom}_{\text{Ab}}(A, B)$, and $h \in \text{Hom}_{\text{Ab}}(B, C)$ one has

$$h \circ (f + g) = h \circ f + h \circ g$$

because

$$h((f + g)(a)) = h(f(a) + g(a)) = h(f(a)) + h(g(a)).$$

Similarly, for $k \in \text{Hom}_{\text{Ab}}(A, B)$, and $f, g \in \text{Hom}_{\text{Ab}}(B, C)$, one has

$$(f + g) \circ k = f \circ k + g \circ k.$$

Therefore the composition is additive in both variables, making Ab preadditive (see [34, Ch. 1, §8, p. 28]).

In preadditive categories, the Hom -sets carry an abelian group structure compatible with composition. This allows one to form sums and differences of morphisms, and in particular to express relations. However, this structure is confined to the level of morphisms and does not yet impose any corresponding structure on the objects themselves.

Definition 1.6.5 (Biproduct). Let $\mathcal{C}$ be a preadditive category, and let $(A_k)_{k \in \{1, \dots, n\}}$ be a family of objects in $\mathcal{C}$. A **biproduct** of $A_1, \dots, A_n$ is an object

$$\bigoplus_{k=1}^n A_k$$

together with morphisms

$$\begin{aligned} \iota_i : A_i &\rightarrow \bigoplus_{k=1}^n A_k \\ \text{pr}_i : \bigoplus_{k=1}^n A_k &\rightarrow A_i, \end{aligned}$$

for $i \in \{1, \dots, n\}$ satisfying

$$\text{pr}_i \circ \iota_j = \begin{cases} \text{id}_{A_i}, & i = j \\ 0_{A_j, A_i}, & i \neq j \end{cases}$$

and

$$\sum_{i=1}^n \iota_i \circ \text{pr}_i = \text{id}_{\bigoplus_{i=1}^n A_i}$$

for all $i, j \in \{1, \dots, n\}$. A preadditive category admitting biproducts of all finite families of objects is said to have **finite biproducts** (see [34, Ch. 8, §2, p. 194 ff.]).

The biproduct is closely related to limits and colimits. In fact, if a category has a biproduct, it coincides with both, finite products and finite coproducts.

Lemma 1.6.6. Let $\mathcal{C}$ be a preadditive category, and let $A_1, A_2, B \in \text{Ob}(\mathcal{C})$, with $B := A_1 \oplus A_2$, and with canonical morphisms

$$\iota_i : A_i \rightarrow B, \quad \text{pr}_i : B \rightarrow A_i, \quad i \in \{1, 2\}.$$

Then $(B, (\text{pr}_i)_{i \in \{1, 2\}})$ is a product of A_1 and A_2 , and $(B, (\iota_i)_{i \in \{1, 2\}})$ is a coproduct of A_1 and A_2 (see [34, Ch. 8, §2, p. 194 ff.]).

Proof. Claim: $(B, (\text{pr}_i)_{i \in \{1, 2\}})$ is a product.

Given any $X \in \text{Ob}(\mathcal{C})$ and morphisms

$$\begin{aligned} f &: X \rightarrow A_1 \\ g &: X \rightarrow A_2 \end{aligned}$$

define

$$\langle f, g \rangle := \iota_1 \circ f + \iota_2 \circ g : X \rightarrow B.$$

Then

$$\text{pr}_1 \circ \langle f, g \rangle = (\text{pr}_1 \circ \iota_1) \circ f + (\text{pr}_1 \circ \iota_2) \circ g = f$$

and

$$\text{pr}_2 \circ \langle f, g \rangle = g.$$

If $h \in \text{Hom}_{\mathcal{C}}(X, B)$ also satisfies

$$\begin{aligned} \text{pr}_1 \circ h &= f \\ \text{pr}_2 \circ h &= g \end{aligned}$$

then by the biproduct identity,

$$\begin{aligned} h &= \text{id}_B \circ h \\ &= (\iota_1 \circ \text{pr}_1 + \iota_2 \circ \text{pr}_2) \circ h \\ &= \iota_1 \circ f + \iota_2 \circ g \\ &= \langle f, g \rangle \end{aligned}$$

making $(B, (\text{pr}_i)_{i \in \{1,2\}})$ a product.

Claim: $(B, (\iota_i)_{i \in \{1,2\}})$ is a coproduct.

Given two morphisms

$$f : A_1 \rightarrow X$$

$$g : A_2 \rightarrow X$$

define

$$[f, g] := f \circ \text{pr}_1 + g \circ \text{pr}_2 : B \rightarrow X.$$

Then

$$[f, g] \circ \iota_1 = f \circ \text{pr}_1 \circ \iota_1 + g \circ \text{pr}_2 \circ \iota_1 = f$$

and

$$[f, g] \circ \iota_2 = g.$$

If $h \in \text{Hom}_{\mathcal{C}}(B, X)$ also satisfies

$$h \circ \iota_1 = f$$

$$h \circ \iota_2 = g$$

then by the biproduct identity,

$$\begin{aligned} h &= h \circ \text{id}_B \\ &= h \circ (\iota_1 \circ \text{pr}_1 + \iota_2 \circ \text{pr}_2) \\ &= f \circ \text{pr}_1 + g \circ \text{pr}_2 \\ &= [f, g] \end{aligned}$$

making $(B, (\iota_i)_{i \in \{1,2\}})$ a coproduct (see [34, Ch. 8, §2, pp. 194 ff.]). □

The preceding lemma shows that the algebraic identities defining a biproduct recover both the universal product property and the universal coproduct property. Thus, in a preadditive category, the compatibility between finite products and coproducts can be expressed entirely in terms of addition and composition of morphisms. This motivates the study of preadditive categories in which biproducts exist for all finite families of objects.

1.6.2 Additive Categories

To obtain a categorical analogue of direct sums of modules or abelian groups, one additionally requires the existence of suitable products and coproducts. This leads to the notion of an additive category.

Definition 1.6.7 (Additive category). A preadditive category $\mathcal{C}$ is **additive** if it has a zero object and all finite biproducts (compare [42, Ch. 3, §. 2, Def. 3.17, Rem. 3.18]).

Morphisms between additive categories should preserve not only composition but also the additive structure. This motivates the following notion.

Definition 1.6.8 (Additive functor). Let $\mathcal{C}, \mathcal{D}$ be preadditive categories. A functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is **additive** if the morphisms

$$\begin{aligned} \text{Hom}_{\mathcal{C}}(A, B) &\rightarrow \text{Hom}_{\mathcal{D}}(F(A), F(B)) \\ f &\mapsto F(f) \end{aligned}$$

are group homomorphisms for every pair of objects $A, B \in \text{Ob}(\mathcal{C})$ (see [42, Ch. 3, §4, Def. 3.41]).

The definition of an additive functor can be reformulated in a number of equivalent ways. The following characterisation is particularly useful in practice.

Theorem 1.6.9 (Characterisation of additive functors). The following properties are equivalent for a functor $F : \mathcal{C} \rightarrow \mathcal{D}$ between additive categories:

1. F is additive.
2. F preserves finite biproducts, that is
 - $F(0)$ is a zero object in $\mathcal{D}$
 - for all $A_1, A_2 \in \text{Ob}(\mathcal{C})$, if

$$\left(A_1 \oplus A_2, (\iota_i), (\text{pr}_i) \right)_{i \in \{1,2\}}$$

is a biproduct in $\mathcal{C}$, then

$$\left(F(A_1 \oplus A_2), (F(\iota_i)), (F(\text{pr}_i)) \right)_{i \in \{1,2\}}$$

is a biproduct of $F(A_1)$ and $F(A_2)$ in $\mathcal{D}$

(see [34, Ch. 8, §2, Prop. 4], [42, Ch. 3, §4, Prop. 3.44]).

Proof. $\Rightarrow$: Assume F is additive, and let 0 be the zero object in $\mathcal{C}$. Then $\text{Hom}_{\mathcal{C}}(0, 0)$ is the trivial group, so $\text{id}_0 = 0_{0,0}$. Applying F yields

$$\text{id}_{F(0)} = F(\text{id}_0) = F(0_{0,0}) = 0_{F(0), F(0)}$$

since F is assumed to be additive. This implies that every morphism with domain or codomain $F(0)$ is zero. Thus, $F(0)$ is a zero object in $\mathcal{D}$.

Let

$$\left(A_1 \oplus A_2, (\iota_i), (\text{pr}_i) \right)_{i \in \{1,2\}}$$

be a biproduct in $\mathcal{C}$, then Lemma 1.6.6 holds in $\mathcal{C}$. By the functoriality of F , applying F preserves identities and compositions, and additivity gives preservation of 0 and $+$. Hence in $\mathcal{D}$

$$\begin{aligned} F(\text{pr}_i) \circ F(\iota_i) &= F(\text{pr}_i \circ \iota_i) \\ &= \begin{cases} F(\text{id}_{A_i}) & i = j \\ 0_{F(A_j), F(A_i)} & i \neq j \end{cases} \end{aligned}$$

and

$$\begin{aligned} F(\iota_1) \circ F(\text{pr}_1) + F(\iota_2) \circ F(\text{pr}_2) &= F(\iota_1 \circ \text{pr}_1) + F(\iota_2 \circ \text{pr}_2) \\ &= F((\iota_1 \circ \text{pr}_1) + (\iota_2 \circ \text{pr}_2)) \\ &= F(\text{id}_{A_1 \oplus A_2}) \\ &= \text{id}_{F(A_1 \oplus A_2)}. \end{aligned}$$

Thus, the corresponding biproduct identity holds for $\left(F(A_1 \oplus A_2), (F(\iota_i)), (F(\text{pr}_i)) \right)$. By Lemma 1.6.6, that is a biproduct of $F(A_1)$ and $F(A_2)$.

$\Leftarrow$: Assume F preserves 0 and binary biproducts.

Define the diagonal

$$\blacktriangle_{A_1} : A_1 \rightarrow A_1 \oplus A_1$$

as the unique morphism such that

$$\text{pr}_i \circ \blacktriangle_{A_1} = \text{id}_{A_1}$$

and the codiagonal

$$\blacktriangledown_{A_2} : A_2 \oplus A_2 \rightarrow A_2$$

as the unique morphism such that

$$\blacktriangledown_{A_2} \circ \iota_i = \text{id}_{A_2}.$$

The addition on $\text{Hom}(A_1, A_2)$ is defined as

$$f + g := \blacktriangledown_{A_2} \circ (f \oplus g) \circ \blacktriangle_{A_1} \quad (1.6.10)$$

where

$$f \oplus g : A_1 \oplus A_1 \rightarrow A_2 \oplus A_2$$

is the unique morphism with

$$(f \oplus g) \circ \iota_1 = \iota_1 \circ f,$$

$$(f \oplus g) \circ \iota_2 = \iota_2 \circ g.$$

Since F preserves the biproduct structure maps, it preserves the canonical morphisms

$$F(\blacktriangle_{A_1}) = \blacktriangle_{F(A_1)}$$

$$F(\blacktriangledown_{A_2}) = \blacktriangledown_{F(A_2)}$$

$$F(f \oplus g) = F(f) \oplus F(g).$$

Applying F to (1.6.10) gives

$$\begin{aligned} F(f + g) &= F(\nabla_{A_2} \circ (f \oplus g) \circ \Delta_{A_1}) \\ &= F(\nabla_{A_2}) \circ F(f \oplus g) \circ F(\Delta_{A_1}) \\ &= \nabla_{F(A_2)} \circ (F(f) \oplus F(g)) \circ \Delta_{F(A_1)} \\ &= F(f) + F(g). \end{aligned}$$

Thus F preserves addition. In particular,

$$F(0_{A_1, A_2}) = F(0_{A_1, A_2} + 0_{A_1, A_2}) = F(0_{A_1, A_2}) + F(0_{A_1, A_2}),$$

and cancellation yields

$$F(0_{A_1, A_2}) = 0_{F(A_1), F(A_2)}.$$

Hence every induced map on Hom-groups is a group homomorphism.

Since $F(0_C) = 0_D$, it follows that $F(0_{A_1, A_2}) = 0_{F(A_1), F(A_2)}$. Therefore each F_{A_1, A_2} is a group homomorphism and F is additive (see [34, Ch. 8, §2, Prop. 4], [42, Ch. 3, §4, Prop. 3.44]). $\square$

Example 1.6.11 (Ab is additive). By Example 1.6.4, Ab is preadditive. The zero object in Ab is the trivial group 0 by Example 1.1.7.

Claim: Ab admits finite biproducts.

By Example 1.5.17, Ab admits small coproducts, and by Example 1.5.5, all small products. For $A_1, A_2 \in \text{Ob}(\text{Ab})$, consider the binary product $A_1 \times A_2$. By Lemma 1.6.6 it suffices to show that the biproduct identities hold.

Define injections

$$\iota_1(a) := (a, 0), \quad \iota_2(b) := (0, b)$$

and projections

$$\text{pr}_1(a, b) = a, \quad \text{pr}_2(a, b) = b.$$

Note that these are precisely the injections and projections from the coproduct and product respectively. Then

$$\text{pr}_i \circ \iota_j = \begin{cases} \text{id}_{A_i} & i = j, \\ 0_{A_j, A_i} & i \neq j. \end{cases}$$

and

$$\begin{aligned} (\iota_1 \circ \text{pr}_1 + \iota_2 \circ \text{pr}_2)(a, b) &= (\iota_1 \circ \text{pr}_1)(a, b) + (\iota_2 \circ \text{pr}_2)(a, b) \\ &= (a, 0) + (0, b) \\ &= (a, b). \end{aligned}$$

Hence (BI1) and (BI2) are satisfied, and $A_1 \times A_2$ is a biproduct. Thus, $\mathbf{Ab}$ has all finite biproducts, since they can be computed as iteration of binary biproducts. Therefore $\mathbf{Ab}$ is an additive category (see [19, Ch. 1, Ex. E]).

While additivity allows one to form sums and differences of morphisms, it does not provide a way to isolate or measure the vanishing of a morphism.

1.6.3 Abelian Categories

The existence of biproducts alone does not ensure that subobjects and quotient objects behave as they do in categories of abelian groups and modules. To recover the familiar behaviour, one introduces kernels and cokernels, which ultimately lead to the notion of an abelian category.

Definition 1.6.12 (Kernel, Cokernel). Let $\mathcal{C}$ be a category with a zero object 0 , and let $f : A \rightarrow B$ be a morphism. Denote by $0_{A,B} : A \rightarrow B$ the unique morphism factoring through the zero object.

If the equaliser $(E, \text{eq}(f, 0_{A,B}))$ exists, then E is called the **kernel** of f , and is denoted by K_f . The corresponding equaliser morphism is denoted by

$$\ker(f) := \text{eq}(f, 0_{A,B}) : K_f \rightarrow A.$$

Dually, if the coequaliser $(E^{\text{co}}, \text{coeq}(f, 0_{A,B}))$ exists, then E^{co} is called the **cokernel** of f , and is denoted by K_f^{co} . The corresponding coequaliser morphism is denoted by

$$\text{coker}(f) := \text{coeq}(f, 0_{A,B}) : B \rightarrow K_f^{\text{co}}$$

(see [8, Ch. 6, §3, Def. 6.3.14]).

Kernels and cokernels need not exist in arbitrary categories, as they rely on the existence of a zero object as well as equalisers and coequalisers respectively. Whenever kernels and cokernels exist, one can form two canonical factorisations of a morphism via its image and coimage.

Definition 1.6.13 (Image, Coimage). Let $\mathcal{C}$ be a category that has kernels and cokernels, and let $f \in \text{Hom}_{\mathcal{C}}(A, B)$.

The **image object of f** is the kernel object of $\text{coker}(f)$ and is denoted by

$$I_f := K_{\text{coker}(f)}.$$

The corresponding kernel morphism is called the **image morphism** of f and is denoted by

$$\text{im}(f) := \ker(\text{coker}(f)) : I_f \rightarrow B.$$

Dually, the **coimage object of f** is the cokernel object of $\ker(f)$, and is denoted by

$$I_f^{\text{co}} := K_{\ker(f)}^{\text{co}}.$$

The corresponding cokernel morphism is called the **coimage morphism of f** and is denoted by

$$\text{coim}(f) := \text{coker}(\ker(f)) : A \rightarrow I_f^{\text{co}}$$

(see [42, Ch. 1, §2, Def. 1.17]).

The image and the coimage give rise to the canonical factorisation

$$A \twoheadrightarrow I_f^{\text{co}} \xrightarrow{\gamma_f} I_f \hookrightarrow B.$$

There is a canonical morphism

$$\gamma_f : I_f^{\text{co}} \rightarrow I_f$$

is the canonical morphism from the coimage to the image.

Proposition 1.6.14 (Canonical image-coimage factorisation). Let $\mathcal{C}$ be a category with kernels and cokernels, and let $f : A \rightarrow B$ be a morphism. Let

$$\text{coim}(f) : A \rightarrow I_f^{\text{co}}$$

be the cokernel of $\ker(f)$, and let

$$\mathrm{im}(f) : I_f \rightarrow B$$

be the kernel of $\mathrm{coker}(f)$. Then there exists a unique morphism

$$\gamma_f : I_f^{\mathrm{co}} \rightarrow I_f$$

such that

$$f = \mathrm{im}(f) \circ \gamma_f \circ \mathrm{coim}(f).$$

The morphism γ_f is called the canonical morphism from the coimage to the image (see [52, Ch. 12, §3, Lemma 12.3.12]).

Proof. Let $\ker(f) : K_f \rightarrow A$ be the kernel of f , and let

$$\mathrm{coim}(f) : A \rightarrow I_f^{\mathrm{co}}$$

be its cokernel. Since

$$f \circ \ker(f) = 0,$$

the universal property of the cokernel yields a unique morphism

$$\bar{f} : I_f^{\mathrm{co}} \rightarrow B$$

such that

$$\bar{f} \circ \mathrm{coim}(f) = f.$$

Let $\mathrm{coker}(f) : B \rightarrow K_f^{\mathrm{co}}$ be the cokernel of f , and let

$$\mathrm{im}(f) : I_f \rightarrow B$$

be its kernel. Since

$$\mathrm{coker}(f) \circ \bar{f} \circ \mathrm{coim}(f) = \mathrm{coker}(f) \circ f = 0,$$

and $\mathrm{coim}(f)$ is an epimorphism (being a cokernel), it follows that

$$\mathrm{coker}(f) \circ \bar{f} = 0.$$

Therefore, by the universal property of the kernel, there exists a unique morphism

$$\gamma_f : I_f^{\text{co}} \rightarrow I_f$$

such that

$$\text{im}(f) \circ \gamma_f = \bar{f}.$$

Combining these two equalities gives

$$f = \bar{f} \circ \text{coim}(f) = \text{im}(f) \circ \gamma_f \circ \text{coim}(f).$$

The uniqueness of γ_f follows from the universal property of the kernel applied to $\bar{f}$ (see [52, Ch. 12, §3, Lemma 12.3.12], [6, Ch. 1, §5, Thm 1.5.5]). $\square$

The preceding factorisation distinguishes epimorphic and monomorphic parts on an arbitrary morphism. Before investigating when the canonical morphism γ_f between them is an isomorphism, it is useful to consider the more fundamental questions of when monomorphisms and epimorphisms together already characterise isomorphisms. This leads to the notion of a balanced category.

Definition 1.6.15 (Normal, Conormal). A monomorphism is **normal** if it is the kernel of some morphism.

An epimorphism is **conormal** if it is the cokernel of some morphism. A category is normal if every monomorphism is normal, and a category is conormal if every epimorphism is conormal.

Finally, a category is **binormal** if it is normal and conormal (see [19, Ch. 2, p. 35]).

The notions of normal and conormal morphisms express whether monomorphisms and epimorphisms arise from kernels and cokernels, respectively.

Lemma 1.6.16. Let $\mathcal{C}$ be a category with kernels and cokernels, and let $m : A \hookrightarrow B$ be a normal monomorphism. Then m is a kernel of $\text{coker}(m)$. Dually, let $e : A \twoheadrightarrow B$ be a conormal epimorphism. Then e is a cokernel of $\text{ker}(e)$ (see [13, Thm. 2.2.3]).

Proof. Since m is normal, there exists a morphism $f : B \rightarrow C$ such that m is a kernel of f . Let

$$\text{coker}(m) : B \rightarrow K_m^{\text{co}}$$

be the cokernel of m . Since $f \circ m = 0$, the universal property of the cokernel yields a unique morphism

$$\bar{f} : K_m^{\text{co}} \rightarrow C$$

such that

$$f = \bar{f} \circ \text{coker}(m).$$

Let $t : T \rightarrow B$ satisfy

$$\text{coker}(m) \circ t = 0.$$

Then

$$f \circ t = \bar{f} \circ \text{coker}(m) \circ t = 0.$$

Since m is a kernel of f , there exists a unique morphism

$$u : T \rightarrow A$$

such that

$$m \circ u = t.$$

Thus m satisfies the universal property of the kernel of $\text{coker}(m)$.

The remaining proof is dual (see [13, Thm. 2.2.3]). □

This leads to the notion of abelian categories.

Definition 1.6.17 (Abelian category). A preadditive category $\mathcal{C}$ is called **abelian** if the following properties are satisfied:

- A1 it has a zero object
- A2 it has all binary biproducts
- A3 it has all kernels and cokernels
- A4 it is binormal

(see [34, Ch. 8, §3, p. 198]).

Example 1.6.18 ($\mathbf{Ab}$ is abelian). By Example 1.1.7, $\mathbf{Ab}$ has a zero object, hence it satisfies (A1), and by Example 1.6.11, $\mathbf{Ab}$ has all binary biproducts, hence satisfies (A2). Since kernels are equalisers, and cokernels are coequalisers, by Example 1.5.10 and Example 1.5.21, $\mathbf{Ab}$ admits all kernels and cokernels.

It remains to show that $\mathbf{Ab}$ is binormal. Let $f \in \mathbf{Hom}_{\mathbf{Ab}}(A, B)$ be a monomorphism. Since monomorphisms are injective in $\mathbf{Ab}$, $f(A) \subseteq B$. Consider the quotient morphism

$$q : B \rightarrow B/f(A).$$

Then

$$q \circ f = 0.$$

If $h \in \mathbf{Hom}_{\mathbf{Ab}}(T, B)$ satisfies

$$q \circ h = 0,$$

then $h(T) \subseteq f(A)$. Since

$$f : A \rightarrow f(A)$$

is an isomorphism, h factors uniquely through f . Thus $f : A \rightarrow B$ satisfies the universal property of the kernel of q . Hence f is a kernel of q , and therefore every monomorphism in $\mathbf{Ab}$ is normal.

Suppose $g : A \twoheadrightarrow B$ and $h : A \rightarrow T$ with

$$\ker(g) : K_g \hookrightarrow A, \quad h \circ \ker(g) = 0.$$

Define

$$\bar{h} : B \rightarrow T, \quad \bar{h}(g(a)) := h(a).$$

This is well-defined, as h vanishes on K_g . Since epimorphisms in $\mathbf{Ab}$ are surjective, every element of B has the form $g(a)$. The morphism $\bar{h}$ is unique because g is epimorphic. Thus g is the cokernel of

$$\ker(g) : K_g \hookrightarrow A.$$

Hence g is the cokernel of $\ker(g)$. Thus every epimorphism in $\mathbf{Ab}$ is conormal. As $\mathbf{Ab}$ is normal and conormal, it is binormal, and $\mathbf{Ab}$ satisfies (A4).

Hence $\mathbf{Ab}$ is an abelian category (compare [34, Ch. 8, §3, p. 199]).

The preceding example shows that $\mathbf{Ab}$ satisfies the axioms of an abelian category. The next result establishes one of the fundamental consequences of the axioms: in every abelian category, the canonical comparison morphism from the coimage to the image is an isomorphism.

Proposition 1.6.19 (Image-Coimage Theorem). Let $f : A \rightarrow B$ be a morphism in an abelian category. Then

$$A \twoheadrightarrow I_f^{\text{co}} \xrightarrow{\sim} I_f \hookrightarrow B$$

is the canonical epimorphism-monomorphism factorisation (see [34, Ch. 8, §3, Prop. 1]).

Proof. Let

$$\text{coker}(f) : B \rightarrow K_f^{\text{co}}$$

be the cokernel of f , and let

$$\text{im}(f) : I_f \hookrightarrow B$$

be its kernel. Since

$$\text{coker}(f) \circ f = 0,$$

the universal property of the kernel yields a unique morphism

$$e : A \rightarrow I_f$$

such that

$$f = \text{im}(f) \circ e.$$

Claim: e is an epimorphism.

Let

$$h : I_f \rightarrow T$$

satisfy

$$h \circ e = 0,$$

and let

$$k : M \hookrightarrow I_f$$

be the kernel of h . Since $h \circ e = 0$, the universal property of the kernel gives a morphism

$$e' : A \rightarrow M$$

such that

$$e = k \circ e'.$$

Therefore

$$f = \text{im}(f) \circ k \circ e'.$$

The composite

$$\text{im}(f) \circ k : M \hookrightarrow B$$

is a monomorphism. Since $\mathcal{C}$ is abelian, it is normal, so there exists a morphism

$$p : B \rightarrow P$$

such that $\text{im}(f) \circ k$ is a kernel of p . Since

$$p \circ f = p \circ \text{im}(f) \circ k \circ e' = 0,$$

the universal property of the cokernel $\text{coker}(f)$ yields a morphism

$$r : K_f^{\text{co}} \rightarrow P$$

such that

$$p = r \circ \text{coker}(f).$$

Hence

$$p \circ \text{im}(f) = r \circ \text{coker}(f) \circ \text{im}(f) = 0,$$

because $\text{im}(f)$ is the kernel of $\text{coker}(f)$. Since $\text{im}(f) \circ k$ is a kernel of p , there exists a unique morphism

$$s : I_f \rightarrow M$$

such that

$$\text{im}(f) \circ k \circ s = \text{im}(f).$$

As $\text{im}(f)$ is monomorphic, it follows that

$$k \circ s = \text{id}_{I_f}.$$

Thus k is both monomorphic and split epimorphic, hence an isomorphism. Since

$$h \circ k = 0,$$

one obtains $h = 0$.

Therefore every morphism annihilating e is zero. If

$$u, v : I_f \rightarrow T$$

satisfy

$$u \circ e = v \circ e,$$

then

$$(u - v) \circ e = 0,$$

and hence $u - v = 0$. Thus e is an epimorphism.

Because $\text{im}(f)$ is monomorphic and

$$f = \text{im}(f) \circ e,$$

a morphism

$$t : T \rightarrow A$$

satisfies

$$f \circ t = 0$$

if and only if

$$e \circ t = 0.$$

Consequently,

$$\ker(f) : K_f \rightarrow A$$

is also a kernel of e .

Since e is an epimorphism in an abelian category, it is conormal. By the preceding lemma, e is therefore a cokernel of its kernel. Hence e is a cokernel of

$$\ker(f) : K_f \rightarrow A.$$

By definition,

$$\text{coim}(f)A \rightarrow I_f^{\text{co}}$$

is also a kernel of $\ker(f)$. Since both e and $\text{coim}(f)$ are cokernels of $\ker(f)$, unique-

ness of cokernels yields a unique isomorphism

$$\gamma_f : I_f^{\text{co}} \xrightarrow{\sim} I_f$$

such that

$$e = \gamma_f \circ \text{coim}(f).$$

Therefore

$$f = \text{im}(f) \circ e = \text{im}(f) \circ \gamma_f \circ \text{coim}(f),$$

and γ_f is precisely the canonical comparison morphism

(see [34, Ch. 8, §3, Prop. 1]). □

Abelian categories share many of the familiar structural properties of the category of abelian groups. The following results are used repeatedly throughout the development of homological algebra.

Proposition 1.6.20. Every abelian category is balanced (see [46, Ch. 2, §1, Thm. 2.1.2]).

Proof. Let $f : A \rightarrow B$ be both monomorphic and epimorphic. Since $\mathcal{C}$ is abelian, f is normal. Hence there exists a morphism $g : B \rightarrow C$ such that f is a kernel of g . Since

$$g \circ f = 0$$

and f is an epimorphism, it follows that $g = 0_{B,C}$. Therefore f is a kernel of the zero morphism

$$0_{B,C} : B \rightarrow C.$$

The identity id_B is also a kernel of $0_{B,C}$. By uniqueness of kernels, there exists an isomorphism

$$A \xrightarrow{\sim} B$$

whose composite with id_B is f . Hence f is an isomorphism (see [46, Ch. 2, §1, Thm. 2.1.2]). □

There exists an equivalent characterisation which is often more convenient in practice. Rather than checking normality and conormality directly, it suffices to compare images and coimages.

Theorem 1.6.21. An additive category is abelian if and only if it has all kernels and cokernels, and for every morphism $f : A \rightarrow B$, the canonical morphism

$$\gamma_f : I_f^{\text{co}} \rightarrow I_f$$

is an isomorphism (see [52, Definition 12.5.1]).

Proof. Let $\mathcal{C}$ be an additive category that admits all kernels and cokernels, and suppose that for every morphism $f : A \rightarrow B$, the canonical morphism

$$\gamma_f : I_f^{\text{co}} \rightarrow I_f$$

is an isomorphism.

Since $\mathcal{C}$ is additive, (A1) and (A2) are satisfied, and by the assumption of the existence of kernels and cokernels, (A3) is satisfied.

It remains to show that $\mathcal{C}$ is binormal. Let $m : A \rightarrow B$ be a monomorphism. Then its kernel is the zero morphism

$$\ker(m) : 0 \rightarrow A.$$

Hence its coimage morphism

$$\text{coim}(m) = A \rightarrow I_m^{\text{co}}$$

is a cokernel of $0 \rightarrow A$ and therefore an isomorphism.

Consider the canonical factorisation

$$A \xrightarrow{\text{coim}(m)} I_m^{\text{co}} \xrightarrow{\gamma_m} I_m \xrightarrow{\text{im}(m)} B,$$

so that

$$m = \text{im}(m) \circ \gamma_m \circ \text{coim}(m).$$

By assumption, the canonical comparison morphism

$$\gamma_m : I_m^{\text{co}} \xrightarrow{\sim} I_m$$

is also an isomorphism. Consequently,

$$\gamma_m \circ \text{coim}(m) : A \xrightarrow{\sim} I_m$$

is an isomorphism. Since

$$\text{im}(m) : I_m \rightarrow B$$

is the kernel of $\text{coker}(m)$, and precomposition of a kernel of $\text{coker}(m)$. Thus every monomorphism is normal.

Dually, let $e : A \rightarrow B$ be an epimorphism. Then its cokernel is the zero morphism

$$\text{coker}(e) : B \rightarrow 0.$$

Hence its image morphism

$$\text{im}(e) = I_e \rightarrow B$$

is a kernel of $B \rightarrow 0$ and therefore an isomorphism.

Consider the canonical factorisation

$$A \xrightarrow{\text{coim}(e)} I_e^{\text{co}} \xrightarrow{\gamma_e} I_e \xrightarrow{\text{im}_e} B.$$

By assumption, γ_e is an isomorphism. Hence

$$\text{im}(e) \circ \gamma_e : I_e^{\text{co}} \xrightarrow{\sim} B$$

is an isomorphism, and

$$e = (\text{im}(e) \circ \gamma_e) \circ \text{coim}(e).$$

Since

$$\text{coim}(e) : A \rightarrow I_e^{\text{co}}$$

is the cokernel of $\ker(e)$, and postcomposition of a cokernel with an isomorphism is again a cokernel, e is the cokernel of its kernel. Thus every epimorphism is conormal.

Therefore $\mathcal{C}$ is binormal, satisfying (A4), and therefore $\mathcal{C}$ is an abelian category.

Conversely, suppose that $\mathcal{C}$ is abelian. Then it satisfies (A1)–(A4). In particular, it

admits all kernels and cokernels. Proposition 1.6.19 shows that for every morphism $f : A \rightarrow B$ the canonical comparison morphism

$$\gamma_f : I_f^{\text{co}} \xrightarrow{\sim} I_f$$

is an isomorphism. □

In particular, a monomorphism whose cokernel vanishes must already be invertible.

Lemma 1.6.22. Let $m : A \hookrightarrow B$ be a monomorphism in an abelian category. Then the following are equivalent:

1. m is an isomorphism,
2. $K_m^{\text{co}} \cong 0$

(see [19, Ch. 2, §1, Thm 2.17, Thm. 2.17*]).

Proof. " $\Leftarrow$ ": Since $K_m^{\text{co}} \cong 0$, every morphism $\varphi : B \rightarrow T$ with

$$\varphi \circ m = 0$$

factors uniquely through the zero object. Hence $\varphi = 0$. Let $g_1, g_2 : B \rightarrow T$ satisfy

$$g_1 \circ m = g_2 \circ m.$$

Then

$$(g_1 - g_2) \circ m = 0.$$

Hence $g_1 - g_2 = 0$, making $g_1 = g_2$. Thus m is an epimorphism. Since m was a monomorphism by assumption, and as every abelian category is balanced (Proposition 1.6.20), m is an isomorphism.

" $\Rightarrow$ ": If m is an isomorphism, then it is in particular an epimorphism. Hence its cokernel is the zero object by the universal property of the cokernel (see [19, Ch. 2, §1, Thm 2.17, Thm. 2.17*]). □

By the preceding lemma one can characterise isomorphisms in terms of monomorphisms with trivial cokernel. The following reformulation is useful when one

works with quotient objects. The preceding lemma justifies the interpretation of cokernels as quotient objects: a quotient is trivial precisely when the corresponding subobject equals the whole object (up to isomorphism). This leads to the following definition.

Definition 1.6.23 (Quotient object). Let $\mathcal{C}$ be an abelian category, let $A \in \text{Ob}(\mathcal{C})$, and let $\iota : B \hookrightarrow A$ be a monomorphism representing a subobject of A . The **quotient object** is defined as

$$A/B := K_{\iota}^{\text{co}},$$

together with the canonical morphism $\pi : A \rightarrow A/B$ (see [8, Ch. 6, §7, Def. 6.7.15]).

Different representatives of the same subobject yield canonically isomorphic quotient objects. By construction, quotient objects satisfy the universal property of cokernels, and thereby of coequalisers.

Having introduced quotient objects, one can consider constructions relating different subobjects of a fixed object. In an abelian category, intersections and unions of subobjects are obtained from pullbacks and pushouts.

Definition 1.6.24 (Intersection, union). Let $A, B \hookrightarrow C$ be subobjects in a category with pullbacks. Their intersection $A \cap B$ is defined as the pullback

$$\begin{array}{ccc} A \cap B & \xrightarrow{p_B} & B \\ \downarrow p_A & & \downarrow \iota_B \\ A & \xhookrightarrow{\iota_A} & C. \end{array}$$

If the category is abelian, the **sum**, also called **union**, $A \cup B$ is defined as the pushout of the intersection diagram

$$\begin{array}{ccc} A \cap B & \longrightarrow & B \\ \downarrow & & \downarrow \\ A & \longrightarrow & A + B. \end{array}$$

In an abelian category this pushout is canonically isomorphic to the smallest subobject of C containing both A and B , usually denoted $A + B$ (see [8, Ch. 5, §7, p. 126]).

With quotients, intersections, and sums available, the classical isomorphism theorems admit a categorical formulation.

Theorem 1.6.25 (Isomorphism theorem). Let $\mathcal{C}$ be an abelian category, and let $A, B \hookrightarrow C$ be subobjects in $\mathcal{C}$. Then there is a canonical isomorphism

$$\frac{A + B}{B} \cong \frac{A}{A \cap B}$$

(see [6, Ch. 1, §10, Lemma 1.10.7]).

Proof. Let

$$\iota_A : A \rightarrow C, \quad \iota_B : B \hookrightarrow C$$

denote the inclusions, and let

$$A + B \hookrightarrow C$$

be the image of the morphism

$$[\iota_A, \iota_B] : A \oplus B \rightarrow C.$$

Then both A and B factor through $A + B$. Let

$$\kappa_A : A \rightarrow A + B, \quad \kappa_B : B \rightarrow A + B$$

be the induced morphisms, and let

$$q : A + B \rightarrow (A + B)/B$$

be the cokernel of κ_B . Since

$$q \circ \kappa_B = 0,$$

it follows that

$$q \circ \kappa_A \circ p_A = q \circ \kappa_B \circ p_B = 0.$$

Therefore

$$q \circ \kappa_A : A \rightarrow (A + B)/B$$

vanishes on $A \cap B$. Hence, by the universal property of the cokernel of

$$p_A : A \cap B \rightarrow A,$$

there exists a unique morphism

$$\bar{\alpha} : A/(A \cap B) \rightarrow (A + B)/B$$

such that

$$\bar{\alpha} \circ \pi = q \circ \kappa_A,$$

where $\pi : A \rightarrow A/(A \cap B)$ is the canonical quotient morphism.

It remains to show that $\bar{\alpha}$ is an isomorphism. Consider the morphism

$$q \circ [\kappa_A, \kappa_B].$$

Since $q \circ \kappa_B = 0$, this morphism is $q \circ \kappa_A$ on the A -summand and zero on the B -summand. Since $A + B$ is defined as the image of $[\iota_A, \iota_B]$, the induced morphism

$$[\kappa_A, \kappa_B] : A \oplus B \rightarrow A + B$$

is the canonical epimorphism onto the image, and therefore an epimorphism. Moreover, q is an epimorphism because it is a cokernel. One shows that $q \circ \kappa_A$ is an epimorphism.

Let

$$a, b : (A + B)/B \rightarrow T$$

satisfy

$$a \circ q \circ \kappa_A = b \circ q \circ \kappa_A.$$

It is to show that $a = b$.

The morphisms

$$a \circ q \circ [\kappa_A, \kappa_B], \quad b \circ q \circ [\kappa_A, \kappa_B]$$

are equal, because on the A -summand they agree by assumption, and on the B -summand, they are both zero since

$$q \circ \kappa_B = 0.$$

Since

$$[\kappa_A, \kappa_B] : A \oplus B \rightarrow A + B$$

is an epimorphism, it follows that

$$a \circ q = b \circ q.$$

Finally, q is itself an epimorphism, yielding $a = b$. Therefore

$$q \circ \kappa_A : A \rightarrow (A + B)/B$$

is an epimorphism.

Since

$$q \circ \kappa_A = \bar{\alpha} \circ \pi$$

and $q \circ \kappa_A$ is an epimorphism, it follows that $\bar{\alpha}$ is an epimorphism.

One computes the kernel of $q \circ \kappa_A$.

Let $j : A + B \hookrightarrow C$ denote the inclusion. Since $A \cap B$ is the pullback,

$$\iota_A \circ p_A = \iota_B \circ p_B.$$

Using $\iota_A = j \circ \kappa_A$ and $\iota_B = j \circ \kappa_B$, one obtains

$$j \circ \kappa_A \circ p_A = j \circ \kappa_B \circ p_B.$$

Since j is monomorphic,

$$\kappa_A \circ p_A = \kappa_B \circ p_B.$$

Therefore

$$q \circ \kappa_A \circ p_A = q \circ \kappa_B \circ p_B = 0,$$

because q is the cokernel of κ_B . Thus, $p_A : A \cap B \rightarrow A$ is annihilated by $q \circ \kappa_A$.

Suppose that $t : T \rightarrow A$ satisfies

$$q \circ \kappa_A \circ t = 0. \tag{1.6.26}$$

Since κ_B is a monomorphism and $\mathcal{C}$ is abelian, κ_B is the kernel of its cokernel q . Hence the equality (1.6.26) and the universal property of the kernel yield a unique morphism

$$s : T \rightarrow B$$

such that

$$\kappa_A \circ t = \kappa_B \circ s.$$

The universal property of the pullback gives a unique morphism

$$u : T \rightarrow A \cap B$$

such that

$$p_A \circ u = t,$$

which is exactly the universal property of the kernel. Thus,

$$p_A : A \cap B \hookrightarrow A$$

is a kernel of $q \circ \kappa_A$. Because the category is abelian, every epimorphism is the cokernel of its kernel. Therefore $q \circ \kappa_A$ is a cokernel of p_A . But

$$\pi : A \rightarrow A/(A \cap B)$$

is also a cokernel of p_A . Hence, by the uniqueness of cokernels, there is a unique isomorphism

$$\bar{\alpha} : A/(A \cap B) \xrightarrow{\sim} (A + B)/B$$

such that

$$\bar{\alpha} \circ \pi = q \circ \kappa_A.$$

(see [6, Ch. 1, §10, Lemma 1.10.7]). □

The preceding constructions describe how individual subobjects and quotient objects interact. In practice, however, one is often interested not merely in single morphisms but in entire chains of morphisms. This leads naturally to the notion of sequences.

1.7 Sequences and Exactness

Abelian categories provide an abstract setting in which kernels, cokernels, images, and coimages behave as they do in the category of abelian groups. This makes it

possible to speak not only about individual morphisms, but about chains of morphisms in which the image of one map is precisely the kernel of the next. Such chains are called exact sequences. Exact sequences are one of the main bookkeeping devices of homological algebra. They encode when a morphism is an epimorphism, a monomorphism, and most importantly, when an object can be regarded as an extension of one object by another. The most basic instance is a short exact sequence.

Definition 1.7.1 (Short exact sequence). A short sequence of morphisms

$$0 \rightarrow A \rightarrow B \rightarrow C$$

is **left-exact** if $A \rightarrow B$ is the kernel of $B \rightarrow C$.

Dually, a sequence

$$A \rightarrow B \rightarrow C \rightarrow 0$$

is **right-exact** if $B \rightarrow C$ is the cokernel of $A \rightarrow B$.

Finally, a sequence

$$0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$$

is **exact** if it is both right- and left-exact (see [8, Ch. 6, §3, Def. 6.3.14]).

A sequence

$$A^{n-1} \rightarrow A^n \rightarrow A^{n+1}$$

is exact at n if

$$\text{im}(A^{n-1} \rightarrow A^n) = \ker(A^n \rightarrow A^{n+1})$$

(see [35, Ch. 1, §2, p. 11]).

Short exact sequences describe how an object may be obtained as an extension of one object by another. In general, the middle object need not be determined by the outer terms alone. The simplest situation occurs when the extension is trivial, that is, when the middle object decomposes as a direct sum of the outer objects.

Definition 1.7.2 (Split-exact sequence). A short exact sequence

$$0 \rightarrow A \xrightarrow{\iota} A \oplus B \xrightarrow{\text{pr}} B \rightarrow 0$$

is called **split-exact** where ι and pr are the canonical inclusion and projection of the biproduct $A \oplus B$ (see [35, Ch. 1, §4, Prop. 4.3]).

Exactness of a sequence is a condition that involves all morphisms simultaneously. In practice, however, exactness is verified locally: a sequence is exact if and only if it is exact at every position.

Remark 1.7.3 (Exactness in abelian categories). In abelian categories, monomorphisms are kernels, and epimorphisms are cokernels. Thus, if

$$0 \rightarrow A \xrightarrow{f} B$$

is exact in an abelian category, it implies that f is a monomorphism. If

$$B \xrightarrow{g} A \rightarrow 0$$

is exact in an abelian category, then g is an epimorphism (compare [22, Ch. 1, §1, Def. 1.5]).

The exactness conditions encountered in short and split-exact sequences extend naturally to longer chains of morphisms. Such sequences arise throughout homological algebra, for example as the outcome of applying functors to short exact sequences, or from the cohomology of complexes. This motivates the study of long exact sequences.

Definition 1.7.4 (Long exact sequence). A long exact sequence is a diagram of the form $\dots \longrightarrow A_{n-1} \xrightarrow{d_{n-1}} A_n \xrightarrow{d_n} A_{n+1} \xrightarrow{d_{n+1}} \dots$ such that

$$\text{im}(d_{n-1}) = \ker(d_n)$$

(see [35, Ch. 1, §2, p.11]).

Since these sequences often occur after applying a functor to a sequence of objects and morphisms, it becomes important to understand which aspects of exactness are preserved by a given functor.

Definition 1.7.5 (Exact functor). An additive, covariant functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is called

1. **half-exact** if $F(A) \rightarrow F(B) \rightarrow F(C)$ is exact
2. **left-exact** if $0 \rightarrow F(A) \rightarrow F(B) \rightarrow F(C)$ is exact
3. **right-exact** if $F(A) \rightarrow F(B) \rightarrow F(C) \rightarrow 0$ is exact
4. **exact** if $0 \rightarrow F(A) \rightarrow F(B) \rightarrow F(C) \rightarrow 0$ is exact

for all short exact sequences

$$0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$$

in $\mathcal{C}$. The notion of half-exact functors in this thesis is distinct from the notion of topological half-exact functors (see [34, Ch. 8, §3, p. 201]).

To identify naturally occurring examples of left and right-exact functors, it is useful to study adjoint functors. Their interaction with limits and colimits leads directly to the exactness properties.

Theorem 1.7.6. Let $\mathcal{C}$ and $\mathcal{D}$ be abelian categories and let $F : \mathcal{C} \rightarrow \mathcal{D}$ be an additive functor. Then the following statements hold:

1. If F is right-adjoint, then F preserves limits. In particular, F preserves kernels.
2. If F is right-adjoint, F is left-exact.
3. If F is left-adjoint, then F preserves colimits. In particular, F preserves cokernels.
4. If F is left-adjoint, then F is right-exact.

In particular, for a small category I , and if $\mathcal{C}$ admits all I -indexed limits, then the limit functor $\lim_I : [I, \mathcal{C}] \rightarrow \mathcal{C}$ is left-exact. Dually, if $\mathcal{C}$ admits all I -indexed colimits, then $\operatorname{colim}_I : [I, \mathcal{C}] \rightarrow \mathcal{C}$ is right-exact (compare [8, Ch. 7, §3, Thm. 7.3.1], [34, Ch. 5, §5, Thm 1], [45, Ch. 2, §3, Prop. 2.3.5]).

Proof. Let

$$L : \mathcal{C} \rightleftarrows \mathcal{D} : R$$

be an adjunction with $L \dashv R$, and let R and L be additive.

1. **Claim:** R preserves limits.

Let $\mathbf{D} : I \rightarrow \mathcal{D}$ be a diagram, and assume that the limit exists in $\mathcal{D}$. It is to show that

$$R\left(\lim_{i \in I} \mathbf{D}(i)\right) \cong \lim_{i \in I} R(\mathbf{D}(i)).$$

For every $X \in \text{Ob}(\mathcal{C})$ there exist natural bijections

$$\text{Hom}_{\mathcal{C}}\left(X, R\left(\lim_{i \in I} \mathbf{D}(i)\right)\right) \cong \text{Hom}_{\mathcal{D}}\left(L(X), \lim_{i \in I} \mathbf{D}(i)\right)$$

by adjunction.

By the definition of limits,

$$\text{Hom}_{\mathcal{D}}\left(L(X), \lim_{i \in I} \mathbf{D}(i)\right) \cong \text{Cone}\left(L(X), \mathbf{D}\right).$$

A cone from $L(X)$ to $\mathbf{D}$ is the same as a family of morphisms

$$\left(L(X) \rightarrow \mathbf{D}(i)\right)_{i \in I}$$

compatible with the diagram $\mathbf{D}$. By adjunction, this is the same as a compatible family of morphisms

$$\left(X \rightarrow R(\mathbf{D}(i))\right)_{i \in I}.$$

Thus

$$\text{Cone}\left(L(X), \mathbf{D}\right) \cong \text{Cone}\left(X, R \circ \mathbf{D}\right).$$

Cones from X to $R \circ \mathbf{D}$ are exactly morphisms from X into the limit of $R \circ \mathbf{D}$. Therefore $R\left(\lim_{i \in I} \mathbf{D}(i)\right)$ satisfies the universal property of $\lim (R \circ \mathbf{D})$. Hence

$$R\left(\lim_{i \in I} \mathbf{D}(i)\right) \cong \lim_{i \in I} R(\mathbf{D}(i)).$$

Thus R preserves limits. Furthermore, as kernels are equalisers by definition, R preserves kernels.

2. **Claim:** R is left-exact.

Consider a short left-exact sequence

$$0 \rightarrow A_1 \xrightarrow{f} A_2 \xrightarrow{g} A_3.$$

Then $f : A_1 \rightarrow A_2$ is a kernel of g . Applying R yields

$$R(0) \rightarrow R(A_1) \xrightarrow{R(f)} R(A_2) \xrightarrow{R(g)} R(A_3).$$

Since R is additive, $R(0)$ is a zero object in $\mathcal{D}$. By the first statement, R preserves kernels. Thus

$$R(f) : R(A_1) \rightarrow R(A_2)$$

is a kernel of $R(g)$. Hence

$$0 \rightarrow R(A_1) \xrightarrow{R(f)} R(A_2) \xrightarrow{R(g)} R(A_3)$$

is left-exact.

3. **Claim:** L preserves colimits.

The proof works dually to the proof of (1). Let $\mathbf{D} : I \rightarrow \mathcal{C}$ be a diagram, and assume that colimits exist in $\mathcal{C}$. It is to show that

$$L(\operatorname{colim}_{i \in I} \mathbf{D}(i)) \cong \operatorname{colim}_{i \in I} L(\mathbf{D}(i)).$$

For every $B \in \operatorname{Ob}(\mathcal{D})$ there exist natural bijections

$$\operatorname{Hom}_{\mathcal{D}} \left(L(\operatorname{colim}_{i \in I} \mathbf{D}(i)), B \right) \cong \operatorname{Hom}_{\mathcal{C}} (\operatorname{colim}_{i \in I} \mathbf{D}(i), R(B))$$

by adjunction.

By the definition of colimits,

$$\operatorname{Hom}_{\mathcal{C}} (\operatorname{colim}_{i \in I} \mathbf{D}(i), R(B)) \cong \operatorname{Cocone} (\mathbf{D}, R(B)).$$

A cocone from D to $R(B)$ is the same as a family of morphisms

$$(\mathbf{D}(i) \rightarrow R(B))_{i \in I}$$

compatible with the diagram $\mathbf{D}$. By adjunction, this is the same as a compatible family of morphisms

$$\left(L(\mathbf{D}(i)) \rightarrow B \right)_{i \in I}.$$

Thus

$$\text{Cocone}(\mathbf{D}, R(B)) \cong \text{Cocone}(L \circ \mathbf{D}, B).$$

Cocones $L \circ \mathbf{D}$ to B are exactly morphisms from the colimit $\text{colim}(L \circ \mathbf{D})$ into B . Therefore $L(\text{colim}_{i \in I} \mathbf{D}(i))$ satisfies the universal property of $\text{colim}(L \circ \mathbf{D})$. Hence

$$L(\text{colim}_{i \in I} \mathbf{D}(i)) \cong \text{colim}_{i \in I} L(\mathbf{D}(i)).$$

Thus L preserves colimits. Since cokernels are particular colimits, it follows that L preserves cokernels.

4. **Claim:** L is right-exact.

Consider a short right-exact sequence

$$B_1 \xrightarrow{f} B_2 \xrightarrow{g} B_3 \rightarrow 0.$$

Applying L yields

$$L(B_1) \xrightarrow{L(f)} L(B_2) \xrightarrow{L(g)} L(B_3) \rightarrow L(0).$$

As L is additive, $L(0) \cong 0$ in $\mathcal{D}$. Moreover,

$$g : B_2 \rightarrow B_3$$

is a cokernel of f . By the preceding part, L preserves cokernels. Therefore

$$L(g) : L(B_2) \rightarrow L(B_3)$$

is a cokernel of $L(f)$. Since L is additive, $L(0)$ is a zero object in $\mathcal{D}$. Hence

$$L(B_1) \xrightarrow{L(f)} L(B_2) \xrightarrow{L(g)} L(B_3) \rightarrow 0$$

is right-exact.

Claim: The limit functor is right-adjoint to the diagonal functor $\mathfrak{D}$.

By definition of the diagonal functor,

$$\mathfrak{D}(B)(i) = B, \quad \mathfrak{D}(B)(\alpha) = \text{id}_B.$$

Let

$$\lim : [I, \mathcal{C}] \rightarrow \mathcal{C}$$

be the limit functor. In order to prove that $\mathfrak{D} \dashv \lim$, one constructs a natural bijection

$$\text{Hom}_{[I, \mathcal{C}]}(\mathfrak{D}(B), \mathbf{D}) \cong \text{Hom}_{\mathcal{C}}(B, \lim \mathbf{D}).$$

On the left-hand side, a morphism $\mathfrak{D}(B) \rightarrow \mathbf{D}$ in $[I, \mathcal{C}]$ is a natural transformation. Since $\mathfrak{D}(B)$ is constant, this consists of morphisms

$$f_i : B \rightarrow \mathbf{D}(i)$$

such that for every morphism $\alpha : i \rightarrow j$,

$$\mathbf{D}(\alpha) \circ f_i = f_j,$$

which is precisely a cone from B to $\mathbf{D}$. Therefore

$$\text{Hom}_{[I, \mathcal{C}]}(\mathfrak{D}(B), \mathbf{D}) \cong \text{Cone}(B, \mathbf{D}).$$

By the universal property of the limit,

$$\text{Cone}(B, \mathbf{D}) \cong \text{Hom}_{\mathcal{C}}(B, \lim \mathbf{D}).$$

Combining the two bijections yields

$$\text{Hom}_{[I, \mathcal{C}]}(\mathfrak{D}(B), \mathbf{D}) \cong \text{Hom}_{\mathcal{C}}(B, \lim \mathbf{D}),$$

which is natural in B and in $\mathbf{D}$. Hence $\mathfrak{D} \dashv \lim$. Thus $\lim$ is a right-adjoint functor, and hence preserves left-exact sequences.

Dually, if $\mathcal{C}$ has I -indexed colimits, then

$$\text{colim} : [I, \mathcal{C}] \rightarrow \mathcal{C}$$

is left-adjoint to the diagonal functor. Indeed,

$$\text{Hom}_{\mathcal{C}}(\text{colim } \mathbf{D}, B) \cong \text{Cocone}(\mathbf{D}, B) \cong \text{Hom}_{[I, \mathcal{C}]}(\mathbf{D}, \mathfrak{D}(B))$$

(compare [8, Ch. 7, §3, Thm. 7.3.1], [34, Ch. 5, §5, Thm 1], [45, Ch. 2, §3, Prop. 2.3.5]). $\square$

Having established important sources of left and right-exact functors, one can investigate which categorical structures and constructions are preserved by left-exact functors.

Theorem 1.7.7. Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a functor between abelian categories. If F is left-exact, then it preserves all finite limits (see [52, Prop. 4.23.2]).

Proof. In an abelian category, finite limits may be constructed as equalisers of finite products. Since F is additive by definition, it preserves finite biproducts by Theorem 1.6.9, hence finite products. Moreover, the equaliser of two morphisms $f, g \in \text{Hom}_{\mathcal{C}}(A, B)$ is the kernel of $f - g$. Since F is left-exact, it preserves kernels. Thus

$$F(\text{eq}(f, g)) \cong F(\ker(f - g)) \cong \ker(F(f) - F(g)) \cong \text{eq}(F(f), F(g))$$

(see [52, Prop. 4.23.2]). $\square$

Thus, right-adjoint functors preserve finite limits. Informally, the preceding two theorems give

$$\text{right-adjoint} \Rightarrow \text{preserves limits} \Rightarrow \text{preserves finite limits} \Leftrightarrow \text{left-exact}.$$

Since for Theorem 1.7.7, the dual statement is also true, that is, an additive right-exact functor between abelian categories preserves all finite colimits, one has

$$\text{left-adjoint} \Rightarrow \text{preserves colimits} \Rightarrow \text{preserves finite colimits} \Leftrightarrow \text{right-exact}.$$

Since finite products are finite limits, every left-exact functor preserves finite products. In an abelian category, finite products and finite coproducts coincide as biproducts. Consequently, left-exact functors are compatible with the additive structure of abelian categories.

The preceding results describe how exactness behaves under functors between abelian categories. In many applications, however, exactness is encoded not merely in individual sequences but in commutative diagrams relating several sequences simultaneously. The following lemma provides a fundamental tool for extracting information from such diagrams.

The Snake Lemma A word of caution: The notation convention of kernels and cokernels will be ignored in this proof, to make it more readable.

Theorem 1.7.8 (Snake lemma). Let $\mathcal{C}$ be an abelian category. Consider a commutative diagram

$$\begin{array}{ccccccccc}
 & & 0 & & & & & & \\
 & & \downarrow & & & & & & \\
 A & \longrightarrow & B & \longrightarrow & C & \longrightarrow & D & \longrightarrow & E \\
 \downarrow a & & \downarrow b & & \downarrow c & & \downarrow d & & \downarrow e \\
 A' & \longrightarrow & B' & \longrightarrow & C' & \longrightarrow & D' & \longrightarrow & E' \\
 & & & & & & & & \downarrow \\
 & & & & & & & & 0
 \end{array} \tag{1.7.9}$$

with exact rows. Then there exists a connecting morphism

$$\partial : \ker(d) \rightarrow \operatorname{coker}(b)$$

such that the sequence

$$\begin{array}{ccccccc}
 0 & \longrightarrow & \ker(b) & \xrightarrow{f} & \ker(c) & \xrightarrow{g} & \ker(d) \\
 & & & & & & \searrow \\
 & & & & & & \xrightarrow{\partial} \\
 & & \operatorname{coker}(b) & \xrightarrow{f'} & \operatorname{coker}(c) & \xrightarrow{g'} & \operatorname{coker}(d) \longrightarrow 0
 \end{array}$$

is exact (see [28, Ch. 1, §1, Lemma 1.6]).

Proof. **Argument 1:**

Consider the exact commutative diagram

$$\begin{array}{ccccccc}
 A & \longrightarrow & B & \longrightarrow & C & \longrightarrow & D \\
 \downarrow a & & \downarrow b & & \downarrow c & & \downarrow d \\
 A' & \longrightarrow & B' & \longrightarrow & C' & \longrightarrow & D' \\
 \downarrow & & & & & & \\
 0 & & & & & &
 \end{array} .$$

The induced sequence

$$\ker(b) \rightarrow \ker(c) \rightarrow \ker(d)$$

is exact.

Set

$$E := \operatorname{im}(B \rightarrow C) = \ker(C \rightarrow D)$$

and

$$E' := \operatorname{im}(B' \rightarrow C') = \ker(C' \rightarrow D').$$

Commutativity induces a morphism

$$e : E \rightarrow E'.$$

Let $E \hookrightarrow C$ and $E' \hookrightarrow C'$ denote the subobjects

$$E = \operatorname{im}(B \rightarrow C) = \ker(C \rightarrow D), \quad E' = \operatorname{im}(B' \rightarrow C') = \ker(C' \rightarrow D').$$

By commutativity, c induces a morphism $e : E \rightarrow E'$.

Consider the diagram

$$\begin{array}{ccccccc}
 0 & \longrightarrow & E & \longrightarrow & C & \longrightarrow & D \\
 \downarrow & & \downarrow e & & \downarrow c & & \downarrow d \\
 0 & \longrightarrow & E' & \longrightarrow & C' & \longrightarrow & D'
 \end{array}$$

where the rows are exact. Then

$$0 \rightarrow \ker(e) \rightarrow \ker(c) \rightarrow \ker(d) \tag{1.7.10}$$

is exact. Indeed, the induced morphism

$$\ker(e) \rightarrow \ker(c)$$

is a kernel of the induced morphism $\ker(c) \rightarrow D$:

Then the composite

$$\ker(e) \rightarrow \ker(c) \rightarrow D$$

is zero, since the composite

$$E \rightarrow C \rightarrow D$$

is zero. Now let

$$f : T \rightarrow \ker(c)$$

satisfy

$$(\ker(c) \rightarrow D) \circ f = 0.$$

After composing with the kernel inclusion $\ker(c) \hookrightarrow C$, the resulting morphism $T \rightarrow C$ factors uniquely through

$$E = \ker(C \rightarrow D).$$

Let $\tilde{f} : T \rightarrow E$ denote the induced morphism. Since f factors through $\ker(c)$, commutativity gives

$$e \circ \tilde{f} = 0.$$

Thus $\tilde{f}$ factors uniquely through $\ker(e)$. Therefore

$$\ker(e) \rightarrow \ker(c)$$

has the universal property of the kernel, proving the exactness of

$$0 \rightarrow \ker(e) \rightarrow \ker(c) \rightarrow \ker(d).$$

By commutativity, the morphism $\ker(c) \rightarrow D$ factors uniquely through the kernel inclusion $\ker(d) \hookrightarrow D$, yielding the induced morphism

$$\ker(c) \rightarrow \ker(d).$$

Since $\ker(d) \hookrightarrow D$ is a monomorphism, a morphism into $\ker(c)$ is annihilated by $\ker(c) \rightarrow \ker(d)$ if and only if it is annihilated by the composite $\ker(c) \rightarrow D$. Consequently, $\ker(e) \rightarrow \ker(c)$ is also the kernel of

$$\ker(c) \rightarrow \ker(d).$$

It remains to prove that the induced morphism

$$\ker(b) \rightarrow \ker(e)$$

is an epimorphism.

Consider the commutative diagram

$$\begin{array}{ccccccc} A & \longrightarrow & B & \longrightarrow & E & \longrightarrow & 0 \\ \downarrow a & & \downarrow b & & \downarrow e & & \\ A' & \longrightarrow & B' & \longrightarrow & E' & \longrightarrow & 0 \\ \downarrow & & & & & & \\ 0 & & & & & & \end{array}$$

with exact rows. Applying the dual of the kernel argument proved above gives an exact sequence

$$\operatorname{coker}(a) \rightarrow \operatorname{coker}(b) \rightarrow \operatorname{coker}(e) \rightarrow \operatorname{coker}(0 \rightarrow 0).$$

Since the column

$$A \xrightarrow{a} A' \rightarrow 0$$

is exact, a is an epimorphism and hence

$$\operatorname{coker}(a) = 0.$$

Moreover,

$$\operatorname{coker}(0 \rightarrow 0) = 0.$$

Consequently,

$$0 \rightarrow \operatorname{coker}(b) \rightarrow \operatorname{coker}(e) \rightarrow 0$$

is exact, and therefore the canonical morphism

$$\operatorname{coker}(b) \xrightarrow{\sim} \operatorname{coker}(e) \quad (1.7.11)$$

is an isomorphism. Now consider the commutative diagram

$$\begin{array}{ccccccc} A' & \longrightarrow & B' & \longrightarrow & E' & \longrightarrow & 0 \\ \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \operatorname{coker}(b) & \xrightarrow{\sim} & \operatorname{coker}(e) & \longrightarrow & 0 \end{array}$$

The kernels of the three vertical morphisms are respectively

$$A', \quad \operatorname{im}(b), \quad \operatorname{im}(e).$$

Applying the first part of Argument 1 to this diagram shows that

$$A' \rightarrow \operatorname{im}(b) \rightarrow \operatorname{im}(e) \quad (1.7.12)$$

is exact. Let

$$b = B \xrightarrow{b'} \operatorname{im}(b) \hookrightarrow B'$$

and

$$e = E \xrightarrow{e'} \operatorname{im}(e) \hookrightarrow E'$$

be the canonical image factorisations. Then one obtains a commutative diagram with exact rows

$$\begin{array}{ccccccc} A & \longrightarrow & B & \longrightarrow & E & \longrightarrow & 0 \\ \downarrow a & & \downarrow b' & & \downarrow e' & & \\ A' & \longrightarrow & \operatorname{im}(b) & \longrightarrow & \operatorname{im}(e) & \longrightarrow & 0 \\ \downarrow & & \downarrow & & \downarrow & & \\ 0 & & 0 & & 0 & & \end{array}$$

Then e' is a cokernel of the induced morphism

$$\ker(b') \rightarrow E.$$

The composite

$$\ker(b') \rightarrow E \xrightarrow{e'} \operatorname{im}(e)$$

is zero by commutativity. Let

$$q : E \rightarrow Q$$

be a morphism which vanishes on $\ker(b')$. Since b' is a cokernel of its kernel in the abelian category $\mathcal{C}$, the composite

$$B \rightarrow E \xrightarrow{q} Q$$

factors uniquely through

$$b' : B \twoheadrightarrow \operatorname{im}(b),$$

yielding a morphism

$$\tilde{q} : \operatorname{im}(b) \rightarrow Q.$$

Let

$$u : A \rightarrow B, \quad p : B \rightarrow E, \quad r : A' \rightarrow \operatorname{im}(b)$$

denote the corresponding morphisms. By commutativity,

$$r \circ a = b' \circ u.$$

Therefore

$$\begin{aligned} \tilde{q} \circ r \circ a &= \tilde{q} \circ b' \circ u \\ &= q \circ p \circ u \\ &= 0, \end{aligned}$$

since $A \rightarrow B \rightarrow E$ is zero by exactness of the upper row. As a is an epimorphism, it follows that

$$\tilde{q} \circ r = 0.$$

By exactness of (1.7.12), the morphism $\operatorname{im}(b) \rightarrow \operatorname{im}(e)$ is a cokernel of

$$r : A' \rightarrow \operatorname{im}(b).$$

Hence $\tilde{q}$ factors uniquely through

$$\text{im}(b) \rightarrow \text{im}(e).$$

Since

$$e' : E \twoheadrightarrow \text{im}(e)$$

is an epimorphism, this factorisation is unique as a factorisation of q through e' . Thus e' is a cokernel of

$$\ker(b') \rightarrow E.$$

Consequently,

$$\text{im}(\ker(b') \rightarrow E) = \ker(e').$$

Under the canonical identifications

$$\ker(b') \cong \ker(b), \quad \ker(e') \cong \ker(e),$$

the morphism

$$\ker(b') \rightarrow \ker(e')$$

induced by $B \rightarrow E$ identifies with the canonical morphism

$$\ker(b) \rightarrow \ker(e).$$

Since e' is a cokernel of $\ker(b') \rightarrow E$, one has

$$\text{im}(\ker(b') \rightarrow E) = \ker(e').$$

Hence the induced morphism

$$\ker(b') \rightarrow \ker(e')$$

is an epimorphism, and therefore so is

$$\ker(b) \rightarrow \ker(e).$$

Combining this with (1.7.10), one obtains the exact sequence

$$\ker(b) \rightarrow \ker(c) \rightarrow \ker(d).$$

The dual argument shows that, for every exact commutative diagram

$$\begin{array}{ccccccc} & & & & 0 & & \\ & & & & \downarrow & & \\ A & \longrightarrow & B & \longrightarrow & C & \longrightarrow & D \\ \downarrow a & & \downarrow b & & \downarrow c & & \downarrow d \\ A' & \longrightarrow & B' & \longrightarrow & C' & \longrightarrow & D' \end{array}$$

the induced sequence

$$\operatorname{coker}(a) \rightarrow \operatorname{coker}(b) \rightarrow \operatorname{coker}(c)$$

is exact.

Argument 2:

Now consider the exact commutative Diagram (1.7.9). Since the lower row is exact, one has

$$\ker(D' \rightarrow E') = \operatorname{im}(C' \rightarrow D').$$

Hence the morphism $C' \rightarrow D'$ admits its canonical image factorisation

$$C' \xrightarrow{f} \ker(D' \rightarrow E') \rightarrow D',$$

where f is an epimorphism.

Claim: $\ker(C \rightarrow D') \rightarrow \ker(d)$ is an epimorphism, and $\operatorname{coker}(b) \rightarrow \operatorname{coker}(B \rightarrow C')$ is a monomorphism.

Consider the morphism

$$f : C' \rightarrow \ker(D' \rightarrow E'),$$

and consider the exact commutative diagram

$$\begin{array}{ccccccccc}
 A & \longrightarrow & B & \longrightarrow & C & \longrightarrow & D & \longrightarrow & E \\
 \downarrow & & \downarrow & & \downarrow f \circ c & & \downarrow d & & \downarrow e \\
 0 & \longrightarrow & 0 & \longrightarrow & \ker(D' \rightarrow E') & \longrightarrow & D' & \longrightarrow & E' \\
 \downarrow & & \downarrow & & & & & & \\
 0 & & 0 & & & & & &
 \end{array}$$

Applying Argument 1 to the relevant four-term subdiagram gives the exact sequence

$$B \rightarrow \ker(C \rightarrow D) \rightarrow \ker(d) \rightarrow 0.$$

Dually, applying the cokernel version of Argument 1 gives the exact sequence

$$0 \rightarrow \operatorname{coker}(b) \rightarrow \operatorname{coker}(B \rightarrow C') \rightarrow D'.$$

The second part of the claim follows from similar arguments.

Claim: There exists a unique map

$$\partial : \ker(d) \rightarrow \operatorname{coker}(b)$$

such that the two composites

$$\begin{aligned}
 &\ker(C \rightarrow D') \rightarrow C \rightarrow C' \rightarrow \operatorname{coker}(B \rightarrow C') \\
 &\ker(C \rightarrow D') \rightarrow \ker(d) \xrightarrow{\partial} \operatorname{coker}(b) \rightarrow \operatorname{coker}(B \rightarrow C')
 \end{aligned}$$

are the same.

The first part of the previous claim shows that

$$\ker(C \rightarrow D') \rightarrow \ker(d)$$

is an epimorphism, while the second part shows that

$$\operatorname{coker}(b) \rightarrow \operatorname{coker}(B \rightarrow C')$$

is a monomorphism. The composite

$$\ker(C \rightarrow D') \rightarrow C \rightarrow C' \rightarrow \operatorname{coker}(B \rightarrow C')$$

vanishes on the image of

$$B \rightarrow \ker(C \rightarrow D'),$$

since

$$B \rightarrow C \rightarrow C'$$

factors through $B \rightarrow C'$, which becomes zero in $\operatorname{coker}(B \rightarrow C')$. Hence it factors uniquely through the epimorphism

$$\ker(C \rightarrow D') \twoheadrightarrow \ker(d).$$

The composite

$$\ker(C \rightarrow D') \rightarrow C \rightarrow C' \rightarrow \operatorname{coker}(B \rightarrow C')$$

vanishes on the image of

$$B \rightarrow \ker(C \rightarrow D'),$$

since the composite

$$B \rightarrow C \rightarrow C'$$

factors through $B \rightarrow C'$, which becomes zero in $\operatorname{coker}(B \rightarrow C')$.

Since

$$B \rightarrow \ker(C \rightarrow D') \rightarrow \ker(d) \rightarrow 0$$

is exact, the morphism

$$\ker(C \rightarrow D') \twoheadrightarrow \ker(d)$$

is a cokernel of

$$B \rightarrow \ker(C \rightarrow D').$$

Hence there exists a unique morphism

$$\bar{u} : \ker(d) \rightarrow \operatorname{coker}(B \rightarrow C')$$

such that

$$\bar{u} \circ (\ker(C \rightarrow D') \rightarrow \ker(d))$$

equals the composite

$$\ker(C \rightarrow D') \rightarrow C \rightarrow C' \rightarrow \operatorname{coker}(B \rightarrow C').$$

The composite

$$\ker(d) \xrightarrow{\bar{u}} \operatorname{coker}(B \rightarrow C') \rightarrow D'$$

is zero. Indeed, after precomposition with the epimorphism

$$\ker(C \rightarrow D') \rightarrow C \rightarrow D',$$

which is zero by the definition of $\ker(C \rightarrow D')$.

Since

$$0 \rightarrow \operatorname{coker}(b) \rightarrow \operatorname{coker}(B \rightarrow C') \rightarrow D'$$

is exact, the morphism

$$\operatorname{coker}(b) \rightarrow \operatorname{coker}(B \rightarrow C')$$

is a kernel of

$$\operatorname{coker}(B \rightarrow C') \rightarrow D'.$$

Therefore $\bar{u}$ factors uniquely through

$$\operatorname{coker}(b) \rightarrow \operatorname{coker}(B \rightarrow C'),$$

yielding a unique morphism

$$\partial : \ker(d) \rightarrow \operatorname{coker}(b),$$

which satisfies the required commutativity.

Claim: The six-term sequence

$$\ker(b) \rightarrow \ker(c) \rightarrow \ker(d) \xrightarrow{\partial} \operatorname{coker}(b) \rightarrow \operatorname{coker}(c) \rightarrow \operatorname{coker}(d)$$

is exact.

By Argument 1 and its dual statement, it suffices to prove that

$$\ker(c) \rightarrow \ker(d) \rightarrow \operatorname{coker}(b)$$

is exact.

Consider the diagram

$$\begin{array}{ccccccc}
 B & \longrightarrow & \ker(C \rightarrow D') & \longrightarrow & \ker(d) & \longrightarrow & 0 \\
 \downarrow \text{id} & & \downarrow & & \downarrow & & \downarrow \\
 B & \longrightarrow & C' & \longrightarrow & \text{coker}(B \rightarrow C') & \longrightarrow & 0 \\
 \downarrow & & & & & & \\
 0 & & & & & &
 \end{array}$$

Applying argument 1 yields the exactness of

$$\ker(c) \rightarrow \ker(d) \rightarrow \text{coker}(B \rightarrow C').$$

Since

$$\text{coker}(b) \rightarrow \text{coker}(B \rightarrow C')$$

is a monomorphism, the kernel of

$$\ker(d) \rightarrow \text{coker}(b)$$

coincides with the kernel of the composite

$$\ker(d) \rightarrow \text{coker}(B \rightarrow C'),$$

and therefore

$$\text{im}(\ker(c) \rightarrow \ker(d)) = \ker(\partial).$$

This proved the exactness at $\ker(d)$ (see [28, Ch. 1, §1, pp. 1 ff.]). □

The snake lemma is one of the fundamental tools of homologic algebra. It allows one to construct long exact sequences from short exact sequences and is used to derive the long exact cohomology associated to a short exact sequence of sheaves.

Another important result of this kind is the Five Lemma, which allows one to deduce that a morphism is an isomorphism from the surrounding morphisms in a commutative diagram of exact sequences.

Lemma 1.7.13 (Five Lemma). Let

$$\begin{array}{ccccccccc} A_1 & \longrightarrow & A_2 & \longrightarrow & A_3 & \longrightarrow & A_4 & \longrightarrow & A_5 \\ \downarrow f_1 & & \downarrow f_2 & & \downarrow f_3 & & \downarrow f_4 & & \downarrow f_5 \\ B_1 & \longrightarrow & B_2 & \longrightarrow & B_3 & \longrightarrow & B_4 & \longrightarrow & B_5 \end{array}$$

be a commutative diagram in an abelian category, where the rows are exact. If f_1, f_2, f_4 and f_5 are isomorphisms, then f_3 is an isomorphism (see [32, Ch. 8, §3, Lemma 8.3.13], [28, Ch. 1, §1, Lemma 1.7]).

Proof. This is a direct consequence of the snake lemma (Theorem 1.7.8) applied to the exact rows (see [32, Ch. 8, §3, Lemma 8.3.13], [28, Ch. 1, §1, Lemma 1.7]). $\square$

While every abelian category admits finite limits and finite colimits, the existence and exactness properties of infinite products, coproducts, and directed colimits vary considerably between examples.

Definition 1.7.14 (Classification of abelian categories). An abelian category $\mathcal{C}$ can be classified as follows:

AB3 $\mathcal{C}$ has all small coproducts.

*AB3** $\mathcal{C}$ has all small products.

AB4 $\mathcal{C}$ is *AB3*, and arbitrary coproducts are exact.

*AB4** $\mathcal{C}$ is *AB3**, and arbitrary products are exact.

AB5 $\mathcal{C}$ is *AB3*, and filtered colimits are exact.

*AB5** $\mathcal{C}$ is *AB3**, and filtered limits are exact

(compare [54, Appendix A, p. 426 ff.]).

Remark 1.7.15. The above classification yields the following implications:

$$AB5 \Rightarrow AB4 \Rightarrow AB3,$$

and dually

$$AB5^* \Rightarrow AB4^* \Rightarrow AB3^*.$$

As coproducts are filtered colimits of finite coproducts, and dually, as products are filtered limits of finite products, this yields the first implication in both cases. The last implication follows straight from the definition of $AB4$, and $AB4^*$ respectively.

A central example of a category that is $AB5$ is $R - \text{Mod}$ (see [54, Appendix A, Ex. A.4.7]).

The axioms $AB3-AB5$ describe increasingly strong compatibility properties between exact sequences and infinite categorical constructions. Categories satisfying these conditions occur naturally in particular in the theory of sheaves.

Definition 1.7.16 (Grothendieck category). An abelian category $\mathcal{C}$ is a **Grothendieck category** if it has the following properties:

GC1 $\mathcal{C}$ has all small colimits,

GC2 $\mathcal{C}$ is $AB5$,

GC3 $\mathcal{C}$ has a generator

(see [32, Ch. 8, p. 167]).

Example 1.7.17 (Ab is $AB5$, $AB4^*$, and Grothendieck). By Example 1.5.24, Ab is cocomplete. Hence Ab has all small coproducts, making it $AB3$. By Example 1.5.5, Ab has all small products, making it $AB3^*$.

Claim: Filtered colimits are exact.

Consider a filtered diagram of short exact sequences

$$0 \rightarrow A_i \rightarrow B_i \rightarrow C_i \rightarrow 0,$$

indexed by a filtered category I . Let $[c_i] \in \text{colim}_I C_i$. Since $B_i \twoheadrightarrow C_i$ is surjective, one can choose a $b_i \in B_i$ mapping to c_i , making

$$\text{colim}_I B_i \rightarrow \text{colim}_I C_i$$

surjective.

Assume that $[a_i] \in \text{colim}_I A_i$ maps to 0 in B_i . By filteredness, there exists a $j \geq i$ such that the image of a_i in B_j is zero. Since $A_j \rightarrow B_j$ is injective, the image of a_i

in A_j is already 0. Hence $[a_i] = 0$ in $\text{colim}_I A_i$. Thus

$$\text{colim}_I A_i \rightarrow \text{colim}_I B_i$$

is injective.

It remains to show exactness at $\text{colim}_I B_i$. Let

$$[b_i] \in \text{colim}_I B_i$$

lie in the kernel of the morphism

$$\text{colim}_I B_i \rightarrow \text{colim}_I C_i.$$

Then the image of $g_i(b_i)$ represents the zero element in $\text{colim}_I C_i$. By the definition of the filtered colimits, there exists an index $j \geq i$ such that the image b_j of b_i in B_j satisfies $g_j(b_j) = 0$. Since

$$0 \rightarrow A_j \xrightarrow{f_j} B_j \xrightarrow{g_j} C_j \rightarrow 0$$

is exact, it follows that

$$b_j \in \ker(g_j) = \text{im}(f_j).$$

Hence there exists $a_j \in A_j$ such that

$$f_j(a_j) = b_j.$$

Therefore $[b_i] = [b_j]$ lies in the image of

$$\text{colim}_I A_i \rightarrow \text{colim}_I B_i.$$

Conversely, every element in the image of

$$\text{colim}_I A_i \rightarrow \text{colim}_I B_i$$

maps to zero in $\text{colim}_I C_i$, since

$$g_i \circ f_i = 0$$

for every $i \in I$. Thus

$$\ker(\operatorname{colim}_I B_i \rightarrow \operatorname{colim}_I C_i) = \operatorname{im}(\operatorname{colim}_I A_i \rightarrow \operatorname{colim}_I B_i).$$

Together,

$$0 \rightarrow \operatorname{colim}_I A_i \rightarrow \operatorname{colim}_I B_i \rightarrow \operatorname{colim}_I C_i \rightarrow 0$$

is exact. Thus, $\mathbf{Ab}$ is $AB5$, satisfying (GC2).

Claim: $\mathbf{Ab}$ has all small colimits.

This follows from the cocompleteness of $\mathbf{Ab}$, as seen in Example 1.5.24. Thus, $\mathbf{Ab}$ satisfies (GC1).

Claim: $\mathbf{Ab}$ has a generator.

Let $A \in \operatorname{Ob}(\mathbf{Ab})$, and consider the morphism

$$\Phi : \prod_{a \in A} \mathbb{Z} \rightarrow A, \quad e_a \mapsto a,$$

where e_a is the element that has 1 in the a -th coordinate, and 0 in all other coordinates. Since every element of A is one of the chosen generators, it is surjective. Thus, $\mathbf{Ab}$ has a generator, hence satisfies (GC3).

Therefore, $\mathbf{Ab}$ is a Grothendieck category (compare [54, Appendix A, §4, §5]).

Claim: All small products in $\mathbf{Ab}$ are exact.

Consider a family of short exact sequences

$$0 \rightarrow A_i \xrightarrow{f_i} B_i \xrightarrow{g_i} C_i \rightarrow 0.$$

Suppose

$$(a_i)_{i \in I} \in \prod_{i \in I} A_i$$

maps to zero. Then

$$f_i(a_i) = 0$$

for all i . Since each f_i is injective, $a_i = 0$ for every i . Thus

$$(a_i)_{i \in I} = 0,$$

making $\prod_i f_i$ injective.

Let

$$(b_i)_{i \in I} \in \prod_{i \in I} B_i$$

lie in the kernel of $\prod_i g_i$. Then

$$g_i(b_i) = 0$$

for every $i \in I$. Since

$$\ker(g_i) = \text{im}(f_i),$$

for each $i \in I$ there exists $a_i \in A_i$ with

$$f_i(a_i) = b_i.$$

Thus

$$(a_i)_{i \in I} \in \prod_i A_i$$

maps to $(b_i)_{i \in I}$. Hence

$$\ker \left(\prod_{i \in I} g_i \right) \cong \text{im} \left(\prod_{i \in I} f_i \right).$$

Let

$$(c_i)_{i \in I} \in \prod_i C_i.$$

Since every g_i is surjective, one can choose $b_i \in B_i$ with

$$g_i(b_i) = c_i.$$

Then $(b_i)_{i \in I}$ maps to $(c_i)_{i \in I}$. Hence $\prod_i g_i$ is surjective. Therefore

$$0 \rightarrow \prod_{i \in I} A_i \xrightarrow{\prod_i f_i} \prod_{i \in I} B_i \xrightarrow{\prod_i g_i} \prod_{i \in I} C_i \rightarrow 0$$

is exact. Hence Ab is $AB4^*$ (compare [56]).

The concepts introduced in this chapter provide the categorical framework for the remainder of this thesis. In particular, abelian categories, exact sequences, exact

functors, diagram results, and Grothendieck categories furnish the tools required for the study of sheaf-theoretic constructions and their cohomology.

2 Sheaf-theoretic Foundations

This chapter develops the sheaf-theoretic language on which the remainder of the thesis is based. After introducing presheaves, Grothendieck topologies, and sheaves on sites, global sections are studied from a categorical perspective. The chapter concludes with torsors, which provide a geometric interpretation of the obstruction to extending local solutions to global ones.

2.1 Presheaves

Sheaf theory is based on the principle that global information can often be recovered from compatible local data. Before formulating this principle precisely, one first studies arbitrary assignments of data to objects of a category. These assignments are modelled by presheaves.

Definition 2.1.1 (Presheaf). Let $\mathcal{C}$ and $\mathcal{D}$ be categories. A $\mathcal{D}$ -valued presheaf F is a functor

$$F : \mathcal{C}^{\text{op}} \rightarrow \mathcal{D}.$$

For $A \in \text{Ob}(\mathcal{C})$ the object $F(A)$ is called the **object of sections over A** .

Given a morphism $f : A \rightarrow B$ in $\mathcal{C}$, the corresponding morphism

$$f^{\text{op}} : B \rightarrow A$$

in $\mathcal{C}^{\text{op}}$ induces a morphism

$$F(f^{\text{op}}) : F(B) \rightarrow F(A),$$

called the **restriction morphism** (see [3, Ch. 14, §4, Def. 14.4.1]).

Remark 2.1.2. When $\mathcal{C} = \text{Open}(X)$ and f is the inclusion, it is customary to write

$$F(f^{\text{op}})(s) = s|_A.$$

The difference between contravariant functors and presheaves is mainly one of interpretation: a presheaf is viewed as assigning data to the objects of the underlying category together with restriction morphisms induced by morphisms of that category. In the classical setting, the domain category is $\text{Open}(X)$.

Remark 2.1.3 (The category of presheaves). Together with natural transformations, presheaves form a category, denoted by

$$\text{PSh}(\mathcal{C}, \mathcal{D}) := [\mathcal{C}^{\text{op}}, \mathcal{D}].$$

If the domain is $\text{Open}(X)$, the category of $\mathcal{D}$ -valued presheaves on X is denoted by $\text{PSh}(X, \mathcal{D})$, or simply by $\text{PSh}(X)$, if the codomain is understood.

Since many constructions in sheaf theory are first performed at the level of presheaves, it is useful to understand which categorical properties of the target category are inherited by the category of presheaves.

Theorem 2.1.4. Let $\mathcal{D}$ be complete. Then $\text{PSh}(\mathcal{C}, \mathcal{D})$ is complete, and limits are computed componentwise. Dually, if $\mathcal{D}$ is cocomplete, then $\text{PSh}(\mathcal{C}, \mathcal{D})$ is cocomplete, and colimits are computed componentwise (see [34, Ch. 5, §3, p. 116]).

Proof. Limits in a functor category are computed componentwise. If $\mathcal{D}$ is complete, the required limits exist in $\mathcal{D}$ at every object $A \in \text{Ob}(\mathcal{C})$.

Consider a diagram $\mathbf{D} : I \rightarrow \text{PSh}(\mathcal{C}, \mathcal{D})$. Given a morphism $f : A \rightarrow B$ in $\mathcal{C}$, the universal property of $\lim_{i \in I} \mathbf{D}(i)(A)$ induces a morphism

$$\lim_{i \in I} \mathbf{D}(i)(B) \rightarrow \lim_{i \in I} \mathbf{D}(i)(A)$$

compatible with the restriction morphisms of all $\mathbf{D}(i)$. These morphisms make $\lim_{i \in I} \mathbf{D}(i)$ a presheaf, and the objectwise limit projections form a universal cone in $\text{PSh}(\mathcal{C}, \mathcal{D})$. Thus $\text{PSh}(\mathcal{C}, \mathcal{D})$ is complete.

The proof for colimits is dual (see [34, Ch. 5, §3, p. 116]). □

In particular, kernels, cokernels, products, coproducts, images, and coimages of presheaves may be computed objectwise whenever they exist in the codomain category.

Consequently, $\mathbf{PSh}(\mathcal{C}, \mathcal{D})$ is complete (respectively cocomplete) whenever the codomain category $\mathcal{D}$ is complete (respectively cocomplete). Beyond limits and colimits, one is often interested in kernels, cokernels, and exact sequences.

Lemma 2.1.5. If $\mathcal{D}$ is an abelian category, then $\mathbf{PSh}(\mathcal{C}, \mathcal{D})$ is an abelian category (see [54, Ex. 1.6.4]).

Proof. Since $\mathcal{D}$ is abelian, it has a zero object, binary biproducts, kernels, and cokernels. Since limits and colimits in $\mathbf{PSh}(\mathcal{C}, \mathcal{D})$ are computed componentwise, it follows that

1. the constant presheaf which takes the value of the zero object in $\mathcal{D}$ is a zero object of $\mathbf{PSh}(\mathcal{C}, \mathcal{D})$,
2. binary biproducts are computed componentwise,
3. kernels and cokernels are computed componentwise.

It remains to show that $\mathbf{PSh}(\mathcal{C}, \mathcal{D})$ is binormal.

Let $\alpha : \mathcal{F}^{\text{pre}} \rightarrow \mathcal{G}^{\text{pre}}$ be a monomorphism. Since kernels are computed componentwise, and a morphism in a preadditive category with kernels is monomorphic if and only if its kernel is the zero object, each component

$$\alpha_A : \mathcal{F}^{\text{pre}}(A) \rightarrow \mathcal{G}^{\text{pre}}(A)$$

is monomorphic in $\mathcal{D}$.

Since $\mathcal{D}$ is abelian, each α_A is a kernel of its cokernel. Since kernels and cokernels are computed componentwise, α is a kernel of $\text{coker}(\alpha)$. Thus every monomorphism in $\mathbf{PSh}(\mathcal{C}, \mathcal{D})$ is normal.

Dually, let $\beta : \mathcal{F}^{\text{pre}} \rightarrow \mathcal{G}^{\text{pre}}$ be an epimorphism. Since cokernels are computed componentwise, and an epimorphism in a preadditive category with cokernels is epimorphic if and only if its cokernel is the zero object, each component

$$\beta_A : \mathcal{F}^{\text{pre}}(A) \rightarrow \mathcal{G}^{\text{pre}}(A)$$

is epimorphic in $\mathcal{D}$. Since the cokernels are computed componentwise, β is a cokernel of $\ker(\beta)$. Thus, every epimorphism in $\text{PSh}(\mathcal{C}, \mathcal{D})$ is conormal. Hence $\text{PSh}(\mathcal{C}, \mathcal{D})$ is abelian. $\square$

Although presheaves inherit many favourable categorical properties, they do not yet encode locality.

2.2 Sheaves

Presheaves assign data to the objects of a category together with compatible restriction morphisms, but they impose no condition on how locally defined data should be reconstructed from compatible local pieces. To formulate such a gluing condition categorically, one first equips the underlying category with a Grothendieck topology. Sheaves are then precisely those presheaves that satisfy the corresponding locality and gluing axioms.

2.2.1 Grothendieck Topologies and Sites

In the category of open subsets of a topological space, locality is described by open inclusions. However, in many geometric situations the relevant local information is not given by subobjects, but by more general morphisms. Grothendieck topologies formalise this idea by specifying which families of morphisms can be regarded as covers. In order to define Grothendieck topologies, a more general notion is needed.

Definition 2.2.1 (Sieve). Let $\mathcal{C}$ be a category, and let $A \in \text{Ob}(\mathcal{C})$. A **sieve** S on A is a subfunctor of the representable functor $S \subseteq \text{Hom}_{\mathcal{C}}(-, A)$ (see [32, Ch. 16, §1, Def. 16.1.1]).

Remark 2.2.2. In essence if S is a sieve on A in a category $\mathcal{C}$, then for every object $B \in \text{Ob}(\mathcal{C})$, $S(B)$ is a set of morphisms $B \rightarrow A$.

If $f : C \rightarrow B$ and $g : B \rightarrow A$ is in $S(B)$, then the composite

$$g \circ f : C \rightarrow A$$

is in $S(C)$. Thus, a sieve is closed under precomposition (see [32, Ch. 16, §1, pp. 389 ff.]).

A sieve on A is a subpresheaf $F \subseteq \mathcal{Y}A$, where $\mathcal{Y}$ denotes the *Yoneda functor*. Sieves are not merely a generalisation of topological covers. A sieve on $A \in \text{Ob}(\text{Open}(X^{\text{op}}))$ is a downward-closed family of open subsets of A .

Definition 2.2.3 (Grothendieck topology). A **Grothendieck topology** J on a category $\mathcal{C}$ assigns to every $A \in \text{Ob}(\mathcal{C})$ a collection $J(A)$ of sieves on A , called **covering sieves**, satisfying the following axioms:

GT1 Maximal Sieve Covers

The maximal sieve $\text{Hom}_{\mathcal{C}}(-, A)$ is a covering sieve on every object A .

GT2 Stability under Pullback

If $S \in J(A)$ is a covering sieve and $f : B \rightarrow A$ is an arbitrary morphism, then the **pullback sieve**

$$f^*S := \{g : C \rightarrow B \mid f \circ g \in S\}$$

is a covering sieve on B .

GT3 Transitivity

Let $S \in J(A)$. If T is a sieve on A such that for every morphism $f : B \rightarrow A$ in S , the pullback f^*T is a covering sieve on B , then T itself is a covering sieve on A

(see [32, Ch. 16, §1, Def. 16.1.2]).

The Grothendieck topology isolates the minimal structure required to speak about locality and descent in a way that is stable under categorical constructions. Without the Grothendieck topology, these notions would depend on ad hoc choices of covers and would fail to be functorial. Although it is called topology, it is not restricted to categories like Top or $\text{Open}(X)$.

Definition 2.2.4 (Cover of a Grothendieck topology). Let τ be a Grothendieck topology on a category $\mathcal{C}$, and let $U \in \text{Ob}(\mathcal{C})$. A family $\{U_i \rightarrow U\}_{i \in I}$ is a covering family if the sieve generated by this family is a sieve (see [32, Ch. 16, §1, Def. 16.1.5]).

Example 2.2.5. As an example, one may consider the Zariski topology on the category of schemes. A family $\{U_i \rightarrow X\}_{i \in I}$ is a cover if each $U_i \rightarrow X$ is an open immersion and $X = \bigcup_{i \in I} U_i$ (see [32, Ch. 16, §1, Ex. 16.1.10, iii]).

A category equipped with a Grothendieck topology forms a basic ambient structure in which sheaves can be defined.

Definition 2.2.6 (Site). A **site** is a pair $(\mathcal{C}, J)$ where $\mathcal{C}$ is a category and J is a Grothendieck topology on $\mathcal{C}$ (see [32, Ch. 17, §2, Def. 17.2.1, Ch. 17, §1, Def. 17.1.1]).

A site records both the underlying category, and the chosen notion of a cover. In the canonical Grothendieck topology on $\text{Open}(X)$, covering sieves are precisely the sieves generated by open covers. Hence $\text{Open}(X)$, equipped with this Grothendieck topology, becomes a site. More details on this can be found in Appendix D.1.

Definition 2.2.7 (Separated presheaf). Let $(\mathcal{C}, J)$ be a site, and let $F : \mathcal{C}^{\text{op}} \rightarrow \mathcal{D}$ be a presheaf. Then F is **separated** if for every object $U \in \text{Ob}(\mathcal{C})$ and every covering sieve $S \subseteq \text{Hom}_{\mathcal{C}}(-, U)$, the canonical morphism

$$F(U) \rightarrow \prod_i F(U_i)$$

is monomorphic, or equivalently if

$$F(U) \rightarrow \text{eq} \left(\prod_i F(U_i) \rightrightarrows \prod_{i,j} F(U_i \times_U U_j) \right)$$

is a monomorphism (see [32, Ch. 17, §3, Def. 17.3.1]).

Separated presheaves enforce uniqueness of gluing, but not existence. Requiring both existence and uniqueness leads to the notion of sheaves.

2.2.2 Sheaves

The central idea of sheaf theory is that local data that agree on overlaps should be glued uniquely to global data, leading to the notion of sheaves on sites.

Sheaves on Sites

Definition 2.2.8 (Sheaf on a site). Let $(\mathcal{C}, J)$ be a site. A **sheaf** $\mathcal{F} : \mathcal{C}^{\text{op}} \rightarrow \mathcal{D}$ is a contravariant functor satisfying the equaliser condition, that is, for every covering family $\{U_i \rightarrow U\}$,

$$\mathcal{F}(U) \longrightarrow \prod_i \mathcal{F}(U_i) \rightrightarrows \prod_{i,j} \mathcal{F}(U_i \times_U U_j).$$

The parallel morphisms are restrictions, that is, for a given morphism $f : V \rightarrow U$ in $\mathcal{C}$, the **restriction along** f is the induced morphism

$$\mathcal{F}(f) : \mathcal{F}(U) \rightarrow \mathcal{F}(V).$$

For any $U \in \text{Ob}(\mathcal{C})$, the object $\mathcal{F}(U)$ is called the **object of sections of $\mathcal{F}$ over U** . The object

$$\prod_i \mathcal{F}(U_i)$$

may be interpreted as the object of families of local sections on the cover $\{U_i \rightarrow U\}$ (see [32, Ch. 17, §3, 17.3.1]).

Remark 2.2.9. One can construct sheaves from presheaves. This construction is called *sheafification*. Details can be found in Appendix D.2.

The category $\mathcal{D}$ has to admit products indexed by covering families and equalisers. However, if $\mathcal{D}$ does not admit at least one of these criteria, then the notion of sheaves becomes unavailable.

The equaliser condition expresses that families of local sections whose restrictions to all fibre products $U_i \times_U U_j$ agree arise uniquely from a section over U . If $\mathcal{D}$ admits elements, this means that a family

$$(s_i) \in \prod_i \mathcal{F}(U_i)$$

is compatible whenever

$$\mathcal{F}(\text{pr}_1)(s_i) = \mathcal{F}(\text{pr}_2)(s_j)$$

for all i, j .

Remark 2.2.10. The classical definition of sheaves on a topological space can be recovered from the definition of sheaves on sites. The object has to be terminal, since a sheaf cannot be reconstructed from only local data (see [33, Ch. 3, §2, pp. 110 ff]).

Remark 2.2.11 (The category of sheaves). The sheaves on a site $(\mathcal{C}, J)$ into $\mathcal{D}$ form a category $\text{Sh}((\mathcal{C}, J), \mathcal{D})$ with morphisms given by natural transformations (see [33, Ch. 2, §2, p. 71]).

Remark 2.2.12 (Sheafification and (Co-) Completeness). If $\mathcal{D}$ is complete, then $\text{Sh}((\mathcal{C}, J), \mathcal{D})$ is complete, since $\text{PSh}((\mathcal{C}, J), \mathcal{D})$ admits all limits, and limits commute with limits.

Since every sheaf is a presheaf, there exists an inclusion $\text{Sh}((\mathcal{C}, J), \mathcal{D}) \hookrightarrow \text{PSh}(\mathcal{C}, \mathcal{D})$. While there always exists the inclusion functor,

$$\iota : \text{Sh}(\mathcal{C}, J) \hookrightarrow \text{PSh}(\mathcal{C})$$

it is not guaranteed that it admits a left-adjoint, called **sheafification**. This needs additional hypotheses on the site and on the target category. Furthermore, the category $\mathcal{D}$ might not have filtered colimits, and their preservation by limits might fail (see [33, Ch. 2, §2, pp. 70 ff, Ch. 3, §5]).

Sheafification provides a universal procedure for turning presheaves into sheaves. It appears naturally in several constructions.

A concrete instance of sheaves on sites are sheaves on topological spaces.

Sheaves on Topological Spaces One recurring example throughout Chapter 5 is the trivial group sheaf.

Example 2.2.13 (Trivial group sheaf). Let X be a topological space. Then

$$1 \in \text{Sh}(X, \text{Grp})$$

is the constant sheaf associated to the trivial group consisting only of the neutral element e (see [48, Ch. 2, §2, Ex. 2.12]).

Moreover, sheaves on topological spaces provide the natural setting for many algebraic concepts, such as locally ringed spaces.

Definition 2.2.14 (Stalk and germ). Let $\mathcal{F} \in \text{Ob}(\text{Sh}(X, \mathcal{C}))$ be a sheaf on a topological space X . The directed colimit

$$\mathcal{F}_x := \text{colim}_{x \in U} \mathcal{F}(U)$$

over the directed poset $(U(x), \supseteq)$ of open neighbourhoods of x is the **stalk**.

An element of $\mathcal{F}_x$ is called a **germ** of a section at x , denoted by $[s]_x$, where $s \in \mathcal{F}(U)$ for some neighbourhood U of x . Two sections $s \in \mathcal{F}(U)$ and $t \in \mathcal{F}(V)$ define the same germ if there exists an open neighbourhood $W \subseteq U \cap V$ of x such that

$$s|_W = t|_W$$

(see [8, Ch. 6, §6, Ex. 6.6.17]).

Stalks capture the local behaviour of a sheaf around a point. They play a fundamental role in the theory of locally ringed spaces and schemes, where local properties are formulated in terms of the stalks of the structure sheaf.

Stalks admit a natural generalisation to stalks of sites, once a notion of points of a site is available. The familiar neighbourhood-based picture of germs is essentially a topological-space phenomenon.

Remark 2.2.15. According to Theorem 1.5.32, the stalk functor

$$\begin{aligned} \text{PSh}(X) &\rightarrow \text{Set} \\ \mathcal{F} &\mapsto \mathcal{F}_x \end{aligned}$$

preserves finite limits (see [8, Ch. 6, §6, Ex. 6.6.17]).

While stalks describe the local behaviour of a sheaf around a point, the sheaf condition also imposes strong constraints on how sections behave with respect to decompositions of open sets.

Proposition 2.2.16. Let $\mathcal{F} : \text{Open}(X)^{\text{op}} \rightarrow \mathcal{D}$ be a sheaf into a category $\mathcal{D}$ with a terminal object and products. Let

$$V = \dot{\bigcup}_{i \in I} U_i$$

be a disjoint union of open subsets $U_i \subseteq X$. Then the canonical morphism

$$\mathcal{F}(V) \rightarrow \prod_{i \in I} \mathcal{F}(U_i)$$

is an isomorphism (see [52, Ex. 7.7.3]).

Proof. Since $\mathcal{F}$ is a sheaf, for any open cover $\mathcal{U} = \{U_i\}_{i \in I}$ of V , the usual sheaf condition

$$\mathcal{F}(V) \rightarrow \prod_{i \in I} \mathcal{F}(U_i) \rightrightarrows \prod_{(i,j) \in I \times I} \mathcal{F}(U_i \cap U_j)$$

is satisfied. Since (U_i) is pairwise disjoint, and since $\mathcal{F}(\emptyset)$ is terminal in $\mathcal{D}$,

$$\prod_{(i,j) \in I \times I} \mathcal{F}(U_i \cap U_j) \cong \left(\prod_{i \in I} \mathcal{F}(U_i) \right) \times T.$$

Since T is a product of terminal objects, it is itself terminal by Corollary 1.5.30. Under this identification, the two parallel morphisms

$$\prod_{i \in I} \mathcal{F}(U_i) \rightrightarrows \prod_{(i,j) \in I \times I} \mathcal{F}(U_i \cap U_j)$$

agree on each component; therefore they are equal, and by the equaliser of two morphisms, this yields

$$\mathcal{F}(V) \cong \prod_{i \in I} \mathcal{F}(U_i)$$

(see [52, Ex. 7.7.3]). □

This proposition illustrates how the sheaf condition reconstructs sections from compatible local data. This reconstruction principle reappears in the discussion of Čech cohomology.

So far, sheaves were considered on a fixed topological space. If one wishes to com-

pare sheaves on different topological spaces, morphisms in $\mathbf{Top}$ provide a natural way to transfer sheaves between topological spaces.

Definition 2.2.17 (Direct image sheaf). Let $\mathcal{C}$ be a category. Let X and Y be topological spaces and let $f : X \rightarrow Y$ be a continuous function. Let $\mathcal{O}_X$ be a $\mathcal{C}$ -valued sheaf on X . The **direct image sheaf** (also called **pushforward** sheaf) on Y is defined as

$$(f_*\mathcal{O}_X)(V) := \mathcal{O}_X(f^{-1}(V))$$

for every open subset $V \subseteq Y$ (see [8, Ch. 9, §5, Ex. 9.5.12]).

Remark 2.2.18. 1. $f_*\mathcal{O}_X$ is a sheaf on Y .

2. The functor $f_* : \mathbf{Sh}(X) \rightarrow \mathbf{Sh}(Y)$ is left-exact and preserves limits.

3. Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be continuous functions between topological spaces. Then $(g \circ f)_* = g_* \circ f_*$

(see [8, Ch. 9, §5, Ex. 9.5.12]).

Two important examples of sheaves used throughout this thesis are constant and locally constant sheaves. They not only allow to make claims on their cohomology on suitable underlying topological sets, but also arise naturally in Chapter 5.

Definition 2.2.19 (Constant sheaf). Let X be a topological space, and let $A \in \mathbf{Ob}(\mathcal{C})$. The **constant presheaf** A^{pre} is defined by

$$A^{\text{pre}}(U) = A$$

for every open subset $U \subseteq X$, with all restriction morphisms equal to id_A . The constant sheaf A^{cst} is the sheafification of A^{pre} (see [48, Ch. 3, §3, Rem. 3.19, Ch. 4, §3 Ex. 5.15]).

Remark 2.2.20. In a category that admits elements, such as $\mathbf{Set}$, a constant presheaf usually fails to be a sheaf. Suppose that $U = U_1 \dot{\cup} U_2$, and take sections

$$s_1 = a_1 \in A \text{ on } U_1; \quad s_2 = a_2 \in A \text{ on } U_2$$

with $a_1 \neq a_2$.

As $U_1 \cap U_2 = \emptyset$, the compatibility condition is vacuous, so the sheaf axiom would require a section on U restricting to both a_1 and a_2 , which is impossible unless $a_1 = a_2$.

If X is locally connected, the constant sheaf admits a very concrete description:

$$A^{\text{cst}}(U) = \{f : U \rightarrow A \mid f \text{ is locally constant}\}.$$

Equivalently, sections are functions that are constant on each connected component of U . Thus, if

$$U = \bigcup_{i \in I} U_i$$

is the decomposition into connected components, then

$$A^{\text{cst}}(U) \cong \prod_{i \in I} A$$

(see [48, Ch. 2, §2, Ex. 2.10]).

Constant sheaves model globally constant data. In many geometric situations, however, data need only be constant locally, leading to a more flexible notion of locally constant sheaves.

Definition 2.2.21 (Locally constant sheaf). Let X be a topological space. A sheaf $\mathcal{F} \in \text{Ob}(\text{Sh}(X, \mathcal{D}))$ is called **locally constant** if for every $x \in X$ there exists an open neighbourhood $U \subseteq X$ and an object $D \in \text{Ob}(\mathcal{D})$ such that

$$\mathcal{F}|_U \cong D_U^{\text{cst}}$$

where D_U^{cst} denotes the constant sheaf on U associated to D (see [48, Ch. 3, §3, Ex. 3.18, Rem. 3.19]).

2.2.3 Categorical Properties of Sh

To perform homological constructions with sheaves, it is important to understand which categorical properties are inherited from the target category.

Remark 2.2.22 (Limits and product sheaves). By Remark 1.5.46, algebraic categories are complete and cocomplete. Limits of sheaves are computed objectwise whenever the corresponding limits exist in $\mathcal{D}$. Colimits, by contrast, need not be computed objectwise: when sheafification exists, a colimit of sheaves is obtained by forming the colimit in the presheaf category and then sheafifying it. Consequently, completeness follows from suitable completeness assumptions on $\mathcal{D}$, whereas cocompleteness additionally requires the existence of sheafification. Let $\mathcal{D}$ be a complete category, and let $(\mathcal{C}, J)$ be a site. Consider a family of sheaves $\mathcal{F}_i : \mathcal{C}^{\text{op}} \rightarrow \mathcal{D}$. The product sheaf

$$\mathcal{F} := \prod_{i \in I} \mathcal{F}_i$$

is computed sectionwise, as products commute with equalisers by Theorem 1.5.29.

As certain limits can be computed sectionwise, this leads to the question to what extent algebraic structures present in the target category are inherited by the category of sheaves.

Theorem 2.2.23. Let $(\mathcal{C}, J)$ be a site and let $\mathcal{D}$ be an abelian category. Then $\text{Sh}((\mathcal{C}, J), \mathcal{D})$ has

- a zero object
- all binary biproducts
- all kernels

(see [33, Ch. 2, §7], [32, Ch. 8, §3, Rem. 8.3.6]).

Proof. Since $\mathcal{D}$ is preadditive, the set of natural transformations between any two $\mathcal{D}$ -valued sheaves is an abelian group under componentwise addition. Thus $\text{Sh}((\mathcal{C}, J), \mathcal{D})$ is preadditive.

Claim: $\text{Sh}((\mathcal{C}, J), \mathcal{D})$ admits a zero object.

Since $\mathcal{D}$ is abelian, it admits a zero object 0 . Define the constant presheaf

$$\begin{aligned} \mathcal{N} : \mathcal{C}^{\text{op}} &\rightarrow \mathcal{D} \\ A &\mapsto 0 \end{aligned}$$

for all $A \in \text{Ob}(\mathcal{C}^{\text{op}})$.

For every morphism $A \rightarrow B$ in $\mathcal{C}$, the restriction morphism

$$\mathcal{N}(A) = 0 \rightarrow \mathcal{N}(B) = 0$$

is the unique morphism $0 \rightarrow 0$. The sheaf condition is satisfied, since every object and every morphism occurring in the corresponding matching diagram is the zero morphism or its unique morphism. Equivalently, the canonical morphism from 0 to the relevant matching object is an isomorphism.

For every sheaf $\mathcal{F}$, there is a unique natural transformation $\mathcal{F} \rightarrow \mathcal{N}$, and a unique natural transformation $\mathcal{N} \rightarrow \mathcal{F}$, because 0 is both initial and terminal in $\mathcal{D}$. Hence $\mathcal{N}$ is a zero object in $\text{Sh}((\mathcal{C}, J), \mathcal{D})$.

Claim: $\text{Sh}((\mathcal{C}, J), \mathcal{D})$ admits all binary biproducts.

Let $\mathcal{F}$ and $\mathcal{G}$ be sheaves. Since limits of sheaves are computed objectwise, the sheaf defined by

$$(\mathcal{F} \times \mathcal{G})(A) := \mathcal{F}(A) \times \mathcal{G}(A).$$

is a sheaf. Because $\mathcal{D}$ is abelian, every binary product in $\mathcal{D}$ is also a binary biproduct. The canonical injections and projections in $\mathcal{D}$ are natural in A , and hence assemble into natural transformations

$$\iota_{\mathcal{F}} : \mathcal{F} \rightarrow \mathcal{F} \times \mathcal{G}, \quad \iota_{\mathcal{G}} : \mathcal{G} \rightarrow \mathcal{F} \times \mathcal{G},$$

and

$$\text{pr}_{\mathcal{F}} : \mathcal{F} \times \mathcal{G} \rightarrow \mathcal{F}, \quad \text{pr}_{\mathcal{G}} : \mathcal{F} \times \mathcal{G} \rightarrow \mathcal{G}.$$

Their biproduct identities hold componentwise:

$$\text{pr}_{\mathcal{F}} \circ \iota_{\mathcal{F}} = \text{id}_{\mathcal{F}}, \quad \text{pr}_{\mathcal{G}} \circ \iota_{\mathcal{G}} = \text{id}_{\mathcal{G}},$$

$$\text{pr}_{\mathcal{F}} \circ \iota_{\mathcal{G}} = 0, \quad \text{pr}_{\mathcal{G}} \circ \iota_{\mathcal{F}} = 0,$$

and

$$\iota_{\mathcal{F}} \circ \text{pr}_{\mathcal{F}} + \iota_{\mathcal{G}} \circ \text{pr}_{\mathcal{G}} = \text{id}_{\mathcal{F} \times \mathcal{G}}.$$

Therefore $\mathcal{F} \times \mathcal{G}$ is a biproduct of $\mathcal{F}$ and $\mathcal{G}$.

Claim: $\text{Sh}((\mathcal{C}, J), \mathcal{D})$ admits all kernels.

Let $\alpha : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of sheaves. Define a presheaf M objectwise by

$$M(A) := \ker(\alpha_A : \mathcal{F}(A) \rightarrow \mathcal{G}(A)).$$

For every morphism $f : B \rightarrow A$ in $\mathcal{C}$, naturality of α gives a commutative square

$$\begin{array}{ccc} \mathcal{F}(A) & \xrightarrow{\alpha_A} & \mathcal{G}(A) \\ \downarrow \mathcal{F}(f) & & \downarrow \mathcal{G}(f) \\ \mathcal{F}(B) & \xrightarrow{\alpha_B} & \mathcal{G}(B) \end{array}$$

Hence

$$\alpha_B \circ \mathcal{F}(f) \circ \iota_A = \mathcal{G}(f) \circ \alpha_A \circ \iota_A = 0,$$

where

$$\iota_A : M(A) \hookrightarrow \mathcal{F}(A)$$

is the kernel morphism. By the universal property of $\ker(\alpha_B)$, there exists a unique morphism

$$M(f) : M(A) \rightarrow M(B)$$

satisfying

$$\iota_B \circ M(f) = \mathcal{F}(f) \circ \iota_A.$$

These morphisms make M a presheaf and the morphisms ι_A assemble into a natural transformation $\iota : M \rightarrow \mathcal{F}$.

Since kernels are limits and limits of sheaves are computed objectwise, M is a sheaf. Moreover, the universal property of ι is verified objectwise: if

$$\beta : \mathcal{C} \rightarrow \mathcal{F}$$

is a morphism of sheaves satisfying

$$\alpha \circ \beta = 0,$$

then each component β_A factors uniquely through $M(A)$. The uniqueness of these componentwise factorisations implies their naturality, so they assemble into a unique

morphism

$$\tilde{\beta} : \mathcal{E} \rightarrow M$$

satisfying

$$\iota \circ \tilde{\beta} = \beta.$$

Hence $\iota : M \rightarrow \mathcal{F}$ is the kernel of α in the category of sheaves (see [33, Ch. 2, §7], [32, Ch. 8, §3, Rem. 8.3.6]). $\square$

Cokernels need not be computed objectwise: the objectwise cokernel of a morphism of sheaves is generally only a presheaf. If the sheafification functor

$$s^\# : \text{PSh}((\mathcal{C}, J), \mathcal{D}) \rightarrow \text{Sh}((\mathcal{C}, J), \mathcal{D})$$

exists as a left-adjoint to the inclusion, then the cokernel in the category of sheaves is obtained by sheafifying the objectwise cokernel. To conclude that the category of sheaves is abelian, one additionally needs sufficient exactness of the sheafification.

Lemma 2.2.24. Let X be a site, and let $\mathcal{A}$ be a category satisfying the following conditions:

1. $\mathcal{A}$ admits small limits and small colimits;
2. small filtered colimits are exact;
3. $\mathcal{A}$ satisfies the IPC-property, that is, for every small family of small filtered categories $(I_s)_{s \in S}$, and every family of functors $\alpha_s : I_s \rightarrow \mathcal{A}$, the canonical morphism

$$\text{colim}_{(i_s)_{s \in S} \in \prod_{s \in S} I_s} \prod_{s \in S} \alpha_s(i_s) \rightarrow \prod_{s \in S} \text{colim}_{i_s \in I_s} \alpha_s(i_s)$$

is an isomorphism.

Then the inclusion

$$\iota : \text{Sh}(X, \mathcal{A}) \rightarrow \text{PSh}(X, \mathcal{A})$$

admits a left-adjoint

$$a : \text{PSh}(X, \mathcal{A}) \rightarrow \text{Sh}(X, \mathcal{A}).$$

For $F \in \text{Ob}(\text{PSh}(X, \mathcal{A}))$, the sheaf $F^a := a(F)$ is called the **sheaf associated with F** , or the **sheafification of F** . If, in addition, $\mathcal{A}$ is abelian, then the functor a is exact (see [32, Ch. 17, §4], [33, Ch. 2, §5, Corollary 4, Ch. 5 §3, Thm. 1]).

Proof. The proof can be found in Appendix D.2. □

In general, sheafification need not exist, and when it exists it need not be exact. Under the stronger hypotheses of the preceding lemma, sheafification exists and is exact. The preceding lemma provides exactly the additional structure missing from Theorem 2.2.23. Since sheafification exists and behaves well with respect to exact sequences, cokernels can be constructed inside the category of sheaves itself.

Theorem 2.2.25. Let $(\mathcal{C}, J)$ be a small site and let $\mathcal{D}$ be a Grothendieck abelian category. Then $\text{Sh}((\mathcal{C}, J), \mathcal{D})$ is an abelian category (see [28, Ch. 2, §2, Thm. 2.5, Thm. 2.6]).

Proof. By Theorem 2.2.23, $\text{Sh}((\mathcal{C}, J), \mathcal{D})$ has a zero object, binary biproducts, and all kernels. It remains to construct cokernels and to verify the image–coimage condition.

Let

$$\varphi : \mathcal{F} \rightarrow \mathcal{G}$$

be a morphism of sheaves. Since the presheaf category $\text{PSh}((\mathcal{C}, J), \mathcal{D})$ is abelian, the cokernel

$$Q := \text{coker}_{\text{PSh}}(\varphi)$$

exists and is computed objectwise. In general, Q need not be a sheaf. Define

$$\text{coker}_{\text{Sh}}(\varphi) := s^\#(Q),$$

where $s^\#$ is the sheafification functor from Lemma 2.2.24. Let

$$q : \mathcal{G} \rightarrow Q$$

be the cokernel morphism in the presheaf category, and let

$$\eta_Q : Q \rightarrow \iota s^\#(Q)$$

be the unit of the sheafification adjunction, where

$$\iota : \mathbf{Sh} \rightarrow \mathbf{PSh}$$

denotes the inclusion. Then

$$\eta_Q \circ q : \mathcal{G} \rightarrow s^\#(Q)$$

is a cokernel of φ in the sheaf category. Indeed, if

$$\psi : \mathcal{G} \rightarrow \mathcal{E}$$

is a morphism of sheaves satisfying

$$\psi \circ \varphi = 0,$$

then, by the universal property of Q , there exists a unique presheaf morphism

$$\overline{\psi} : Q \rightarrow \iota(\mathcal{E})$$

such that

$$\overline{\psi} \circ q = \psi.$$

By the adjunction $s^\# \dashv \iota$, this morphism corresponds uniquely to a sheaf morphism

$$\tilde{\psi} : s^\#(Q) \rightarrow \mathcal{E}$$

satisfying

$$\tilde{\psi} \circ \eta_Q \circ q = \psi.$$

Thus all cokernels exist in the sheaf category.

It remains to verify that for every morphism φ the canonical morphism

$$\mathrm{coim}_{\mathbf{Sh}}(\varphi) \rightarrow \mathrm{im}_{\mathbf{Sh}}(\varphi)$$

is an isomorphism.

Since kernels of sheaves are computed in the presheaf category and sheafification

is exact, one has

$$\begin{aligned}\mathrm{coim}_{\mathrm{Sh}}(\varphi) &= \mathrm{coker}_{\mathrm{Sh}}(\ker_{\mathrm{Sh}}(\varphi)) \\ &\cong s^{\#}(\ker_{\mathrm{PSh}}(\varphi)) \\ &= s^{\#}(\mathrm{coim}_{\mathrm{PSh}}(\varphi)).\end{aligned}$$

Likewise,

$$\begin{aligned}\mathrm{im}_{\mathrm{Sh}}(\varphi) &= \ker_{\mathrm{Sh}}(\mathrm{coker}_{\mathrm{Sh}}(\varphi)) \\ &\cong s^{\#}\left(\ker_{\mathrm{PSh}}(\mathrm{coker}_{\mathrm{PSh}}(\varphi))\right) \\ &= s^{\#}(\mathrm{im}_{\mathrm{PSh}}(\varphi)).\end{aligned}$$

Since the presheaf category is abelian, the canonical isomorphism

$$\mathrm{coim}_{\mathrm{PSh}}(\varphi) \xrightarrow{\sim} \mathrm{im}_{\mathrm{PSh}}(\varphi)$$

is an isomorphism. Applying $s^{\#}$ therefore gives an isomorphism

$$\mathrm{coim}_{\mathrm{Sh}}(\varphi) \xrightarrow{\sim} \mathrm{im}_{\mathrm{Sh}}(\varphi).$$

Hence by Theorem 1.6.21, $\mathrm{Sh}((\mathcal{C}, J), \mathcal{D})$ is abelian (see [28, Ch. 2, §2, Thm. 2.5, Thm. 2.6]). $\square$

Under the hypotheses above, the category of $\mathcal{D}$ -valued sheaves admits all notions of homological algebra: kernels, cokernels, and exact sequences, among others. Once $\mathrm{Sh}((\mathcal{C}, J), \mathcal{D})$ is known to be abelian, the methods of homological algebra become available. In particular, one may study injective objects and resolutions, which are used to define sheaf cohomology via derived functors.

Theorem 2.2.26 (Sheaves over Grothendieck categories). Let $\mathcal{D}$ be an abelian Grothendieck category, and let $(\mathcal{C}, J)$ be a small site. Then the category $\mathrm{Sh}((\mathcal{C}, J), \mathcal{D})$ is again a Grothendieck abelian category (compare [52, Ch. 6, §18, Lemma 6.18.2, Ch. 6, §29]).

Proof. Claim: The category of presheaves $[\mathcal{C}^{\mathrm{op}}, \mathcal{D}]$ is Grothendieck.

Limits, colimits, and exactness of filtered colimits are computed pointwise. A generator is obtained from a generator of $\mathcal{D}$ by placing copies of it on representable presheaves.

The sheafification functor is exact by Theorem 2.2.24 and left-adjoint to the inclusion

$$\mathbf{PSh}((\mathcal{C}, J), \mathcal{D}) \hookrightarrow \mathbf{Sh}((\mathcal{C}, J), \mathcal{D}).$$

Thus $\mathbf{Sh}((\mathcal{C}, J), \mathcal{D})$ is an exact reflective localisation of a Grothendieck category. Exact localisations of Grothendieck categories are again Grothendieck. Hence $\mathbf{Sh}((\mathcal{C}, J), \mathcal{D})$ is Grothendieck. Details can be found in Appendix D. $\square$

The preceding results establish the internal algebraic structure of the category of sheaves. A complementary question is how individual sections of a sheaf can be recovered categorically.

2.2.4 Representability of $\mathbf{ev}$

Many constructions in the category of sheaves are defined by evaluating a sheaf on a fixed object. Indeed, one can connect it to the Yoneda philosophy by describing the fixed object through morphisms from a suitable representing object. Understanding evaluation functors allows one to view global sections as a special case and prepare the categorical framework underlying torsors and cohomological algebra.

Lemma 2.2.27. Let $(\mathcal{C}, J)$ be a site, let $U \in \mathbf{Ob}(\mathcal{C})$, and let $\mathcal{D}$ be a category admitting a free-forgetful adjunction

$$L : \mathbf{Set} \rightleftarrows \mathcal{D} : R$$

Assume that the category of $\mathcal{D}$ -valued presheaves admits a sheafification functor

$$s_{\mathcal{D}}^{\#} : \mathbf{PSh}(\mathcal{C}, \mathcal{D}) \rightarrow \mathbf{Sh}((\mathcal{C}, J), \mathcal{D}).$$

Let

$$h_U : \mathbf{Hom}_{\mathcal{C}}(-, U)$$

be the representable presheaf. Then the sheaf

$$s_D^\#(L(h_U))$$

represents the underlying $\mathbf{Set}$ -valued evaluation functor at U . More precisely, for every

$$\mathcal{F} \in \mathbf{Ob} \left(\mathbf{Sh}((\mathcal{C}, J), \mathcal{D}) \right)$$

there is a natural bijection

$$\mathbf{Hom}_{\mathbf{Sh}((\mathcal{C}, J), \mathcal{D})} (L(h_U), \mathcal{F}) \cong R(\mathcal{F}(U)).$$

Consequently, the functor

$$\mathbf{Sh}((\mathcal{C}, J), \mathcal{D}) \rightarrow \mathbf{Set}, \quad \mathcal{F} \mapsto R(\mathcal{F}(U)),$$

is representable (see [4, Ch. 1, §9, 1.9.1]).

Proof. Let

$$\iota : \mathbf{Sh}((\mathcal{C}, J), \mathcal{D}) \hookrightarrow \mathbf{PSh}(\mathcal{C}, \mathcal{D})$$

denote the inclusion. Since sheafification is left-adjoint to ι , one has a natural bijection

$$\mathbf{Hom}_{\mathbf{Sh}((\mathcal{C}, J), \mathcal{D})} \left(s_D^\#(L(h_U)), \mathcal{F} \right) \cong \mathbf{Hom}_{\mathbf{PSh}(\mathcal{C}, \mathcal{D})} (L(h_U), \iota \circ \mathcal{F}).$$

Applying the adjunction $L \dashv R$ objectwise yields a natural bijection

$$\mathbf{Hom}_{\mathbf{PSh}(\mathcal{C}, \mathcal{D})} (L(h_U), \iota \circ \mathcal{F}) \cong \mathbf{Hom}_{\mathbf{PSh}(\mathcal{C}, \mathbf{Set})} (h_U, R \circ \iota \circ \mathcal{F}).$$

By the Yoneda lemma,

$$\mathbf{Hom}_{\mathbf{PSh}(\mathcal{C}, \mathbf{Set})} (h_U, R \circ \iota(\mathcal{F})) \cong (R \circ \iota(\mathcal{F}))(U).$$

Since $\iota(\mathcal{F})$ is simply $\mathcal{F}$ regarded as presheaf,

$$(R \circ \iota(\mathcal{F}))(U) = R(\mathcal{F}(U)).$$

Combining these natural bijections gives

$$\mathrm{Hom}_{\mathrm{Sh}((\mathcal{C}, J), \mathcal{D})} \left(s_{\mathcal{D}}^{\#}(L(h_U)), \mathcal{F} \right),$$

naturally in $\mathcal{F}$ (see [4, Ch. 1, §9, 1.9.1]). □

Remark 2.2.28. In case $\mathcal{D} = \mathbf{Ab}$, this says

$$\mathrm{Hom}_{\mathrm{Sh}((\mathcal{C}, J), \mathbf{Ab})} (\mathbb{Z}[h_U], \mathcal{F}) \cong \mathcal{F}(U),$$

where $\mathbb{Z}[h_U]$ denotes the free abelian sheaf generated by h_U . More generally, the statement applies to concrete algebraic categories with a free-object functor. It is not a statement about arbitrary abelian categories.

The representability of evaluation functors provides a categorical description of sections. Among all evaluation functors, evaluation at a terminal object is of particular importance.

2.2.5 Global Sections

So far, the main focus was on arbitrary covering morphisms, or, in the case of sheaves on a topological space X , on arbitrary open subsets $U \subseteq X$. If one chooses $U = X$, or generally speaking, a terminal object, this gives global sections.

Definition 2.2.29 (Global section of sheaves on a site). Let $\mathcal{F}$ be a sheaf on a site $(\mathcal{C}, J)$. If $\mathcal{C}$ has a terminal object $\star$, the **object of global sections** of $\mathcal{F}$ is

$$\Gamma(\mathcal{F}) := \mathcal{F}(\star).$$

When $\mathcal{D}$ is concrete, an element of the underlying set of $\mathcal{F}(\star)$ is called a global section of $\mathcal{F}$ (see [48, Ch. 1, §1, Def. 1.4]).

Global sections are sections defined over terminal objects of the site. For a sheaf $\mathcal{F}$ on a topological space X , a global section is $\mathcal{F}(X)$. The global section functor extracts precisely the global data.

Definition 2.2.30 (Global section functor for sheaves on sites). Let $\mathcal{F}$ be a sheaf on a site $(\mathcal{C}, J)$ and let $\star$ be a terminal object in $\mathcal{C}$. The **global section functor on a site** is defined as

$$\begin{aligned}\Gamma &:= \text{ev}_\star : \text{Sh}((\mathcal{C}, J), \mathcal{D}) \rightarrow \mathcal{D} \\ \mathcal{F} &\mapsto \mathcal{F}(\star)\end{aligned}$$

(see [33, Ch. 1, §6, p. 48]).

Global section functors bridge the gap between local properties and global invariants of underlying sites and of sheaves. The only assumption that has to be made on the site is that it has a terminal object.

Theorem 2.2.31. Let $(\mathcal{C}, J)$ be a small site with a terminal object $\star$, and let $\mathcal{D}$ be a category for which the sheafification functor

$$s^\# : \text{PSh}((\mathcal{C}, J), \mathcal{D}) \rightarrow \text{Sh}((\mathcal{C}, J), \mathcal{D})$$

exists. Let

$$(-)^{\text{cst}} : \mathcal{D} \rightarrow \text{Sh}((\mathcal{C}, J), \mathcal{D})$$

send $D \in \text{Ob}(\mathcal{D})$ to the sheafification of the constant presheaf with value D . Then there is a natural bijection

$$\text{Hom}_{\text{Sh}}(D^{\text{cst}}, \mathcal{F}) \cong \text{Hom}_{\mathcal{D}}(D, \Gamma(\mathcal{F})),$$

natural in D and $\mathcal{F}$. Hence $(-)^{\text{cst}} \dashv \Gamma$ (see [33, Ch. 2, §5, Thm 2, Corollary 4], [24, Ch. 2, Lemma 2.0.3]).

Proof. The sheafification

$$s^\# : \text{PSh}((\mathcal{C}, J), \mathcal{D}) \rightarrow \text{Sh}((\mathcal{C}, J), \mathcal{D})$$

is left-adjoint to the inclusion

$$i : \text{Sh}((\mathcal{C}, J), \mathcal{D}) \hookrightarrow \text{PSh}((\mathcal{C}, J), \mathcal{D}).$$

Let $D^{\text{cst}} := s^\# \circ D^{\text{pre}} : \mathcal{D} \rightarrow \text{Sh}((\mathcal{C}, J), \mathcal{D})$ be the constant sheaf functor. Since

$s^\# \dashv i$, there exists a natural bijection

$$\mathrm{Hom}_{\mathrm{Sh}((\mathcal{C}, J), \mathcal{D})}(s^\# \circ D^{\mathrm{pre}}, \mathcal{F}) \cong \mathrm{Hom}_{\mathrm{PSh}((\mathcal{C}, J), \mathcal{D})}(D^{\mathrm{pre}}, i \circ \mathcal{F}).$$

Since $i \circ \mathcal{F}$ is just $\mathcal{F}$ regarded as a presheaf,

$$\mathrm{Hom}_{\mathrm{PSh}((\mathcal{C}, J), \mathcal{D})}(D^{\mathrm{pre}}, i \circ \mathcal{F}) \cong \mathrm{Hom}_{\mathrm{PSh}((\mathcal{C}, J), \mathcal{D})}(D^{\mathrm{pre}}, \mathcal{F}).$$

Thus it is sufficient to show that

$$\mathrm{Hom}_{\mathrm{PSh}((\mathcal{C}, J), \mathcal{D})}(D^{\mathrm{pre}}, \mathcal{F}) \cong \mathrm{Hom}_{\mathcal{D}}(D, \mathcal{F}(\star)).$$

Let $\eta : D^{\mathrm{pre}} \rightarrow \mathcal{F}$ be a natural transformation of presheaves. Its component at the terminal object $\star$ is a morphism

$$\eta_\star : D^{\mathrm{pre}}(\star) = D \rightarrow \mathcal{F}(\star)$$

in $\mathcal{D}$. This gives a morphism

$$\begin{aligned} \Phi : \mathrm{Hom}_{\mathrm{PSh}((\mathcal{C}, J), \mathcal{D})}(D^{\mathrm{pre}}, \mathcal{F}) &\rightarrow \mathrm{Hom}_{\mathcal{D}}(D, \mathcal{F}(\star)) \\ \eta &\mapsto \eta_\star. \end{aligned} \tag{2.2.32}$$

Conversely, let $\alpha : D \rightarrow \mathcal{F}(\star)$ be a morphism in $\mathcal{D}$. For each $A \in \mathrm{Ob}(\mathcal{C})$, let $\mathrm{res}_A : A \rightarrow \star$ be the unique morphism to the terminal object. Define

$$\eta_A^\alpha : D^{\mathrm{pre}}(A) = D \rightarrow \mathcal{F}(A)$$

by

$$\eta_A^\alpha := \mathcal{F}(\mathrm{res}_A) \circ \alpha.$$

Claim: $\eta^\alpha = (\eta_A^\alpha)_A : D^{\mathrm{pre}} \rightarrow \mathcal{F}$ is a natural transformation.

For any morphism $a : B \rightarrow A$ in $\mathcal{C}$, naturality requires

$$\mathcal{F}(a) \circ \eta_A^\alpha = \eta_B^\alpha \circ D^{\mathrm{pre}}(a).$$

But $D^{\mathrm{pre}}(a) = \mathrm{id}_D$, so it suffices to show that

$$\mathcal{F}(a) \circ \eta_A^\alpha = \eta_B^\alpha.$$

Compute

$$\begin{aligned}
 \mathcal{F}(a) \circ \eta_A^\alpha &= \mathcal{F}(a) \circ \mathcal{F}(\text{res}_A) \circ \alpha \\
 &= \mathcal{F}(\text{res}_A \circ a) \circ \alpha \\
 &= \mathcal{F}(\text{res}_B) \circ \alpha \\
 &= \eta_B^\alpha
 \end{aligned}$$

using the functoriality of $\mathcal{F}$ and the equality $\text{res}_A \circ a = \text{res}_B$. Thus η^α is natural. Hence one has a morphism

$$\begin{aligned}
 \Psi : \text{Hom}_{\mathcal{D}}(D, \mathcal{F}(\star)) &\rightarrow \text{Hom}_{\text{PSh}((\mathcal{C}, J), D)}(D^{\text{pre}}, \mathcal{F}) \\
 \alpha &\mapsto \eta^\alpha.
 \end{aligned}$$

Claim: Ψ and Φ are inverse.

For $\alpha : D \rightarrow \mathcal{F}(\star)$,

$$\begin{aligned}
 \Phi(\Psi(\alpha)) &= \Phi(\eta^\alpha) \\
 &= (\eta^\alpha)_\star \\
 &= \mathcal{F}(\text{res}_\star) \circ \alpha \\
 &= \mathcal{F}(\text{id}_\star) \circ \alpha \\
 &= \text{id}_{\mathcal{F}(\star)} \circ \alpha \\
 &= \alpha,
 \end{aligned}$$

since $\text{res}_\star : \star \rightarrow \star$ is $\text{id}_\star$. Thus $\Phi(\Psi(\alpha)) = \alpha$.

For $\eta : D^{\text{pre}} \rightarrow \mathcal{F}$, one shows that $\Psi(\Phi(\eta)) = \eta$.

Let $\alpha = \eta_\star$. Then for any A ,

$$(\Psi(\alpha))_A = \mathcal{F}(\text{res}_A) \circ \alpha = \mathcal{F}(\text{res}_A) \circ \eta_\star.$$

By naturality of η with respect to $\text{res}_A : A \rightarrow \star$,

$$\mathcal{F}(\text{res}_A) \circ \eta_\star = \eta_A \circ D^{\text{pre}}(\text{res}_A) = \eta_A \circ \text{id}_D = \eta_A.$$

So $(\Psi(\Phi(\eta)))_A = \eta_A$ for all A , hence $\Psi(\Phi(\eta)) = \eta$. Therefore Φ is a bijection with inverse Ψ , proving (2.2.32).

Together, these considerations give

$$\begin{aligned} \mathrm{Hom}_{\mathrm{Sh}((\mathcal{C}, J), \mathcal{D})}(D^{\mathrm{cst}}, \mathcal{F}) &\cong \mathrm{Hom}_{\mathrm{PSh}((\mathcal{C}, J), \mathcal{D})}(D^{\mathrm{pre}}, \mathcal{F}) \\ &\cong \mathrm{Hom}_{\mathcal{D}}(D, \Gamma(\mathcal{F})). \end{aligned}$$

Naturality in D and $\mathcal{F}$ follow from the constructions. Hence $D^{\mathrm{cst}} \dashv \Gamma$ (see [33, Ch. 2, §5, Thm 2, Corollary 4], [24, Ch. 2, Lemma 2.0.3]). $\square$

If one considers a sequence of sheaves, it is not self-evident that a global section of the quotient sheaf lifts to a global section of the original sheaf. Consequently, the global section functor need not preserve epimorphisms: a section of a quotient sheaf may fail to lift to a global section of the original sheaf.

Theorem 2.2.33. Let $(\mathcal{C}, J)$ be a small site with terminal object, and let $\mathcal{D}$ be an abelian category that admits the sheafification functor. Assume that $\mathrm{Sh}((\mathcal{C}, J), \mathcal{D})$ is abelian and that the constant sheaf functor exists. Then the global section functor

$$\Gamma((\mathcal{C}, J), -) : \mathrm{Sh}((\mathcal{C}, J), \mathcal{D}) \rightarrow \mathcal{D}$$

is left-exact (see [49, Ch. 1, §6, Prop. 1.6.6]).

Proof. The global section functor is right-adjoint to the constant sheaf functor by Theorem 2.2.31. Since right-adjoint functors preserve finite limits by Theorem 1.7.6, Γ preserves finite limits. In an abelian category, preserving finite limits implies left-exactness by Lemma 1.7.6, $\Gamma((\mathcal{C}, J), -)$ is left-exact. $\square$

Since the global section functor is right-adjoint to the constant sheaf functor, it preserves limits and forms the starting point for sheaf cohomology via its derived functors. The existence of local sections does not imply the existence of global sections. Torsors provide a natural framework for studying this discrepancy: they are locally trivial objects whose global triviality is equivalent to the existence of global sections.

2.3 Global Sections and Torsors

The geometry of topological spaces is encoded by its open subsets. However, in some situations one wishes to keep track of continuous functions on a topological

space. The collection of continuous functions on an open subset of a topological space forms a commutative ring under pointwise addition and multiplication. The assignment of continuous functions to open subsets, together with corresponding restrictions, defines a sheaf, motivating the notion of *ringed spaces*.

Definition 2.3.1 (Locally ringed space). A **locally ringed space** is a pair $(X, \mathcal{O}_X)$, where X is a topological space and $\mathcal{O}_X$ is a sheaf

$$\mathcal{O}_X : \text{Open}(X)^{\text{op}} \rightarrow \text{CRing}$$

such that all stalks $\mathcal{O}_{X,x}$ are local rings (see [8, Ch. 7, §5, Ex. 7.5.5], [14, Ch. 1, §1, Def. 1.1]).

Remark 2.3.2 (Category of locally ringed spaces LRS). Locally ringed spaces form a category. The morphisms between locally ringed spaces can be defined as follows:

Let $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ be locally ringed spaces and let $f \in \text{Hom}_{\text{Top}}(X, Y)$. A morphism between two locally ringed spaces $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ is a pair of morphisms

$$(f, f^\#) : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y),$$

where $f^\#$ is a morphism between sheaves of CRing

$$f^\# : \mathcal{O}_Y \rightarrow f_* \mathcal{O}_X. \quad (2.3.3)$$

For any $x \in X$, the sheaf morphism $f^\#$ induces a ring homomorphism on stalks

$$f_x^\# : \mathcal{O}_{Y, f(x)} \rightarrow \mathcal{O}_{X, x}. \quad (2.3.4)$$

The pair $(f, f^\#)$ is a morphism of locally ringed spaces if every induced stalk morphism $f_x^\#$ is local, that is,

$$(f_x^\#)^{-1}(\mathfrak{m}_{(X, x)}) = \mathfrak{m}_{(Y, f(x))}$$

denote the respective maximal ideals.

The ring homomorphism induced by $f^\#$ is characterised by the following commut-

ative diagram:

$$\begin{array}{ccc} \mathcal{O}_Y(V) & \longrightarrow & f_*\mathcal{O}_X(V) \\ \downarrow & & \downarrow \\ \mathcal{O}_{Y,f(x)} & \xrightarrow{\exists! f_x^\#} & \mathcal{O}_{X,x} \end{array} \quad (2.3.5)$$

for all open subsets $V \subseteq Y$ with $f(x) \in V$ (see [8, Ch. 7, §5, Ex. 7.5.5]).

In essence, for any point $x \in X$ and open subsets $V \subseteq Y$ such that $f(x) \in V$, the compositions

$$\mathcal{O}_Y(V) \xrightarrow{f^\#(V)} \mathcal{O}_X(f^{-1}(V)) \longrightarrow \mathcal{O}_{X,x}$$

are compatible with restriction morphisms of $\mathcal{O}_Y$. Hence, they induce a ring homomorphism

$$f_x^\# : \mathcal{O}_{Y,f(x)} \rightarrow \mathcal{O}_{X,x}$$

as claimed (see [7, Ch. 6 §6, Rem. 7], [14, Ch. 1, §1, Def. 1.4]).

The spectrum construction associates to every commutative ring R the topological space $\text{spec}(R)$ of prime ideals. Equipped with its structure sheaf, it becomes a locally ringed space.

Definition 2.3.6 (Scheme). An **affine scheme** is a locally ringed space isomorphic to the spectrum $\text{spec}(R)$ of a commutative ring R .

A **scheme** is a locally ringed space $(X, \mathcal{O}_X)$ admitting a cover by open sets U_i such that each locally ringed space $(U_i, \mathcal{O}_X|_{U_i})$ is an affine scheme. In particular, X comes with a sheaf $\mathcal{O}_X$, which assigns to every open subset U a commutative ring $\mathcal{O}_X(U)$ called the **ring of regular functions** on U (see [16, Ch. 15, §5, p. 742]).

Remark 2.3.7 (Morphisms of schemes and the category of schemes). The morphisms of schemes are precisely the morphisms between locally ringed spaces. Thus schemes form a category, denoted by Sch . In fact, the category of schemes is a full subcategory of LRS (see [16, Ch. 15, §5, p. 743]).

Since torsors are defined in terms of group actions, it is necessary to formulate the notion of a group in more general categories.

Definition 2.3.8 (Group object). Let $\mathcal{C}$ be a category with finite products, and denote its terminal object by T . A **group object** in $\mathcal{C}$ is an object $G \in \text{Ob}(\mathcal{C})$ with morphisms

GM1 $m : G \times G \rightarrow G$, informally thought of as "*group multiplication*" such that m is associative, that is,

$$m \circ (m \times \text{id}_G) = m \circ (\text{id}_G \times m)$$

GM2 an "*inclusion of the identity morphism*" $e : T \rightarrow G$ such that e is a two-sided unit of m , that is, the diagrams

$$\begin{array}{ccc} G \times T & \xrightarrow{\text{id}_G \times e} & G \times G \\ & \searrow \cong & \downarrow m \\ & & G, \end{array} \quad \begin{array}{ccc} T \times G & \xrightarrow{e \times \text{id}_G} & G \times G \\ & \searrow \cong & \downarrow m \\ & & G \end{array}$$

commute, and

GM3 a two-sided inverse $\text{inv} : G \rightarrow G$ of m , that is,

$$m \circ (\text{id}_G \times \text{inv}) \circ \blacktriangle_G = e_G = m \circ (\text{inv} \times \text{id}_G) \circ \blacktriangle_G,$$

where $e_G : G \rightarrow G$ is the morphism obtained by composing e with the unique morphism into the terminal object, that is,

$$G \rightarrow T \xrightarrow{e} G$$

(see [32, Ch. 8, §1, Def. 8.1.1]).

Example 2.3.9. Over a base scheme S , one can define

$$\mu_{2,S} := \text{spec}_S(\mathcal{O}_S[t]/(t^2 - 1)),$$

where the group structure comes from three morphisms:

1. Multiplication

On coordinate rings, one has

$$t \mapsto t \otimes t \text{ in } (\mathcal{O}_S[t]/(t^2 - 1)) \otimes (\mathcal{O}_S[t]/(t^2 - 1)).$$

Geometrically,

$$\mu : \mu_2 \times_S \mu_2 \rightarrow \mu_2$$

corresponds to multiplying two roots of unity. On U -points, $(a, b) \mapsto ab$.

2. Identity $e : S \rightarrow \mu_2$ corresponds to the element 1. On coordinate rings, this is $t \mapsto 1$.
3. Inverse $\text{inv} : \mu_2 \rightarrow \mu_2$. Since every element satisfies $a^2 = 1$, $a^{-1} = a$, making inv the identity morphism on μ_2 .

Thus, μ_2 is a group object.

Applied to the category of schemes, this motivates the notion of group schemes.

Definition 2.3.10 (Group scheme). Let S be a scheme. A **group scheme over S** is a group object in the category Sch/S of schemes over S . Thus it consists of an S -scheme G , equipped with a triple of S -morphisms

$$m : G \times_S G \rightarrow G$$

$$e : T \rightarrow G$$

$$\text{inv} : G \rightarrow G$$

satisfying the associativity of m , the identity, and the inverse axiom (see [52, Ch. 39, §4, Def. 39.4.1]).

Remark 2.3.11. Equivalently, a group scheme over S represents a group-valued functor

$$F : (\text{Sch}/S)^{\text{op}} \rightarrow \text{Grp}.$$

Conversely, if a group-valued functor has a representable underlying Set -valued functor, then its natural multiplication, identity, and inverse transformations correspond by the Yoneda lemma to a group-scheme structure on the representing S -scheme (see [48, Ch. 7, §1, Def. 7.14]).

Example 2.3.12. One can define a group scheme

$$\mu_{2,S}(U) := \{f \in \Gamma(U, \mathcal{O}_U^\times) \mid f^2 = 1\}.$$

Over a connected complex space, its local sections are locally constant with values in $\{\pm 1\}$.

Analogous to group actions, one can define actions on group schemes.

Definition 2.3.13 (Actions on schemes). Let G be a group scheme over T , and let P be a T -scheme. A **right action** of G on P is a T -morphism

$$\sigma : P \times_T G \rightarrow P$$

such that

$$\sigma \circ (\sigma \times \text{id}_G) = \sigma \circ (\text{id}_P \times m)$$

and

$$\sigma \circ (\text{id}_P \times e) = \text{id}_P,$$

under the canonical identifications.

A **left action** of G on P is a T -morphism

$$\sigma : G \times_T P \rightarrow P$$

such that

1. Associativity: $\sigma \circ (1_G \times \sigma) = \sigma \circ (m \times 1_S)$ where $m : G \times_T G \rightarrow G$ is the group law.
2. Unitality: $\sigma \circ (e \times 1_S) = 1_S$ where $e : T \rightarrow G$ is the identity section of G .

A **right action** of G on S is defined analogously (see [52, Ch. 39, §10, Def. 39.10.1]).

Although the following example is naturally sheaf-theoretic rather than scheme-theoretic, it illustrates the action underlying a torsor considered in Chapter 5.

Example 2.3.14. Consider a sheaf of local square roots

$$\mathcal{F}(U) := \{g \in \mathcal{O}_S(U)^\times \mid g^2 = f\},$$

where f is a nowhere vanishing function. The sheaf of groups

$$\mu_2(U) = \{\varepsilon \in \mathcal{O}_S(U)^\times \mid \varepsilon^2 = 1\}$$

acts on $\mathcal{F}(U)$ by multiplication, that is,

$$(g, \varepsilon) \mapsto g \cdot \varepsilon.$$

Similarly to group actions, one can define the notions of free and transitive actions on group schemes.

Definition 2.3.15 (Free and transitive actions). Let G be a group scheme over S acting on an S -scheme P . The action is called **free** if the morphism

$$P \times_S G \rightarrow P \times_S P, \quad (p, g) \mapsto (p, p \cdot g),$$

is a monomorphism. This implies that for every S -scheme U , the induced action of $G(U)$ on $P(U)$ is free.

The action is called **locally transitive** if for every scheme U and all sections $p_1, p_2 \in P(U)$ there exists a covering family $\{U_i \rightarrow U\}_{i \in I}$ such that for every $i \in I$ there exists $g_i \in G(U_i)$ with

$$p_1|_{U_i} = p_2|_{U_i} \cdot g_i$$

(see [52, Ch. 39, §10, Def. 39.10.2]).

Example 2.3.16. For the square-root torsor, the action of μ_2 is given by $\phi \cdot \varepsilon$. Locally, the two square roots are ϕ and $-\phi$. The element $1 \in \mu_2$ fixes ϕ , while $-1 \in \mu_2$ sends ϕ to $-\phi$. Thus locally any two square roots differ by a unique element of μ_2 .

Example 2.3.17 (Trivial torsor). Let G be a group scheme over S , and let U be an S -scheme. The projection

$$\mathrm{pr}_1 : U \times_S G \rightarrow U$$

together with the action

$$(u, g) \cdot h = (u, gh)$$

is called the **trivial G -torsor over U** (see [64]).

In the example of μ_2 , if f admits a square root ϕ on U , every local square root is uniquely of the form $\varepsilon \cdot \phi$ with $\varepsilon \in \mu_2(U)$. Hence the square-root torsor becomes isomorphic to the trivial μ_2 -torsor over U .

Definition 2.3.18 (Torsor). Let τ be a Grothendieck topology on the category Sch/S of schemes over S , and let G be a group scheme over S . A **G -torsor** over S is an S -scheme

$$f : P \rightarrow S,$$

equipped with a right action of G such that there exists a τ -covering family $\{U_i \rightarrow S\}_{i \in I}$ for which

$$P \times_S U_i \cong U_i \times_S G$$

as G -spaces over U_i for every $i \in I$ (see [27, Def. 4.1]).

Example 2.3.19. Let $f \in \Gamma(S, \mathcal{O}_S^\times)$, and define

$$P_f := \text{spec}_S(\mathcal{O}_S[t]/(t^2 - f)).$$

The group scheme $\mu_{2,S}$ acts on P_f by multiplication in the coordinate t . For every S -scheme U ,

$$P_f(U) = \{g \in \Gamma(U, \mathcal{O}_U^\times) \mid g^2 = f_U\}.$$

If f admits a square root ϕ_i over a covering $U_i \rightarrow S$, then multiplication by ϕ_i gives a $\mu_{2,S}$ -equivariant isomorphism

$$\mu_{2,S} \times_S U_i \xrightarrow{\sim} P_f \times_S U_i.$$

Hence P_f is a $\mu_{2,S}$ -torsor.

One can define morphisms between torsors. For the purpose of torsors in this thesis, it is sufficient to focus on isomorphisms, as torsors are classified up to isomorphism.

Definition 2.3.20 (Isomorphism of torsors). Let $\mathcal{T}_1$ and $\mathcal{T}_2$ be G -torsors over S . An **isomorphism of G -torsors** is an isomorphism of S -schemes $\varphi : \mathcal{T}_1 \rightarrow \mathcal{T}_2$ such that

$$\varphi(t \cdot g) = \varphi(t) \cdot g$$

for all sections t of $\mathcal{T}_1$ and g of G . Equivalently, if σ_1 and σ_2 denote the respective actions, then the diagram

$$\begin{array}{ccc} \mathcal{T}_1 \times_S G & \xrightarrow{\sigma_1} & \mathcal{T}_1 \\ \downarrow \varphi \times \text{id}_G & & \downarrow \varphi \\ \mathcal{T}_2 \times_S G & \xrightarrow{\sigma_2} & \mathcal{T}_2 \end{array}$$

commutes (see [52, Ch. 20, §4, Def. 20.4.1]).

Remark 2.3.21. Torsors will be considered up to G -equivariant isomorphism over S .

Locally, torsors are isomorphic to the trivial torsor.

Proposition 2.3.22. Let P be a G -torsor over S . Then the action of G on P is locally free and transitive (see [52, Ch. 39, §11]).

Proof. By definition, there exists a cover $\{U_i \rightarrow S\}$ such that

$$P \times_S U_i \cong U_i \times_S G$$

as G -spaces over U_i . Thus it suffices to check the assertion for the trivial torsor

$$U_i \times_S G \rightarrow U_i.$$

On the trivial torsor the right action is given by

$$(u, g) \cdot h = (u, g \cdot h).$$

If

$$(u, g) \cdot h = (u, g) \quad \Rightarrow \quad \underbrace{g^{-1} \cdot g \cdot h}_{=e \cdot h = h} = \underbrace{g^{-1} \cdot g}_{=e} \quad \Rightarrow \quad h = e,$$

since G is a group. Thus the action is free. Moreover, given two points $(u, g_1), (u, g_2) \in U_i \times_S G$, there exists a unique element

$$h = g_1^{-1} \cdot g_2$$

such that

$$(u, g_1) \cdot h = (u, g_2),$$

making the action transitive. Since it holds after passing to the covering family $\{U_i \rightarrow S\}$, the original torsor action is locally free and transitive (see [52, Ch. 39, §11]). $\square$

Torsors are closely connected to sheaves, not only by means of locally ringed spaces and schemes, but also through global sections, as far they exist. The following criterion relates the existence of global sections to triviality of the torsor.

Theorem 2.3.23. Let $\mathcal{T}$ be a G -torsor over S . Then $\mathcal{T}$ is trivial if and only if it admits a global section (see [52, Ch. 21, §4, Lemma 21.4.2]).

Proof. Assume that $\mathcal{T}$ is isomorphic, as a G -torsor over S , to the trivial torsor $G \rightarrow S$. Then the identity section $e \in G(S)$ yields a global section of $\mathcal{T}$.

Assume that $\mathcal{T}$ admits a global section $s \in \mathcal{T}(S)$. For every S -scheme U , define

$$\phi_U : G(U) \rightarrow \mathcal{T}(U), \quad g \mapsto s|_U \cdot g.$$

Since $\mathcal{T}$ is a G -torsor, there exists a covering family $\{U_i \rightarrow S\}$ such that

$$\mathcal{T} \times_S U_i \cong U_i \times_S G.$$

After restricting to a trivialising cover $U_i \rightarrow S$, choose a G -equivariant isomorphism

$$\theta_i : G \times_S U_i \xrightarrow{\sim} \mathcal{T} \times_S U_i.$$

The section $s|_{U_i}$ corresponds under θ_i to a section $g_i \in G(U_i)$. Composing θ_i with translation by g_i^{-1} produces a G -equivariant trivialisation sending the identity section

to $s|_{U_i}$. Under this adjusted trivialisation, the morphism

$$\phi_{U_i} : G \times_S U_i \rightarrow \mathcal{T} \times_S U_i, \quad g \mapsto s|_{U_i} \cdot g,$$

is an isomorphism. Since all local morphisms ϕ_{U_i} are induced by the restriction of the same global section s , they are compatible on overlaps and therefore glue to an isomorphism

$$\phi : G \rightarrow \mathcal{T}$$

of G -torsors. □

Thus, a torsor may be viewed as a space that is locally indistinguishable from the group itself, but lacks a distinguished identity morphism. While torsors tell what the obstruction is, one needs something else to organise and compute these obstructions.

3 Homological Algebra

Homological algebra provides a systematic framework for studying complexes, exactness, and the failure of functors to preserve exactness. The central objects of study are chain complexes, whose homology and cohomology objects measure the failure of cycles to be boundaries. Homology and cohomology objects encode information that is invisible at the level of individual morphisms and frequently admit geometric or topological interpretations.

A second fundamental problem arises from functors that preserve only part of the exact structure. Many naturally occurring functors preserve kernels but not cokernels, or conversely. For example, the global section functor is left exact, but a global section of a quotient object need not lift to a global section of the original object. Derived functors measure this failure of exactness.

This chapter introduces the foundational machinery of homological algebra used throughout the remainder of the thesis. After introducing chain and cochain complexes, the chapter develops injective resolutions and derived functors, discusses acyclic objects and resolutions, and concludes with the basic formalism of spectral sequences.

3.1 Chains and Cochains

Definition 3.1.1 (Chain complex, cochain complex). Let $\mathcal{C}$ be an additive category. A **chain complex** $(A_\bullet, \delta_\bullet)$ is a sequence of objects $(A_n)_{n \in \mathbb{Z}}$ in $\mathcal{C}$ together with morphisms $\delta_n : A_n \rightarrow A_{n-1}$ such that the composition of any consecutive maps satisfies

$$\delta_{n-1} \circ \delta_n = 0$$

for all $n \in \mathbb{Z}$. These morphisms are called **boundary operators** or **differentials**. The chain complex may be written as

$$\dots \xleftarrow{\delta_0} A_0 \xleftarrow{\delta_1} A_1 \xleftarrow{\delta_2} A_2 \xleftarrow{\delta_3} A_3 \xleftarrow{\delta_4} A_4 \xleftarrow{\delta_5} \dots$$

A **cochain complex** $(A^\bullet, \delta^\bullet)$ consists of a sequence of objects $(A^n)_{n \in \mathbb{Z}}$ in $\mathcal{C}$ together with morphisms $\delta^n : A^n \rightarrow A^{n+1}$ such that the composition of any consecutive maps satisfies

$$\delta^{n+1} \circ \delta^n = 0$$

for all $n \in \mathbb{Z}$. These maps are called **coboundary operators**. The cochain complex may be written as

$$\dots \xrightarrow{\delta^{-1}} A^0 \xrightarrow{\delta^0} A^1 \xrightarrow{\delta^1} A^2 \xrightarrow{\delta^2} A^3 \xrightarrow{\delta^3} A^4 \xrightarrow{\delta^4} \dots$$

(see [35, Ch. 2, §2, pp. 39 ff.]).

If $\mathcal{C}$ is an abelian category, a **cochain subcomplex** $(B^\bullet, \delta^\bullet|_{B^\bullet})$ of a cochain complex $(A^\bullet, \delta^\bullet)$ consists of monomorphisms

$$\iota^n : B^n \hookrightarrow A^n$$

such that

$$\delta_A^n \circ \iota^n = \iota^{n+1} \circ \delta_B^n$$

for every $n \in \mathbb{Z}$. Equivalently, the morphisms

$$\delta_A^n \circ \iota^n$$

factor through the subobjects

$$B^{n+1} \hookrightarrow A^{n+1}$$

(see [35, Ch. 2, §2, pp. 39 ff.]).

In chain complexes and in cochain complexes, the index n is referred to as the **degree**. The difference is that in chain complexes, the differential decreases the degree, while in cochain complexes, the differential increases the degree. A complex is therefore a sequence in which consecutive morphisms compose to zero. While every exact sequence is a complex, a complex need not be exact.

Definition 3.1.2 (Chain map). Let $(A_\bullet, \delta_\bullet^{(A)})$ and $(B_\bullet, \delta_\bullet^{(B)})$ be chain complexes in an additive category $\mathcal{C}$. A **chain map**

$$f_\bullet : A_\bullet \rightarrow B_\bullet$$

consists of morphisms

$$f_n : A_n \rightarrow B_n$$

such that for every $n \in \mathbb{Z}$,

$$\delta_n^{(B)} \circ f_n = f_{n-1} \circ \delta_n^{(A)}$$

(see [35, Ch. 2, §2, pp. 39 ff.]).

Remark 3.1.3. Equivalently, the diagram

$$\begin{array}{ccc} A_n & \xrightarrow{\delta_n^{(A)}} & A_{n-1} \\ \downarrow f_n & & \downarrow f_{n-1} \\ B_n & \xrightarrow{\delta_n^{(B)}} & B_{n-1} \end{array}$$

commutes. Moreover, chain complexes in $\mathcal{C}$, together with chain maps, form a category denoted by $\text{Ch}(\mathcal{C})$ (see [35, Ch. 2, §2, pp. 39 ff.]).

Definition 3.1.4 (Cochain map). Let $(A^\bullet, \delta_A^\bullet)$, $(B^\bullet, \delta_B^\bullet)$ be cochain complexes. A **cochain map**

$$f^\bullet : (A^\bullet, \delta_A^\bullet) \rightarrow (B^\bullet, \delta_B^\bullet)$$

is a collection of morphisms

$$f^n : A^n \rightarrow B^n$$

in $\mathcal{C}$ such that

$$\delta_B^n \circ f^n = f^{n+1} \circ \delta_A^n$$

for every $n \in \mathbb{Z}$ (see [35, Ch. 2, §2, pp. 39 ff.]).

Remark 3.1.5. Cochain complexes in $\mathcal{C}$, together with cochain maps, form a category denoted by $\text{Coch}(\mathcal{C})$. Note that this is not the same as $\text{Ch}(\mathcal{C})^{\text{op}}$, as the objects are not chains but cochains.

Chain maps preserve the differential structure of complexes. To compare chain maps themselves, one introduces the notion of chain homotopy, which is the algebraic analogue of homotopy between continuous maps.

Definition 3.1.6 (Chain homotopy). Let $(A_\bullet, \delta_\bullet^{(A)})$, $(B_\bullet, \delta_\bullet^{(B)})$ be chain complexes with chain maps

$$f_\bullet, g_\bullet : A_\bullet \rightarrow B_\bullet.$$

A **chain homotopy** between $f_\bullet$ and $g_\bullet$ is a family of maps

$$M_n : A_n \rightarrow B_{n+1}$$

such that for every $n \in \mathbb{Z}$

$$g_n - f_n = \delta_{n+1}^{(B)} \circ M_n + M_{n-1} \circ \delta_n^{(A)}. \quad (3.1.7)$$

The equation (3.1.7) is called the **chain homotopy identity** (see [35, Ch. 2, §2, pp. 39 ff.]).

A chain map that is chain homotopic to the zero chain map is called **null-homotopic** (see [35, Ch. 2, §2, pp. 39 ff.]).

Dually, let $(A^\bullet, \delta_{(A)}^\bullet)$, $(B^\bullet, \delta_{(B)}^\bullet)$ be cochain complexes with cochain maps

$$f^\bullet, g^\bullet : A^\bullet \rightarrow B^\bullet.$$

A **cochain homotopy** between $f^\bullet$ and $g^\bullet$ is a family of maps

$$M^n : A^n \rightarrow B^{n-1}$$

such that for every $n \in \mathbb{Z}$

$$g^n - f^n = \delta_{(B)}^{n-1} \circ M^n + M^{n+1} \circ \delta_{(A)}^n. \quad (3.1.8)$$

The equation (3.1.8) is called the **cochain homotopy identity**. A cochain map that is cochain homotopic to the zero cochain map is called **null-homotopic** (see [35, Ch. 2, §2, pp. 39 ff.]).

Chain homotopies identify chain maps that differ by a boundary term. They play

the algebraic role of ordinary homotopies in topology and induce the same maps on homology.

Lemma 3.1.9. Additive functors preserve chain homotopies and cochain homotopies (see [35, Ch. 10, §9, p. 301]).

Proof. Since the proof for chain homotopies is analogous, only the proof for cochain homotopies is provided in this thesis.

Let

$$F : \mathcal{C} \rightarrow \mathcal{D}$$

be an additive functor, and let

$$f^\bullet, g^\bullet : A^\bullet \rightarrow B^\bullet$$

be cochain-homotopic cochain maps. Thus there exist morphisms

$$M^n : A^n \rightarrow B^{n-1}$$

such that

$$g^n - f^n = \delta_{(B)}^{n-1} \circ M^n + M^{n+1} \circ \delta_{(A)}^n.$$

Applying F yields

$$F(g^n) - F(f^n) = F(\delta_{(B)}^{n-1}) \circ F(M^n) + F(M^{n+1}) \circ F(\delta_{(A)}^n),$$

and

$$F(\delta_A^{n+1}) \circ F(\delta_A^n) = F(\delta_A^{n+1} \circ \delta_A^n) = F(0_{A^n, A^{n+2}}) = 0,$$

since functors preserve composition, and additive functors preserve sums and zero morphisms. Hence $(F(A^\bullet), F(\delta_A^\bullet))$ and $(F(B^\bullet), F(\delta_B^\bullet))$ are cochain complexes. Thus the family $F(M^n)$ is a cochain homotopy between $F(f^\bullet)$ and $F(g^\bullet)$ (see [35, Ch. 10, §9, p. 301]).

The proof for chain homotopies is analogous. □

A special chain homotopy is one between the identity map of a complex and the zero morphism. Such a homotopy contracts the entire complex.

Definition 3.1.10 (Cone operator). Let $(C_\bullet, \delta_\bullet)$ be a chain complex. A family of morphisms

$$M_n : C_n \rightarrow C_{n+1}$$

is called a **cone operator** or **contracting homotopy** if

$$\delta_{n+1} \circ M_n + M_{n-1} \circ \delta_n = \text{id}_{C_n}$$

for all $n \in \mathbb{Z}$.

If $C_\bullet$ is concentrated in non-negative degrees, then $C_n = 0$ for $n < 0$, and the corresponding components M_n are necessarily zero in those degrees (see [35, Ch. 2, §2, p. 41]).

A cone operator provides a chain homotopy between the identity and the zero morphism. Thus a complex admitting a cone operator is **contractible**, meaning that its identity chain map is null-homotopic. In particular, in an abelian category, every contractible complex has trivial homology.

3.2 Abstract Cohomology

Chain complexes are useful because they allow one to measure failure of exactness. Given a cochain complex, one distinguishes the kernel of each differential from the image of the preceding differential. The discrepancy between these two notions is captured by cohomology.

Construction 3.2.1. Let $(A^\bullet, \delta_A^\bullet)$ be a cochain complex in an abelian category. One can construct the following objects of $\mathcal{C}$ (and subobjects of A^n):

1. $Z_A^n = \ker(\delta_A^n)$
2. $B_A^n = \text{im}(\delta_A^{n-1})$

The object Z_A^n is called the **object of n -cocycles**, and the object B_A^n is called the **object of n -coboundaries**. Since for any $n \in \mathbb{Z}$ it holds that

$$\delta_A^{n+1} \circ \delta_A^n = 0,$$

the morphism

$$\delta_A^{n-1} : A^{n-1} \rightarrow A^n$$

factors through

$$\ker(\delta_A^n) = Z_A^n.$$

Consequently, its image

$$B_A^n = \text{im}(\delta_A^{n-1})$$

is canonically a subobject of Z_A^n (see [35, Ch. 2, §3, pp. 42 ff.]).

Remark 3.2.2. For every $D \in \text{Ob}(\mathcal{C})$, one may call two morphisms

$$f, g : D \rightarrow Z_A^n$$

cohomologous if $f - g$ factors through

$$B_A^n \hookrightarrow Z_A^n.$$

This defines an equivalence relation on $\text{Hom}(D, Z_A^n)$ (see [35, Ch. 2, §3, pp. 42 ff.]).

If it is clear which cochain complex $\delta_A^\bullet$, Z_A^n , and B_A^n refer to, one may denote them by $\delta^\bullet$, Z^n , and B^n .

Each object Z^n , B^n in Definition 3.2.1 comes with a canonical monomorphism into A^n , making the objects Z^n and B^n subobjects of A^n . If

$$Z^n = \ker(\delta^n),$$

then one has a monomorphism

$$\iota_Z : Z^n \rightarrow A^n$$

by the definition of kernels. If

$$B^n = \text{im}(\delta^{n-1}) = \ker(\text{coker}(\delta^{n-1})),$$

then the image is again defined as the kernel of its cokernel, so its structure includes

a monomorphism

$$\iota_{A_2} : B^n \rightarrow A^n,$$

often called **image inclusion**. Thus the inclusions

$$Z^n \hookrightarrow A^n, \quad B^n \hookrightarrow A^n$$

are the natural kernel morphism occurring in the definition of Z^n and B^n (see [35, Ch. 2, §3, pp. 42 ff.]).

Definition 3.2.3 (Cohomology object). Let $(A^\bullet, \delta^\bullet)$ be a cochain complex in an abelian category. The **n -th cohomology object** of $(A^\bullet, \delta^\bullet)$ is defined as the quotient object

$$H^n(A^\bullet, \delta^\bullet) = Z^n / B^n$$

(see [35, Ch. 2, §3, pp. 42 ff.]).

Remark 3.2.4. Note that since B^n is a subobject of Z^n inside an abelian category, the quotient Z^n / B^n exists: If

$$\delta^n \circ \delta^{n-1} = 0,$$

one has

$$B^n = \text{im}(\delta^{n-1}) \subseteq \ker(\delta^n) = Z^n.$$

Therefore the quotient object

$$H^n(A^\bullet, \delta^\bullet) = Z^n / B^n$$

is defined.

Moreover, for every object T , two morphisms $f, g : T \rightarrow Z_A^n$ are called **cohomologous** if $f - g$ factors through B_A^n .

The n -th cohomology object measures the failure of exactness of the complex in degree n . By the universal property of the cokernel, morphisms

$$H^n(A^\bullet, \delta^\bullet) \rightarrow T$$

correspond naturally to morphisms

$$Z^n \rightarrow T$$

whose composite with $B^n \hookrightarrow Z^n$ is zero (see [35, Ch. 2, §3, pp. 42 ff.]).

Since cohomology is defined as a quotient of cocycles and coboundaries, it should detect precisely the failure of exactness. The following theorem makes this interpretation precise.

Theorem 3.2.5. Let $(A^\bullet, \delta^\bullet)$ be a cochain complex in an abelian category. Then

$$H^n(A^\bullet, \delta^\bullet) = 0$$

if and only if $(A^\bullet, \delta^\bullet)$ is exact at n , that is,

$$\text{im}(\delta^{n-1}) = \ker(\delta^n)$$

(see [47, Ch. 6, §1, Ex. 6.7]).

Proof. Since $\delta^n \circ \delta^{n-1} = 0$, the quotient object $H^n(A^\bullet, \delta^\bullet) = Z^n/B^n$ is defined by Remark 3.2.4.

" $\Rightarrow$ ": Suppose that $H^n(A^\bullet, \delta^\bullet) = 0$. Since

$$H^n(A^\bullet, \delta^\bullet) = \text{coker}(B^n \hookrightarrow Z^n)$$

the monomorphism $B^n \hookrightarrow Z^n$ has trivial cokernel. In an abelian category, a monomorphism with trivial cokernel is an isomorphism by Lemma 1.6.22. Hence the canonical morphism $B^n \hookrightarrow Z^n$ is an isomorphism. Therefore B^n and Z^n represent the same subobject of A^n , so

$$\text{im}(\delta^{n-1}) = \ker(\delta^n).$$

Thus

$$\ker(\delta^n) = \text{im}(\delta^{n-1}).$$

making the complex exact at n .

" $\Leftarrow$ ": If the complex is exact at n , then the canonical monomorphism

$$B^n \hookrightarrow Z^n$$

is an isomorphism. Its cokernel is therefore zero, and hence

$$H^n(A^\bullet, \delta^\bullet) = 0$$

(see [47, Ch. 6, §1, Ex. 6.7]). □

Remark 3.2.6. The cohomology measures non-exactness: the n -th cohomology object is zero if and only if the cochain complex is exact in degree n .

Cohomology is not merely an invariant of a single cochain complex. Morphisms of complexes induce morphisms on cohomology, making cohomology a functorial construction.

Proposition 3.2.7. Let $f^\bullet : (A_1^\bullet, \delta_{A_1}^\bullet) \rightarrow (A_2^\bullet, \delta_{A_2}^\bullet)$ be a morphism of cochain complexes in an abelian category $\mathcal{C}$. Then for every $n \in \mathbb{Z}$, there is an induced morphism

$$H^n(f^\bullet) : H^n(A_1^\bullet, \delta_{A_1}^\bullet) \rightarrow H^n(A_2^\bullet, \delta_{A_2}^\bullet).$$

Thus $H^n : \text{Coch}(\mathcal{C}) \rightarrow \mathcal{C}$ is functorial (see [47, Ch. 6, §1, Prop. 6.8]).

Proof. Since $f^\bullet$ is a cochain map, it holds that

$$\delta_{A_2}^n \circ f^n = f^{n+1} \circ \delta_{A_1}^n.$$

Let

$$\iota : Z_{A_1}^n \hookrightarrow A_1^n, \quad \iota_{Z,2}^n : Z_{A_2}^n \hookrightarrow A_2^n$$

denote the respective cocycle inclusions. Then

$$\delta_{A_2}^n \circ f^n \circ \iota_{Z,1}^n = f^{n+1} \circ \delta_{A_1}^n \circ \iota_{Z,1}^n = 0.$$

By the universal property of the kernel $Z_{A_2}^n = \ker(\delta_{A_2}^n)$, there exists a unique morphism

$$Z^n(f^\bullet) : Z_{A_1}^n \rightarrow Z_{A_2}^n$$

such that

$$\iota_{Z,2}^n \circ Z^n(f^\bullet) = f^n \circ \iota_{Z,1}^n.$$

Similarly, $f^\bullet$ induces a morphism on coboundaries. Since

$$f^n \circ \delta_{A_1}^{n-1} = \delta_{A_2}^{n-1} \circ f^{n-1},$$

the morphism

$$f^n \circ \delta_{A_1}^{n-1} : A_1^{n-1} \rightarrow A_2^n$$

factors through $B_{A_2}^n = \text{im}(\delta_{A_2}^{n-1})$. Let

$$\iota_{B,1}^n : B_{A_1}^n \hookrightarrow A_1^n, \quad \iota_{B,2}^n : B_{A_2}^n \hookrightarrow A_2^n$$

denote the respective image inclusions. By the universal property of the image factorisation, there exists a unique morphism

$$B^n(f^\bullet) : B_{A_1}^n \rightarrow B_{A_2}^n$$

such that

$$\iota_{B,2}^n \circ B^n(f^\bullet) = f^n \circ \iota_{B,1}^n.$$

Since $B_{A_i}^n$ is canonically a subobject of $Z_{A_i}^n$, let

$$j_i^n : B_{A_i}^n \hookrightarrow Z_{A_i}^n$$

denote the corresponding monomorphism for $i \in \{1, 2\}$. Then

$$\begin{array}{ccc} B_{A_1}^n & \xrightarrow{B^n(f^\bullet)} & B_{A_2}^n \\ j_1^n \downarrow & & \downarrow j_2^n \\ Z_{A_1}^n & \xrightarrow{Z^n(f^\bullet)} & Z_{A_2}^n \end{array}$$

commutes.

Let

$$\pi_i^n : Z_{A_i}^n \twoheadrightarrow H^n(A_i^\bullet, \delta_{A_i}^\bullet)$$

be the cokernel of j_i^n . Then

$$\pi_2^n \circ Z^n(f^\bullet) \circ j_1^n = \pi_2^n \circ j_2^n \circ B^n(f^\bullet) = 0.$$

Therefore, by the universal property of the cokernel π_1^n , there exists a unique morphism

$$H^n(f^\bullet) : H^n(A_1^\bullet, \delta_{A_1}^\bullet) \rightarrow H^n(A_2^\bullet, \delta_{A_2}^\bullet)$$

such that

$$H^n(f^\bullet) \circ \pi_1^n = \pi_2^n \circ Z^n(f^\bullet).$$

The construction of $Z^n(f^\bullet)$ and $B^n(f^\bullet)$ are compatible with the identity morphisms and compositions by uniqueness in the corresponding universal properties. Consequently,

$$H^n(\text{id}_{A^\bullet}) = \text{id}_{H^n(A^\bullet)}$$

and

$$H^n(g^\bullet \circ f^\bullet) = H^n(g^\bullet) \circ H^n(f^\bullet).$$

Hence

$$H^n : \text{Coch}(\mathcal{C}) \rightarrow \mathcal{C}$$

is a functor (see [47, Ch. 6, §1, Prop. 6.8]). □

Functoriality associates morphisms on cohomology to morphisms of complexes. However, cohomology cannot distinguish between morphisms that differ only by a cochain homotopy. The following lemma shows that cochain-homotopic morphisms induce identical maps on cohomology.

Lemma 3.2.8. Cochain homotopic cochain maps induce the same morphism on cohomology (see [47, Ch. 6, §1, Thm. 6.14]).

Proof. Let $f^\bullet, g^\bullet : A_1^\bullet \rightarrow A_2^\bullet$ be cochain-homotopic, so there exist morphisms

$$M^n : A_1^n \rightarrow A_2^{n-1}$$

with

$$g^n - f^n = \delta_{A_2}^{n-1} \circ M^n + M^{n+1} \circ \delta_{A_1}^n.$$

Let

$$\iota_{Z,1}^n : Z_{A_1}^n \hookrightarrow A_1^n, \quad \iota_{Z,2}^n : Z_{A_2}^n \hookrightarrow A_2^n$$

denote the cocycle inclusions, and let

$$\pi_2^n : Z_{A_2}^n \twoheadrightarrow H^n(A_2^\bullet)$$

be the cokernel of the canonical inclusion

$$B_{A_2}^n \hookrightarrow Z_{A_2}^n.$$

By functoriality of cocycles, $f^\bullet$ and $g^\bullet$ induce morphisms

$$Z^n(f^\bullet), Z^n(g^\bullet) : Z_{A_1}^n \rightarrow Z_{A_2}^n$$

satisfying

$$\iota_{Z,2}^n \circ Z^n(f^\bullet) = f^n \circ \iota_{Z,1}^n$$

and

$$\iota_{Z,2}^n \circ Z^n(g^\bullet) = g^n \circ \iota_{Z,1}^n.$$

Therefore

$$\iota_{Z,2}^n \circ (Z^n(g^\bullet) - Z^n(f^\bullet)) = (g^n - f^n) \circ \iota_{Z,1}^n.$$

Since

$$\delta_{A_1}^n \circ \iota_{Z,1}^n = 0,$$

the cochain–homotopy identity gives

$$\begin{aligned} (g^n - f^n) \circ \iota_{Z,1}^n &= \delta_{A_2}^{n-1} \circ M^n \circ \iota_{Z,1}^n + M^{n+1} \circ \delta_{A_1}^n \circ \iota_{Z,1}^n \\ &= \delta_{A_2}^{n-1} \circ M^n \circ \iota_{Z,1}^n. \end{aligned}$$

The differential

$$\delta_{A_2}^{n-1} : A_2^{n-1} \rightarrow A_2^n$$

factors through its image

$$B_{A_2}^n = \text{im}(\delta_{A_2}^{n-1}),$$

and $B_{A_2}^n$ is canonically a subobject of $Z_{A_2}^n$. Hence the morphism

$$Z^n(g^\bullet) - Z^n(f^\bullet) : Z_{A_1}^n \rightarrow Z_{A_2}^n$$

factors through

$$B_{A_2}^n \hookrightarrow Z_{A_2}^n.$$

Since π_2^n is the cokernel of this inclusion,

$$\pi_2^n \circ (Z^n(g^\bullet) - Z^n(f^\bullet)) = 0.$$

Therefore

$$\pi_2^n \circ Z^n(g^\bullet) = \pi_2^n \circ Z^n(f^\bullet).$$

By the construction of the induced morphisms on cohomology,

$$H^n(g^\bullet) \circ \pi_1^n = \pi_2^n \circ Z^n(g^\bullet)$$

and

$$H^n(f^\bullet) \circ \pi_1^n = \pi_2^n \circ Z^n(f^\bullet),$$

where

$$\pi_1^n : Z_{A_1}^n \twoheadrightarrow H^n(A_1^\bullet)$$

is the canonical quotient morphism. Thus

$$H^n(g^\bullet) \circ \pi_1^n = H^n(f^\bullet) \circ \pi_1^n.$$

Since π_1^n is an epimorphism,

$$H^n(g^\bullet) = H^n(f^\bullet).$$

Hence cochain-homotopic maps induce the same morphism on cohomology (see [47, Ch. 6, §1, Thm. 6.14]). $\square$

Since cohomology is often the primary invariant of interest, it is natural to single out those morphisms of complexes that induce isomorphisms on all cohomology objects.

Definition 3.2.9 (Quasi-isomorphism). Let $f : A^\bullet \rightarrow B^\bullet$ be a morphism of cochain complexes in an abelian category. Then f is a **quasi-isomorphism** if it induces isomorphisms on cohomology in every degree:

$$H^n(f) : H^n(A^\bullet) \xrightarrow{\cong} H^n(B^\bullet)$$

for all $n \in \mathbb{Z}$ (see [47, Ch. 10, §5, p. 651]).

Note that a quasi-isomorphism does not require f itself to be an isomorphism. Cohomology behaves particularly well with respect to short exact sequences of cochain complexes. The resulting relationship between the cohomology objects is expressed by the long exact sequence.

Construction 3.2.10 (Long exact sequence). Let

$$0 \rightarrow A_1^\bullet \xrightarrow{f^\bullet} A_2^\bullet \xrightarrow{g^\bullet} A_3^\bullet \rightarrow 0$$

be a short exact sequence of cochain complexes. Then there exist **connecting morphisms**

$$\partial^n : H^n(A_3^\bullet, \delta_{A_3}^\bullet) \rightarrow H^{n+1}(A_1^\bullet, \delta_{A_1}^\bullet)$$

such that the sequence

$$\begin{array}{ccccccc} \dots & \xrightarrow{\partial^{n-1}} & H^n(A_1^\bullet, \delta_{A_1}^\bullet) & \xrightarrow{f_*^n} & H^n(A_2^\bullet, \delta_{A_2}^\bullet) & \xrightarrow{g_*^n} & H^n(A_3^\bullet, \delta_{A_3}^\bullet) \\ & & & & & & \downarrow \partial^n \\ & & H^{n+1}(A_1^\bullet, \delta_{A_1}^\bullet) & \xrightarrow{f_*^{n+1}} & H^{n+1}(A_2^\bullet, \delta_{A_2}^\bullet) & \xrightarrow{g_*^{n+1}} & H^{n+1}(A_3^\bullet, \delta_{A_3}^\bullet) \xrightarrow{\partial^{n+1}} \dots \end{array}$$

is exact.

The connecting morphisms are obtained from the snake lemma (Theorem 1.7.8): For each n consider the commutative diagram with exact rows

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A_1^n & \xrightarrow{f^n} & A_2^n & \xrightarrow{g^n} & A_3^n & \longrightarrow & 0 \\ & & \downarrow \delta_{A_1}^n & & \downarrow \delta_{A_2}^n & & \downarrow \delta_{A_3}^n & & \\ 0 & \longrightarrow & A_1^{n+1} & \xrightarrow{f^{n+1}} & A_2^{n+1} & \xrightarrow{g^{n+1}} & A_3^{n+1} & \longrightarrow & 0 \end{array}.$$

Applying Theorem 1.7.8 yields a connecting morphism

$$\tilde{\partial}^n : \ker(\delta_{A_3}^n) = Z_{A_3}^n \rightarrow \operatorname{coker}(\delta_{A_1}^n).$$

Since in an abelian category the cokernel of a morphism is canonically isomorphic to the cokernel of its image inclusion,

$$\operatorname{coker}(\delta_{A_1}^n) \cong A_1^{n+1}/B_{A_1}^{n+1}.$$

By the construction of the snake morphism, the morphism

$$\tilde{\partial}^n : Z_{A_3}^n \rightarrow A_1^{n+1}/B_{A_1}^{n+1}$$

vanishes on the subobject $B_{A_3}^n \subseteq Z_{A_3}^n$, hence it factors through the quotient

$$Z_{A_3}^n \twoheadrightarrow Z_{A_3}^n/B_{A_3}^n = H^n(A_3^\bullet, \delta_{A_3}^\bullet),$$

yielding

$$\hat{\partial}^n : H^n(A_3^\bullet, \delta_{A_3}^\bullet) \rightarrow A_1^{n+1}/B_{A_1}^{n+1}.$$

It remains to show that $\hat{\partial}^n$ factors through the kernel of $\bar{\delta}_{A_1}^{n+1}$.

Since

$$\bar{\delta}_{A_1}^{n+1} \circ \hat{\partial}^n = 0,$$

the universal property of the kernel yields a unique factorisation

$$\partial^n : H^n(A_3^\bullet) \rightarrow H^{n+1}(A_1^\bullet).$$

By the construction of the snake morphism,

$$\bar{\delta}_{A_1}^{n+1} \circ \hat{\partial}^n = 0.$$

Therefore

$$\operatorname{im}(\hat{\partial}^n) \subseteq \ker(\bar{\delta}_{A_1}^{n+1}) \cong Z_{A_1}^{n+1}/B_{A_1}^{n+1} = H^{n+1}(A_1^\bullet, \delta_{A_1}^\bullet)$$

by theorem 1.6.25. Thus ∂^n is a well-defined morphism

$$\partial^n : H^n(A_3^\bullet, \delta_{A_3}^\bullet) \rightarrow H^{n+1}(A_1^\bullet, \delta_{A_1}^\bullet).$$

Applying the Snake Lemma 1.7.8 in each degree, and identifying the resulting kernels and cokernels with the corresponding cohomology objects yields the following long exact sequence:

$$\cdots \rightarrow H^n(A_1^\bullet) \rightarrow H^n(A_2^\bullet) \rightarrow H^n(A_3^\bullet) \xrightarrow{\partial^n} H^{n+1}(A_1^\bullet) \rightarrow \cdots$$

whose exactness follows from the Snake Lemma 1.7.8 (see [47, Ch. 6, §1, Prop. 6.9, Thm. 6.10]).

The preceding construction is entirely categorical and relies on the snake lemma (Theorem 1.7.8).

Remark 3.2.11 (Element-level construction of the connecting morphism). In concrete abelian categories such as $\mathbf{Ab}$ or $R - \mathbf{Mod}$, the connecting morphism admits the following description:

Given a class $[a_3] \in H^n(A_3^\bullet)$, one can choose a cocycle representative a_3 , lift it to $a_2 \in A_2^\bullet$, apply the differential to obtain $\delta(a_2)$, and lift this uniquely through f^{n+1} to a cocycle $a_1 \in A_1^{n+1}$.

The connecting morphism is then $\partial^n([a_3]) = [a_1]$. A diagram chase shows that this construction is independent of all choices and agrees with the categorical connecting morphism constructed above (see [47, Ch. 6, §1, Prop. 6.9, Thm. 6.10]).

Cohomology is defined for cochain complexes. To associate cohomological invariants to individual objects, one replaces an object by a suitable chosen cochain complex.

3.3 Injective Resolutions

Many functors, such as the global section functor, are not exact. Thus, applying such a functor to a short exact sequence need not preserve its exactness. However, there are distinguished objects on which left-exact functors exhibit particularly favourable exactness properties.

Definition 3.3.1 (Injective object). Let $\mathcal{C}$ be a category. An object $A_1 \in \mathbf{Ob}(\mathcal{C})$ is called **injective** if for every monomorphism $f : A_2 \rightarrow A_3$ and every morphism

$g : A_2 \rightarrow A_1$ there exists a morphism $h : A_3 \rightarrow A_1$ extending g to A_3 , that is, g factors through an arbitrary monomorphism $f : A_2 \hookrightarrow A_3$ as

$$h \circ f = g.$$

The category $\mathcal{C}$ has **enough injectives** if for every object $A_1 \in \text{Ob}(\mathcal{C})$, there exists a monomorphism from A_1 to an injective object (see [47, Ch. 5, §5, p.314]).

Diagrammatically,

$$\begin{array}{ccc} A_2 & \xhookrightarrow{f} & A_3 \\ & \searrow g & \downarrow \exists h \\ & & A_1 \end{array}$$

In particular, every morphism into an injective object extends across monomorphisms, which are precisely injections in categories like Set . The dual notion is that of a *projective object*.

Injectivity is preserved by some categorical constructions but not by all of them.

Lemma 3.3.2. Let $(\mathcal{C}_i)_{i \in I}$ be a family of abelian categories. An object

$$A = (A_i)_{i \in I} \in \prod_{i \in I} \mathcal{C}_i$$

is injective if and only if each component A_i is injective in $\mathcal{C}_i$ (see [49, Ch. 2, §2, Prop. 2.2.7], [32, Ch. 1, §2, Def. 1.2.13]).

Proof. Recall that morphisms in the product category are given componentwise. Assume that $A = (A_i)_{i \in I}$ is injective in $\prod_{i \in I} \mathcal{C}_i$.

To show that A_j is injective in $\mathcal{C}_j$ for a fixed $j \in I$, let $B \hookrightarrow C$ be a monomorphism in $\mathcal{C}_j$, and let $B \rightarrow A_j$ be an arbitrary morphism. Define objects $\tilde{B}, \tilde{C} \in \text{Ob}(\prod_{i \in I} \mathcal{C}_i)$ by

$$\tilde{B}_i := \begin{cases} B, & i = j \\ 0, & i \neq j \end{cases}$$

$$\tilde{C}_i := \begin{cases} C, & i = j \\ 0, & i \neq j \end{cases}$$

and let $\tilde{B} \rightarrow \tilde{C}$ be induced componentwise by $B \rightarrow C$ in the j -th component and by taking the identity $0 \rightarrow 0$ in all other components. Since monomorphisms in a product category are componentwise monomorphisms, this is a monomorphism in $\prod_{i \in I} \mathcal{C}_i$.

The given map $B \rightarrow A_j$ defines a morphism $\tilde{B} \rightarrow A$ in the product category by taking the given morphism in the j -th component and the zero morphism in all other components. Since A is injective, there exists a lift $\tilde{C} \rightarrow A$ making the diagram commute.

Taking the j -th component yields a morphism $C \rightarrow A_j$ whose composite with $B \hookrightarrow C$ is the original morphism $B \rightarrow A_j$. Hence A_j is injective.

Conversely, assume that each A_i is injective in $\mathcal{C}_i$.

Let

$$X = (X_i)_{i \in I} \rightarrow Y = (Y_i)_{i \in I}$$

be a monomorphism in $\prod_{i \in I} \mathcal{C}_i$, and let $X \rightarrow A$ be any morphism. Since monomorphisms in the product category are componentwise monomorphisms, each $X_i \rightarrow Y_i$ is a monomorphism in $\mathcal{C}_i$. Since A_i is injective, the component morphism $X_i \rightarrow A_i$ extends to a morphism $Y_i \rightarrow A_i$.

These componentwise extensions assemble to a morphism $Y \rightarrow A$ in the product category extending $X \rightarrow A$. Therefore A is injective in $\prod_{i \in I} \mathcal{C}_i$ (see [49, Ch. 2, §2, Prop. 2.2.7], [32, Ch. 1, §2, Def. 1.2.13]). $\square$

Thus injectivity in a product category is detected componentwise. Closely related to products are biproducts. Indeed, the binary biproduct of injective objects is again an injective object.

Lemma 3.3.3. Let $\mathcal{C}$ be an abelian category, and let $A, B \in \text{Ob}(\mathcal{C})$ be injective objects. Then the biproduct $A \oplus B$ is injective in $\mathcal{C}$ (see [32, Ch. 8, §4, Lemma 8.4.5]).

Proof. Take a monomorphism $m : C \hookrightarrow D$ and a morphism $f : C \rightarrow A \oplus B$. By the universal property of the biproduct, this is equivalent to the pair

$$\begin{aligned} f_A : C &\rightarrow A \\ f_B : C &\rightarrow B. \end{aligned}$$

Since A and B are injective, there exist extensions

$$g_A : D \rightarrow A$$

$$g_B : D \rightarrow B$$

such that

$$g_A \circ m = f_A$$

$$g_B \circ m = f_B.$$

By the universal property of the biproduct, there exists a unique morphism

$$g : D \rightarrow A \oplus B$$

with components g_A, g_B . Then

$$g \circ m = f.$$

Hence $A \oplus B$ is injective (see [32, Ch. 8, §4, Lemma 8.4.5]). □

Injective objects are particularly useful when arranged into resolutions. Such resolutions allow left-exact functors to be studied through complexes whose terms are injective.

Definition 3.3.4 (Injective resolution). Let $\mathcal{C}$ be an abelian category, and let $A \in \text{Ob}(\mathcal{C})$. An **injective resolution** of A is an exact sequence

$$0 \longrightarrow A \xrightarrow{\varepsilon} I^0 \xrightarrow{\delta^0} I^1 \xrightarrow{\delta^1} I^2 \xrightarrow{\delta^2} \dots$$

such that each I^n is an injective object of $\mathcal{C}$ (see [45, Ch. 3, §2, Def 3.2.7]).

Injective resolutions of objects in a short exact sequence interact with each other. Indeed, one can construct an injective resolution of the middle object from injective resolutions of the outer objects.

Theorem 3.3.5 (Horseshoe lemma). Let $\mathcal{C}$ be an abelian category, and let

$$0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0$$

be a short exact sequence in $\mathcal{C}$. Let

$$0 \rightarrow A \xrightarrow{\varepsilon_A} I_A^0 \xrightarrow{\delta_A^0} I_A^1 \xrightarrow{\delta_A^1} I_A^2 \rightarrow \dots$$

and

$$0 \rightarrow C \xrightarrow{\varepsilon_C} I_C^0 \xrightarrow{\delta_C^0} I_C^1 \xrightarrow{\delta_C^1} I_C^2 \rightarrow \dots$$

be injective resolutions of A and of C . Then there exists an injective resolution

$$0 \rightarrow B \xrightarrow{\varepsilon_B} I_B^0 \xrightarrow{\delta_B^0} I_B^1 \xrightarrow{\delta_B^1} I_B^2 \rightarrow \dots$$

of B and morphisms $I_A^n \rightarrow I_B^n$ and $I_B^n \rightarrow I_C^n$ for all $n \geq 0$ such that

1. for every $n \geq 0$ the sequence

$$0 \rightarrow I_A^n \rightarrow I_B^n \rightarrow I_C^n \rightarrow 0$$

is a short exact sequence

2. the resulting diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & A & \xrightarrow{f} & B & \xrightarrow{g} & C \longrightarrow 0 \\ & & \downarrow \varepsilon_A & & \downarrow \varepsilon_B & & \downarrow \varepsilon_C \\ 0 & \longrightarrow & I_A^0 & \longrightarrow & I_B^0 & \longrightarrow & I_C^0 \longrightarrow 0 \\ & & \downarrow \delta_A^0 & & \downarrow \delta_B^0 & & \downarrow \delta_C^0 \\ 0 & \longrightarrow & I_A^1 & \longrightarrow & I_B^1 & \longrightarrow & I_C^1 \longrightarrow 0 \\ & & \downarrow \delta_A^1 & & \downarrow \delta_B^1 & & \downarrow \delta_C^1 \\ & & \vdots & & \vdots & & \vdots \end{array}$$

commutes, and

3. each I_B^n is injective.

Moreover, one may choose

$$I_B^n \cong I_A^n \oplus I_C^n$$

for all n , which makes each row split-exact (see [47, Ch. 6, §2, Prop. 6.24]).

Proof. For each $n \geq 0$ set

$$I_B^n := I_A^n \oplus I_C^n,$$

which is injective by Lemma 3.3.3.

Since I_A^0 is injective and $f : A \hookrightarrow B$ is monomorphic, there exists

$$u^0 : B \rightarrow I_A^0$$

with

$$u^0 \circ f = \varepsilon_A.$$

Define

$$h^0 := \varepsilon_C \circ g : B \rightarrow I_C^0$$

and

$$\varepsilon_B := (u^0, h^0) : B \rightarrow I_A^0 \oplus I_C^0.$$

The identities

$$\varepsilon_B \circ f = (\varepsilon_A, 0), \quad \text{pr}_2 \circ \varepsilon_B = \varepsilon_C \circ g$$

are immediate. Define the differentials of $I_B^\bullet$ as

$$\delta_B^n := \begin{pmatrix} \delta_A^n & t^n \\ 0 & \delta_C^n \end{pmatrix} : I_A^n \oplus I_C^n \rightarrow I_A^{n+1} \oplus I_C^{n+1},$$

where the morphisms

$$t^n : I_C^n \rightarrow I_A^{n+1}$$

are chosen inductively:

Since

$$\delta_A^0 \circ u^0 \circ f = \delta_A^0 \circ \varepsilon_A = 0,$$

the morphism

$$\delta_A^0 \circ u^0 : B \rightarrow I_A^1$$

vanishes on the image of f . Hence it factors uniquely through

$$g : B \twoheadrightarrow C.$$

Since g is the cokernel of f , the universal property of the cokernel yields a unique

morphism

$$\alpha : C \rightarrow I_A^1$$

with

$$\alpha \circ g = \delta_A^0 \circ u^0.$$

Since I_A^1 is injective and

$$\varepsilon_C : C \hookrightarrow I_C^0$$

is monomorphic, $-\alpha$ extends to

$$t^0 : I_C^0 \rightarrow I_A^1.$$

Consequently

$$\delta_A^0 \circ u^0 + t^0 \circ h^0 = 0,$$

so

$$\delta_B^0 \circ \varepsilon_B = 0.$$

For $n = 1$, since the resolution of C is exact,

$$\ker(\delta_C^0) = \operatorname{im}(\varepsilon_C).$$

Moreover,

$$\delta_A^1 \circ t^0 \circ \varepsilon_C = -\delta_A^1 \circ \alpha.$$

Since

$$\delta_A^1 \circ \alpha \circ g = \delta_A^1 \circ \delta_A^0 \circ u^0 = 0$$

and g is an epimorphism, it follows that

$$\delta_A^1 \circ \alpha = 0.$$

Hence $\delta_A^1 \circ t^0$ vanishes on $\ker(\delta_C^0)$, so by the universal property of the cokernel it factors through

$$\operatorname{coim}(\delta_C^0) = \operatorname{coker}(\ker(\delta_C^0)),$$

and therefore factors through $\operatorname{im}(\delta_C^0)$.

As I_A^2 is injective, the negative of the induced morphism on $\text{im}(\delta_C^0)$ extends across

$$\text{im}(\delta_C^0) \hookrightarrow I_C^1$$

to a morphism

$$t^1 : I_C^1 \rightarrow I_A^2$$

satisfying

$$\delta_A^1 \circ t^0 + t^1 \circ \delta_C^0 = 0.$$

Now let $n \geq 2$, and assume that $t^0, \dots, t^{n-1}$ have been constructed such that

$$\delta_A^k \circ t^{k-1} + t^k \circ \delta_C^{k-1} = 0$$

for $1 \leq k \leq n-1$. Consider $\delta_A^n \circ t^{n-1}$. Since

$$\delta_A^n \circ t^{n-1} \circ \delta_C^{n-2} = -\delta_A^n \circ \delta_A^{n-1} \circ t^{n-2} = 0,$$

this morphism vanishes on

$$\ker(\delta_C^{n-1}) = \text{im}(\delta_C^{n-2}),$$

and therefore factors through $\text{im}(\delta_C^{n-1})$. Since I_A^{n+1} is injective, the negative of this induced morphism extends across the monomorphism

$$\text{im}(\delta_C^{n-1}) \hookrightarrow I_C^n,$$

yielding

$$t^n : I_C^n \rightarrow I_A^{n+1}$$

such that

$$\delta_A^n \circ t^{n-1} + t^n \circ \delta_C^{n-1} = 0.$$

A direct matrix computation gives

$$\begin{aligned}\delta_B^n \circ \delta_B^{n-1} &= \begin{pmatrix} \delta_A^n & t^n \\ 0 & \delta_C^n \end{pmatrix} \begin{pmatrix} \delta_A^{n-1} & t^{n-1} \\ 0 & \delta_C^{n-1} \end{pmatrix} \\ &= \begin{pmatrix} \delta_A^n \circ \delta_A^{n-1} & \delta_A^n \circ t^{n-1} + t^n \circ \delta_C^{n-1} \\ 0 & \delta_C^n \circ \delta_C^{n-1} \end{pmatrix} \\ &= 0.\end{aligned}$$

Hence $(I_B^\bullet, \delta_B^\bullet)$ is a cochain complex.

The canonical inclusions and projections

$$I_A^n \rightarrow I_B^n \rightarrow I_C^n$$

form split-exact sequences in every degree, and the matrix shows that they are cochain maps. Hence

$$0 \rightarrow I_A^\bullet \rightarrow I_B^\bullet \rightarrow I_C^\bullet \rightarrow 0$$

is a short exact sequence of cochain complexes. Applying the long exact cohomology sequence gives

$$\dots \rightarrow H^n(I_A^\bullet) \rightarrow H^n(I_B^\bullet) \rightarrow H^n(I_C^\bullet) \rightarrow H^{n+1}(I_A^\bullet) \rightarrow \dots$$

Since $I_A^\bullet$ and $I_C^\bullet$ are injective resolutions,

$$H^0(I_A^\bullet) \cong A, \quad H^0(I_C^\bullet) \cong C,$$

and

$$H^n(I_A^\bullet) = H^n(I_C^\bullet) = 0$$

for every $n > 0$. Hence $H^n(I_B^\bullet) = 0$ for all $n > 0$, and

$$0 \rightarrow A \rightarrow H^0(I_B^\bullet) \rightarrow C \rightarrow 0$$

is exact.

Thus it remains to identify $H^0(I_B^\bullet)$ with B .

Since the augmentations induce the canonical morphisms into the degree-zero

cocycles, one obtains a commutative diagram

$$\begin{array}{ccccccc}
 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C \longrightarrow 0 \\
 & & \downarrow \overline{\varepsilon}_A & & \downarrow \overline{\varepsilon}_B & & \downarrow \overline{\varepsilon}_C \\
 0 & \longrightarrow & Z^0(I_A^\bullet) & \longrightarrow & Z^0(I_B^\bullet) & \longrightarrow & Z^0(I_C^\bullet) \longrightarrow 0,
 \end{array}$$

Since $I_A^\bullet$ and $I_C^\bullet$ are injective resolutions, the induced morphisms

$$\overline{\varepsilon}_A : A \rightarrow Z^0(I_A^\bullet), \quad \overline{\varepsilon}_C : C \rightarrow Z^0(I_C^\bullet)$$

are isomorphisms. By Lemma 1.7.13, the middle vertical morphism is also an isomorphism. Since the complexes are concentrated in non-negative degrees,

$$H^0(I_B^\bullet) = Z^0(I_B^\bullet),$$

it follows that

$$H^0(I_B^\bullet) \cong B.$$

Therefore

$$0 \rightarrow B \rightarrow I_B^0 \rightarrow I_B^1 \rightarrow \dots$$

is an injective resolution of B .

(see [47, Ch. 6, §2, Prop. 6.24]). □

Injective resolutions by injective sheaves are used to compute derived functors, including sheaf cohomology.

Definition 3.3.6 (Injective sheaf). Let $(\mathcal{C}, J)$ be a site, and let $\mathcal{D}$ be a category such that $\text{Sh}((\mathcal{C}, J), \mathcal{D})$ is abelian. A sheaf $\mathcal{I} \in \text{Ob}(\text{Sh}((\mathcal{C}, J), \mathcal{D}))$ is called an **injective sheaf** if it is an injective object of this abelian category.

If one considers an arbitrary category $\mathcal{C}$ with enough injectives, there is no reason why the category of sheaves with values in $\mathcal{C}$ should have enough injectives too. Moreover, not every category has enough injectives. Grothendieck categories provide an answer to both problems.

Theorem 3.3.7. A Grothendieck category has enough injectives (see [23, Ch. 1, §10, Thm. 1.10.1]).

Proof. The proof uses the existence of a generator, together with the exactness of filtered colimits. One constructs an injective cogenerator and shows that every object admits a monomorphism into a suitable product of copies of this injective object. Details can be found in Appendix D.3.1. $\square$

Remark 3.3.8. Since $\mathbf{Ab}$ is a Grothendieck category by Example 1.7.17, $\mathbf{Sh}((\mathcal{C}, J), \mathbf{Ab})$ is a Grothendieck category if $(\mathcal{C}, J)$ is a small site by Theorem 2.2.26. Thus $\mathbf{Sh}((\mathcal{C}, J), \mathbf{Ab})$ has enough injectives.

3.4 Derived Functors

Having developed cohomology for cochain complexes, one can attach cohomological invariants to objects by replacing them with suitable resolutions. For left-exact functors, injective resolutions give rise to the right derived functors.

Definition 3.4.1 (Right derived functors). Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a left-exact functor between abelian categories, and assume that $\mathcal{C}$ has enough injectives. For $A \in \mathbf{Ob}(\mathcal{C})$ choose an injective resolution

$$0 \longrightarrow A \xrightarrow{\varepsilon} I^0 \xrightarrow{\delta^0} I^1 \xrightarrow{\delta^1} I^2 \xrightarrow{\delta^2} \dots$$

Since F is additive by definition, it preserves zero morphisms. Hence

$$F(\delta^{n+1}) \circ F(\delta^n) = F(\delta^{n+1} \circ \delta^n) = F(0) = 0,$$

so applying F termwise yields a cochain complex

$$0 \longrightarrow F(I^0) \longrightarrow F(I^1) \longrightarrow F(I^2) \longrightarrow \dots$$

The n -th right derived functor of F is defined by

$$R^n F(A) := H^n(F(I^\bullet))$$

(see [47, Ch. 6, §2, pp. 364 ff]).

Remark 3.4.2. The term $F(A)$ is not included in the cochain complex used to define $R^n F(A)$. Since F is left-exact, applying F to

$$0 \rightarrow A \rightarrow I^0 \rightarrow I^1$$

yields an exact sequence

$$0 \rightarrow F(A) \rightarrow F(I^0) \rightarrow F(I^1).$$

Consequently,

$$R^0 F(A) = H^0(F(I^\bullet)) \cong F(A).$$

The definition appears to depend on the chosen injective resolution. To show that the construction is well-defined and functorial, one can compare two injective resolutions.

Theorem 3.4.3 (Standard Comparison Theorem). Let $\mathcal{C}$ be an abelian category. Let

$$0 \rightarrow A \xrightarrow{\varepsilon_A} I^\bullet$$

and

$$0 \rightarrow B \xrightarrow{\varepsilon_B} J^\bullet$$

be injective resolutions of A and B , respectively. Then every morphism $f : A \rightarrow B$ extends to a **cochain morphism**

$$f^\bullet : I^\bullet \rightarrow J^\bullet$$

such that

$$\begin{array}{ccccc} 0 & \longrightarrow & A & \xrightarrow{\varepsilon_A} & I^\bullet \\ & & \downarrow f & & \downarrow f^\bullet \\ 0 & \longrightarrow & B & \xrightarrow{\varepsilon_B} & J^\bullet \end{array}$$

commutes. Moreover, any two such cochain morphisms are cochain homotopic. Thus, the homotopy class of $f^\bullet$ depends only on f (see [47, Ch. 6, §2, Thm. 6.16]).

Proof. The construction is inductive and uses the key fact that injectives allow extensions along monomorphisms.

One has

$$\varepsilon_A : A \rightarrow I^0, \quad \varepsilon_B : B \rightarrow J^0.$$

Since J^0 is injective and

$$\varepsilon_A : A \hookrightarrow I^0$$

is a monomorphism, the composite

$$A \xrightarrow{f} B \xrightarrow{\varepsilon_B} J^0$$

extends along ε_A , yielding a morphism

$$f^0 : I^0 \rightarrow J^0$$

such that

$$f^0 \circ \varepsilon_A = \varepsilon_B \circ f.$$

Assume

$$f^n : I^n \rightarrow J^n$$

is constructed and compatible with differentials. One needs to construct f^{n+1} . Consider the diagram

$$\begin{array}{ccc} I^n & \xrightarrow{\delta_I^n} & I^{n+1} \\ f^n \downarrow & & \downarrow f^{n+1} \\ J^n & \xrightarrow{\delta_J^n} & J^{n+1}. \end{array}$$

It is to show that

$$\delta_J^n \circ f^n = f^{n+1} \circ \delta_I^n.$$

For $n = 0$, one has

$$\delta_J^0 \circ f^0 \circ \varepsilon_A = \delta_J^0 \circ \varepsilon_B \circ f = 0.$$

Since the resolution of A is exact,

$$\ker(\delta_I^0) = \operatorname{im}(\varepsilon_A).$$

Thus $\delta_J^0 \circ f^0$ vanishes on $\ker(\delta_I^0)$.

For $n \geq 1$, the induction hypothesis gives

$$\delta_J^n \circ f^n \circ \delta_I^{n-1} = \underbrace{\delta_J^n \circ \delta_J^{n-1}}_{=0} \circ f^{n-1} = 0.$$

Since the resolution $I^\bullet$ is exact, $\ker(\delta_I^n) = \text{im}(\delta_I^{n-1})$, so

$$\delta_J^n \circ f^n : I^n \rightarrow J^{n+1}$$

vanishes on $\ker(\delta_I^n)$.

Let

$$I^n \xrightarrow{q_I^n} \text{im}(\delta_I^n) \xrightarrow{\iota_I^n} I^{n+1}$$

be the canonical image factorisation of δ_I^n . Since $\delta_J^n \circ f^n$ vanishes on

$$\ker(\delta_I^n) = \ker(q_I^n),$$

it factors uniquely through q_I^n . Thus there exists a morphism

$$\alpha^n : \text{im}(\delta_I^n) \rightarrow J^{n+1}$$

such that

$$\alpha^n \circ q_I^n = \delta_J^n \circ f^n.$$

Since

$$\iota_I^n : \text{im}(\delta_I^n) \hookrightarrow I^{n+1}$$

is a monomorphism and J^{n+1} is injective, α^n extends to a morphism

$$f^{n+1} : I^{n+1} \rightarrow J^{n+1}.$$

Then

$$\begin{aligned} f^{n+1} \circ \delta_I^n &= f^{n+1} \circ \iota_I^n \circ q_I^n \\ &= \alpha^n \circ q_I^n \\ &= \delta_J^n \circ f^n. \end{aligned}$$

Suppose that

$$f^\bullet, g^\bullet : I^\bullet \rightarrow J^\bullet$$

are two such lifts. One can inductively construct a homotopy $h^\bullet$ such that

$$f^n - g^n = \delta_J^{n-1} \circ h^n + h^{n+1} \circ \delta_I^n$$

using the injectivity of J^n to extend morphisms.

Set

$$u^\bullet := f^\bullet - g^\bullet.$$

Then

$$u^\bullet : I^\bullet \rightarrow J^\bullet$$

is a cochain morphism extending the zero morphism $A \rightarrow B$.

Claim: $u^\bullet$ is cochain homotopic to zero.

Construct morphisms

$$h^{n+1} : I^{n+1} \rightarrow J^n$$

inductively such that

$$u^n = \delta_J^{n-1} \circ h^n + h^{n+1} \circ \delta_I^n$$

for all $n \geq 0$, where the term $\delta_J^{-1} \circ h^0$ is omitted.

For $n = 0$, the morphism

$$u^0 : I^0 \rightarrow J^0$$

satisfies

$$u^0 \circ \varepsilon_A = 0,$$

since both $f^\bullet$ and $g^\bullet$ extend the same morphism $f : A \rightarrow B$. Since the resolution

$$0 \rightarrow A \rightarrow I^0 \rightarrow \dots$$

is exact, one has

$$\ker(\delta_I^0) = \operatorname{im}(\varepsilon_A).$$

Hence u^0 vanishes on $\ker(\delta_I^0)$, and therefore factors through $\operatorname{im}(\delta_I^0)$. Since

$$\operatorname{im}(\delta_I^0) \hookrightarrow I^1$$

is a monomorphism and J^0 is injective, this morphism extends to a morphism

$$h^1 : I^1 \rightarrow J^0$$

such that

$$h^1 \circ \delta_I^0 = u^0.$$

Thus the homotopy identity holds in degree 0.

Assume that $h^1, \dots, h^n$ have been constructed so that the homotopy identities hold in all degrees $< n$.

Define

$$\begin{aligned} t^n &:= u^n - \delta_J^{n-1} \circ h^n \\ t^n &: I^n \rightarrow J^n. \end{aligned}$$

Claim: t^n vanishes on $\ker(\delta_I^n)$.

Since the resolution $I^\bullet$ is exact, $\ker(\delta_I^n) = \text{im}(\delta_I^{n-1})$. Therefore it suffices to compute $t^n \circ \delta_I^{n-1}$. Using that $u^\bullet$ is a cochain morphism and the induction hypothesis, one obtains

$$\begin{aligned} t^n \circ \delta_I^{n-1} &= u^n \circ \delta_I^{n-1} - \delta_J^{n-1} \circ h^n \circ \delta_I^{n-1} \\ &= \delta_J^{n-1} \circ u^{n-1} - \delta_J^{n-1} \circ h^n \circ \delta_I^{n-1} \\ &= \delta_J^{n-1} (\delta_J^{n-2} \circ h^{n-1} + h^n \circ \delta_I^{n-1}) - \delta_J^{n-1} \circ h^n \circ \delta_I^{n-1} \\ &= 0. \end{aligned}$$

Let

$$q_I^n : I^n \twoheadrightarrow \text{im}(\delta_I^n)$$

be the canonical epimorphism. Since t^n vanishes on

$$\ker(\delta_I^n) = \ker(q_I^n),$$

there exists a unique morphism

$$\bar{t}^n : \text{im}(\delta_I^n) \rightarrow J^n$$

such that

$$\bar{t}^n \circ q_I^n = t^n.$$

Since

$$\text{im}(\delta_I^n) \hookrightarrow I^{n+1}$$

is a monomorphism and J^n is injective, $\bar{t}^n$ extends to a morphism

$$h^{n+1} : I^{n+1} \rightarrow J^n$$

such that

$$h^{n+1} \circ \delta_I^n = t^n.$$

Hence

$$u^n = \delta_J^{n-1} \circ h^n + h^{n+1} \circ \delta_I^n.$$

By induction,

$$u^\bullet = f^\bullet - g^\bullet$$

is null-homotopic. Therefore $f^\bullet$ and $g^\bullet$ are cochain homotopic (see [47, Ch. 6, §2, Thm. 6.16]). $\square$

This implies that any two injective resolutions of the same object are homotopy equivalent. Since cohomology is invariant under cochain homotopy, it is natural to ask whether right derived functors are independent of the chosen injective resolution.

Theorem 3.4.4. Let $\mathcal{C}$ and $\mathcal{D}$ be abelian categories, assume that $\mathcal{C}$ has enough injectives, and let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a left-exact functor. For an object $A \in \text{Ob}(\mathcal{C})$ define

$$R^n F(A) := H^n(F(I^\bullet)),$$

where

$$A \rightarrow I^\bullet$$

is an injective resolution of A . Then the objects $R^n F(A)$ are well-defined up to canonical isomorphism, that is, they do not depend on the chosen injective resolution (see [47, Ch. 6, §2, Prop. 6.40]).

Proof. Let

$$0 \rightarrow A \xrightarrow{\varepsilon} I^\bullet$$

and

$$0 \rightarrow A \xrightarrow{\varepsilon'} J^\bullet$$

be two injective resolutions of A . One wants to show that

$$H^n(F(I^\bullet)) \cong H^n(F(J^\bullet))$$

canonically for every $n \in \mathbb{N}_0$.

By the comparison theorem 3.4.3, the identity morphism id_A extends to the cochain morphism

$$f^\bullet : I^\bullet \rightarrow J^\bullet$$

such that

$$f^0 \circ \varepsilon = \varepsilon'.$$

Similarly, id_A extends to a cochain morphism

$$g^\bullet : J^\bullet \rightarrow I^\bullet.$$

The composition $g^\bullet \circ f^\bullet$ and $\text{id}_{I^\bullet}$ are both cochain morphisms $I^\bullet \rightarrow I^\bullet$ extending id_A .

By uniqueness up to homotopy in the comparison theorem 3.4.3,

$$g^\bullet \circ f^\bullet \simeq \text{id}_{I^\bullet}.$$

Likewise,

$$f^\bullet \circ g^\bullet \simeq \text{id}_{J^\bullet}.$$

Thus $f^\bullet$ and $g^\bullet$ are homotopy inverse cochain morphisms.

By definition, the left-exact functor F is additive. Applying F yields cochain morphisms

$$F(f^\bullet) : F(I^\bullet) \rightarrow F(J^\bullet)$$

and

$$F(g^\bullet) : F(J^\bullet) \rightarrow F(I^\bullet).$$

Since F preserves cochain homotopies, from

$$g^\bullet \circ f^\bullet \simeq \text{id}_{I^\bullet},$$

and

$$f^\bullet \circ g^\bullet \simeq \text{id}_{J^\bullet}$$

one obtains

$$F(g^\bullet) \circ F(f^\bullet) \simeq \text{id}_{F(I^\bullet)}$$

and

$$F(f^\bullet) \circ F(g^\bullet) \simeq \text{id}_{F(J^\bullet)}.$$

Therefore, $F(f^\bullet)$ and $F(g^\bullet)$ are homotopy inverse cochain morphisms.

The comparison morphism $f^\bullet$ is unique up to cochain homotopy. Since F preserves cochain homotopies, and cochain homotopic maps induce the same map on cohomology, the resulting isomorphism

$$H^n(F(I^\bullet)) \xrightarrow{\sim} H^n(F(J^\bullet))$$

is independent of all choices. Hence $R^n F(A)$ is well-defined up to canonical isomorphism (see [28, Ch. 1, §7], [47, Ch. 6, §2, Prop. 6.40]). $\square$

Remark 3.4.5. The isomorphism is canonical, because the comparison morphism $f^\bullet$ extending id_A is unique up to cochain homotopy. Since cohomology does not see cochain homotopies, the induced isomorphism

$$H^n(F(I^\bullet)) \rightarrow H^n(F(J^\bullet))$$

is independent of all choices.

Moreover, let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a left-exact functor between abelian categories, and let $R^n F(A)$ be the n -th right derived functor. A morphism $f \in \text{Hom}_{\mathcal{C}}(A, B)$ induces morphisms $R^n F(A) \rightarrow R^n F(B)$, making $R^n F : \mathcal{C} \rightarrow \mathcal{D}$ a functor.

Right derived functors vary functorially with the object and interact naturally with short exact sequences.

Theorem 3.4.6. Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a left-exact functor between abelian categories,

and assume that $\mathcal{C}$ has enough injectives. Then every short exact sequence

$$0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0$$

in $\mathcal{C}$ induces a long exact sequence

$$\dots \rightarrow R^n F(A) \rightarrow R^n F(B) \rightarrow R^n F(C) \xrightarrow{\partial^n} R^{n+1} F(A) \rightarrow \dots$$

which is natural with respect to morphisms of short exact sequences. The morphisms ∂^n are called the connecting morphisms (see [47, Ch. 6, §2, Thm. 6.43], [28, Ch. 1, §7]).

Proof. Choose injective resolutions $A \rightarrow I_A^\bullet$ and $C \rightarrow I_C^\bullet$. By the injective horseshoe lemma 3.3.5 there exists an injective resolution $B \rightarrow I_B^\bullet$ such that

$$0 \rightarrow I_A^\bullet \rightarrow I_B^\bullet \rightarrow I_C^\bullet \rightarrow 0$$

is a short exact sequence of cochain complexes, and, moreover,

$$I_B^n \cong I_A^n \oplus I_C^n$$

for all n . Hence each degree sequence is split-exact. Since each degree sequence

$$0 \rightarrow I_A^n \rightarrow I_B^n \rightarrow I_C^n \rightarrow 0$$

is split-exact, and since F is additive by definition, applying F yields a split-exact sequence

$$0 \rightarrow F(I_A^n) \rightarrow F(I_B^n) \rightarrow F(I_C^n) \rightarrow 0$$

for every n . Hence

$$0 \rightarrow F(I_A^\bullet) \rightarrow F(I_B^\bullet) \rightarrow F(I_C^\bullet) \rightarrow 0$$

is a short exact sequence of cochain complexes. Taking cohomology gives the associated long exact sequence

$$\dots \rightarrow H^n(F(I_A^\bullet)) \rightarrow H^n(F(I_B^\bullet)) \rightarrow H^n(F(I_C^\bullet)) \xrightarrow{\partial^n} H^{n+1}(F(I_A^\bullet)) \rightarrow \dots$$

By definition $H^n(F(I_X^\bullet)) = R^n F(X)$ for $X \in \{A, B, C\}$, hence one obtains

$$\dots \rightarrow R^n F(A) \rightarrow R^n F(B) \rightarrow R^n F(C) \xrightarrow{\partial^n} R^{n+1} F(A) \rightarrow \dots$$

The construction is independent of the chosen resolutions and is natural with respect to morphisms of short exact sequences. This follows from the comparison theorem together with the standard naturality of the connecting morphism (see [28, Ch. 1, §7]). $\square$

Although derived functors are defined using injective resolutions, in practice one often computes derived functors using more flexible resolutions.

3.5 Acyclicity

Definition 3.5.1 (F -acyclic object). Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a left-exact functor between abelian categories and assume $\mathcal{C}$ has enough injectives. An object $A \in \text{Ob}(\mathcal{C})$ is called **F -acyclic** if

$$R^n F(A) = 0$$

for all $n \geq 1$ (see [47, Ch. 10, §6, p. 656]).

Remark 3.5.2 (Equivalent formulations of F -acyclic objects). Fix an injective resolution $A \rightarrow I^\bullet$. Then the following are equivalent:

1. A is F -acyclic.
2. $H^n(F(I^\bullet)) = 0$ for all $n \geq 1$.
3. The complex

$$0 \rightarrow F(I^0) \rightarrow F(I^1) \rightarrow F(I^2) \rightarrow \dots$$

is exact in positive degrees.

(see [47, Ch. 10, §6, p. 656]).

In the special case of sheaf cohomology, acyclicity is usually expressed in terms of the vanishing of higher cohomology.

Definition 3.5.3 (Acyclic sheaf). Let X be a topological space and suppose that the abelian category of sheaves under consideration has enough injectives. A sheaf $\mathcal{F}$ on X is called Γ -**acyclic** (or simply **acyclic**), if

$$H^n(X, \mathcal{F}) = R^n\Gamma(X, \mathcal{F}) = 0$$

for all $n \in \mathbb{N}$.

The collection of F -acyclic objects is generally much larger than the collection of injective objects. Nevertheless, injective objects provide fundamental examples of acyclic objects.

Lemma 3.5.4. Let $\mathcal{C}$ and $\mathcal{D}$ be abelian categories, and let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a left-exact functor. Then every injective object $I \in \text{Ob}(\mathcal{C})$ is F -acyclic (see [47, Ch. 6, §2, Corollary 6.41]).

Proof. Since I is injective, the augmented complex

$$0 \rightarrow I \xrightarrow{\text{id}_I} I \rightarrow 0 \rightarrow 0 \rightarrow \dots$$

is an injective resolution of I . After omitting the augmented term and applying F , one obtains a cochain complex

$$0 \rightarrow F(I) \rightarrow 0 \rightarrow \dots$$

Its cohomology vanishes in every positive degree. Hence

$$R^n F(I) = 0$$

for every $n \in \mathbb{N}$, so I is F -acyclic. □

Since injective objects are F -acyclic, one may ask whether injectivity is really necessary for computing derived functors. This leads to the more general notion of an acyclic resolution.

Definition 3.5.5 (Acyclic resolution). Let $\mathcal{C}$ and $\mathcal{D}$ be abelian categories, assume that $\mathcal{C}$ has enough injectives, and let

$$F : \mathcal{C} \rightarrow \mathcal{D}$$

be a left-exact functor. An F -**acyclic resolution** of $A \in \text{Ob}(\mathcal{C})$ is an exact sequence

$$0 \rightarrow A \rightarrow Q^0 \rightarrow Q^1 \rightarrow Q^2 \rightarrow \dots$$

such that each Q^p is F -acyclic, that is, $R^n F(Q^p) = 0$ for all $n \in \mathbb{N}$ and $p \in \mathbb{N}_0$ (see [47, Ch. 6, §1, p. 326]).

An object A is F -acyclic if and only if all higher derived functors on F vanish on A . The vanishing of the higher derived functors means that A contributes no higher F -cohomology. This makes F -acyclic objects suitable terms for resolutions used to compute derived functors.

Theorem 3.5.6 (Acyclic resolution theorem). Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a left-exact functor between abelian categories, and assume that $\mathcal{C}$ has enough injectives. Let

$$0 \rightarrow A \rightarrow Q^\bullet$$

be a resolution of A such that Q^p is F -acyclic, that is, $R^q F(Q^p) = 0$ for all $q \in \mathbb{N}$ and all $p \geq 0$. Then there is a canonical isomorphism

$$R^n F(A) \cong H^n(F(Q^\bullet))$$

for all $n \in \mathbb{N}_0$ (see [47, Ch. 6, §2, pp. 364 ff.]).

Proof. Set $Z^0 := A$. For every $p \in \mathbb{N}$, let

$$Z^p := \ker(\delta^p : Q^p \rightarrow Q^{p+1}).$$

Since the augmented complex is exact,

$$Z^p = \text{im}(\delta^{p-1} : Q^{p-1} \rightarrow Q^p)$$

for every $p \in \mathbb{N}$. For every $p \in \mathbb{N}_0$, let

$$q^p : Q^p \twoheadrightarrow Z^{p+1}$$

denote the canonical epimorphism induced by δ^p . The resolution therefore decomposes into short exact sequences

$$0 \rightarrow Z^p \rightarrow Q^p \rightarrow Z^{p+1} \rightarrow 0. \quad (3.5.7)$$

Applying the long exact sequence of derived functors to (3.5.7) gives

$$\dots \rightarrow R^q F(Q^p) \rightarrow R^q F(Z^{p+1}) \rightarrow R^{q+1} F(Z^p) \rightarrow R^{q+1} F(Q^p) \rightarrow \dots$$

Since Q^p is F -acyclic, one has $R^q F(Q^p) = 0$ for all $q \in \mathbb{N}$. Hence for $q \in \mathbb{N}$ the connecting morphism gives an isomorphism

$$R^q F(Z^{p+1}) \cong R^{q+1} F(Z^p).$$

For every $n \in \mathbb{N}$, repeated dimension shifting gives

$$\begin{aligned} R^n F(A) &= R^n F(Z^0) \\ &\cong R^{n-1} F(Z^1) \\ &\vdots \\ &\cong R^1 F(Z^{n-1}), \end{aligned}$$

where for $n = 1$, this is simply the identity

$$R^1 F(A) = R^1 F(Z^0).$$

It remains to identify $R^1 F(Z^{n-1})$ with the n -th cohomology of $F(Q^\bullet)$. Consider the short exact sequence

$$0 \rightarrow Z^{n-1} \rightarrow Q^{n-1} \rightarrow Z^n \rightarrow 0.$$

The beginning of the associated long exact sequence is

$$0 \rightarrow F(Z^{n-1}) \rightarrow F(Q^{n-1}) \rightarrow F(Z^n) \rightarrow R^1 F(Z^{n-1}) \rightarrow R^1 F(Q^{n-1}).$$

Since Q^{n-1} is F -acyclic,

$$R^1 F(Q^{n-1}) = 0.$$

Therefore

$$0 \rightarrow F(Z^{n-1}) \rightarrow F(Q^{n-1}) \rightarrow F(Z^n) \rightarrow R^1 F(Z^{n-1}) \rightarrow 0$$

is exact. Hence

$$R^1 F(Z^{n-1}) \cong \operatorname{coker} (F(Q^{n-1}) \rightarrow F(Z^n)).$$

Now $Z^n \subseteq Q^n$ is the kernel of $\delta^n : Q^n \rightarrow Q^{n+1}$. Since F is left-exact, the sequence

$$0 \rightarrow F(Z^n) \rightarrow F(Q^n) \rightarrow F(Q^{n+1})$$

is exact. Hence

$$F(Z^n) \cong \ker (F(\delta^n) : F(Q^n) \rightarrow F(Q^{n+1})).$$

Thus

$$\operatorname{coker} (F(Q^{n-1}) \rightarrow F(Z^n)) \cong \frac{\ker (F(\delta^n))}{\operatorname{im} (F(\delta^{n-1}))}.$$

Let

$$q^{n-1} : Q^{n-1} \twoheadrightarrow Z^n$$

be the canonical factor of δ^{n-1} , and let

$$\iota^n : Z^n \hookrightarrow Q^n$$

be the kernel inclusion. Then

$$\delta^{n-1} = \iota^n \circ q^{n-1}.$$

Since F is left-exact, $F(\iota^n)$ is monomorphic. Consequently,

$$\operatorname{im} (F(\delta^{n-1})) \cong \operatorname{im} (F(q^{n-1}))$$

as subobjects of

$$F(Z^n) \cong \ker (F(\delta^n)).$$

Thus, the right-hand side is precisely

$$R^1 F(Z^{n-1}) \cong \frac{\ker(F(\delta^n))}{\operatorname{im}(F(\delta^{n-1}))} = H^n(F(Q^\bullet)).$$

Hence

$$R^n F(A) \cong H^n(F(Q^\bullet))$$

for all $n \in \mathbb{N}$. For $n = 0$, since F is left-exact and

$$0 \rightarrow A \rightarrow Q^0 \rightarrow Q^1$$

is exact, one has

$$F(A) \cong \ker(F(Q^0) \rightarrow F(Q^1)) = H^0(F(Q^\bullet)).$$

Since $R^0 F = F$, this gives

$$R^0 F(A) \cong H^0(F(Q^\bullet)).$$

Therefore

$$R^n F(A) \cong H^n(F(Q^\bullet))$$

for every $n \in \mathbb{N}_0$ (see [47, Ch. 6, §2, pp. 364 ff.]). □

The preceding theorem provides one of the fundamental tools for computing derived functors. Many naturally occurring constructions involve several cohomological directions simultaneously. In such a situation, the resulting cohomology is often difficult to compute directly. One therefore seeks a method that decomposes the computation into successive approximations that can be analysed one step at a time.

3.6 Bicomplexes, Filtrations, and Spectral Sequences

Definition 3.6.1 (Bicomplex). A **bicomplex** is a collection of objects $C^{p,q}$ in an abelian category equipped with two differentials: a **horizontal differential**

$$\delta_h : C^{p,q} \rightarrow C^{p+1,q}$$

and a **vertical differential**

$$\delta_v : C^{p,q} \rightarrow C^{p,q+1}$$

such that $\delta_h^2 = \delta_v^2 = 0$ and the differentials *anticommute*, that is $\delta_v \circ \delta_h = -\delta_h \circ \delta_v$.

In a bicomplex $C^{p,q}$, the pair (p, q) is called the **bidegree**. The index p is the horizontal degree (column index), and q is the vertical degree (row index).

The **bottom row** refers to the row with vertical degree $q = 0$.

A **first-quadrant bicomplex** is a bicomplex where $C^{p,q} = 0$ whenever $q < 0$ or $p < 0$ (see [47, Ch. 10, §1, p. 612]).

A **morphism of bicomplexes**

$$f : C^{\bullet,\bullet} \rightarrow D^{\bullet,\bullet}$$

is a family of morphisms

$$f^{p,q} : C^{p,q} \rightarrow D^{p,q}$$

such that for all p, q the following holds:

$$\begin{aligned} f^{p+1,q} \circ \delta_h^{(C)} &= \delta_h^{(D)} \circ f^{p,q} \\ f^{p,q+1} \circ \delta_v^{(C)} &= \delta_v^{(D)} \circ f^{p,q} \end{aligned}$$

where $\delta^{(C)}$ denotes the differentials in $C^{\bullet,\bullet}$, and $\delta^{(D)}$ denotes the differentials in $D^{\bullet,\bullet}$ (see [47, Ch. 10, §1, p. 611]).

A **vertical complex** is obtained by fixing the horizontal degree p and considering only the vertical differential δ_v . Analogously, the **horizontal complex** is obtained by fixing the vertical degree q and considering only the horizontal differential δ_h .

A bicomplex encodes two independent cohomological directions. However, cohomology is defined for single complexes. To extract cohomological information from a bicomplex, one therefore needs to combine both gradings into a single complex. This is achieved by passing to the associated total complex.

Definition 3.6.2 (Total complex). Consider a first-quadrant bicomplex $C^{p,q}$ in an abelian category with differentials

$$\begin{aligned}\delta_h : C^{p,q} &\rightarrow C^{p+1,q}, \\ \delta_v : C^{p,q} &\rightarrow C^{p,q+1},\end{aligned}$$

such that

$$\delta_v^2 = 0, \quad \delta_h^2 = 0,$$

and

$$\delta_h \circ \delta_v + \delta_v \circ \delta_h = 0.$$

The **total complex** is

$$\mathrm{Tot}^n(C) := \bigoplus_{p+q=n} C^{p,q}$$

with differential

$$\delta|_{C^{p,q}} := \delta_h + \delta_v$$

on $C^{p,q}$ (see [47, Ch. 10, §1, p. 613]).

The **total cohomology** of a bicomplex is the cohomology of its associated total complex.

If $f : B^{\bullet,\bullet} \rightarrow C^{\bullet,\bullet}$ is a morphism of bicomplexes, it induces a morphism $\mathrm{Tot}(f)$ between total complexes, that is,

$$\begin{aligned}\mathrm{Tot}(f) : \mathrm{Tot}(B) &\rightarrow \mathrm{Tot}(C) \\ \mathrm{Tot}(f)^n &= \bigoplus_{p+q=n} f^{p,q}\end{aligned}$$

(see [47, Ch. 10, §1, p. 613 ff.]).

The total complex is obtained by collapsing the bicomplex along diagonals. The anticommutation relation ensures that $\delta^2 = 0$. Note that the definition above gives the

direct sum totalisation. For more general unbounded bicomplexes, there exists also the notion of product totalisation, provided that coproducts exist, which is usually used when infinitely many terms contribute in each total degree, for example in unbounded bicomplexes. In the context of Čech cohomology, one usually considers a bicomplex like $C^{p,q} = \check{C}^p(\mathcal{U}, \mathcal{I}^q)$, where $\mathcal{U}$ is a covering and $\mathcal{F} \rightarrow \mathcal{I}^\bullet$ is an injective resolution. Then p gives the Čech direction, and q the resolution direction.

While the total complex allows one to reduce a bicomplex to a single cochain complex, its cohomology is typically difficult to compute directly. However, the underlying double structure still provides additional information. By filtering the total complex according to one of the two gradings, one obtains a spectral sequence, which computes the cohomology of the total complex step by step.

Definition 3.6.3 (Separated filtration). Let $C^\bullet$ be a cochain complex. A **(decreasing) filtration** on $C^\bullet$ is a family of subcomplexes

$$\dots \supseteq F^r C^\bullet \supseteq F^{r+1} C^\bullet \supseteq \dots$$

such that

1. **Nestedness:** $F^{r+1} C^n \subseteq F^r C^n$ for all r, n
2. **Compatibility with differential:** $\delta(F^r C^n) \subseteq F^r C^{n+1}$.

So each $F^r C^\bullet$ is a subcomplex. The pair $(A, F^\bullet A)$ is called **filtered object** (see [47, Ch. 10, §2, pp. 616 ff.]).

A morphism of filtered objects

$$f : (A, F^\bullet A) \rightarrow (B, F^\bullet B)$$

is a morphism $f : A \rightarrow B$ such that $f(F^p A) \subseteq F^p B$ for every p .

Let $C^\bullet$ be a cochain complex with decreasing filtration

$$\dots \supseteq F^r C^\bullet \supseteq F^{r+1} C^\bullet \supseteq \dots$$

The filtration is **separated in degree** n if

$$\bigcap_r F^r C^n = 0,$$

and **separated** if this holds for every degree.

Moreover, it is **exhaustive in degree** n if

$$\bigcup_{p \in \mathbb{Z}} F^p C^n = C^n,$$

and exhaustive if it is so in every degree (see [52, Ch. 12, §19, Def. 12.19.1]).

Let A be an object in an abelian category equipped with decreasing filtration

$$A = F^0 A \supseteq F^1 A \supseteq F^2 A \supseteq \cdots \supseteq F^N A \supseteq F^{N+1} A = 0$$

The p -th **graded piece** is defined as

$$\mathrm{gr}_F^p A := \frac{F^p A}{F^{p+1} A}.$$

The **associated graded object** is

$$\mathrm{gr}_F A := \bigoplus_p \mathrm{gr}_F^p A$$

(see [47, Ch. 10, §1, pp. 609 ff.]).

Remark 3.6.4. Formally, a subcomplex is a collection of subobjects $D^n \subseteq C^n$ such that they are preserved under the differential, that is, $\delta(D^n) \subseteq D^{n+1}$ for all $n \in \mathbb{N}_0$. Equivalently, the differentials restrict to morphisms

$$\delta|_{D^n} : D^n \rightarrow D^{n+1}$$

so that $(D^\bullet, \delta|_{D^\bullet})$ itself is a cochain complex (see [47, Ch. 10, §1]).

Exhaustiveness ensures that the filtration captures the entire complex, so that the spectral sequence can recover the full cohomology. Boundedness guarantees stabilisation of the spectral sequence, while separateness ensures that distinct filtration levels do not collapse, so that no information is lost in passing to the associated graded pieces.

In the spectral sequences arising from first-quadrant bicomplexes, these finiteness conditions are automatically satisfied. The following lemma records the properties

of the column filtration that are needed for convergence.

Lemma 3.6.5. Let $C^{\bullet,\bullet}$ be a first-quadrant bicomplex in an abelian category, and let its total complex $\text{Tot}(C)$ be equipped with the column filtration

$$F^r \text{Tot}^n(C) := \bigoplus_{\substack{p+q=n \\ p \geq r}} C^{p,q}.$$

Then for every n , the filtration $F^\bullet \text{Tot}^n(C)$ is exhaustive, separated, and bounded (see [47, Ch. 10, §2 & §3], [54, Ch. 5, §5], [36]).

Proof. If $r \leq 0$, every occurring summand satisfies $p \geq 0 \geq r$, hence

$$F^r \text{Tot}^n(C) = \text{Tot}^n(C).$$

Thus the filtration is exhaustive in degree n .

If $r > n$, then there is no pair (p, q) with $p + q = n$, $p \geq r$ and $p, q \geq 0$. Therefore

$$F^r \text{Tot}^n(C) = 0.$$

Hence the filtration is separated in degree n , since

$$\bigcap_r F^r \text{Tot}^n(C) \subseteq F^{n+1} \text{Tot}^n(C) = 0.$$

This also shows boundedness: for each fixed n ,

$$F^r \text{Tot}^n(C) = \text{Tot}^n(C)$$

whenever $r \leq 0$, and

$$F^r \text{Tot}^n(C) = 0$$

for $r > n$. Thus the column filtration is exhaustive, separated, and bounded in every degree (see [47, Ch. 10, §2 & §3], [54, Ch. 5, §5], [36]).

□

Since the spectral sequence is constructed from a filtration on the total complex, it is

important to understand how morphisms of bicomplexes interact with this filtration, and whether they preserve it.

Lemma 3.6.6. Let $f : K^{\bullet,\bullet} \rightarrow L^{\bullet,\bullet}$ be a morphism of bicomplexes. Equip the total complexes with the column filtration

$$F^r \operatorname{Tot}^n(K) := \bigoplus_{\substack{p+q=n \\ p \geq r}} K^{p,q}$$

$$F^r \operatorname{Tot}^n(L) := \bigoplus_{\substack{p+q=n \\ p \geq r}} L^{p,q}.$$

Then the induced morphism

$$\operatorname{Tot}(f) : \operatorname{Tot}(K) \rightarrow \operatorname{Tot}(L)$$

is filtration-preserving, that is,

$$\operatorname{Tot}(f)(F^r \operatorname{Tot}^n(K)) \subseteq F^r \operatorname{Tot}^n(L)$$

for all r, n (see [47, Ch. 10, §2-3]).

Proof. Since

$$F^r \operatorname{Tot}^n(K) = \bigoplus_{\substack{p+q=n \\ p \geq r}} K^{p,q},$$

it is enough to verify the claim on each summand $K^{p,q}$, by the universal property of the direct sum.

Consider the canonical inclusions

$$\iota_{p,q} : K^{p,q} \rightarrow F^r \operatorname{Tot}^n(K).$$

By construction of $\operatorname{Tot}(f)$,

$$\operatorname{Tot}(f) \circ \iota_{p,q} = \iota'_{p,q} \circ f^{p,q},$$

where

$$\iota'_{p,q} : L^{p,q} \rightarrow F^r \operatorname{Tot}^n(L)$$

is the corresponding inclusion. Hence $\text{Tot}(f)$ factors through $F^r \text{Tot}^n(L)$, proving

$$\text{Tot}(f)(F^r \text{Tot}^n(K)) \subseteq F^r \text{Tot}^n(L).$$

□

A morphism of bicomplexes respects the column filtration on total complexes because it acts componentwise and preserves bidegree (p, q) . This is essential to the comparison of associated spectral sequences in Theorem 3.6.13. Having established the filtration properties of total complexes, one can construct the spectral sequence associated to a filtered complex. Spectral sequences encode the successive approximations to the cohomology obtained from the filtration.

Definition 3.6.7 (Spectral sequence, page of a spectral sequence). Let $\mathcal{C}$ be an abelian category. A **cohomological spectral sequence** consists of

- a family of objects $E_r = \{E_r^{p,q}\}_{r \geq r_0, p, q \in \mathbb{Z}}$ is called the **r -th page**, and
- **differentials** $\delta_r^{p,q} : E_r^{p,q} \rightarrow E_r^{p+r, q-r+1}$ satisfying $\delta_r \circ \delta_r = 0$ for all r .

For each r , define **cocycles**

$$Z_r^{p,q} := \ker(\delta_r^{p,q}),$$

and **coboundaries**

$$B_r^{p,q} := \text{im}(\delta_r^{p-r, q+r-1}).$$

Then the next page is obtained by taking cohomology:

$$E_{r+1}^{p,q} \cong \frac{Z_r^{p,q}}{B_r^{p,q}}.$$

The **total degree** is $n := p + q$, where p is the **filtration degree**, and q is the **complementary degree**.

If, for fixed (p, q) , the objects $E_r^{p,q}$ stabilise for $r \gg 0$, the stable value is denoted by $E_\infty^{p,q}$ (see [47, Ch. 10, §2, p. 622]).

Spectral sequences provide a method for computing the cohomology of a complex by successively refining approximations through the pages E_r .

The spectral sequence does not directly give H^n . Instead, it gives the graded pieces of a filtration on H^n , which then is recovered via successive extensions of graded pieces $E_\infty^{p,n-p}$ using short exact sequences.

In particular, if $H^n(C^\bullet)$ is equipped with the induced filtration from a filtered complex, one obtains the graded pieces $\text{gr}_F^p H^n(C^\bullet)$ appearing in the limit of the associated spectral sequence.

The abstract definition does not yet explain how spectral sequences arise from bicomplexes. The standard construction proceeds by filtering the total complex and examining the induced filtration on cohomology.

Construction 3.6.8. A spectral sequence arises from the filtered total complex associated to a bicomplex as follows:

Let $C^\bullet = \text{Tot}(K)$ be a complex with a decreasing filtration

$$\dots \supseteq F^p C^\bullet \supseteq F^{p+1} C^\bullet \supseteq \dots$$

- for example for the column filtration

$$F^p \text{Tot}^n(K) := \bigoplus_{\substack{i+j=n \\ i \geq p}} K^{i,j}.$$

Define

$$\text{gr}_F^p C^n := F^p C^n / F^{p+1} C^n$$

(for example $\text{gr}_F^p \text{Tot}^{p+q}(K) \cong K^{p,q}$). Then

$$E_0^{p,q} = \text{gr}_F^p \text{Tot}^{p+q}(K) \cong K^{p,q}.$$

The first differential on E_0 is induced by the differential of $\text{Tot}(K)$; since the horizontal differential increases the filtration degree, it vanishes on the associated graded object, so $\delta_0 = \delta_v$. Hence

$$(E_0^{p,q}, \delta_0) = (K^{p,q}, \delta_v).$$

The first page is defined as

$$E_1^{p,q} = H^q(K^{p,\bullet})$$

which is precisely the cohomology of the columns with respect to the vertical differential. The E_2 -page is defined as

$$E_2^{p,q} = H^p(E_1^{\bullet,q})$$

with respect to the differential induced by the horizontal differential δ_h (see [47, Ch. 10, §2, Thm. 10.11]).

More generally, for the spectral sequence of a filtered complex one can describe the r -th page directly inside the filtered complex. Define

$$\overline{Z}_r^{p,q} := \ker \left(F^p C^{p+q} \rightarrow C^{p+q+1} / F^{p+r} C^{p+q+1} \right)$$

and

$$\overline{B}_r^{p,q} := \delta(F^{p-r+1} C^{p+q-1}) \cap F^p C^{p+q}.$$

Then the r -th page is given by

$$E_r^{p,q} \cong \frac{\overline{Z}_r^{p,q}}{\overline{B}_r^{p,q} + \overline{Z}_{r-1}^{p+1,q-1}}.$$

Since $\overline{Z}_r^{p,q}$ and $\overline{B}_r^{p,q}$ are subobjects of C^{p+q} , one may also define

$$\overline{Z}_\infty^{p,q} := \bigcap_r \overline{Z}_r^{p,q}, \quad \overline{B}_\infty^{p,q} := \bigcup_r \overline{B}_r^{p,q}.$$

If the filtration is bounded in each degree, these subobjects stabilise for $r \gg 0$. In that case the stabilised page satisfies

$$E_\infty^{p,q} \cong \frac{\overline{Z}_\infty^{p,q}}{\overline{B}_\infty^{p,q} + \overline{Z}_\infty^{p+1,q-1}}.$$

The differential

$$\delta_r : E_r^{p,q} \rightarrow E_r^{p+r,q-r+1}$$

is induced by the differential of $C^\bullet$, namely

$$\delta_r([x]) = [\delta x]$$

(see [47, Ch. 10, §2, Thm. 10.11]).

For a filtered total complex arising from a bicomplex, this construction yields a spectral sequence whose first page is obtained by taking vertical cohomology (in q -direction), and whose second page is obtained by taking horizontal cohomology (in p -direction). The successive pages E_r provide increasingly refined approximations to the cohomology of the total complex. To make this precise, one must understand the limiting behaviour of the spectral sequence.

Definition 3.6.9 (Convergence of spectral sequences). Let $C^\bullet$ be a cochain complex with a decreasing filtration $F^\bullet$ that is exhaustive, separated, and bounded in each degree, and let $E_r^{p,q}$ be the spectral sequence associated to $(C^\bullet, F^\bullet)$. The spectral sequence **converges** to $H^n(C^\bullet)$ if the filtration $F^\bullet C^\bullet$ induces a filtration $F^\bullet H^n(C^\bullet)$ on cohomology such that

$$E_\infty^{p,q} \cong \text{gr}_F^p H^{p+q}(C^\bullet).$$

If the induced filtration on $H^n(C^\bullet)$ is finite for each n , the spectral sequence **converges strongly** (see [47, Ch. 10, §3, p. 626]).

For spectral sequences arising from filtered complexes, convergence is guaranteed under suitable boundedness assumptions. Indeed, one can identify the stabilised page with the graded pieces of the induced filtration on cohomology.

Theorem 3.6.10 (Standard convergence theorem). Let $C^\bullet$ be a cochain complex with a decreasing filtration by subcomplexes

$$\dots \supseteq F^p C^\bullet \supseteq F^{p+1} C^\bullet \supseteq \dots$$

such that for every n , $F^p C^n = 0$ for $p \gg 0$ and $F^p C^n = C^n$ for $p \ll 0$. Then the spectral sequence associated to $(C^\bullet, F^\bullet)$ satisfies

1. **Stabilisation:** For each (p, q) , the terms $E_r^{p,q}$ stabilise for $r \gg 0$, defining $E_\infty^{p,q}$.

2. **Identification of the limit:** There is a natural isomorphism

$$E_\infty^{p,q} \cong \text{gr}_F^p H^{p+q}(C^\bullet)$$

where

$$\mathrm{gr}_F^p H^n(C^\bullet) = \frac{F^p H^n(C^\bullet)}{F^{p+1} H^n(C^\bullet)}.$$

In particular, the spectral sequence converges strongly to $H^\bullet(C^\bullet)$ (see [54, Ch. 5, §5, Theorem 5.5.1], [47, Ch. 10, §3, Thm. 10.14]).

Proof. The proof follows three steps: (1) the stabilisation, (2) the identification of the stable value, (3) the comparison with the induced filtration on cohomology.

(1) **Stabilisation:**

Recall the standard definitions

$$\begin{aligned}\overline{Z}_r^{p,q} &:= \ker \left(F^p C^{p+q} \rightarrow C^{p+q+1} / F^{p+r} C^{p+q+1} \right) \\ \overline{B}_r^{p,q} &:= \delta(F^{p-r+1} C^{p+q-1}) \cap F^p C^{p+q}.\end{aligned}$$

Then

$$E_r^{p,q} \cong \frac{\overline{Z}_r^{p,q}}{\overline{B}_r^{p,q} + \overline{Z}_{r-1}^{p+1,q-1}}.$$

Fix (p, q) and let $n = p + q$.

Since the filtration is bounded in each degree, there exist integers a, b such that

$$F^a C^{n-1} = C^{n-1} \text{ and } F^b C^{n+1} = 0.$$

Hence for r sufficiently large one has $p - r + 1 \leq a$ and $p + r \geq b$. For such r ,

$$F^{p-r+1} C^{n-1} = C^{n-1} \text{ and } F^{p+r} C^{n+1} = 0.$$

Therefore

$$\begin{aligned}\overline{B}_r^{p,q} &= \delta(C^{n-1}) \cap F^p C^n \\ \overline{Z}_r^{p,q} &= \ker(\delta : C^n \rightarrow C^{n+1}) \cap F^p C^n.\end{aligned}$$

Both $\overline{Z}_r^{p,q}$ and $\overline{B}_r^{p,q}$ stabilise for sufficiently large r , and thus also $\overline{B}_r^{p,q} + \overline{Z}_{r-1}^{p+1,q-1}$, since from some r onward, the summands are both constant. Hence the groups $E_r^{p,q}$ stabilise, and their value is denoted by $E_\infty^{p,q}$.

(2) **Identification of the stable value:**

It follows from (1) that for $r \gg 0$, the stabilised terms satisfy

$$E_{\infty}^{p,q} \cong \frac{Z^n(C^{\bullet}) \cap F^p C^n}{(B^n(C^{\bullet}) \cap F^p C^n) + (Z^n(C^{\bullet}) \cap F^{p+1} C^n)}.$$

It remains to show that this quotient is exactly

$$\frac{F^p H^n(C^{\bullet})}{F^{p+1} H^n(C^{\bullet})}.$$

(3) Comparison with the induced filtration on cohomology:

By definition,

$$F^p H^n(C^{\bullet}) = \text{im} (H^n(F^p C^{\bullet}) \rightarrow H^n(C^{\bullet})),$$

which is the filtration on cohomology induced by $F^{\bullet} C^{\bullet}$. An element of $H^n(F^p C^{\bullet})$ is represented by a cocycle $x \in Z^n(C^{\bullet}) \cap F^p C^n$, and its image in $H^n(C^{\bullet})$ is the class of x modulo $B^n(C^{\bullet})$. Hence

$$F^p H^n(C^{\bullet}) = \frac{(Z^n(C^{\bullet}) \cap F^p C^n) + B^n(C^{\bullet})}{B^n(C^{\bullet})},$$

and similarly,

$$F^{p+1} H^n(C^{\bullet}) = \frac{(Z^n(C^{\bullet}) \cap F^{p+1} C^n) + B^n(C^{\bullet})}{B^n(C^{\bullet})}.$$

Therefore

$$\frac{F^p H^n(C^{\bullet})}{F^{p+1} H^n(C^{\bullet})} = \text{gr}_F^p H^n(C^{\bullet}) \cong \frac{Z^n(C^{\bullet}) \cap F^p C^n + B^n(C^{\bullet})}{(Z^n(C^{\bullet}) \cap F^{p+1} C^n) + B^n(C^{\bullet})}.$$

By the isomorphism theorem 1.6.25, this quotient is naturally isomorphic to

$$\frac{Z^n(C^{\bullet}) \cap F^p C^n}{(B^n(C^{\bullet}) \cap F^p C^n) + (Z^n(C^{\bullet}) \cap F^{p+1} C^n)},$$

which coincides with the stabilised expression for $E_{\infty}^{p,q}$. Comparing with the description of $E_{\infty}^{p,q}$ from (2) yields

$$E_{\infty}^{p,q} \cong \text{gr}_F^p H^{p+q}(C^{\bullet}),$$

which proves the claim (see [54, Ch. 5, §5, Theorem 5.5.1], [47, Ch. 10, §3, Thm.

10.14)). □

The theorem ensures that a spectral sequence coming from a bounded filtration actually computes the cohomology of the complex via its associated graded pieces. To recover the cohomology objects themselves, one needs a comparison result showing that an isomorphism on all graded pieces implies an isomorphism of the filtered objects.

Lemma 3.6.11 (Comparison of finite filtrations). Let

$$\begin{aligned} F^\bullet A &:= F^0 A \supseteq F^1 A \supseteq F^2 A \supseteq \cdots \supseteq F^N A \supseteq F^{N+1} A = 0 \\ F^\bullet B &:= F^0 B \supseteq F^1 B \supseteq F^2 B \supseteq \cdots \supseteq F^N B \supseteq F^{N+1} B = 0 \end{aligned}$$

be finite filtrations, and let

$$f : F^\bullet A \rightarrow F^\bullet B$$

be a morphism of filtered objects. If

$$\mathrm{gr}_F^p(f) : \mathrm{gr}_F^p(A) \rightarrow \mathrm{gr}_F^p(B)$$

is an isomorphism for every p , then f is an isomorphism (see [47, Ch. 10, §3, Ex. 10.6]).

Proof. For each p there exists a commutative diagram with exact rows

$$\begin{array}{ccccccc} 0 & \longrightarrow & F^{p+1} A & \longrightarrow & F^p A & \longrightarrow & \mathrm{gr}_F^p A \longrightarrow 0 \\ & & \downarrow f^{p+1} & & \downarrow f^p & & \downarrow \mathrm{gr}_F^p(f) \\ 0 & \longrightarrow & F^{p+1} B & \longrightarrow & F^p B & \longrightarrow & \mathrm{gr}_F^p B \longrightarrow 0 \end{array}$$

By descending induction on p one can prove that

$$f_p : F^p A \rightarrow F^p B$$

is an isomorphism.

For $p = N + 1$, both $F^{N+1} A$ and $F^{N+1} B$ are zero, so f_{N+1} is an isomorphism.

Assume that f_{p+1} is an isomorphism. By assumption,

$$\mathrm{gr}_F^p(f) : \mathrm{gr}_F^p A \rightarrow \mathrm{gr}_F^p B$$

is an isomorphism. Applying Lemma 1.7.13 to short exact sequences padded by zeros, it follows that

$$f_p : F^p A \rightarrow F^p B$$

is an isomorphism.

By induction, $f_0 : F^0 A \rightarrow F^0 B$ is an isomorphism. Since $F^0 A = A$ and $F^0 B = B$, it follows that $f : A \rightarrow B$ is an isomorphism. $\square$

The spectral sequence constructed above provides successive approximations to the cohomology of the total complex. To make this precise, one needs to relate its limit to the actual cohomology objects. In the case of first-quadrant bicomplexes, the boundedness of the induced filtration ensures that the spectral sequence converges strongly, and its limit recovers the associated graded pieces of the total cohomology.

Corollary 3.6.12. Let $K^{\bullet,\bullet}$ be a first-quadrant bicomplex, and equip $\mathrm{Tot}(K)$ with the column filtration

$$F^p \mathrm{Tot}^n(K) := \bigoplus_{\substack{i+j=n \\ i \geq p}} K^{i,j}.$$

Then the *associated spectral sequence* satisfies

$$E_{\infty}^{p,q} \cong \mathrm{gr}_F^p H^{p+q}(\mathrm{Tot}(K))$$

(see [47, Ch. 10, §4, Prop. 10.21]).

Proof. Fix $n \in \mathbb{Z}$. Since $K^{\bullet,\bullet}$ is a first-quadrant bicomplex, one has $K^{i,j} = 0$ whenever $i < 0$ or $j < 0$. Hence in total degree n , only summands with $i + j = n$, and $i, j \geq 0$ occur. In particular, $0 \leq i \leq n$. Therefore, if $p > n$, there is no pair (i, j) with $i + j = n$ and $i \geq p$. Thus

$$F^p \mathrm{Tot}^n(K) = \bigoplus_{\substack{i+j=n \\ i \geq p}} K^{i,j} = 0.$$

On the other hand, if $p \leq 0$, then every summand $K^{i,j}$ occurring in $\text{Tot}^n(K)$ satisfies $i \geq 0 \geq p$, hence

$$F^p \text{Tot}^n(K) = \bigoplus_{\substack{i+j=n \\ i \geq p}} K^{i,j} = \bigoplus_{i+j=n} K^{i,j} = \text{Tot}^n(K).$$

Thus for every n ,

$$F^p \text{Tot}^n(K) = 0$$

for $p \gg 0$, and

$$F^p \text{Tot}^n(K) = \text{Tot}^n(K)$$

for $p \ll 0$. Hence the filtration is bounded in every degree. Hence Theorem 3.6.10 applies and yields

$$E_{\infty}^{p,q} \cong \text{gr}_F^p H^{p+q}(\text{Tot}(K)),$$

proving the claim (see [47, Ch. 10, §4, Prop. 10.21]). $\square$

The corollary ensures that the spectral sequence actually computes the cohomology of the total complex via a finite filtration, which is used in the proof of Theorem 4.3.26. The convergence theorem allows one to compare total cohomology by comparing the associated spectral sequences. This yields the following criterion for quasi-isomorphisms.

Theorem 3.6.13. Let $f : K^{\bullet,\bullet} \rightarrow L^{\bullet,\bullet}$ be a morphism of first-quadrant bicomplexes in an abelian category. Assume that either

1. for every $q \geq 0$, the induced morphism of row complexes

$$f^{\bullet,q} : K^{\bullet,q} \rightarrow L^{\bullet,q}$$

is a quasi-isomorphism, or

2. for every $p \geq 0$, the induced morphism of column complexes

$$f^{p,\bullet} : K^{p,\bullet} \rightarrow L^{p,\bullet}$$

is a quasi-isomorphism.

Then the induced morphism on total complexes

$$\mathrm{Tot}(f) : \mathrm{Tot}(K) \rightarrow \mathrm{Tot}(L)$$

is a quasi-isomorphism, with

$$\mathrm{Tot}^n(K) := \bigoplus_{p+q=n} K^{p,q}$$

$$\delta_{\mathrm{Tot}} = \delta_h + \delta_v$$

on $K^{p,q}$, and analogously for L (see [47, Ch. 10, §3, Ex. 10.6]).

Proof. The two cases are analogous. In the first case, one filters the total complex by rows, while in the second case, one filters by columns. One proves the second case as follows:

For $r \in \mathbb{Z}$, define

$$F^r \mathrm{Tot}^n(K) := \bigoplus_{\substack{p+q=n \\ p \geq r}} K^{p,q}$$

$$F^r \mathrm{Tot}^n(L) := \bigoplus_{\substack{p+q=n \\ p \geq r}} L^{p,q}.$$

Since L and K are first-quadrant bicomplexes, these filtrations are bounded, exhaustive, and separated in each total degree. Moreover, $\mathrm{Tot}(f)$ preserves filtrations by Lemma 3.6.6. The spectral sequence associated with the filtered complex $\mathrm{Tot}(K)$ has

$$E_0^{p,q}(K) = K^{p,q}$$

with differential induced by the vertical differential δ_v , hence

$$E_1^{p,q}(K) \cong H^q(K^{p,\bullet}).$$

Analogously,

$$E_1^{p,q}(L) \cong H^q(L^{p,\bullet}).$$

Since each column map

$$f^{p,\bullet} : K^{p,\bullet} \rightarrow L^{p,\bullet}$$

is a quasi-isomorphism, the induced maps

$$H^q(K^{p,\bullet}) \rightarrow H^q(L^{p,\bullet})$$

are isomorphisms for all p, q . Therefore the morphism of spectral sequences induced by $\text{Tot}(f)$ is an isomorphism on the E_1 -page:

$$E_1^{p,q}(K) \xrightarrow{\sim} E_1^{p,q}(L).$$

Hence there is an isomorphism on every subsequent page E_r , $r \geq 1$. Because the filtrations are bounded in each total degree, both spectral sequences converge to the cohomology of the total complexes:

$$\begin{aligned} E_\infty^{p,q}(K) &\cong \text{gr}_F^p H^{p+q}(\text{Tot}(K)) \\ E_\infty^{p,q}(L) &\cong \text{gr}_F^p H^{p+q}(\text{Tot}(L)) \end{aligned}$$

by Corollary 3.6.12. Since the induced map is an isomorphism on E_∞ , it is an isomorphism on the associated graded objects of the filtrations on total cohomology:

$$\text{gr}_F H^n(\text{Tot}(K)) \xrightarrow{\sim} \text{gr}_F H^n(\text{Tot}(L))$$

for every n . Finally, since in each total degree the filtrations are finite, Lemma 3.6.11 implies that the induced morphism

$$H^n(\text{Tot}(K)) \xrightarrow{\sim} H^n(\text{Tot}(L))$$

is an isomorphism for all n . Therefore $\text{Tot}(f)$ is a quasi-isomorphism. $\square$

The theorem shows that a quasi-isomorphism in one direction of a bicomplex induces a quasi-isomorphism on the associated total complexes. It allows one to pass from local or directional quasi-isomorphisms to global ones, which is essential in spectral sequence arguments and in comparing resolutions. In particular in Proposition 4.3.25, it justifies replacing a bicomplex by a columnwise quasi-isomorphic one without changing the resulting cohomology.

Up to this point, cohomology has been defined abstractly for arbitrary cochain complexes in an abelian category. To obtain topological invariants, one must associate a

suitable cochain complex to a topological space. The classical example is provided by singular cohomology, which is constructed from singular simplices and their associated cochains.

4 Cohomology

This chapter provides a brief introduction to singular cohomology in Section 4.1, on which several foundational arguments used in the computations of Chapter 5 rely. Section 4.2 briefly develops sheaf cohomology. Finally, Section 4.3 introduces Čech cohomology, which provides a concrete method for computing sheaf cohomology under suitable conditions.

4.1 Singular Cohomology

Singular cohomology associates algebraic invariants to a topological space by studying cochains on its singular chain complex. Like singular homology, it detects topological features of a space, including phenomena related to connected components, loops, cavities, and higher-dimensional holes. Rather than studying singular chains directly, singular cohomology considers cochains, which assign coefficients to them. This perspective is used in Section 4.2 to relate singular cohomology to the cohomology of constant sheaves. In order to study topological spaces by algebraic methods, one seeks elementary geometric objects whose continuous images in an arbitrary topological space can be organised algebraically.

Definition 4.1.1 (Singular simplex). Let X be a topological space. A **singular n -simplex** in X is a continuous map $\sigma : \Delta^n \rightarrow X$. The collection of singular n -simplices is denoted by $\text{Sing}_n(X)$ (see [47, Ch. 1, §3, p. 29]).

In contrast to the standard simplex, a singular simplex need not be an embedding, it may self-intersect, collapse dimensions, and be completely degenerate.

Singular simplices serve as the basic building blocks of singular homology. By taking formal linear combinations of singular simplices, one obtains chains. The

face structure of the standard simplex induces boundary operators on these chains, leading to a chain complex whose homology captures topological information about the underlying space.

Definition 4.1.2 (Singular (co-)chains). Let X be a topological space. The **singular n -chain group** is a free abelian group

$$C_n^{\text{sing}}(X) := \mathbb{Z}[\text{Sing}_n(X)],$$

and its elements are called **singular n -chains**.

For an abelian group G , the **singular n -cochain group** is

$$C_{\text{sing}}^n(X, G) := \text{Hom}_{\text{Ab}}(C_n^{\text{sing}}(X), G),$$

and its elements are called **singular n -cochains** (see [47, Ch. 1, §3, pp. 29 ff.]).

To obtain a chain complex, one must define morphisms between the singular chain groups. The boundary of an n -simplex consists of its $(n-1)$ -dimensional faces. As every singular simplex is a continuous map from a standard simplex, this notion of boundary extends naturally to singular simplices.

Definition 4.1.3 (Singular (co-)boundary operators). Consider singular chain groups $C_{\bullet}^{\text{sing}}(X)$. For a singular simplex $\sigma : \Delta^n \rightarrow X$, the **singular boundary operator** or **singular differential** is defined by

$$\delta_n^{\text{sing}}(\sigma) := \sum_{i=0}^n (-1)^i \sigma \circ \iota_i,$$

where

$$\iota_i : \Delta^{n-1} \hookrightarrow \Delta^n$$

is the inclusion of the i -th face. The **singular chain complex** is

$$\dots \rightarrow C_2^{\text{sing}}(X) \xrightarrow{\delta_2^{\text{sing}}} C_1^{\text{sing}}(X) \xrightarrow{\delta_1^{\text{sing}}} C_0^{\text{sing}}(X) \rightarrow 0.$$

The **singular coboundary operator** or **singular codifferential**

$$\delta_{\text{sing}}^n : C_{\text{sing}}^n(X, G) \rightarrow C_{\text{sing}}^{n+1}(X, G)$$

between two singular cochain groups $C_{\text{sing}}^n(X, G)$ and $C_{\text{sing}}^{n+1}(X, G)$ is defined by

$$\delta_{\text{sing}}^n(\varphi) := \varphi \circ \delta_{n+1}^{\text{sing}}.$$

The **singular cochain complex** becomes

$$0 \rightarrow C_{\text{sing}}^0(X, G) \xrightarrow{\delta_{\text{sing}}^0} C_{\text{sing}}^1(X, G) \xrightarrow{\delta_{\text{sing}}^1} C_{\text{sing}}^2(X, G) \rightarrow \dots$$

(see [47, Ch. 1, §3, pp. 29 ff.]).

While singular homology is defined from the singular chain complex, singular cohomology arises from the corresponding cochain complex. The resulting cohomology groups provide a dual perspective on the topology of the underlying space.

Definition 4.1.4 (Singular cohomology). Consider a singular cochain complex

$$(C_{\text{sing}}^\bullet(X, G), \delta_{\text{sing}}^\bullet).$$

For each degree $n \in \mathbb{N}_0$ define the group of **singular n -cocycles**

$$Z_{\text{sing}}^n(C_{\text{sing}}^\bullet(X, G)) := \ker(\delta_{\text{sing}}^n),$$

and the group of **singular n -coboundaries**

$$B_{\text{sing}}^n(C_{\text{sing}}^\bullet(X, G)) := \text{im}(\delta_{\text{sing}}^{n-1}),$$

with $B_{\text{sing}}^0 := 0$. The **n -th singular cohomology group** of X with coefficients in G is

$$H_{\text{sing}}^n(X, G) = \frac{\ker(\delta_{\text{sing}}^n : C_{\text{sing}}^n(X, G) \rightarrow C_{\text{sing}}^{n+1}(X, G))}{\text{im}(\delta_{\text{sing}}^{n-1} : C_{\text{sing}}^{n-1}(X, G) \rightarrow C_{\text{sing}}^n(X, G))}$$

(see [47, Ch. 1, §3, pp. 30 ff.]).

An **augmented singular cochain complex** is

$$0 \rightarrow G \xrightarrow{\varepsilon} C_{\text{sing}}^0(X, G) \xrightarrow{\delta_{\text{sing}}^0} C_{\text{sing}}^1(X, G) \xrightarrow{\delta_{\text{sing}}^1} C_{\text{sing}}^2(X, G) \rightarrow \dots$$

where

$$\varepsilon : G \rightarrow C_{\text{sing}}^0(X, G)$$

is defined by sending $g \in G$ to the constant 0-cochain

$$\varepsilon(g)(\sigma) = g$$

for every singular 0-simplex $\sigma : \Delta^0 \rightarrow X$.

Remark 4.1.5. Since $C_n^{\text{sing}}(X)$ is the free abelian group on $\text{Sing}_n(X)$, every singular n -chain can be written uniquely in the form

$$\sum_i a_i \sigma_i,$$

with $a_i \in \mathbb{Z} \setminus \{0\}$, and the singular simplices

$$\sigma_1, \dots, \sigma_k : \Delta^n \rightarrow X$$

are pairwise distinct.

Chain homotopic chain morphisms induce cochain homotopic maps on singular cochains.

Lemma 4.1.6. Let $f_\bullet, g_\bullet : C_\bullet^{\text{sing}}(X) \rightarrow C_\bullet^{\text{sing}}(Y)$ be chain maps. Suppose there exist morphisms

$$M_n : C_n^{\text{sing}}(X) \rightarrow C_{n+1}^{\text{sing}}(Y)$$

such that

$$g_n - f_n = \delta_{n+1}^{\text{sing}} \circ M_n + M_{n-1} \circ \delta_n^{\text{sing}}.$$

Then the induced cochain maps

$$f^\bullet, g^\bullet : C_{\text{sing}}^\bullet(Y, G) \rightarrow C_{\text{sing}}^\bullet(X, G)$$

are cochain homotopic (see [47, Ch. 6, §1, Thm. 6.14]).

Proof. Set $M^0 = 0$. For $n \in \mathbb{N}$, define

$$M^n : C_{\text{sing}}^n(Y, G) \rightarrow C_{\text{sing}}^{n-1}(X, G)$$

by

$$M^n(\varphi)(c) = \varphi(M_{n-1}).$$

for $c \in C_{n-1}^{\text{sing}}(X)$. Let $\varphi \in C_{\text{sing}}^n(Y, G)$ and $c \in C_n^{\text{sing}}(X)$. Then

$$\begin{aligned} ((g^n - f^n)(\varphi))(c) &= (\varphi \circ g_n)(c) - (\varphi \circ f_n)(c) \\ &= \varphi((g_n - f_n)(c)) \\ &= \varphi(\delta_{n+1}^{\text{sing}} \circ M_n(c) + M_{n-1} \circ \delta_n^{\text{sing}}(c)) \\ &= \varphi(\delta_{n+1}^{\text{sing}} \circ M_n(c)) + \varphi(M_{n-1} \circ \delta_n^{\text{sing}}(c)). \end{aligned}$$

On the other hand,

$$\left(M^{n+1}(\delta_{\text{sing}}^n(\varphi)) \right)(c) = (\delta_{\text{sing}}^n(\varphi))(M_n(c)) = \varphi(\delta_{n+1}^{\text{sing}} \circ M_n(c))$$

and

$$(\delta_{\text{sing}}^{n-1} \circ M^n(\varphi))(c) = (M^n(\varphi))(\delta_n^{\text{sing}}(c)) = \varphi(M_{n-1} \circ \delta_n^{\text{sing}}(c)).$$

Hence

$$g^n - f^n = M^{n+1} \circ \delta_{\text{sing}}^n + \delta_{\text{sing}}^{n-1} \circ M^n$$

in degree n . Therefore $f^\bullet$ and $g^\bullet$ are cochain homotopic (see [47, Ch. 6, §1, Thm. 6.14]). $\square$

This shows that chain homotopies induce cochain homotopies after applying $\text{Hom}(-, G)$. One can construct a prism operator, which yields a chain homotopy between suitable chain maps. A question that naturally arises is whether singular cohomology depends on the homotopy type of the underlying space. To prove homotopy invariance, one must construct a chain homotopy between chain maps,

To prove homotopy invariance, one must construct a chain homotopy between the chain maps induced by homotopic continuous maps. Prism operators provide precisely such a chain homotopy.

Definition 4.1.7 (Prism). A **prism** on an n -simplex Δ^n is $\Delta^n \times [0, 1]$ (see [26, Ch. 2, §1, p. 112]).

For $n = 0$, the prism gives a line segment, for $n = 1$ a rectangle, and for $n = 2$ a triangular prism. A prism itself is not a simplex, but it can be decomposed into $(n + 1)$ simplices.

Definition 4.1.8 (Prism operator). Let $\sigma : \Delta^n \rightarrow X$ be a singular simplex, and let $H : X \times [0, 1] \rightarrow Y$ be a continuous map between topological spaces X and Y . The prism

$$\Delta^n \times [0, 1] \xrightarrow{\sigma \times \text{id}} X \times [0, 1] \xrightarrow{H} Y$$

admits a canonical decomposition into $(n + 1)$ simplices. The alternating sum of the resulting singular $(n + 1)$ -simplices in Y is denoted by

$$P_n(\sigma) \in C_{n+1}^{\text{sing}}(Y).$$

Extending linearly defines the **prism operator**

$$P_n : C_n^{\text{sing}}(X) \rightarrow C_{n+1}^{\text{sing}}(Y)$$

(see [26, Ch. 2, §1, p. 112]).

Prism operators provide the chain homotopies needed for the proof of homotopy invariance. Their construction is based on a subdivision of the prism $\Delta^n \times [0, 1]$ into $(n + 1)$ -simplices.

Lemma 4.1.9. Let $f, g : X \rightarrow Y$ be homotopic maps with homotopy

$$G : X \times [0, 1] \rightarrow Y.$$

Then there exists a family of homomorphisms

$$P_n : C_n^{\text{sing}}(X) \rightarrow C_{n+1}^{\text{sing}}(Y)$$

such that

$$\delta_{n+1}^{\text{sing}} \circ P_n + P_{n-1} \circ \delta_n^{\text{sing}} = g_n - f_n$$

for all $n \geq 0$ and $P_{-1} = 0$ (see [26, Ch. 2, §1, pp. 112 ff.]).

Proof. Let $\sigma : \Delta^n \rightarrow X$ be a singular n -simplex. Then

$$G \circ (\sigma \times \text{id}_{[0,1]}) : \Delta^n \times [0, 1] \rightarrow Y$$

is the singular prism determined by G and σ . Subdivide $\Delta^n \times [0, 1]$ into $(n + 1)$

simplices. If $v_0, \dots, v_n$ are vertices of Δ^n , define

$$Q_i : \Delta^{n+1} \rightarrow \Delta^n \times [0, 1],$$

where Q_i denotes the affine singular $(n+1)$ -simplex of $\Delta^n \times [0, 1]$ with ordered vertices $(v_0, 0), \dots, (v_i, 0), (v_i, 1), \dots, (v_n, 1)$. Define

$$P_n(\sigma) = \sum_{i=0}^n (-1)^i G \circ (\sigma \times \text{id}_{[0,1]}) \circ Q_i.$$

Since $C_n^{\text{sing}}(X)$ is the free abelian group generated by singular n -simplices, every assignment $\sigma \mapsto P_n(\sigma)$ extends uniquely by linearity to a homomorphism

$$P_n : C_n^{\text{sing}}(X) \rightarrow C_{n+1}^{\text{sing}}(Y).$$

The boundary of each Q_i consists of faces where one vertex is omitted. In the alternating sum

$$\delta_{n+1}^{\text{sing}} \circ P_n(\sigma) = \sum_{i=0}^n (-1)^i \delta_{n+1}^{\text{sing}} (G \circ (\sigma \times \text{id}_{[0,1]}) \circ Q_i)$$

the internal faces cancel each other out in a telescoping sum, that is, for $0 \leq i \leq n-1$, the face of Q_i obtained by deleting $(v_i, 1)$ coincides with the face of Q_{i+1} , obtained by deleting $(v_{i+1}, 0)$, as

$$Q_i \circ \iota_{i+1} = Q_{i+1} \circ \iota_{i+1}.$$

Both faces have ordered vertices $(v_0, 0), \dots, (v_i, 0), (v_{i+1}, 1), \dots, (v_n, 1)$, and their coefficients are

$$(-1)^i (-1)^{i+1} = -1, \quad (-1)^{i+1} (-1)^{i+1} = 1;$$

hence the faces cancel.

The only remaining faces under the map $G \circ (\sigma \times \text{id}_{[0,1]})$ are the top face $[(v_0, 1), \dots, (v_n, 1)]$ with coefficient

$$(-1)^0 (-1)^0 = 1,$$

the bottom face $[(v_0, 0), \dots, (v_n, 0)]$ with coefficient

$$(-1)^n (-1)^{n+1} = -1,$$

and the side faces. The top face contributes $g_n(\sigma)$, the bottom face contributes $-f_n(\sigma)$, and the side faces together contribute $-P_{n-1}(\delta_n^{\text{sing}}(\sigma))$.

Indeed, the side faces are precisely the prisms over the faces $\sigma \circ \iota_k$ of σ . Since

$$\delta_n^{\text{sing}}(\sigma) = \sum_{k=0}^n (-1)^k \sigma \circ \iota_k,$$

the linearity of P_{n-1} gives

$$P_{n-1}(\delta_n^{\text{sing}}(\sigma)) = \sum_{k=0}^n (-1)^k P_{n-1}(\sigma \circ \iota_k).$$

With the boundary signs of the side faces, these occur in $\delta_{n+1}^{\text{sing}} \circ P_n(\sigma)$ with the opposite sign, hence the total side contribution is

$$-P_{n-1}(\delta_n^{\text{sing}}(\sigma)).$$

Hence

$$\delta_{n+1}^{\text{sing}} \circ P_n(\sigma) = g_n(\sigma) - f_n(\sigma) - P_{n-1} \circ \delta_n^{\text{sing}}(\sigma).$$

Equivalently,

$$\delta_{n+1}^{\text{sing}} \circ P_n(\sigma) + P_{n-1} \circ \delta_n^{\text{sing}}(\sigma) = g_n(\sigma) - f_n(\sigma).$$

Since singular chains are free abelian groups on singular simplices, this identity extends by linearity to all chains. Thus

$$\delta_{n+1}^{\text{sing}} \circ P_n + P_{n-1} \circ \delta_n^{\text{sing}} = g_{\bullet} - f_{\bullet}.$$

(see [26, Ch. 2, §1, pp. 112 ff.]). □

The prism operators therefore define a chain homotopy between the chain maps induced by homotopic continuous maps. As a consequence, homotopic maps induce identical homomorphisms on singular homology and singular cohomology.

Theorem 4.1.10 (Homotopy invariance of singular cohomology). Let $A \in \text{Ob}(\text{Ab})$. If $f, g : X \rightarrow Y$ are homotopic, then the induced maps

$$f^*, g^* : H_{\text{sing}}^n(Y, A) \rightarrow H_{\text{sing}}^n(X, A)$$

are equal for all $n \in \mathbb{N}_0$. In particular, if $X \simeq Y$, then

$$H_{\text{sing}}^n(X, A) \cong H_{\text{sing}}^n(Y, A)$$

for all $n \in \mathbb{N}_0$ (see [26, Ch. 2, §1, pp. 112 ff.]).

Proof. Since $f \simeq g$ by assumption, there exists a homotopy

$$G : X \times [0, 1] \rightarrow Y$$

such that $G(x, 0) = f(x)$ and $G(x, 1) = g(x)$. By Lemma 4.1.9, the induced chain maps

$$f_{\bullet}, g_{\bullet} : C_{\text{sing}\bullet}(X, \mathbb{Z}) \rightarrow C_{\text{sing}\bullet}(Y, \mathbb{Z})$$

are chain homotopic. Hence by Lemma 4.1.6 the induced cochain maps

$$f^*, g^* : C^{\bullet}(Y, A) \rightarrow C^{\bullet}(X, A)$$

are cochain homotopic. Therefore they induce the same map on cohomology, that is, $f^* = g^*$ as morphisms

$$H_{\text{sing}}^n(Y, A) \rightarrow H_{\text{sing}}^n(X, A).$$

This proves the first statement.

If $X \simeq Y$, choose maps

$$\phi : X \rightarrow Y, \quad \gamma : Y \rightarrow X$$

such that

$$\phi \circ \gamma \simeq \text{id}_Y, \quad \gamma \circ \phi \simeq \text{id}_X.$$

Applying the first part to the homotopies

$$\gamma \circ \phi \simeq \text{id}_X, \quad \phi \circ \gamma \simeq \text{id}_Y$$

gives

$$\phi^* \circ \gamma^* = (\gamma \circ \phi)^* = \text{id}_{H_{\text{sing}}^n(X, A)},$$

and similarly

$$\gamma^* \circ \phi^* = (\phi \circ \gamma)^* = \text{id}_{H_{\text{sing}}^n(Y, A)}.$$

Hence

$$H_{\text{sing}}^n(X, A) \cong H_{\text{sing}}^n(Y, A)$$

for all $n \in \mathbb{N}_0$ (see [26, Ch. 2, §1, pp. 112 ff.]). □

The result can be generalised in the following sense: If A is an object in an arbitrary abelian category $\mathcal{C}$, then the sets $\text{Hom}_{\mathcal{C}}(M, A)$ are always abelian groups. If one wants to define cohomology inside the category $\mathcal{C}$, one must replace the ordinary singular chain complex by a chain complex taking values inside $\mathcal{C}$. Homotopy-invariance holds for any additive functor $F : \text{Ab} \rightarrow \mathcal{C}$. Since the chain homotopy P is a purely additive relation, it is preserved under F .

Homotopy invariance allows one to compare the cohomology of homotopy equivalent spaces. Since every contractible space is homotopy equivalent to a one-point space, one expects its cohomology to be as simple as possible. The following theorem makes this observation precise.

Theorem 4.1.11 (Singular cohomology of contractible spaces). Let $U \subseteq X$ be a contractible subspace of a topological space X , and let $G \in \text{Ob}(\text{Ab})$. Then

$$H_{\text{sing}}^0(U, G) \cong G, \quad H_{\text{sing}}^n(U, G) = 0$$

for all $n > 0$ (see [26, Ch. 2, §1, pp. 112 ff.]).

Proof. First consider the one-point space $*$. For each $n \geq 0$, there exists exactly one singular simplex $\sigma_n : \Delta^n \rightarrow *$. Hence $\text{Sing}_n(*) = \{\sigma_n\}$, and

$$C_n^{\text{sing}}(*) := \mathbb{Z}[\text{Sing}_n(*)].$$

As $\text{Sing}_n(*)$ has exactly one element, the free abelian group on it is $C_n^{\text{sing}}(*) \cong \mathbb{Z}$ with generator σ_n . Therefore

$$C^n(*, G) = \text{Hom}_{\text{Ab}}(C_n^{\text{sing}}(*), G) \cong \text{Hom}_{\text{Ab}}(\mathbb{Z}, G) \cong G.$$

The boundary maps satisfy

$$\delta_n^{\text{sing}}(\sigma_n) = \sum_{i=0}^n (-1)^i \sigma_{n-1}.$$

Since all faces coincide, this becomes

$$\delta_n^{\text{sing}}(\sigma_n) = \left(\sum_{i=0}^n (-1)^i \right) \sigma_{n-1}.$$

Thus

$$\delta_n^{\text{sing}} = \begin{cases} 0 & \text{if } n \text{ is odd,} \\ \text{id} & \text{if } n \text{ is even.} \end{cases}$$

The coboundary maps

$$\delta_{\text{sing}}^n : C_{\text{sing}}^n(*, G) \rightarrow C_{\text{sing}}^{n+1}(*, G)$$

are induced by precomposition with $\delta_{n+1}^{\text{sing}}$:

$$\delta_{\text{sing}}^n(\varphi)(c) = \varphi(\delta_{n+1}^{\text{sing}}(c)),$$

yielding

$$\delta_{\text{sing}}^n(\varphi)(c) = \begin{cases} \varphi(0), & \text{for } n \text{ even} \\ \varphi(\text{id } c), & \text{for } n \text{ odd.} \end{cases}$$

Hence

$$\delta_{\text{sing}}^n = \begin{cases} 0 & \text{if } n \text{ is even,} \\ \text{id} & \text{if } n \text{ is odd.} \end{cases}$$

Therefore the singular cochain complex has cohomology

$$H_{\text{sing}}^0(*, G) = \ker(\delta_{\text{sing}}^0) \cong G,$$

and for $n > 0$

$$H_{\text{sing}}^n(*, G) = \frac{\ker(\delta_{\text{sing}}^n)}{\text{im}(\delta_{\text{sing}}^{n-1})} = 0.$$

Since U is contractible, it is homotopy equivalent to the one-point space $*$. Singular

cohomology is homotopy invariant by Theorem 4.1.10. Thus there are isomorphisms

$$H_{\text{sing}}^0(U, G) \cong H_{\text{sing}}^0(*, G) \cong G$$

and

$$H_{\text{sing}}^n(U, G) \cong H_{\text{sing}}^n(*, G) = 0$$

for $n \in \mathbb{N}$. This proves the theorem (see [26, Ch. 2, §1, pp. 112 ff.]). $\square$

While singular cohomology captures global topological information about a space, it does not directly encode geometric or analytical data. When one studies objects that are defined locally but are required to satisfy compatibility conditions on overlaps, one needs a theory which provides information on whether local data extends globally. Nevertheless, singular cohomology provides important invariants such as Betti numbers. Moreover, the vanishing of higher singular cohomology on contractible spaces plays a central role in the computation of the cohomology of constant sheaves.

4.2 Sheaf Cohomology

Sheaf cohomology provides a systematic way to measure the failure of local data to extend globally. While sheaves encode local-to-global principles through restriction and gluing, the global section functor

$$\Gamma(X, -) : \text{Sh}(X, \mathcal{C}) \rightarrow \mathcal{C}$$

is generally left-exact but not exact, as seen in Theorem 2.2.33. The resulting obstructions are captured by the right derived functors of global sections. In the sheaf-theoretic setting, the general notions of cochain complexes, injective resolutions, and derived functors specialise as follows. Injective and, more generally, acyclic resolutions then allow sheaf cohomology to be computed concretely from global sections.

Although the theory is usually developed for sheaves of abelian groups or modules, the abstract construction only requires the category of sheaves under consideration to be abelian and have enough injectives. For the comparison with singular

cohomology stated at the end of this section, the target category will be specialised to $\mathbf{Ab}$.

Definition 4.2.1 (Sheaf cochain complex). Let $\mathbf{Sh}(X, \mathcal{C})$ be an abelian category. A **sheaf cochain complex** or **cochain complex of sheaves** on a topological space X is a sequence

$$\mathcal{F}^\bullet = (\mathcal{F}^0 \xrightarrow{\delta^0} \mathcal{F}^1 \xrightarrow{\delta^1} \mathcal{F}^2 \xrightarrow{\delta^2} \dots)$$

of sheaves and morphisms of sheaves such that

$$\delta^{n+1} \circ \delta^n = 0$$

for all $n \in \mathbb{N}_0$. The morphisms

$$\delta^n : \mathcal{F}^n \rightarrow \mathcal{F}^{n+1}$$

are called **sheaf coboundary morphisms** or **sheaf differentials**. Sheaf cochain complexes are denoted by $(\mathcal{F}^\bullet, \delta^\bullet)$. Equivalently, since equality of natural transformations is determined componentwise, for every open subset $U \subseteq X$ the sequence

$$\mathcal{F}^0(U) \xrightarrow{\delta_U^0} \mathcal{F}^1(U) \xrightarrow{\delta_U^1} \mathcal{F}^2(U) \rightarrow \dots$$

is a **cochain complex** in $\mathcal{C}$ (see [47, Ch. 6, §3, pp. 377 ff.]).

The condition $\delta^{n+1} \circ \delta^n = 0$ implies that images are contained in kernels. This allows one to define cocycles, coboundaries, and cohomology objects.

Definition 4.2.2 (Cocycles and coboundaries (of sheaves)). Let $(\mathcal{F}^\bullet, \delta^\bullet)$ be a cochain complex of sheaves. For each degree $n \in \mathbb{N}_0$ define

$$\mathcal{Z}^n(\mathcal{F}^\bullet) := \ker(\delta^n)$$

called the **sheaf of n -cocycles**, and for $n \in \mathbb{N}$,

$$\mathcal{B}^n(\mathcal{F}^\bullet) := \operatorname{im}(\delta^{n-1})$$

called the **sheaf of n -coboundaries**. For $n = 0$, $\mathcal{B}^0(\mathcal{F}^\bullet) := 0$. Since

$$\delta^n \circ \delta^{n-1} = 0,$$

there is a canonical inclusion

$$\mathcal{B}^n(\mathcal{F}^\bullet) \subseteq \mathcal{Z}^n(\mathcal{F}^\bullet)$$

(see [47, Ch. 6, §2, pp. 370 ff.]).

Since every coboundary is a cocycle, one may form the quotient of cocycles by coboundaries. This measures the failure of cocycles to arise from lower-degree data and leads to the cohomology sheaf of the complex.

Definition 4.2.3 (Cohomology sheaf). The n -th cohomology sheaf of the complex $\mathcal{F}^\bullet$ is

$$\mathcal{H}^n(\mathcal{F}^\bullet) := \mathcal{Z}^n(\mathcal{F}^\bullet) / \mathcal{B}^n(\mathcal{F}^\bullet) = \frac{\ker(\delta^n : \mathcal{F}^n \rightarrow \mathcal{F}^{n+1})}{\operatorname{im}(\delta^{n-1} : \mathcal{F}^{n-1} \rightarrow \mathcal{F}^n)}.$$

The complex $\mathcal{F}^\bullet$ is called **exact in degree n** if $\mathcal{H}^n(\mathcal{F}^\bullet) = 0$, and **exact** if it is exact in every degree (see [32, Ch. 12, §2, p. 300]).

Applying the global section functor to a cochain complex of sheaves produces an ordinary cochain complex in the target category. Its cohomology should be distinguished from the cohomology sheaves defined above.

Remark 4.2.4 (Global sections and ordinary cochain complexes). Applying the global section functor $\Gamma(X, -)$ to a sheaf cochain complex yields a cochain complex

$$\Gamma(X, \mathcal{F}^\bullet) : \Gamma(X, \mathcal{F}^0) \xrightarrow{\Gamma(\delta^0)} \Gamma(X, \mathcal{F}^1) \xrightarrow{\Gamma(\delta^1)} \dots$$

The resulting cocycles are

$$Z^n(\Gamma(X, \mathcal{F}^\bullet)) := \ker(\Gamma(\delta^n)),$$

for $n \in \mathbb{N}_0$, and the coboundaries are

$$B^n(\Gamma(X, \mathcal{F}^\bullet)) := \operatorname{im}(\Gamma(\delta^{n-1}))$$

for $n \in \mathbb{N}$. Their quotient

$$H^n(\Gamma(X, \mathcal{F}^\bullet)) = \frac{Z^n(\Gamma(X, \mathcal{F}^\bullet))}{B^n(\Gamma(X, \mathcal{F}^\bullet))}$$

is the ordinary cohomology of the global section complex (see [47, Ch. 6, §3, pp. 378 ff.]).

The cohomology of the global section complex depends on the chosen complex of sheaves. To obtain an intrinsic cohomological invariant of a sheaf itself, one derives the global section functor using injective resolutions.

Definition 4.2.5 (Sheaf cohomology). Assume that $\text{Sh}(X, \mathcal{C})$ is an abelian category with enough injectives, and let

$$\Gamma(X, -) : \text{Sh}(X, \mathcal{C}) \rightarrow \mathcal{C}$$

be the global section functor. The n -th **sheaf cohomology object** of a sheaf $\mathcal{F}$ is the right derived functor

$$\mathcal{H}^n(X, \mathcal{F}) := R^n\Gamma(X, \mathcal{F}).$$

Equivalently, if

$$0 \rightarrow \mathcal{F} \rightarrow \mathcal{I}^\bullet$$

is an injective resolution, then

$$\mathcal{H}^n(X, \mathcal{F}) \cong H^n(\Gamma(X, \mathcal{I}^\bullet))$$

(see [47, Ch. 6, §3, pp. 378 ff.]).

Note that if $\mathcal{C}$ is a Grothendieck abelian category, then $\text{Sh}(X, \mathcal{C})$ is a Grothendieck abelian category, and therefore has enough injectives.

Remark 4.2.6 (Expanded form of sheaf cohomology). Let

$$0 \rightarrow \mathcal{F} \rightarrow \mathcal{I}^0 \xrightarrow{\delta^0} \mathcal{I}^1 \xrightarrow{\delta^1} \mathcal{I}^2 \rightarrow \dots$$

be an injective resolution of $\mathcal{F}$. For $n \geq 1$,

$$H^n(X, \mathcal{F}) = \frac{\ker(\Gamma(\delta^n))}{\text{im}(\Gamma(\delta^{n-1}))}.$$

For $n = 0$, one has

$$\mathcal{H}^0(X, \mathcal{F}) = \ker (\Gamma(\delta^0) : \Gamma(X, \mathcal{I}^0) \rightarrow \Gamma(X, \mathcal{I}^1)) \cong \Gamma(X, \mathcal{F})$$

(see [47, Ch. 6, §3, pp. 378 ff.]).

As sheaf cohomology in its expanded form derives from injective resolutions, one might expect that injective sheaves play a distinguished role in sheaf cohomology. Indeed, injective sheaves have trivial cohomology, as the following lemma shows.

Lemma 4.2.7. Let $\text{Sh}(X, \mathcal{C})$ be an abelian category with enough injectives, and let

$$\mathcal{I} : \text{Open}(X)^{\text{op}} \rightarrow \mathcal{C}$$

be an injective sheaf. Then

$$\mathcal{H}^0(X, \mathcal{I}) = \Gamma(X, \mathcal{I}), \quad \mathcal{H}^n(X, \mathcal{I}) = 0$$

for all $n \in \mathbb{N}$ (compare [28, Ch. 3, §5, Thm. 3.5]).

Proof. Since $\text{Sh}(X, \mathcal{C})$ has enough injectives, one can choose an injective resolution of $\mathcal{I}$. This resolution may be arbitrary by Theorem 3.4.4.

Choose the injective resolution

$$0 \rightarrow \mathcal{I} \xrightarrow{\text{id}_{\mathcal{I}}} \mathcal{I} \rightarrow 0 \rightarrow 0 \rightarrow \dots$$

Since the zero sheaf takes the value $0 \in \text{Ob}(\mathcal{C})$ on every open subset, $\Gamma(X, 0) = 0$. The associated unaugmented injective complex is

$$\mathcal{I} \rightarrow 0 \rightarrow 0 \rightarrow \dots,$$

with $\mathcal{I}$ placed in degree 0. Applying $\Gamma(X, -)$ yields

$$\Gamma(X, \mathcal{I}) \rightarrow 0 \rightarrow 0 \rightarrow \dots$$

Thus,

$$\mathcal{H}^0(X, \mathcal{I}) = R^0\Gamma(X, \mathcal{I}) = \Gamma(X, \mathcal{I}),$$

and $\mathcal{H}^n(X, \mathcal{F}) = 0$ for all $n > 0$ (compare [28, Ch. 3, §5, Thm. 3.5]). $\square$

Although sheaf cohomology is defined abstractly via derived functors, it is often computed using suitable resolutions. If one considers a Grothendieck abelian category $\mathcal{C}$, the category $\text{Sh}(X, \mathcal{C})$ becomes abelian too, making $\Gamma(X, -) : \text{Sh}(X, \mathcal{C}) \rightarrow \mathcal{C}$ a left-exact functor between abelian categories. Applying Theorem 3.5.6 yields the standard computation theorem for sheaf cohomology.

Theorem 4.2.8 (Computation of sheaf cohomology via acyclic resolutions). Let $\mathcal{F} \in \text{Ob}(\text{Sh}(X, \mathcal{C}))$, where $\mathcal{C}$ is a Grothendieck abelian category. Let

$$0 \rightarrow \mathcal{F} \xrightarrow{\varepsilon} \mathcal{A}^0 \xrightarrow{\delta^0} \mathcal{A}^1 \xrightarrow{\delta^1} \mathcal{A}^2 \rightarrow \dots$$

be a resolution of $\mathcal{F}$ through Γ -acyclic sheaves, that is,

$$\mathcal{H}^n(X, \mathcal{A}^p) = 0$$

for all $p \in \mathbb{N}_0$ and $n \in \mathbb{N}$. Then there exists a canonical isomorphism

$$\mathcal{H}^n(X, \mathcal{F}) \cong H^n(\Gamma(X, \mathcal{A}^\bullet))$$

for all $n \geq 0$ (see [28, Ch. 2, §7, p. 115]).

Proof. This is the standard acyclic resolution theorem for right derived functors of the left-exact functor $\Gamma(X, -)$.

Choose a Γ -acyclic resolution

$$0 \rightarrow \mathcal{F} \rightarrow \mathcal{A}^\bullet.$$

Since $\Gamma(X, -)$ is left-exact and $\mathcal{A}^\bullet$ is a Γ -acyclic resolution of $\mathcal{F}$, Theorem 3.5.6 yields canonical isomorphisms

$$R^n \Gamma(X, \mathcal{F}) \cong H^n(\Gamma(X, \mathcal{A}^\bullet)).$$

By definition,

$$\mathcal{H}^n(X, \mathcal{F}) = R^n \Gamma(X, \mathcal{F}) \cong H^n(\Gamma(X, \mathcal{A}^\bullet))$$

(see [28, Ch. 2, §7, p. 115]). □

The preceding theorem reduces the computation of sheaf cohomology to the construction of suitable acyclic resolutions. It is therefore natural to ask which classes of sheaves are acyclic with respect to the global section functor. A fundamental example is given by flabby sheaves, whose higher sheaf cohomology vanishes.

Flabby Sheaves

Definition 4.2.9 (Flabby sheaf). A sheaf $\mathcal{F} : \text{Open}(X)^{\text{op}} \rightarrow \mathcal{C}$ on a topological space X is **flabby** if for every inclusion of open subsets $U \subseteq V \subseteq X$, the restriction morphism

$$\mathcal{F}(V) \rightarrow \mathcal{F}(U)$$

is an epimorphism in $\mathcal{C}$ (see [47, Ch. 6, §3, p. 379/380]).

For sheaves of abelian groups, modules, or other concrete algebraic structures in which epimorphisms are precisely the surjective morphisms, this means that every section on an open subset extends to every larger open subset.

The importance of flabby sheaves in sheaf cohomology stems from their tight connection to injective sheaves.

Theorem 4.2.10. Let X be a topological space, and let $\mathcal{C}$ be an abelian category such that, for every open subset $U \subseteq X$ and every $A \in \text{Ob}(\mathcal{C})$, there exists a sheaf $\mathcal{A}_U \in \text{Ob}(\text{Sh}(X, \mathcal{C}))$ representing the evaluation functor at U in the sense that

$$\text{Hom}_{\text{Sh}(X, \mathcal{C})}(\mathcal{A}_U, \mathcal{F}) \cong \text{Hom}_{\mathcal{C}}(A, \mathcal{F}(U))$$

naturally in $\mathcal{F} \in \text{Ob}(\text{Sh}(X, \mathcal{C}))$. Assume moreover that for every inclusion $V \subseteq U$ the induced morphism

$$\mathcal{A}_V \rightarrow \mathcal{A}_U$$

is a monomorphism in $\text{Sh}(X, \mathcal{C})$. If $\mathcal{I} \in \text{Ob}(\text{Sh}(X, \mathcal{C}))$ is injective, then $\mathcal{I}$ is flabby (see [47, Ch. 6, §3, Corollary 6.74], [62]).

Proof. Let $V \subseteq U$ be open subsets of X . Then the restriction morphism

$$\text{res}_V^U : \mathcal{I}(U) \rightarrow \mathcal{I}(V)$$

is an epimorphism in $\mathcal{C}$. Let $A \in \text{Ob}(\mathcal{C})$. By assumption, the inclusion $V \subseteq U$ induces a monomorphism

$$\mathcal{A}_V \rightarrow \mathcal{A}_U$$

in $\text{Sh}(X, \mathcal{C})$. Since $\mathcal{I}$ is injective, applying $\text{Hom}_{\text{Sh}(X, \mathcal{C})}(-, \mathcal{I})$ gives a surjective map

$$\text{Hom}_{\text{Sh}(X, \mathcal{C})}(\mathcal{A}_U, \mathcal{I}) \rightarrow \text{Hom}_{\text{Sh}(X, \mathcal{C})}(\mathcal{A}_V, \mathcal{I}).$$

Using the representing property of $\mathcal{A}_U$ and $\mathcal{A}_V$, this map identifies naturally with

$$\text{Hom}_{\mathcal{C}}(A, \mathcal{I}(U)) \rightarrow \text{Hom}_{\mathcal{C}}(A, \mathcal{I}(V))$$

induced by the restriction morphism

$$\text{res}_V^U : \mathcal{I}(U) \rightarrow \mathcal{I}(V).$$

Hence, for every $A \in \text{Ob}(\mathcal{C})$, the map

$$\text{Hom}_{\mathcal{C}}(A, \mathcal{I}(U)) \rightarrow \text{Hom}_{\mathcal{C}}(A, \mathcal{I}(V))$$

is surjective. Taking $A = \mathcal{I}(V)$, the identity morphism $\text{id}_{\mathcal{I}(V)}$ in $\mathcal{C}$ lifts to a morphism

$$s : \mathcal{I}(V) \rightarrow \mathcal{I}(U)$$

such that

$$\text{res}_V^U \circ s = \text{id}_{\mathcal{I}(V)}.$$

Thus res_V^U is a split epimorphism, hence an epimorphism. Since this holds for every inclusion $V \subseteq U$, the sheaf $\mathcal{I}$ is flabby (see [47, Ch. 6, §3, Corollary 6.74], [62], [20, Ch. 2, §4], [9, Ch. 1, §6-§7]). $\square$

Remark 4.2.11. The sheaf $\mathcal{A}_U$ is the representing object for the functor

$$\begin{aligned} \mathrm{ev}_{U,A} : \mathrm{Sh}(X, \mathcal{C}) &\rightarrow \mathrm{Set} \\ \mathcal{F} &\mapsto \mathrm{Hom}_{\mathcal{C}}(A, \mathcal{F}(U)). \end{aligned}$$

The statement that

$$\mathrm{Hom}_{\mathrm{Sh}(X, \mathcal{C})}(\mathcal{A}_U, \mathcal{F}) \cong \mathrm{Hom}_{\mathcal{C}}(A, \mathcal{F}(U))$$

is a representability statement, and the resulting bijection is precisely a Yoneda-type correspondence. This property will be from now on referred to as **representation property**.

The preceding theorem is the abstract form of the statement that every injective sheaf is flabby. The representation property translates restriction morphisms of sections into morphisms between representing objects, allowing injectivity to be applied directly. This allows one to exploit the concrete extension property of flabby sheaves while retaining the homological machinery of injective resolutions. Moreover, flabbiness has strong consequences for exactness properties of the global section functor.

Lemma 4.2.12. Consider a topological space X , and let

$$0 \rightarrow \mathcal{E} \xrightarrow{\iota} \mathcal{F} \xrightarrow{\varphi} \mathcal{G} \rightarrow 0$$

be a short exact sequence in $\mathrm{Sh}(X, \mathcal{C})$, where $\mathcal{C}$ is a small abelian category. If $\mathcal{E}$ is flabby, then the induced sequence of global sections

$$0 \rightarrow \Gamma(X, \mathcal{E}) \rightarrow \Gamma(X, \mathcal{F}) \rightarrow \Gamma(X, \mathcal{G}) \rightarrow 0$$

is exact (see [47, Ch. 6, §3, Prop. 6.75]).

Proof. In general, categories do not need to allow the notion of elements. However, since $\mathcal{C}$ is a small abelian category, the Mitchell Embedding Theorem (see Appendix C) provides a fully faithful exact embedding $\mathcal{C} \hookrightarrow R\text{-Mod}$ for some ring R . Thus one can use the notion of elements.

Left-exactness follows from the left-exactness of the global section functor (see Theorem 2.2.33). It remains to prove that

$$\begin{aligned}\varphi_X : \Gamma(X, \mathcal{F}) &\rightarrow \Gamma(X, \mathcal{G}) \\ f &\mapsto \varphi(f)\end{aligned}$$

is an epimorphism. Let $g \in \mathcal{G}(X) = \Gamma(X, \mathcal{G})$, and define

$$\chi = \left\{ (U, f) : U \subseteq X \text{ open}, f \in \mathcal{F}(U), \varphi(f) = g|_U \right\},$$

equipped with a partial order $(U_1, f_1) \leq (U_2, f_2)$ if $U_1 \subseteq U_2$ and $f_2|_{U_1} = f_1$. Every chain $(U_\lambda, f_\lambda)_\lambda$ in χ has an upper bound: the sections f_λ agree under restriction and therefore glue to a section on $\bigcup_\lambda U_\lambda$. Hence Lemma A.1.2 applies, by which there exists a maximal element (U_0, f_0) .

If $U_0 = X$, then f_0 is a global section and φ_X is epimorphic.

Assume that $U_0 \neq X$. Choose $x \in X$ with $x \notin U_0$.

In the category $R - \text{Mod}$, since

$$\varphi_x : \mathcal{F}_x \rightarrow \mathcal{G}_x$$

is surjective, one can choose a germ $\xi_x \in \mathcal{F}_x$ with

$$\varphi_x(\xi_x) = g_x.$$

By definition of stalks, after shrinking to a neighbourhood V of x , the germ ξ_x is represented by a section $f' \in \mathcal{F}(V)$ satisfying

$$\varphi(f') = g|_V.$$

Regard $\iota : \mathcal{E} \rightarrow \mathcal{F}$ as inclusion. Then

$$\varphi(f_0 - f') = g|_{U_0 \cap V} - g|_{U_0 \cap V} = 0.$$

Since

$$0 \rightarrow \mathcal{E} \xrightarrow{\iota} \mathcal{F} \xrightarrow{\varphi} \mathcal{G}$$

is exact, $\ker(\varphi) = \operatorname{im}(\iota) = \mathcal{E}$. Then

$$f_0 - f' \in \mathcal{E}(U_0 \cap V),$$

so that the flabbiness of $\mathcal{E}$ provides a global section $f'' \in \mathcal{E}(X)$ extending $f_0 - f'$. Hence

$$f_0 = f' + f''|_{U_0 \cap V} \in \mathcal{F}(U_0 \cap V).$$

Therefore these sections can be glued: there exists a section $\tilde{f}_0 \in \mathcal{F}(U_0 \cup V)$ with

$$\tilde{f}_0|_{U_0} = f_0$$

and

$$\tilde{f}_0|_V = f' + f''|_V.$$

However,

$$\varphi(\tilde{f}_0) = g|_{U_0 \cup V},$$

contradicting the maximality of (U_0, f_0) . This shows that $U_0 = X$. Therefore, the induced morphism φ_X is an epimorphism, and the induced sequence of global sections is exact (see [47, Ch. 6, §3, Prop. 6.75]). $\square$

Flabby sheaves also permit several constructions, resulting in a flabby sheaf.

Lemma 4.2.13. Let $(\mathcal{F}_i)_{i \in I} \in \operatorname{Ob}(\operatorname{Sh}(X, \mathcal{C}))$ be a family of flabby sheaves on a topological space X , and let $\mathcal{C}$ be an $AB4^*$ category. Then the product sheaf

$$\prod_{i \in I} \mathcal{F}_i$$

is flabby (compare [5, Scholium 3.9]).

Proof. Let $V \subseteq U$ be open subsets of X . Since products of sheaves with values in a complete category are computed sectionwise, the restriction morphism is

$$\prod_{i \in I} \mathcal{F}_i(U) \rightarrow \prod_{i \in I} \mathcal{F}_i(V)$$

For every $i \in I$, the morphism

$$\mathcal{F}_i(U) \rightarrow \mathcal{F}_i(V)$$

is an epimorphism, as every $\mathcal{F}_i$ is flabby. Since products are exact in $AB4^*$ categories, they preserve cokernels, and therefore epimorphisms. Hence the morphism

$$\prod_{i \in I} \mathcal{F}_i(U) \rightarrow \prod_{i \in I} \mathcal{F}_i(V)$$

is an epimorphism. Therefore the product sheaf $\prod_{i \in I} \mathcal{F}_i$ is flabby. $\square$

Whenever the required products exist in $\mathcal{C}$, products of $\mathcal{C}$ -valued sheaves are computed sectionwise, since products commute with the limits appearing in the sheaf condition. An arbitrary abelian category need not admit infinite products; this existence is guaranteed, for example, under the axiom $AB3^*$. Not only are products of flabby sheaves in $AB4^*$ categories flabby, they also arise under certain conditions in short exact sequences of sheaves, as the following lemma shows.

Lemma 4.2.14. Consider a topological space X , an abelian category $\mathcal{C}$, and let

$$0 \rightarrow \mathcal{E} \rightarrow \mathcal{F} \rightarrow \mathcal{G} \rightarrow 0$$

be a short exact sequence of sheaves in $\text{Sh}(X, \mathcal{C})$. If $\mathcal{E}$ and $\mathcal{F}$ are flabby, then $\mathcal{G}$ is flabby (see [47, Ch. 6, §3, Prop. 6.75]).

Proof. Let $U \subseteq V \subseteq X$ be open subsets, and consider the commutative diagram

$$\begin{array}{ccc} \mathcal{F}(V) & \xrightarrow{\varphi_V} & \mathcal{G}(V) \\ \downarrow \rho & & \downarrow \rho'' \\ \mathcal{F}(U) & \xrightarrow{\varphi_U} & \mathcal{G}(U), \end{array}$$

where ρ and ρ'' are restriction morphisms. Since $\mathcal{F}$ is flabby, ρ is an epimorphism. Restricting the short exact sequence to U gives a short exact sequence

$$0 \rightarrow \mathcal{E}|_U \rightarrow \mathcal{F}|_U \rightarrow \mathcal{G}|_U \rightarrow 0.$$

Since $\mathcal{E}|_U$ is flabby, Lemma 4.2.12 implies that

$$\varphi_U : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$$

is an epimorphism. Moreover, since $\mathcal{F}$ is flabby, the restriction

$$\rho : \mathcal{F}(V) \rightarrow \mathcal{F}(U)$$

is an epimorphism. By commutativity,

$$\varphi_U \circ \rho = \rho'' \circ \varphi_V.$$

Therefore $\rho'' \circ \varphi_V$ is an epimorphism. It follows that ρ'' itself is an epimorphism. Hence

$$\mathcal{G}(V) \rightarrow \mathcal{G}(U)$$

is epimorphic. Hence $\mathcal{G}$ is flabby (see [47, Ch. 6, §3, Prop. 6.75]). $\square$

The preceding results show that flabby sheaves are stable under certain constructions appearing in resolutions. In particular, injective sheaves are flabby, making any injective resolution a flabby resolution.

Definition 4.2.15 (Flabby resolution). Let $\mathcal{F} \in \text{Ob}(\text{Sh}(X, \mathcal{C}))$. A **flabby resolution** of $\mathcal{F}$ is an exact sequence

$$0 \rightarrow \mathcal{F} \xrightarrow{\varepsilon} \mathcal{G}^0 \xrightarrow{\delta^0} \mathcal{G}^1 \rightarrow \dots$$

such that each sheaf $\mathcal{G}^n$ is flabby (see [47, Ch. 6, §3, p. 379]).

As injective sheaves are acyclic, the question arises if acyclicity carries over to flabby sheaves.

Theorem 4.2.16. Let X be a topological space. Assume that $\text{Sh}(X, \mathcal{C})$ is a Grothendieck category, and let $\mathcal{E} : \text{Open}(X)^{\text{op}} \rightarrow \mathcal{C}$ be a flabby sheaf such that the representation property 4.2.11 is satisfied. Then

$$\mathcal{H}^n(X, \mathcal{E}) = 0$$

for all $n > 0$ (see [47, Ch. 6, §3, Prop. 6.75]).

Proof. Let $\mathcal{E}$ be flabby. Since there are enough injectives, there exists an exact sequence

$$0 \rightarrow \mathcal{E} \rightarrow \mathcal{F} \rightarrow \mathcal{G} \rightarrow 0,$$

where $\mathcal{F}$ is injective. By Theorem 4.2.10, injective sheaves are flabby, hence $\mathcal{F}$ is flabby, and by Lemma 4.2.14, $\mathcal{G}$ is flabby.

The proof that $\mathcal{H}^n(X, \mathcal{E}) = 0$ is by induction on $n \in \mathbb{N}$.

If $n = 1$, consider the following part of the long exact cohomology sequence:

$$\dots \rightarrow \mathcal{H}^0(X, \mathcal{F}) \rightarrow \mathcal{H}^0(X, \mathcal{G}) \rightarrow \mathcal{H}^1(X, \mathcal{E}) \rightarrow \mathcal{H}^1(X, \mathcal{F}) \rightarrow \dots$$

By Lemma 4.2.7, it holds that $\mathcal{H}^1(X, \mathcal{F}) = 0$. Hence,

$$\mathcal{H}^1(X, \mathcal{E}) = \operatorname{coker} (\Gamma(X, \mathcal{F}) \rightarrow \Gamma(X, \mathcal{G}))$$

By Lemma 4.2.12,

$$\operatorname{coker} (\Gamma(X, \mathcal{F}) \rightarrow \Gamma(X, \mathcal{G})) = 0,$$

hence $\mathcal{H}^1(X, \mathcal{E}) = 0$.

Assume that $\mathcal{H}^n(X, \mathcal{D}) = 0$ for all flabby sheaves $\mathcal{D}$ and a fixed $n \in \mathbb{N}$.

Consider

$$\dots \rightarrow \mathcal{H}^n(X, \mathcal{F}) \rightarrow \mathcal{H}^n(X, \mathcal{G}) \rightarrow \mathcal{H}^{n+1}(X, \mathcal{E}) \rightarrow \mathcal{H}^{n+1}(X, \mathcal{F}) \rightarrow \dots$$

Since $\mathcal{G}$ is flabby, the induction hypothesis applies, making $\mathcal{H}^n(X, \mathcal{G}) = 0$. Again, $\mathcal{H}^{n+1}(X, \mathcal{F}) = 0$ by the injectivity of $\mathcal{F}$. The exactness of the sequence gives $\mathcal{H}^{n+1}(X, \mathcal{E}) = 0$ (see [47, Ch. 6, §3, Prop. 6.75]). $\square$

The acyclicity of flabby sheaves allows one to compute sheaf cohomology from suitable flabby resolutions. To compare sheaf cohomology with singular cohomology, one therefore studies sheaves built from singular cochains.

Singular cohomology gives computable topological information, which can be transferred into sheaf cohomology under sufficient assumptions. As the following the-

orem shows, singular cohomology and sheaf cohomology coincide under the right assumptions.

Theorem 4.2.17 (Comparison theorem for singular cohomology and sheaf cohomology). Let X be a hereditarily paracompact and locally contractible topological space, and let $G^{\text{cst}} \in \text{Ob}(\text{Sh}(X, \text{Ab}))$ be a constant sheaf associated to an abelian group G . Then the sheaf cohomology of G^{cst} agrees with singular cohomology, that is,

$$\mathcal{H}^n(X, G^{\text{cst}}) \cong H_{\text{sing}}^n(X, G)$$

for all $n \geq 0$ (see [50]).

Proof. The proof can be found in Appendix E. □

The comparison theorem identifies sheaf cohomology with singular cohomology under suitable topological assumptions. Since singular cohomology is computable via explicit cochain complexes, this provides concrete access to sheaf cohomology in many situations. However, singular cohomology is fundamentally global in nature, and depends on singular simplices in the underlying topological space. In contrast, sheaves encode local data together with compatibility on overlaps. This suggests constructing a cohomological theory directly from open covers and their intersections. That local viewpoint is particularly well-suited for studying monodromy phenomena and obstructions to gluing local branches of holomorphic functions.

4.3 Čech Cohomology

Čech cohomology provides a concrete model for computing sheaf cohomology under suitable hypotheses. While sheaf cohomology is defined abstractly via derived functors, Čech cohomology is built from explicit local data on open covers. Under suitable hypotheses, both theories agree, as shown by the Leray Theorem 4.3.26.

In this section, I^{n+1} denotes the set of tuples $(i_0, \dots, i_n)$ with values in I , and for $\alpha := (i_0, \dots, i_n) \in I^{n+1}$,

$$U_\alpha := U_{i_0} \cap \dots \cap U_{i_n}.$$

If α is a multiindex $(i_0, \dots, i_n)$, then

$$\widehat{\alpha}_k := (i_0, \dots, i_{k-1}, i_{k+1}, \dots, i_n).$$

A multiindex $\alpha = (i_0, \dots, i_n)$ is called **degenerate** if at least two indices coincide, such as $(i, i) \in I^2$ or $(i, j, i) \in I^3$. The corresponding intersection is called degenerate intersection by abuse of terminology.

Definition 4.3.1 (Čech cochain). Let $\mathcal{C}$ be a complete abelian category, let $\mathcal{F}, \mathcal{G} \in \text{Ob}(\text{Sh}(X, \mathcal{C}))$, and let $\mathcal{U} = \{U_i\}_{i \in I}$ be an open cover of X . The n -th **Čech cochain object** is

$$\check{C}^n(\mathcal{U}, \mathcal{F}) := \prod_{\alpha \in I^{n+2}} \mathcal{F}(U_\alpha)$$

for $n \in \mathbb{N}_0$. The collection of all cochains of $\mathcal{F}$ with respect to the cover $\mathcal{U}$ is denoted by $\check{C}^\bullet(\mathcal{U}, \mathcal{F})$ (see [47, Ch. 6, §3, pp. 384 ff.]).

A morphism of sheaves $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ induces a morphism of Čech cochain complexes

$$\check{C}^m(\mathcal{U}, \varphi) : \check{C}^m(\mathcal{U}, \mathcal{F}) \rightarrow \check{C}^m(\mathcal{U}, \mathcal{G})$$

defined as the product morphism induced by the components

$$\varphi_{U_\alpha} : \mathcal{F}(U_\alpha) \rightarrow \mathcal{G}(U_\alpha).$$

Equivalently, in element notation,

$$(c_{i_0 \dots i_n}) \rightarrow (\varphi(c_{i_0 \dots i_n}))$$

(see [47, Ch. 6, §3, pp. 384 ff.]).

An n -th Čech cochain of $\mathcal{F}$ with respect to $\mathcal{U}$ assigns sections to $(n+1)$ -fold intersections of the cover. At this stage, no compatibility conditions are imposed. The n -th Čech cochain depends on the cover $\mathcal{U}$.

Definition 4.3.2 (Čech!- differential). Let $\check{C}^\bullet(\mathcal{U}, \mathcal{F})$ be the collection of all Čech cochains of $\mathcal{F}$ with respect to a cover $\mathcal{U}$. For every multiindex $\alpha \in I^{n+1}$, define the α -component of

$$\delta^n : \check{C}^n(\mathcal{U}, \mathcal{F}) \rightarrow \check{C}^{n+1}(\mathcal{U}, \mathcal{F})$$

by

$$\mathrm{pr}_\alpha \circ \check{\delta}^n := \sum_{k=0}^{n+1} (-1)^k \mathrm{res}_{U_\alpha}^{U_{\widehat{\alpha_k}}} \circ \mathrm{pr}_{\widehat{\alpha_k}},$$

where pr_β denotes the projection onto the factor $\mathcal{F}(U_\beta)$. By the universal property of the product, these components determine a unique morphism $\check{\delta}^n$. In a concrete abelian category this reads

$$\begin{aligned} \check{\delta}^n : \check{C}^n(\mathcal{U}, \mathcal{F}) &\rightarrow \check{C}^{n+1}(\mathcal{U}, \mathcal{F}) \\ (\check{\delta}^n c)_\alpha &= \sum_{k=0}^{n+1} (-1)^k c_{\widehat{\alpha_k}}|_{U_\alpha}, \end{aligned}$$

with respect to a fixed multiindex $\alpha \in I^{n+2}$ and $c \in \check{C}^n(\mathcal{U}, \mathcal{F})$. These morphisms are called the **Čech differential** (see [47, Ch. 6, §3, pp. 384 ff.]).

The n -th (alternating) **Čech coboundary** is defined as

$$\check{B}^n := \mathrm{im}(\check{\delta}^{n-1}),$$

for $n \in \mathbb{N}$ and $\check{B}^0 := 0$, and the **Čech n -cocycle** is defined as

$$\check{Z}^n := \ker(\check{\delta}^n)$$

for $n \in \mathbb{N}_0$ (see [47, Ch. 6, §3, pp. 384 ff.]).

Remark 4.3.3. Note that $\check{\delta}^{n+1} \circ \check{\delta}^n = 0$: For $\alpha = (i_0, \dots, i_{n+2}) \in I^{n+3}$ and $c \in \check{C}^n(\mathcal{U}, \mathcal{F})$, one has

$$\begin{aligned} (\check{\delta}^{n+1} \circ \check{\delta}^n)(c)_\alpha &= \left(\check{\delta}^{n+1}(\check{\delta}^n(c)) \right)_\alpha \\ &= \sum_{j=0}^{n+2} (-1)^j \check{\delta}^n(c)_{\widehat{\alpha_j}}|_{U_\alpha} \\ &= \sum_{j=0}^{n+2} (-1)^j \left(\sum_{k < j} (-1)^k c_{\widehat{\alpha_{jk}}} |_{U_\alpha} + \sum_{k > j} (-1)^{k-1} c_{\widehat{\alpha_{jk}}} |_{U_\alpha} \right). \end{aligned}$$

Consider $j = p$ and $k = q$. If $p < q$, the sign becomes $(-1)^p(-1)^q = (-1)^{p+q}$.

If $p > q$, the sign becomes $(-1)^p(-1)^{q-1} = -(-1)^{p+q}$. Since every pair of indices

appears once in each setting, the terms cancel out. Thus,

$$(\check{\delta}^{n+1} \circ \check{\delta}^n)(c)_\alpha = 0$$

(see [47, Ch. 6, §3, pp. 384 ff.]).

The geometric intuition here is that one takes the sum over faces of faces twice, but with opposite orientation. To define Čech cohomology, one needs kernels, images, and quotients. This motivates working in abelian categories.

To relate local data to global sections, one considers the augmented Čech complex.

Definition 4.3.4 ((Augmented) Čech complex). A collection of Čech cochains $\check{C}^\bullet(\mathcal{U}, \mathcal{F})$ of $\mathcal{F} \in \text{Ob}(\text{Sh}(X, \mathcal{C}))$ with respect to the cover $\mathcal{U}$, together with the collection of Čech differentials $\check{\delta}^\bullet$ is called **Čech cochain complex**, denoted by $(\check{C}^\bullet(\mathcal{U}, \mathcal{F}), \check{\delta}^\bullet)$.

The **augmented Čech cochain complex** is the complex

$$0 \rightarrow \mathcal{F}(X) \xrightarrow{\varepsilon} \check{C}^0(\mathcal{U}, \mathcal{F}) \xrightarrow{\check{\delta}^0} \check{C}^1(\mathcal{U}, \mathcal{F}) \xrightarrow{\check{\delta}^1} \dots,$$

where

$$\varepsilon(s) := (s|_{U_i})_{i \in I}.$$

The Čech complex $\check{C}^\bullet(\mathcal{U}, \mathcal{F})$ can thus be viewed as the truncation of the augmented complex obtained by omitting the global sections. From this perspective, Čech cohomology measures the failure of the augmented complex to be exact: the sheaf condition states that the augmented Čech complex is exact at $\mathcal{F}(X)$ and $\check{C}^0(\mathcal{U}, \mathcal{F})$, while higher Čech cohomology measures higher-order failures of compatibility and gluing.

In contrast to the Čech cochains, Čech cochain complexes allow to study how structures evolve with respect to the underlying intersections. This is formalised by the Čech cohomology.

Definition 4.3.5 (Čech cohomology). Let $(\check{C}^\bullet(\mathcal{U}, \mathcal{F}), \check{\delta}^\bullet)$ be a Čech cochain com-

plex. The **Čech cohomology** is defined as

$$\begin{aligned}\check{H}^n(\mathcal{U}, \mathcal{F}) &:= \check{Z}^n(\mathcal{U}, \mathcal{F}) / \check{B}^n(\mathcal{U}, \mathcal{F}) \\ \check{H}^n(X, \mathcal{F}) &:= \operatorname{colim}_{\longrightarrow \mathcal{U}} \check{H}^n(\mathcal{U}, \mathcal{F})\end{aligned}$$

for $n \in \mathbb{N}_0$ (see [47, Ch. 6, §3, pp. 384 ff.]).

Let $\mathcal{V} = \{V_j\}_{j \in J}$ be a refinement of $\mathcal{U} = \{U_i\}_{i \in I}$, and choose a refinement map

$$\rho : J \rightarrow I$$

such that $V_j \subseteq U_{\rho(j)}$ for every $j \in J$. Restriction then defines a cochain morphism

$$\tau^* : \check{C}^\bullet(\mathcal{U}, \mathcal{F}) \rightarrow \check{C}^\bullet(\mathcal{V}, \mathcal{F}).$$

Different choices of refinement map induce cochain homotopic morphisms and therefore the same morphism on Čech cohomology. The resulting system over the filtered category of open covers and refinements defines

$$\check{H}^n(X, \mathcal{F}) := \operatorname{colim}_{\longrightarrow \mathcal{U}} \check{H}^n(\mathcal{U}, \mathcal{F}).$$

The open covers of X , ordered by refinement, form a directed set. Hence $\check{H}^n(\mathcal{U}, \mathcal{F})$ form a directed system, and $\check{H}^n(X, \mathcal{F})$ is its directed colimit. Čech cohomology measures the failure of local data, encoded by cochains, to satisfy global compatibility conditions. In degree zero, it recovers global sections, while higher degrees detect obstructions to gluing.

The directed colimit over all covers is taken in the target category. In $AB5$ categories, filtered colimits are exact, ensuring good formal properties of Čech cohomology.

Note that $\check{H}^n(X, \mathcal{F})$ is the directed colimit over all covers of the underlying topological space X . If one considers a short exact sequence of sheaves

$$0 \rightarrow \mathcal{F}_1 \xrightarrow{\alpha} \mathcal{F}_2 \xrightarrow{\beta} \mathcal{F}_3 \rightarrow 0,$$

any local section in $\mathcal{F}_3$ comes from a local section in $\mathcal{F}_2$. However, this section in $\mathcal{F}_2$ is not necessarily a global section, since $\Gamma(X, -)$ is generally not right-exact.

Čech cohomology provides a concrete cohomological theory built from local data. Passing to the directed colimit over all covers gives long exact sequences in Čech cohomology:

Remark 4.3.6. Let $\mathcal{C}$ be a Grothendieck abelian category. Then the right derived functors

$$\mathcal{H}^n(X, -) := R^n\Gamma(X, -)$$

form a categorical ∂ -functor. Hence every short exact sequence

$$0 \rightarrow \mathcal{F}_1 \rightarrow \mathcal{F}_2 \rightarrow \mathcal{F}_3 \rightarrow 0$$

induces a canonical connecting morphism

$$\partial^n : \mathcal{H}^n(X, \mathcal{F}_3) \rightarrow \mathcal{H}^{n+1}(X, \mathcal{F}_1).$$

Whenever the canonical morphism

$$\check{H}^n(X, \mathcal{F}_i) \rightarrow \mathcal{H}^n(X, \mathcal{F}_i)$$

is an isomorphism, the derived connecting morphism induces a connecting morphism on Čech cohomology.

Construction 4.3.7 (Long sequence). Let R be a ring, and consider a short exact sequence of sheaves of R -modules on a topological space X ,

$$0 \rightarrow \mathcal{F}_1 \xrightarrow{\iota} \mathcal{F}_2 \xrightarrow{\pi} \mathcal{F}_3 \rightarrow 0.$$

Then there exist natural connecting morphisms

$$\check{\partial}^n : \check{H}^n(X, \mathcal{F}_3) \rightarrow \check{H}^{n+1}(X, \mathcal{F}_1).$$

Let

$$\xi \in \check{H}^n(X, \mathcal{F}_3)$$

be represented by a cocycle

$$h \in \check{Z}^n(\mathcal{U}, \mathcal{F}_3)$$

with respect to an open cover $\mathcal{U}$ of X . Since

$$\pi : \mathcal{F}_2 \rightarrow \mathcal{F}_3$$

is an epimorphism of sheaves of R -modules, sections of $\mathcal{F}_3$ lift locally to sections of $\mathcal{F}_2$. Consequently, after passing to a sufficiently fine refinement $\mathcal{V}$ of $\mathcal{U}$, the image

$$h_{\mathcal{V}} \in \check{Z}^n(\mathcal{V}, \mathcal{F}_3)$$

of h admits a cochain lift

$$g \in \check{C}^n(\mathcal{V}, \mathcal{F}_2)$$

such that

$$\pi(g) = h_{\mathcal{V}}.$$

Applying the Čech differential gives

$$\pi(\check{\delta}(g)) = \check{\delta}(\pi(g)) = \check{\delta}(h_{\mathcal{V}}) = 0.$$

Thus every component of $\check{\delta}(g)$ lies in the kernel of π . Since

$$\ker(\pi) = \text{im}(\iota),$$

there exists a unique cochain

$$f \in \check{C}^{n+1}(\mathcal{V}, \mathcal{F}_1)$$

satisfying

$$\iota(f) = \check{\delta}(g).$$

Moreover,

$$\iota(\check{\delta}(f)) = \check{\delta}(\iota(f)) = \check{\delta}^2(g) = 0.$$

Since ι is a monomorphism,

$$\check{\delta}(f) = 0.$$

Hence f is a Čech cocycle. Define

$$\check{\partial}^n(\xi) := [f] \in \check{H}^{n+1}(X, \mathcal{F}_1),$$

where $[f]$ denotes the image of the class of f in the directed colimit over all open covers.

This class is independent of the choices made. Indeed, two refinements admit a common refinement. After passing to such a refinement, two choices of lift differ by a cochain with values in $\mathcal{F}_1$, and the corresponding cocycles differ by a Čech coboundary. Similarly, replacing h by a cohomologous cocycle changes f only by a coboundary after passing to a sufficiently fine common refinement. Thus $\check{\partial}^n$ is well-defined. The construction is additive, natural in short exact sequences, and compatible with refinement (see [52, Ch. 12, §13, Lemma 12.13.12, Ch. 20, §10, Lemma 20.10.2]).

For a fixed open cover $\mathcal{U}$, the sequence

$$0 \rightarrow \check{C}^\bullet(\mathcal{U}, \mathcal{F}_1) \rightarrow \check{C}^\bullet(\mathcal{U}, \mathcal{F}_2) \rightarrow \check{C}^\bullet(\mathcal{U}, \mathcal{F}_3) \rightarrow 0$$

need not be exact, since sections of $\mathcal{F}_3$ need not lift over the original intersections of $\mathcal{U}$. Local liftability permits such lifts only after passing to a sufficiently fine refinement. Therefore the connecting morphism is naturally defined on the directed colimit over all open covers rather than on the cohomology associated to a fixed cover.

The connecting morphisms fit into a natural long exact sequence

$$\begin{aligned} 0 \rightarrow \check{H}^0(X, \mathcal{F}_1) \rightarrow \check{H}^0(X, \mathcal{F}_2) \rightarrow \check{H}^0(X, \mathcal{F}_3) \\ \xrightarrow{\check{\partial}^0} \check{H}^1(X, \mathcal{F}_1) \rightarrow \check{H}^1(X, \mathcal{F}_2) \rightarrow \check{H}^1(X, \mathcal{F}_3) \xrightarrow{\check{\partial}^1} \dots \end{aligned}$$

Here filtered colimits are exact because $R - \text{Mod}$ is an $AB5$ category.

Remark 4.3.8. The preceding construction uses the concrete local lifting property of epimorphisms of sheaves of modules. More generally, let $\mathcal{C}$ be a Grothendieck abelian category. Whenever the canonical comparison morphisms

$$\check{H}^n(X, \mathcal{F}) \rightarrow \mathcal{H}^n(X, \mathcal{F})$$

are isomorphisms, the canonical connecting morphisms for derived sheaf cohomology transport to connecting morphisms for Čech cohomology.

Definition 4.3.9 (Čech nerve). Let X be a topological space and let $\mathcal{U} = \{U_i\}_{i \in I}$ be an open cover of X . The **Čech nerve** of the cover $\mathcal{U}$ is the simplicial space

$$\check{C}(\mathcal{U})_\bullet : \Delta^{\text{op}} \rightarrow \text{Top}$$

defined as follows: For each $n \in \mathbb{N}_0$

$$\check{C}(\mathcal{U})_n := \coprod_{\alpha \in I^{n+1}} U_\alpha.$$

The **face maps**

$$\check{d}_m : \check{C}(\mathcal{U})_n \rightarrow \check{C}(\mathcal{U})_{n-1}$$

are induced by omitting the m -th index

$$U_\alpha \rightarrow U_{\widehat{\alpha_m}},$$

for $\alpha \in I^{n+1}$ and the **degeneracy maps**

$$\check{s}_m : \check{C}(\mathcal{U})_n \rightarrow \check{C}(\mathcal{U})_{n+1}$$

are induced by repeating the m -th index (see [47, Ch. 1, §2, Ex. 1.3. (x)]).

The Čech nerve is also called the *Čech nerve diagram*, emphasising its functorial nature. It provides a uniform framework for encoding local data with their compatibility conditions.

Remark 4.3.10. The definition of the Čech nerve is equivalent to the following:

If

$$\text{pr} : \coprod_{i \in I} U_i \rightarrow X$$

is the canonical morphism induced by the inclusions $U_i \hookrightarrow X$, then there is a canonical isomorphism

$$\check{C}(\mathcal{U})_n \cong \underbrace{\left(\coprod_{i \in I} U_i \right) \times_X \cdots \times_X \left(\coprod_{i \in I} U_i \right)}_{n+1 \text{ times}}.$$

Indeed, the fibre product decomposes canonically as

$$\underbrace{\left(\prod_{i \in I} U_i \right) \times_X \cdots \times_X \left(\prod_{i \in I} U_i \right)}_{n+1 \text{ factors}} \cong \coprod_{(i_0, \dots, i_n) \in I^{n+1}} U_{i_0} \cap \cdots \cap U_{i_n}.$$

Thus $\check{C}(\mathcal{U})_\bullet$ is the simplicial object associated to the morphism

$$\coprod_{i \in I} U_i \rightarrow X$$

(see [47, Ch. 6, §3, p. 384]).

Associated to the cover $\mathcal{U}$ is also its **abstract nerve** $\check{N}(\mathcal{U})$. Its vertices are elements of I , and a finite set $\{i_0, \dots, i_n\} \subseteq I$ spans an n -simplex precisely when

$$U_{i_0} \cap \cdots \cap U_{i_n} \neq \emptyset.$$

This abstract simplicial complex should be distinguished from the Čech nerve $\check{C}(\mathcal{U})_\bullet$, which is a simplicial topological space whose n -th level is a coproduct of the actual intersections.

If the cover has no non-empty triple intersections, then the abstract nerve $\check{N}(\mathcal{U})$ is one-dimensional and may therefore be regarded as a graph. For constant coefficients, or more generally under hypotheses identifying the coefficient objects and restriction morphisms on the non-empty intersections, normalised Čech 1-cochains may be interpreted as labellings of the oriented edges of this graph. Since the abstract nerve has no non-degenerate 2-simplices, the degree-two term of its normalised cochain complex vanishes. Thus every normalised 1-cochain is a cocycle. The first Čech cohomology is therefore obtained by quotienting the group of edge labellings by the coboundaries arising from vertex labellings. For constant coefficients, this quotient is determined by the cycle structure of the nerve graph.

Definition 4.3.11 (Cycle space). Let $G = (V, E)$ be a finite graph. Its **cycle space over $\mathbb{F}_2$** is the vector space consisting of all subsets of edges for which every vertex is incident with an even number of selected edges (see [11])

The dimension of the cycle space equals

$$|E| - |V| + c,$$

where c is the number of connected components. When a finite graph is regarded as a topological space, its first Betti number equals the dimension of its cycle space over $\mathbb{F}_2$.

Definition 4.3.12 (Betti number). Let X be a topological space and let $H_n^{\text{sing}}(X, \mathbb{Z})$ denote its singular homology group. The n -th **Betti number** of X is

$$\beta_n(X) := \text{rank } H_n^{\text{sing}}(X, \mathbb{Z})$$

(see [16, Ch. 5, §2, p. 159, Ch. 12, §1, p. 465]).

Intuitively, β_0 counts the components, β_1 measures independent one-dimensional holes, and β_2 measures independent two-dimensional homology classes.

The first Betti number of a graph corresponds to the number of independent cycles, or in terms of graph theory, the cyclomatic number

$$\beta_1(G) = |E| - |V| + c,$$

where c denotes the number of connected components.

The preceding discussion shows that the abstract nerve of a cover records the combinatorial structure of non-empty intersections of a cover. However, the nerve does not only serve as a combinatorial object. The face and degeneracy morphisms of the Čech nerve induce, through the restriction morphisms of $\mathcal{F}$, an associated cosimplicial object of sections. Its n -th term is the product of the values of $\mathcal{F}$ on all $(n + 1)$ -fold intersections, and its structure morphisms encode the corresponding restriction morphisms.

Lemma 4.3.13. Let $\mathcal{C}$ be a complete category, let $\mathcal{U} = \{U_i\}_{i \in I}$ be an open cover of a topological space X , and let $\mathcal{F} : \text{Open}(X)^{\text{op}} \rightarrow \mathcal{C}$ be a sheaf. Define a cosimplicial object

$$\check{C}^\bullet(\mathcal{U}, \mathcal{F}) : \Delta \rightarrow \mathcal{C}$$

by

$$\check{C}^n(\mathcal{U}, \mathcal{F}) := \prod_{(i_0, \dots, i_n) \in I^{n+1}} \mathcal{F}(U_{i_0} \cap \dots \cap U_{i_n}),$$

with coface and codegeneracy morphisms induced by the inclusions and canonical identifications among the intersections $U_{i_0, \dots, i_n}$. Then the canonical morphism

$$\mathcal{F}(X) \rightarrow \lim_{\Delta} \check{C}^\bullet(\mathcal{U}, \mathcal{F})$$

is an isomorphism.

Proof. For every cosimplicial object $A^\bullet : \Delta \rightarrow \mathcal{C}$, its limit is canonically isomorphic to the equaliser

$$\text{eq}(A^0 \rightrightarrows A^1)$$

of two morphisms induced by the two maps $[0] \rightrightarrows [1]$ in Δ .

Here,

$$\check{C}^0(\mathcal{U}, \mathcal{F}) = \prod_{i \in I} \mathcal{F}(U_i)$$

and

$$\check{C}^1(\mathcal{U}, \mathcal{F}) = \prod_{(i,j) \in I^2} \mathcal{F}(U_i \cap U_j).$$

Let

$$d^0, d^1 : \prod_{i \in I} \mathcal{F}(U_i) \rightrightarrows \prod_{(i,j) \in I^2} \mathcal{F}(U_i \cap U_j)$$

denote the two coface morphisms. They are uniquely determined by

$$\text{pr}_{ij} \circ d^0 = \text{res}_{U_i \cap U_j}^{U_i} \circ \text{pr}_i$$

and

$$\text{pr}_{ij} \circ d^1 = \text{res}_{U_i \cap U_j}^{U_j} \circ \text{pr}_j,$$

where $\text{res}_W^V : \mathcal{F}(V) \rightarrow \mathcal{F}(W)$ denotes the restriction morphism associated to an inclusion $W \subseteq V$. Hence $\text{eq}(d^0, d^1)$ is precisely the matching object appearing in the equaliser formulation of the sheaf axiom. Therefore the canonical morphism

$$\mathcal{F}(X) \rightarrow \text{eq}\left(\prod_{i \in I} \mathcal{F}(U_i) \rightrightarrows \prod_{(i,j) \in I^2} \mathcal{F}(U_i \cap U_j)\right)$$

is an isomorphism. Consequently,

$$\mathcal{F}(X) \cong \lim_{\Delta} \check{C}^\bullet(\mathcal{U}, \mathcal{F}).$$

□

Thus the sheaf condition can be interpreted as the statement that global sections are the limit of compatible local data.

Under suitable acyclicity assumptions on a sheaf and an open cover, the sheaf cohomology of the space with coefficients in the sheaf can be computed from the combinatorics of an open cover via the Čech complex. The Nerve Theorem strengthens this connection substantially: under the usual hypotheses on a good cover, the geometric realisation of its Čech nerve is homotopy equivalent to the underlying space. Thus the nerve recovers not only its cohomology, but the entire homotopy type of the space.

4.3.1 The Nerve Theorem

The proof of the nerve theorem relies on the following proposition:

Proposition 4.3.14. Let $\mathcal{S}$ be a simplicial complex, and let $\mathfrak{X} = \{X_\sigma\}_{\sigma \in \mathcal{S}}$ and $\mathfrak{Y} = \{Y_\sigma\}_{\sigma \in \mathcal{S}}$ be complexes of spaces over $\mathcal{S}$. Let

$$f : \mathfrak{X} \rightarrow \mathfrak{Y}$$

be a morphism of complexes of spaces, induced by compatible maps

$$f_\sigma : X_\sigma \rightarrow Y_\sigma.$$

If every f_σ is a homotopy equivalence, then

$$|f| : |\mathfrak{X}| \rightarrow |\mathfrak{Y}|$$

is also a homotopy equivalence (see [26, Ch. 4, §. G, Prop. 4G.1]).

Proof. Applying the mapping cylinder construction to the compatible maps

$$f_\sigma : X_\sigma \rightarrow Y_\sigma$$

yields a complex of spaces $M(f)$ over the same base complex $\mathcal{S}$. Its realisation is naturally homeomorphic to the mapping cylinder of the induced map:

$$|M(f)| \cong M(|f|).$$

Since the mapping cylinder $M(|f|)$ deformation retracts onto its target end $|\mathcal{Y}|$, it remains to show that it also deformation retracts onto its source end $|\mathcal{X}|$. Let $M^n(|f|)$ denote the part of $M(|f|)$ lying over the n -skeleton $\mathcal{S}^{(n)}$ of $\mathcal{S}$.

Claim: For every $n \geq 0$, $M^n(|f|) \cup |\mathcal{X}|$ deformation retracts onto $M^{n-1}(|f|) \cup |\mathcal{X}|$, where $M^{-1}(|f|) := \emptyset$.

It suffices to establish the claim when the base is a single n -simplex. Indeed, the deformation retractions over distinct n -simplices agree on their common faces, since they restrict to the identity on $M^{n-1}(|f|) \cup |\mathcal{X}|$. Therefore they glue to a deformation retraction over the whole n -skeleton.

In this case, the corresponding part of f is a morphism between two sequences

$$(X_0 \rightarrow X_1 \rightarrow \cdots \rightarrow X_n) \rightarrow (Y_0 \rightarrow Y_1 \rightarrow \cdots \rightarrow Y_n).$$

For $n = 0$, the claim reduces to the fact that $M(f_0)$ deformation retracts onto X_0 . This follows from Corollary A.4.56, since f_0 is a homotopy equivalence and the pair $(M(f_0), X_0)$ has the homotopy extension property.

Consider the inclusion

$$j : M^{n-1}(|f|) \cup |\mathcal{X}| \hookrightarrow M(|f|).$$

The pair

$$(M(|f|), M^{n-1}(|f|) \cup |\mathcal{X}|)$$

has the homotopy extension property by Example A.4.55, since $M^{n-1}(|f|) \cup |\mathcal{X}|$ admits a mapping cylinder neighbourhood in $M(|f|)$.

It remains to show that j is a homotopy equivalence. By induction on the dimension

of the base simplex, one may assume that $M^{n-1}(|f|)$ deformation retracts onto the corresponding part of $|\mathfrak{X}|$ lying over $\partial\Delta^n$. Moreover, the inclusion

$$|\mathfrak{X}| \hookrightarrow M(|f|)$$

is homotopy equivalent to the map

$$f_n : X_n \rightarrow Y_n,$$

since both spaces deformation retract onto their terminal vertex spaces. As f_n is a homotopy equivalence by assumption, the inclusion j is a homotopy equivalence.

Corollary A.4.56 therefore implies that $M^n(|f|) \cup |\mathfrak{X}|$ deformation retracts onto $M^{n-1}(|f|) \cup |\mathfrak{X}|$. This proves the claim.

Finally, perform the deformation retraction at the n -th stage during the interval

$$\left[\frac{1}{2^{n+1}}, \frac{1}{2^n} \right].$$

The resulting infinite concatenation is continuous. Indeed, on every finite skeleton only finitely many deformation stages are non-trivial, and the realisation has the weak topology with respect to its skeleta. It gives a deformation retraction

$$M(|f|) \rightarrow |\mathfrak{X}|.$$

Thus $M(|f|)$ deformation retracts onto both $|\mathfrak{X}|$ and $|\mathfrak{Y}|$. Consequently,

$$|f| : |\mathfrak{X}| \rightarrow |\mathfrak{Y}|$$

is a homotopy equivalence (see [26, Ch. 4, §G, Prop. 4G.1]). □

Theorem 4.3.15 (Nerve Theorem). Let X be a paracompact topological space and let $\mathcal{U} = \{U_i\}_{i \in I}$ be an open cover. If every non-empty finite intersection $U_{i_0} \cap \cdots \cap U_{i_k}$ is contractible, then the geometric realisation of the nerve satisfies $|\check{N}(\mathcal{U})| \simeq X$ (see [26, Ch. 4, §G, Corollary 4G.3]).

Proof. For each finite non-empty set $\sigma = \{i_0, \dots, i_p\} \subseteq I$ with

$$U_\sigma := U_{i_0} \cap \dots \cap U_{i_p} \neq \emptyset,$$

let Δ^σ be the simplex spanned by the vertices $i_0, \dots, i_p$.

One constructs the **thickened Čech space** $\check{X}$ over X and over the abstract nerve, that is,

$$\begin{aligned} \check{X} &:= \left(\coprod_{\sigma} U_\sigma \times \Delta^\sigma \right) / \sim \\ &= \left\{ (x, t) \in X \times |\check{N}(\mathcal{U})| \mid t_i > 0 \Rightarrow x \in U_i \right\}, \end{aligned}$$

where (t_i) are the barycentric coordinates of the nerve, and the equivalence relation $\sim$ is generated by

$$(x, t) \sim (\iota_U(x), \iota_\Delta(t))$$

whenever $\tau \subseteq \sigma$ is a face, and for $x \in U_\sigma$ and $t \in \Delta^\tau$. Here,

$$\iota_U : U_\sigma \hookrightarrow U_\tau$$

is the inclusion of intersections, and

$$\iota_\Delta : \Delta^\tau \hookrightarrow \Delta^\sigma$$

is the corresponding face inclusion.

There are projections

$$\begin{aligned} \text{pr}_X : \check{X} &\rightarrow X, & (x, t) &\mapsto x \\ \text{pr}_{\check{N}} : \check{X} &\rightarrow |\check{N}(\mathcal{U})|, & (x, t) &\mapsto t. \end{aligned}$$

The proof strategy is to show that both projections pr_X and $\text{pr}_{\check{N}}$ are homotopy equivalences.

Claim: pr_X is a homotopy equivalence.

Since X is paracompact, one can choose a partition of unity $(\varphi_i)_{i \in I}$ subordinate to

$\mathcal{U}$ by Lemma A.4.36, that is

$$\text{supp}(\varphi_i) = \overline{\{x \in X \mid \varphi_i(x) \neq 0\}} \subseteq U_i$$

for all $i \in I$, and define

$$\begin{aligned} s : X &\rightarrow \check{X} \\ x &\mapsto \left(x, (\varphi_i(x))_i\right). \end{aligned}$$

Since $\varphi_i(x) \neq 0$, $x \in \text{supp}(\varphi_i) \subseteq U_i$. Then

$$\text{pr}_X \circ s = \text{id}_X.$$

Define a homotopy

$$\begin{aligned} H : \check{X} \times [0, 1] &\rightarrow \check{X} \\ H((x, t), u) &= \left(x, (1 - u)t + u \cdot (\varphi_i(x))_i\right). \end{aligned}$$

It is well-defined since for $x \notin U_i$, one has $t_i = 0$ and $\varphi_i(x) = 0$. Thus the convex combination becomes

$$(1 - u)t_i + u \cdot \varphi_i(x) = 0$$

whenever $x \notin U_i$, and the support of the barycentric coordinates does not leave the index set of opens containing x . Hence the image remains in $\check{X}$. Therefore

$$\begin{aligned} H(-, 0) &= \text{id}_{\check{X}} \\ H(-, 1) &= s \circ \text{pr}_X, \end{aligned}$$

and pr_X is a homotopy equivalence, with homotopy inverse s . In fact, $\check{X}$ deformation retracts onto the image of s , which is homeomorphic to X (compare [26, Ch. 4, §G, Prop. 4G.2.]).

Claim: $\text{pr}_{\check{N}}$ is a homotopy equivalence.

The contractible intersections U_σ may be regarded as complex of spaces over the nerve. Replacing each U_σ by a point yields a second complex of spaces whose realisation is the geometric realisation of the nerve.

Apply Proposition 4.3.14 over the simplicial complex $\mathcal{S} = \check{N}(\mathcal{U})$ to the complex of spaces $\mathfrak{X} = (U_\sigma)_{\sigma \in \mathcal{S}}$ and $\mathfrak{Y} = (\{*\})_{\sigma \in \mathcal{S}}$. Then

$$|\mathfrak{X}| = \left(\prod_{\sigma \in \mathcal{S}} U_\sigma \times \Delta^\sigma \right) / \sim = \check{X}$$

is the **thickened Čech space**, whereas

$$|\mathfrak{Y}| = \left(\prod_{\sigma \in \mathcal{S}} \{*\} \times \Delta^\sigma \right) / \sim = |\check{N}(\mathcal{U})|$$

Let $f : \mathfrak{X} \rightarrow \mathfrak{Y}$ be the morphism of complexes of spaces whose components are the unique maps

$$f_\sigma : U_\sigma \rightarrow \{*\}.$$

Since each U_σ is contractible by assumption, every f_σ is a homotopy equivalence. Hence Proposition 4.3.14 implies that

$$|f| : |\mathfrak{X}| \rightarrow |\mathfrak{Y}|$$

is a homotopy equivalence. Under the identifications $|\mathfrak{X}| = \check{X}$ and $|\mathfrak{Y}| \cong |\check{N}(\mathcal{U})|$, this map is exactly

$$\text{pr}_{\check{N}} : \check{X} \rightarrow |\check{N}(\mathcal{U})|, \quad [(x, t)] \mapsto t.$$

Therefore $\text{pr}_{\check{N}}$ is a homotopy equivalence. Thus, as both projections are homotopy equivalences, it follows that

$$X \simeq \check{X} \simeq |\check{N}(\mathcal{U})|$$

(compare [26, Ch. 4, §G, Corollary 4G.3]). □

The thickened Čech space simultaneously records the position $x \in X$ and the combinatorial overlap data encoded by the barycentric coordinates $t \in \Delta^\sigma$. That way, it interpolates between the space X and the geometric realisation of the nerve.

This shows that for a good cover, the homotopy type of a space is completely encoded by the combinatorics of the abstract nerve. Consequently, many topological and cohomological questions can be reduced to explicit computations on simplicial objects associated to suitable covers.

This perspective becomes particularly effective for punctured complex domains, and in particular for $\mathbb{C}^\times$, where one can choose covers whose nerves are combinatorially simple graphs or simplicial complexes. The resulting Čech cocycles encode monodromy data and describe the obstructions to extracting roots of holomorphic functions.

The simplicial structure of the Čech nerve induces, via the restriction morphisms of a sheaf $\mathcal{F}$, an associated cosimplicial object in $\mathcal{C}$. The natural question is whether global sections can be reconstructed from compatible local data, and how this compatibility is organised categorically. This leads to the notion of descent, which formalises the problem of gluing local data into global objects. Lemma 4.3.13 provides the conceptual foundation of the following viewpoint:

Interpretations of Čech Cohomology

Descent. Čech constructions provide a concrete model for descent. The problem of reconstructing global sections on X from compatible local sections on an open cover $\mathcal{U} = \{U_i\}_{i \in I}$ is encoded by the multiple intersections occurring in the Čech nerve.

Let

$$\mathcal{F} : \text{Open}(X)^{\text{op}} \rightarrow \mathcal{C}$$

be a sheaf, where $\mathcal{C}$ admits the required products. Associated to $\mathcal{U}$ and $\mathcal{F}$ is the cosimplicial object

$$\check{C}^\bullet(\mathcal{U}, \mathcal{F}) : \Delta \rightarrow \mathcal{C}$$

whose n -th term is

$$\check{C}^n(\mathcal{U}, \mathcal{F}) := \prod_{(i_0, \dots, i_n) \in I^{n+1}} \mathcal{F}(U_{i_0} \cap \dots \cap U_{i_n})$$

and whose coface and codegeneracy morphisms are induced by the restriction morphisms of $\mathcal{F}$ and by repetition of indices. The **descent object** of $\mathcal{F}$ with respect to $\mathcal{U}$ is defined by

$$\text{Desc}(\mathcal{F}, \mathcal{U}) := \lim_{\Delta} \check{C}^\bullet(\mathcal{U}, \mathcal{F}).$$

One says that **descent holds** if the canonical morphism

$$\mathcal{F}(X) \rightarrow \text{Desc}(\mathcal{F}, \mathcal{U}),$$

is an isomorphism. By Lemma 4.3.13, this is precisely the sheaf gluing condition expressed as a limit condition.

Assume now that $\mathcal{C}$ is a Grothendieck abelian category. Then the functor category $[\Delta, \mathcal{C}]$ is again a Grothendieck abelian category and therefore has enough injectives. The limit functor

$$\lim_{\Delta} : [\Delta, \mathcal{C}] \rightarrow \mathcal{C}$$

is left-exact by Lemma 1.7.6, so its right derived functors $R^n \lim_{\Delta}$ are defined. A standard result, whose proof relies on the Dold–Kan correspondence between cosimplicial objects and cochain complexes, identifies these derived limits with the cohomology of the associated alternating Čech complex. Since a development of the Dold–Kan correspondence would considerably exceed the scope of this thesis, only this interpretation is stated here. Details can be found for example in [52, Ch. 14, §25].

Applied to $\check{C}^{\bullet}(\mathcal{U}, \mathcal{F})$, these derived limits describe the higher derived objects associated to the descent limit.

Thus Čech cohomology provides a derived form of descent.

In degree zero, Čech cohomology recovers the ordinary descent condition:

$$\check{H}^0(\mathcal{U}, \mathcal{F}) = \ker(\check{\delta}^0 : \check{C}^0(\mathcal{U}, \mathcal{F}) \rightarrow \check{C}^1(\mathcal{U}, \mathcal{F})).$$

The kernel is the equaliser of the two degree-zero coface morphisms. Consequently,

$$\check{H}^0(\mathcal{U}, \mathcal{F}) \cong \text{Desc}(\mathcal{F}, \mathcal{U}),$$

so degree-zero Čech cohomology recovers ordinary descent.

Higher Čech cohomology objects are therefore the higher derived objects of this descent limit in the abelian setting. The Grothendieck hypothesis above guarantees the existence of the required derived functors (compare [12]; for the general descent perspective via hypercoverings, see [53]).

Monodromy as Obstruction to Descent. Locally constant sheaves give rise to transport isomorphisms between stalks along paths. For loops, these isomorphisms define the monodromy representation. The same global twisting can also be described by transition cocycles on a suitable open cover.

Let X be a path-connected, locally path-connected topological space, and let $\mathcal{F} \in \text{Ob}(\text{Sh}(X, \mathcal{C}))$ be a locally constant sheaf, where $\mathcal{C}$ admits the filtered colimits required for the formation of stalks. Fix a point $x_0 \in X$, and let $G := \mathcal{F}_{x_0}$. Since $\mathcal{F}$ is locally constant, every path $\gamma : [0, 1] \rightarrow X$ induces an isomorphism of stalks

$$T_\gamma : \mathcal{F}_{\gamma(0)} \xrightarrow{\sim} \mathcal{F}_{\gamma(1)}.$$

The transport map is constructed by covering the compact image $\gamma([0, 1])$ by open subsets on which $\mathcal{F}$ is constant, subdividing $[0, 1]$ accordingly, and composing the induced identification of stalks. Refining the subdivision does not change the resulting isomorphism, and the transport depends only on the homotopy class of γ relative to its endpoints. Hence one obtains the following functor.

Definition 4.3.16 (Monodromy). The assignment $x \mapsto \mathcal{F}_x$ and $[\gamma] \mapsto T_\gamma$ defines a functor

$$\Pi_1(X) \rightarrow \mathcal{C}^\simeq$$

called the **monodromy of $\mathcal{F}$** , where $\mathcal{C}^\simeq$ is the maximal subgroupoid of $\mathcal{C}$ (see [28, Ch. 4, §9, pp. 252 ff.]).

Here, $\Pi_1(X)$ denotes the fundamental groupoid of X , whose objects are points of X , and whose morphisms are homotopy classes of paths relative to their endpoints.

Remark 4.3.17. Functoriality follows from the compatibility of transport with concatenation of paths and invariance under homotopy.

In particular, at the base point x_0 one obtains the monodromy representation

$$\rho_{\mathcal{F}} : \pi_1(X, x_0) \rightarrow \text{Aut}_{\mathcal{C}}(G),$$

where $\text{Aut}_{\mathcal{C}}(G)$ denotes the automorphism group of G in $\mathcal{C}$. Changing the chosen identification of the stalk $\mathcal{F}_{x_0}$ with G conjugates this representation. Hence the isomorphism class of $\mathcal{F}$ determines the conjugacy class of $\rho_{\mathcal{F}}$. The same twisting

can be described by Čech cocycles. Since $\mathcal{F}$ is locally constant, one can choose an open cover $\mathcal{U} = (U_i)_{i \in I}$ such that every U_i is path-connected and

$$\mathcal{F}|_{U_i} \cong G_{U_i}^{\text{cnst}}$$

for a fixed object $G \in \text{Ob}(\mathcal{C})$. Choose trivialisations

$$\varphi_i : \mathcal{F}|_{U_i} \xrightarrow{\sim} G_{U_i}^{\text{cnst}}.$$

If an intersection $U_i \cap U_j$ is disconnected, the corresponding transition automorphism may take different constant values on its different connected components. Then on $U_i \cap U_j$ define

$$g_{ij} := \varphi_i|_{U_i \cap U_j} \circ (\varphi_j|_{U_i \cap U_j})^{-1}.$$

This is an automorphism of the constant sheaf $G_{U_i \cap U_j}^{\text{cnst}}$ and hence corresponds to a locally constant map

$$g_{ij} : U_i \cap U_j \rightarrow \text{Aut}_{\mathcal{C}}(G).$$

On triple intersections they satisfy

$$g_{ij}g_{jk} = g_{ik}.$$

Thus (g_{ij}) is a non-abelian Čech 1-cocycle with values in the sheaf $\text{Aut}_{\mathcal{C}}(G)^{\text{cnst}}$ of locally constant $\text{Aut}_{\mathcal{C}}(G)$ -valued maps. Under the usual connectedness hypotheses, $\mathcal{F}$ is globally isomorphic to the constant sheaf G^{cnst} precisely when its monodromy representation is trivial. Thus monodromy records the obstruction to choosing the local trivialisations compatibly around all loops.

The notion of a locally constant sheaf does not itself require local path-connectedness. Under the standard hypotheses that X is connected locally path-connected, and semilocally simply connected, locally constant sheaves with fibre G are classified, up to isomorphism, by conjugacy classes of representations $\pi_1(X, x_0) \rightarrow \text{Aut}_{\mathcal{C}}(G)$. Equivalently, they may be described by non-abelian Čech 1-cocycles with values in $(\text{Aut}_{\mathcal{C}}(G))^{\text{cnst}}$ (compare [28, Ch. 4, §4, pp. 216 ff.]).

Thus the monodromy representation and the transition cocycle encode the same locally constant twisting: the former records transport along loops, while the latter records the changes between local trivialisations.

Torsors and Čech Cohomology. The notation $\check{H}^1$ is used in two related but distinct senses. For sheaves with values in an abelian category, $\check{H}^1$ is a cohomology object of the Čech cochain complex. For a sheaf of possibly non-abelian groups, non-abelian $\check{H}^1$ is the pointed set of 1-cocycles modulo the corresponding change-of-trivialisation relation.

The interpretation of $\check{H}^1$ in terms of descent can be made precise via torsors. Let $\mathcal{G}$ be a sheaf of groups on X , and let $\mathcal{T}$ be a $\mathcal{G}$ -torsor, that is, a sheaf with a $\mathcal{G}$ -action which is locally $\mathcal{G}$ -equivariantly isomorphic to $\mathcal{G}$ acting on itself by translation.

Let $\mathcal{U} = (U_i)_{i \in I}$ be a cover on which $\mathcal{T}$ is trivial. Choosing sections $t_i \in \mathcal{T}(U_i)$ determines $\mathcal{G}|_{U_i}$ -equivariant trivialisations

$$\mathcal{G}|_{U_i} \xrightarrow{\sim} \mathcal{T}|_{U_i}.$$

on $U_i \cap U_j$, there is a unique section $g_{ij} \in \mathcal{G}(U_i \cap U_j)$ such that

$$t_i = g_{ij} \cdot t_j.$$

On triple intersections, uniqueness implies

$$g_{ij}g_{jk} = g_{ik}.$$

If the local sections are replaced by

$$t'_i = h_i \cdot t_i, \quad h_i \in \mathcal{G}(U_i),$$

then the transition sections are replaced by

$$g'_{ij} = h_i g_{ij} h_j^{-1}.$$

Thus changing local trivialisations changes the cocycle by a non-abelian Čech coboundary. Thus one obtains a well-defined class

$$[g_{ij}] \in \check{H}^1(\mathcal{U}, \mathcal{G}),$$

where $\check{H}^1(\mathcal{U}, \mathcal{G})$ denotes the non-abelian Čech cohomology. It is in general a pointed set rather than a group. The class depends only on the isomorphism class of

the torsor together with the fact that it is trivialised by the cover $\mathcal{U}$.

Conversely, every non-abelian Čech 1-cocycle g_{ij} defines, by gluing the trivial torsors $\mathcal{G}|_{U_i}$, a $\mathcal{G}$ -torsor trivialised by $\mathcal{U}$. This construction induces a bijection

$$\check{H}^1(\mathcal{U}, \mathcal{G}) \xrightarrow{\sim} \left\{ \begin{array}{l} \text{isomorphism classes of } \mathcal{G}\text{-torsors} \\ \text{which are trivialised by } \mathcal{U} \end{array} \right\}.$$

Passing to the colimit over refinements of open covers yields the classification of arbitrary $\mathcal{G}$ -torsors by $\check{H}^1(X, \mathcal{G})$.

Thus, $\check{H}^1$ classifies the twisting of locally trivial objects, making precise the interpretation of cohomology as a measure of descent data (compare [52, Ch. 20, §4]).

The simplest non-trivial case of Čech cohomology arises from a cover by two open subsets. For the alternating, equivalently normalised, Čech complex, only degrees zero and one contribute. The resulting exact sequence is closely related to the classical Mayer–Vietoris sequence.

Theorem 4.3.18 (Mayer–Vietoris). Let X be a topological space, and let U, V be an open two-set cover of X , that is, $X = U \cup V$. Furthermore, let $\mathcal{C}$ be an abelian category for which sheaf cohomology admits computation by flabby resolutions, and let $\mathcal{F} \in \text{Ob}(\text{Sh}(X, \mathcal{C}))$. Then there exists a long exact sequence

$$\begin{aligned} 0 \rightarrow \mathcal{H}^0(X, \mathcal{F}) \rightarrow \mathcal{H}^0(U, \mathcal{F}) \oplus \mathcal{H}^0(V, \mathcal{F}) \rightarrow \mathcal{H}^0(U \cap V, \mathcal{F}) \rightarrow \mathcal{H}^1(X, \mathcal{F}) \rightarrow \dots \\ \dots \rightarrow \mathcal{H}^n(X, \mathcal{F}) \rightarrow \mathcal{H}^n(U, \mathcal{F}) \oplus \mathcal{H}^n(V, \mathcal{F}) \rightarrow \mathcal{H}^n(U \cap V, \mathcal{F}) \rightarrow \mathcal{H}^{n+1}(X, \mathcal{F}) \rightarrow \dots \end{aligned}$$

(see [28, Ch. 2, §5, Theorem 5.10]).

Proof. For the two-set cover $\{U, V\}$, the normalised Čech cochain complex is concentrated in degrees 0 and 1:

$$0 \rightarrow \mathcal{F}(U) \oplus \mathcal{F}(V) \xrightarrow{\check{\delta}^0} \mathcal{F}(U \cap V) \rightarrow 0.$$

The differential is the morphism

$$\check{\delta}^0 := \text{res}_{U \cap V}^U \circ \text{pr}_U - \text{res}_{U \cap V}^V \circ \text{pr}_V,$$

where

$$\text{res}_{U \cap V}^U : \mathcal{F}(U) \rightarrow \mathcal{F}(U \cap V)$$

and

$$\text{res}_{U \cap V}^V : \mathcal{F}(V) \rightarrow \mathcal{F}(U \cap V)$$

are the restriction morphisms. There are no non-degenerate simplices in degrees greater than 1.

Including the global sections gives a short sequence

$$0 \rightarrow \mathcal{F}(X) \rightarrow \mathcal{F}(U) \oplus \mathcal{F}(V) \rightarrow \mathcal{F}(U \cap V) \rightarrow 0$$

which is exact at the first two terms since sheaves satisfy the equaliser condition, but not necessarily exact at $\mathcal{F}(U \cap V) \rightarrow 0$.

Choose a flabby resolution

$$0 \rightarrow \mathcal{F} \rightarrow \mathcal{G}^\bullet$$

of $\mathcal{F}$. For every open subset $W \subseteq X$, the restricted complex

$$0 \rightarrow \mathcal{F}|_W \rightarrow \mathcal{G}^\bullet|_W$$

is again a flabby resolution. Indeed, restriction to an open subset preserves exactness, and the restriction of a flabby sheaf is flabby. Consequently,

$$\mathcal{H}^n(W, \mathcal{F}|_W) \cong H^n(\Gamma(W, \mathcal{G}^\bullet)).$$

For every $q \geq 0$, flabbiness of $\mathcal{G}^q$ implies that the restriction morphism

$$\text{res}_{U \cap V}^U : \mathcal{G}^q(U) \rightarrow \mathcal{G}^q(U \cap V)$$

is an epimorphism. Hence the morphism

$$\delta^q : \mathcal{G}^q(U) \oplus \mathcal{G}^q(V) \rightarrow \mathcal{G}^q(U \cap V),$$

defined by

$$\delta^q := \text{res}_{U \cap V}^U \circ \text{pr}_U - \text{res}_{U \cap V}^V \circ \text{pr}_V,$$

is an epimorphism. By the sheaf axiom, its kernel is canonically isomorphic to

$\mathcal{G}^q(X)$. Thus for every $q \in \mathbb{N}_0$, there is a short exact sequence

$$0 \rightarrow \mathcal{G}^q(X) \rightarrow \mathcal{G}^q(U) \oplus \mathcal{G}^q(V) \xrightarrow{\delta^q} \mathcal{G}^q(U \cap V) \rightarrow 0.$$

The sequences assemble into a short exact sequence of cochain complexes

$$0 \rightarrow \Gamma(X, \mathcal{G}^\bullet) \rightarrow \Gamma(U, \mathcal{G}^\bullet) \oplus \Gamma(V, \mathcal{G}^\bullet) \rightarrow \Gamma(U \cap V, \mathcal{G}^\bullet) \rightarrow 0.$$

The long exact cohomology sequence associated to this short exact sequence of cochain complexes is

$$\begin{aligned} \dots &\rightarrow H^n(\Gamma(X, \mathcal{G}^\bullet)) \rightarrow H^n(\Gamma(U, \mathcal{G}^\bullet) \oplus \Gamma(V, \mathcal{G}^\bullet)) \rightarrow \dots \\ \dots &\rightarrow H^n(\Gamma(U \cap V, \mathcal{G}^\bullet)) \rightarrow H^{n+1}(\Gamma(X, \mathcal{G}^\bullet)) \rightarrow \dots \end{aligned}$$

Thus

$$\dots \rightarrow \mathcal{H}^n(X, \mathcal{F}) \rightarrow \mathcal{H}^n(U, \mathcal{F}) \oplus \mathcal{H}^n(V, \mathcal{F}) \rightarrow \mathcal{H}^n(U \cap V, \mathcal{F}) \rightarrow \mathcal{H}^{n+1}(X, \mathcal{F}) \rightarrow \dots \quad (4.3.19)$$

is the asserted Mayer–Vietoris sequence (see [28, Ch. 2, §5, 5.6]). $\square$

The Mayer–Vietoris sequence illustrates how global cohomological information can be reconstructed from local data and their compatibilities on non-empty intersections. For more general covers, these compatibility relations are encoded combinatorially by the intersections of the covering sets.

The remaining question is under which hypotheses Čech cohomology agrees with sheaf cohomology. The relevant conditions concern the sheaf, the underlying space, and the chosen cover.

Lemma 4.3.20. Let $\mathcal{D}$ be an $AB4^*$ abelian category, and let $\text{Sh}(X, \mathcal{D})$ be Grothendieck, and such that the representation property 4.2.11 is satisfied. Let $\mathcal{F} \in \text{Sh}(X, \mathcal{D})$ be a flabby sheaf, and let $\mathcal{U} = \{U_i\}_{i \in I}$ be an open cover of X . Then the augmented Čech complex with respect to $\mathcal{U}$

$$0 \rightarrow \Gamma(X, \mathcal{F}) \rightarrow \check{C}^0(\mathcal{U}, \mathcal{F}) \rightarrow \check{C}^1(\mathcal{U}, \mathcal{F}) \rightarrow \dots$$

is exact. In particular,

$$\check{H}^n(\mathcal{U}, \mathcal{F}) = 0$$

for all $n \in \mathbb{N}$ (see [28, Ch. 2, §3, Thm. 3.5]).

Proof. Consider the sheafified Čech complex

$$0 \rightarrow \mathcal{F} \rightarrow \mathcal{G}^0(\mathcal{U}, \mathcal{F}) \rightarrow \mathcal{G}^1(\mathcal{U}, \mathcal{F}) \rightarrow \dots,$$

where

$$\mathcal{G}^p(\mathcal{U}, \mathcal{F}) := \prod_{\alpha \in I^{p+1}} (\iota_\alpha)_*(\mathcal{F}|_{U_\alpha})$$

with $\iota_\alpha : U_\alpha \hookrightarrow X$.

One first shows that this augmented complex of sheaves is exact. For each $j \in I$, one restricts it to U_j . On U_j , define morphisms

$$h_j^p : \mathcal{G}^p(\mathcal{U}, \mathcal{F})|_{U_j} \rightarrow \mathcal{G}^{p-1}(\mathcal{U}, \mathcal{F})|_{U_j}, \quad p \geq 0,$$

by insertion of the fixed index j , with

$$\mathcal{G}^{-1}(\mathcal{U}, \mathcal{F}) := \mathcal{F},$$

with the augmentation

$$\mathcal{F} \rightarrow \mathcal{G}^0(\mathcal{U}, \mathcal{F}).$$

More precisely, on sections over an open subset $W \subseteq U_j$, the component indexed by $(i_0, \dots, i_{p-1})$ is the component of the original family indexed by $(j, i_0, \dots, i_{p-1})$. This is well-defined because

$$W \cap U_j \cap U_{i_0} \cap \dots \cap U_{i_{p-1}} = W \cap U_{i_0} \cap \dots \cap U_{i_{p-1}}.$$

Together with the augmentation in degree -1 , these morphisms satisfy

$$\check{\delta}^{p-1} \circ h_j^p + h_j^{p+1} \circ \check{\delta}^p = \text{id}$$

on the restriction of the augmented complex to U_j . Hence the restriction of the augmented complex is contractible as a cochain complex, and therefore exact, after restriction to each member U_j of the cover.

Exactness of complexes of sheaves is local. Since sets U_j cover X , the augmented sheafified Čech complex is exact on X .

Since $\mathcal{F}$ is flabby, each sheaf $(\iota_\alpha)_*(\mathcal{F}|_{U_\alpha})$ is flabby, and products of flabby sheaves are flabby by Lemma 4.2.13. Thus $\mathcal{G}^p(\mathcal{U}, \mathcal{F})$ is flabby for every p . Therefore the complex

$$0 \rightarrow \mathcal{F} \rightarrow \mathcal{G}^0(\mathcal{U}, \mathcal{F}) \rightarrow \mathcal{G}^1(\mathcal{U}, \mathcal{F}) \rightarrow \dots,$$

is a flabby resolution of $\mathcal{F}$. Therefore

$$H^n\left(\Gamma(X, \mathcal{G}^\bullet(\mathcal{U}, \mathcal{F}))\right) \cong \mathcal{H}^n(X, \mathcal{F}).$$

Moreover,

$$\begin{aligned} \Gamma(X, \mathcal{G}^p(\mathcal{U}, \mathcal{F})) &\cong \prod_{\alpha \in I^{p+1}} \Gamma(X, (\iota_\alpha)_*(\mathcal{F}|_{U_\alpha})) \\ &= \check{C}^p(\mathcal{U}, \mathcal{F}). \end{aligned}$$

Since $\mathcal{F}$ is flabby, it is $\Gamma(X, -)$ -acyclic by Theorem 4.2.16, and hence

$$\mathcal{H}^n(X, \mathcal{F}) = 0$$

for $n \in \mathbb{N}$. It follows that the augmented Čech complex is exact (see [28, Ch. 2, §3, Thm. 3.5]). $\square$

Since not every sheaf is flabby, this result alone is not sufficient. Locally constant sheaves for instance are not necessarily flabby. The computability of sheaf cohomology via Čech cohomology relies on good covers.

Definition 4.3.21 (Acyclic cover). Let X be a topological space and $\mathcal{F}$ a sheaf on X . An open cover $\mathcal{U} = \{U_i\}_{i \in I}$ of X is called $\mathcal{F}$ -acyclic if for all finite non-empty intersections

$$R^n \Gamma(U_{i_0} \cap \dots \cap U_{i_k}, \mathcal{F}|_{U_{i_0} \cap \dots \cap U_{i_k}}) = 0$$

for all $n \in \mathbb{N}$ (see [28, Ch. 1, §3, Def. 7.4]).

The definition of acyclic covers uses ordinary sheaf cohomology. It measures the obstructions of all finite intersections, independently of covers. In contrast, Čech

cohomology is initially computed relative to a chosen cover and then globalised by passing to refinements.

Theorem 4.3.22 (Criterion for Acyclicity). Let X be a locally contractible, hereditarily paracompact topological space, and let $\mathcal{F} \in \text{Ob}(\text{Sh}(X, \text{Ab}))$ be a locally constant sheaf. Let $U \subseteq X$ be a non-empty, contractible, open subset. Then $\mathcal{H}^n(U, \mathcal{F}|_U) = 0$ for $n \in \mathbb{N}$ (compare [50]).

Proof. The restriction $\mathcal{F}|_U$ is locally constant. Since U is contractible, it is path-connected and simply connected. Moreover, as an open subset of the locally contractible space X , the space U is locally contractible and hence locally path-connected. Therefore every locally constant sheaf on U is constant. Thus there exists an object G such that

$$\mathcal{F}|_U \cong \mathcal{G}_U^{\text{cst}}$$

with $\mathcal{G}_U^{\text{cst}}(U) \cong G$ for $G \in \text{Ob}(\text{Ab})$.

It remains to compute the cohomology of the constant sheaf $\mathcal{G}_U^{\text{cst}}$. Since U is an open subspace of X , it is locally contractible and hereditarily paracompact. Hence the comparison theorem 4.2.17 between sheaf cohomology and singular cohomology gives

$$\mathcal{H}^n(U, \mathcal{G}_U^{\text{cst}}) \cong H_{\text{sing}}^n(U, G)$$

for all $n \geq 0$. Since U is contractible, its singular cohomology with coefficients in G vanishes in positive degrees, that is,

$$H_{\text{sing}}^n(U, G) = 0$$

for all $n \in \mathbb{N}$ by Theorem 4.1.11. Hence

$$\mathcal{H}^n(U, \mathcal{F}|_U) \cong \mathcal{H}^n(U, \mathcal{G}_U^{\text{cst}}) \cong H_{\text{sing}}^n(U, G) = 0$$

for all $n \in \mathbb{N}$ (compare [50]). □

4.3.2 The Leray Theorem

This subsection aims to prove the Leray Theorem 4.3.26, which bridges sheaf cohomology and Čech cohomology. To compare Čech cohomology with sheaf cohomology, one combines two constructions: an injective resolution $\mathcal{F} \rightarrow \mathcal{I}^\bullet$, which carries a cohomological degree, and the Čech complex $\check{C}^\bullet(\mathcal{U}, -)$ which introduces an additional degree coming from the cover. Applying the Čech construction to each $\mathcal{I}^q$ therefore gives a bicomplex

$$C^{p,q} := \check{C}^p(\mathcal{U}, \mathcal{I}^q) = \prod_{\alpha \in I^{p+1}} \mathcal{I}^q(U_\alpha).$$

Its horizontal differential is the alternating Čech differential, while its vertical differential is induced by the injective resolution.

To identify the terms of the spectral sequence explicitly, one needs to compute the cohomology of this bicomplex. For fixed p , the vertical complex is a product of the complexes $\mathcal{I}^\bullet(U_\alpha)$. The $AB4^*$ assumption ensures that products are exact and therefore commute with taking cohomology. Consequently, the vertical cohomology can be identified with the product of the sheaf cohomology objects of the finite intersections U_α . This provides the E_1 -page of the Čech-to-sheaf-cohomology spectral sequence.

Corollary 4.3.23 (Vertical cohomology in the Čech bicomplex). Let X be a topological space, let $\mathcal{U} = \{U_i\}_{i \in I}$ be an open cover of X , and let

$$0 \rightarrow \mathcal{F} \rightarrow \mathcal{I}^0 \rightarrow \mathcal{I}^1 \rightarrow \dots$$

be an injective resolution in $\text{Sh}(X, \mathcal{D})$. Assume that $\mathcal{D}$ is an $AB4^*$ category and that the hypotheses of Theorem 4.2.10 and Theorem 4.2.16 are satisfied on every open subset of X . For $p, q \in \mathbb{N}_0$, define

$$C^{p,q} := \check{C}^p(\mathcal{U}, \mathcal{I}^q) = \prod_{\alpha \in I^{p+1}} \mathcal{I}^q(U_\alpha).$$

Then

$$\mathcal{H}^q(C^{p,\bullet}) \cong \prod_{\alpha \in I^{p+1}} \mathcal{H}^q(U_\alpha, \mathcal{F}|_{U_\alpha}).$$

(compare [20, Ch. 4, §4, Thm 4.4.4, Ch. 5]).

Proof. Since the representation property is satisfied, every injective sheaf $\mathcal{I}^q$ is flabby by Theorem 4.2.10, its restriction to every open subset U_α is flabby. Hence

$$0 \rightarrow \mathcal{F}|_{U_\alpha} \rightarrow \mathcal{I}^\bullet|_{U_\alpha}$$

is an acyclic resolution by Theorem 4.2.16. By Theorem 4.2.8,

$$H^q(\mathcal{I}^\bullet(U_\alpha)) \cong \mathcal{H}^q(U_\alpha, \mathcal{F}|_{U_\alpha}).$$

Since $\mathcal{D}$ is an $AB4^*$ category, products are exact and therefore commute with taking cohomology. Consequently,

$$\begin{aligned} H^q(C^{p,\bullet}) &= H^q\left(\prod_{\alpha \in I^{p+1}} \mathcal{I}^\bullet(U_\alpha)\right) \\ &\cong \prod_{\alpha \in I^{p+1}} H^q(\mathcal{I}^\bullet(U_\alpha)) \\ &\cong \prod_{\alpha \in I^{p+1}} \mathcal{H}^q(U_\alpha, \mathcal{F}|_{U_\alpha}) \end{aligned}$$

(compare [20, Ch. 4, §4, Thm 4.4.4, Ch. 5]). □

Remark 4.3.24. The preceding product decomposition is closely related to the behaviour of sheaf cohomology under disjoint unions. If

$$Y = \coprod_{i \in I} Y_i$$

is a disjoint union of topological spaces and $\mathcal{F}$ is a sheaf on Y with values in an $AB4^*$ category $\mathcal{D}$, under the usual hypotheses ensuring the existence of sheaf cohomology there are natural isomorphisms

$$\mathcal{H}^q(Y, \mathcal{F}) \cong \prod_{i \in I} \mathcal{H}^q(Y_i, \mathcal{F}|_{Y_i})$$

for all $q \in \mathbb{N}_0$.

Indeed, there is an equivalence

$$\mathrm{Sh}(Y, \mathcal{D}) \sim \prod_{i \in I} \mathrm{Sh}(Y_i, \mathcal{D}),$$

and global sections decompose as

$$\Gamma(Y, \mathcal{F}) \cong \prod_{i \in I} \Gamma(Y_i, \mathcal{F}|_{Y_i}).$$

Since products are exact in an $AB4^*$ category, this decomposition passes to cohomology (compare [20, Ch. 4, /S 4, Thm. 4.4.4, Ch. 5]).

This allows to identify the vertical cohomology of the Čech bicomplex with the local sheaf cohomology of the cover. It is used to identify the E_1 -page of the spectral sequence with Čech-type local cohomology data.

The preceding results allow the computation of the vertical cohomology of the Čech bicomplex explicitly. This identifies the first page of the associated spectral sequence with cohomology groups of the intersections of the cover, thereby linking the spectral sequence to the Čech-type local data.

Proposition 4.3.25. Let X be a topological space, let $\mathcal{D}$ be an $AB4^*$ abelian category such that $\mathrm{Sh}(X, \mathcal{D})$ has enough injectives and the representation property 4.2.11 is satisfied. Let $\mathcal{F} : \mathrm{Open}(X)^{\mathrm{op}} \rightarrow \mathcal{D}$ be a sheaf, and let $\mathcal{U} = (U_i)_{i \in I}$ be an open cover. Then there exists a first-quadrant spectral sequence

$$E_1^{p,q} = \prod_{\alpha \in I^{p+1}} \mathcal{H}^q(U_\alpha, \mathcal{F}|_{U_\alpha}) \rightarrow \mathcal{H}^{p+q}(X, \mathcal{F}),$$

whose differential on the E_1 -page is induced by the alternating Čech differential (compare [20, Ch. 5, §9], [47, Ch. 10, §4, Thm. 10.16]).

Proof. Let $U_{i_0, \dots, i_p} := U_{i_0} \cap \dots \cap U_{i_p}$. Choose an injective resolution

$$0 \rightarrow \mathcal{F} \rightarrow \mathcal{I}^0 \rightarrow \mathcal{I}^1 \rightarrow \dots$$

in $\text{Sh}(X, \mathcal{D})$. Define a first-quadrant bicomplex by

$$C^{p,q} := \mathcal{I}^q(\mathcal{U}, \mathcal{I}^q) = \prod_{\alpha \in I^{p+1}} \mathcal{I}^q(U_\alpha).$$

Its horizontal differential is the alternating Čech differential, while its vertical differential is induced by the differential on the injective resolution.

First one shows that the total complex $\text{Tot}(C^{\bullet,\bullet})$ computes the cohomology of $\mathcal{F}$.

For each fixed q , the sheaf $\mathcal{I}^q$ is injective, hence flabby by Theorem 4.2.10, so its augmented Čech complex with respect to the cover $\mathcal{U}$ is exact by Lemma 4.3.20.

Consider the bicomplex

$$K^{p,q} := \begin{cases} \Gamma(X, \mathcal{I}^q), & p = 0 \\ 0, & p \in \mathbb{N}, \end{cases}$$

together with the morphism of bicomplexes

$$K^{\bullet,\bullet} \rightarrow C^{\bullet,\bullet}$$

induced by the Čech augmentation. For every fixed q , this morphism induces an isomorphism on horizontal cohomology. Hence Theorem 3.6.13 implies that the induced morphism of total complexes

$$\Gamma(X, \mathcal{I}^\bullet) \rightarrow \text{Tot}(C^{\bullet,\bullet})$$

is a quasi-isomorphism. Since $\Gamma(X, \mathcal{I}^\bullet)$ computes sheaf cohomology, it follows that

$$H^n(\text{Tot}(C^{\bullet,\bullet})) \cong \mathcal{H}^n(X, \mathcal{F})$$

for all $n \in \mathbb{N}_0$.

Filtering the bicomplex so that the vertical differential is taken first yields a spectral sequence with

$$E_1^{p,q} = H^q(C^{p,\bullet}).$$

By Corollary 4.3.23,

$$E_1^{p,q} \cong \prod_{\alpha \in I^{p+1}} \mathcal{H}^q(U_\alpha, \mathcal{F}|_{U_\alpha}).$$

Under this identification, the differential $\delta_1^{p,q}$ is induced by the horizontal differential of the bicomplex and hence is the alternating Čech differential on the local cohomology objects.

Since the bicomplex is a first-quadrant bicomplex, the associated spectral sequence converges to the cohomology of the total complex. Therefore

$$E_1^{p,q} \cong \prod_{\alpha \in I^{p+1}} \mathcal{H}^q(U_\alpha, \mathcal{F}|_{U_\alpha}) \rightarrow \mathcal{H}^{p+q}(X, \mathcal{F})$$

(compare [20, Ch. 5, §9], [47, Ch. 10, §4, Thm. 10.16]). □

This proposition provides a spectral sequence relating the cohomology of a sheaf on X to the cohomology of its restrictions to finite intersections of the cover. It forms the key technical tool in the proof of the following theorem:

Theorem 4.3.26 (Leray). Let X be a topological space, let $\mathcal{D}$ be an $AB4^*$ abelian category such that $\text{Sh}(X, \mathcal{D})$ has enough injectives and the representation property 4.2.11 is satisfied. Let $\mathcal{F} : \text{Open}(X)^{\text{op}} \rightarrow \mathcal{D}$ be a sheaf, and let $\mathcal{U} = \{U_i\}_{i \in I}$ be an open cover. If $\mathcal{U}$ is acyclic for $\mathcal{F}$, that is,

$$\mathcal{H}^q(U_\alpha, \mathcal{F}|_{U_\alpha}) = 0$$

for all $q > 0$ and all multiindices $\alpha \in I^{p+1}$, then the natural comparison map

$$\check{H}^n(\mathcal{U}, \mathcal{F}) \rightarrow \mathcal{H}^n(X, \mathcal{F})$$

is an isomorphism for all $n \geq 0$ (compare [20, Ch. 5, §9], [47, Ch. 10, §4, Thm. 10.79]).

Proof. The proof follows the idea that the associated spectral sequence collapses under the acyclicity assumption. Consider the Čech-to-sheaf-cohomology spectral

sequence from Proposition 4.3.25, that is,

$$E_1^{p,q} = \prod_{\alpha \in I^{p+1}} \mathcal{H}^q(U_\alpha, \mathcal{F}|_{U_\alpha}) \rightarrow \mathcal{H}^{p+q}(X, \mathcal{F}).$$

Under the acyclicity hypothesis, one has $\mathcal{H}^q(U_\alpha, \mathcal{F}|_{U_\alpha}) = 0$ for all $q > 0$, and the E_1 -page is concentrated in row $q = 0$. Hence the spectral sequence collapses at the E_2 -page, and

$$E_2^{p,0} = E_\infty^{p,0}.$$

The bottom row of the E_1 -page is

$$E_1^{p,0} = \prod_{\alpha \in I^{p+1}} \mathcal{F}(U_\alpha) = \check{C}^p(\mathcal{U}, \mathcal{F})$$

with differential given by the Čech coboundary. Thus the E_2 -page satisfies

$$E_2^{p,0} = \check{H}^p(\mathcal{U}, \mathcal{F}).$$

Since the spectral sequence converges to $\mathcal{H}^{p+q}(X, \mathcal{F})$ by Corollary 3.6.12, which also identifies

$$E_\infty^{p,q} \cong \text{gr}^p \mathcal{H}^{p+q}(X, \mathcal{F}),$$

one obtains

$$E_\infty^{n,0} \cong \text{gr}^n \mathcal{H}^n(X, \mathcal{F}).$$

Since the spectral sequence is concentrated in the bottom row, the induced filtration on $\mathcal{H}^n(X, \mathcal{F})$ has only one non-zero graded piece, so

$$\text{gr}^n \mathcal{H}^n(X, \mathcal{F}) \cong \mathcal{H}^n(X, \mathcal{F}).$$

Therefore

$$\check{H}^n(\mathcal{U}, \mathcal{F}) \cong E_2^{n,0} = E_\infty^{n,0} \cong \mathcal{H}^n(X, \mathcal{F})$$

(compare [20, Ch. 5, §9], [47, Ch. 10, §4, Thm. 10.79]). □

The acyclicity assumption forces the spectral sequence to collapse and identifies sheaf cohomology with Čech cohomology. In particular, if $\mathcal{U} = \{U_i\}_{i \in I}$ is an open cover of X such that all non-empty finite intersections $U_{(i_0, \dots, i_n)} := U_{i_0} \cap \dots \cap U_{i_n}$

are contractible, then $\mathcal{U}$ satisfies the Leray acyclicity condition for a locally constant sheaf $\mathcal{F}$ with values in $\mathbf{Ab}$.

5 Descent and Cohomological Obstructions to Global Lifts

Global surjectivity is not an intrinsic consequence of surjectivity as a morphism of sheaves. Rather, it depends on the interaction between the sheaves, the topology of the underlying space, and the possibility of gluing local data. In this sense, global surjectivity is a descent problem. For a morphism of sheaves, the relevant question is whether locally defined lifts can be glued to a global lift. The obstruction to such gluing is measured by cohomology; in particular Čech cohomology detects the failure of compatible local data to descend globally.

Consider an open subset $X \subseteq \mathbb{C}$ equipped with the Euclidean topology, and the sheaf

$$\mathcal{O}_X^\times : \text{Open}(X)^{\text{op}} \rightarrow \text{Ab}$$

defined by

$$\mathcal{O}_X^\times(U) := \{f : U \rightarrow \mathbb{C}^\times \text{ holomorphic}\}$$

which is a sheaf of abelian groups with group structure given by pointwise multiplication. Let

$$\mu_2 := \ker((\cdot)^2 : \mathcal{O}_X^\times \rightarrow \mathcal{O}_X^\times).$$

For every open subset $U \subseteq X$, its sections are precisely the locally constant maps

$$U \rightarrow \{\pm 1\}.$$

Thus μ_2 is a locally constant sheaf with stalks isomorphic to $\{\pm 1\}$. The squaring morphism gives rise to a short exact sequence of sheaves of abelian groups

$$1 \rightarrow \mu_2 \rightarrow \mathcal{O}_X^\times \xrightarrow{(\cdot)^2} \mathcal{O}_X^\times \rightarrow 1, \quad (5.0.1)$$

where 1 denotes the trivial sheaf of groups

$$\begin{aligned} 1 &: \text{Open}(X)^{\text{op}} \rightarrow \text{Ab} \\ U &\mapsto \{1\} \end{aligned}$$

for all $U \in \text{Ob}(\text{Open}(X)^{\text{op}})$. Indeed, its kernel is μ_2 , and it is an epimorphism of sheaves because every germ of a nowhere-vanishing holomorphic function admits a holomorphic square root after restriction to a sufficiently small neighbourhood. Nevertheless, the induced morphism on global sections

$$(\cdot)^2 : \Gamma(X, \mathcal{O}_X^\times) \rightarrow \Gamma(X, \mathcal{O}_X^\times)$$

need not be an epimorphism. The associated connecting morphism

$$\partial : \Gamma(X, \mathcal{O}_X^\times) \rightarrow \mathcal{H}^1(X, \mu_2)$$

assigns to each nowhere-vanishing holomorphic function its obstruction to admitting a global square root. By exactness, a function $f \in \Gamma(X, \mathcal{O}_X^\times)$ admits a global holomorphic square root if and only if $\partial(f) = 0$.

The following examples make this obstruction explicit.

5.1 Square Roots on a Simply Connected Domain

Example 5.1.1. Consider a set $X = \mathbb{C} \setminus (-\infty, 0] \subset \mathbb{C}^\times$, equipped with the euclidean topology. Since X is contractible, it is simply connected. Let $f \in \mathcal{O}_X^\times(X)$ be a nowhere-vanishing holomorphic function. Since the squaring morphism

$$(\cdot)^2 : \mathcal{O}_X^\times \rightarrow \mathcal{O}_X^\times$$

is an epimorphism of sheaves, there exists an open cover $\mathcal{U} = \{U_k\}_{k \in I}$ (for example $U_k := ((-k, k) \times (-k, k)) \setminus (-\infty, 0]$) of X and local sections

$$g_k \in \mathcal{O}_X^\times(U_k)$$

such that

$$g_k^2 = f|_{U_k}$$

for every $k \in I$.

On an intersection $U_k \cap U_j$, one has

$$\left(\frac{g_k}{g_j}\right)^2 = 1.$$

Hence

$$\varepsilon_{kj} := \frac{g_k}{g_j} \in \mu_2(U_k \cap U_j).$$

On triple intersections,

$$\varepsilon_{kj}\varepsilon_{jl} = \frac{g_k g_j}{g_j g_l} = \varepsilon_{kl}.$$

Thus $\varepsilon = (\varepsilon_{kj})$ is a Čech 1-cocycle with values in μ_2 . Its cohomology class

$$[\varepsilon] \in \check{H}^1(\mathcal{U}, \mu_2)$$

is the obstruction to gluing the local square roots.

Indeed, suppose that

$$[\varepsilon] = 0.$$

Then there exists a Čech 0-cochain

$$(\eta_k)_{k \in I}, \quad \eta_k \in \mu_2(U_k),$$

such that

$$\varepsilon_{kj} = \eta_k \eta_j^{-1}.$$

Define modified local square roots

$$\tilde{g}_k := \eta_k^{-1} g_k.$$

Then

$$\tilde{g}_k^2 = g_k^2 = f|_{U_k},$$

since $\eta_k^2 = 1$, and on overlaps

$$\frac{\widetilde{g}_k}{\widetilde{g}_j} = \eta_k^{-1} \varepsilon_{kj} \eta_j = 1.$$

Hence the sections $\widetilde{g}_k$ agree on intersections and therefore glue to a global section

$$g \in \mathcal{O}_X^\times(X)$$

satisfying

$$g^2 = f.$$

Conversely, if a global square root g exists, then choosing $g_k = g|_{U_k}$ gives $\varepsilon_{kj} = 1$ on every intersection. Thus the obstruction class vanishes. Since μ_2 is locally constant and X is contractible, the comparison theorem gives

$$\check{H}^1(X, \mu_2) \cong \mathcal{H}^1(X, \mu_2) \cong H_{\text{sing}}^1(X, \{\pm 1\}) = 0.$$

Therefore the image of $[\varepsilon]$ in

$$\check{H}^1(X, \mu_2) = \varinjlim_{\mathcal{V}} \check{H}^1(\mathcal{V}, \mu_2)$$

vanishes. Hence there exists a refinement $\mathcal{V}$ on $\mathcal{U}$ such that the image of $[\varepsilon]$ in $\check{H}^1(\mathcal{V}, \mu_2)$ is zero. Applying the preceding gluing argument to the restrictions of the local square roots to $\mathcal{V}$ yields a global holomorphic square root of f . Consequently, every nowhere-vanishing holomorphic function on X admits a global holomorphic square root.

5.2 A Non-Trivial Square-Root Obstructions on $\mathbb{C}^\times$

Example 5.2.1. Consider $X = \mathbb{C}^\times = \mathbb{C} \setminus \{0\}$, equipped with the Euclidean topology, which is paracompact by Theorem A.4.33, and locally contractible.

Consider the cover $\mathcal{U} := \{U_k\}_{k \in \{1, \dots, m\}}$ of open radial sectors of X with

$$U_k := \left\{ r e^{i\varphi} \mid r \in \mathbb{R}_{>0}, \quad \varphi \in \left(\frac{2\pi k}{m} - \frac{3\pi}{2m}, \frac{2\pi k}{m} + \frac{3\pi}{2m} \right) \right\}$$

for $m \geq 3$ fixed, and indices are taken cyclically, so that $m+1 = 1$. The assumption $m \geq 3$ ensures that the associated Čech nerve is a cycle graph. For each k ,

$$\begin{aligned} U_k \cap U_{k+1} &\neq \emptyset \\ U_k \cap U_l &= \emptyset \text{ if } |(k-l) \bmod m| > 1, \end{aligned}$$

and hence there are no non-empty triple intersections.

Since all U_k and all non-empty finite intersections are contractible, and since μ_2 is a locally constant sheaf, the cover is acyclic by Theorem 4.3.22. By the Leray Theorem 4.3.26,

$$\check{H}^1(\mathcal{U}, \mu_2) \cong \mathcal{H}^1(X, \mu_2).$$

The associated Čech nerve provides a combinatorial model for the computation of the obstruction group.

As there are no non-empty triple intersections, the associated Čech nerve is a 1-dimensional simplicial complex, after passing to the normalised Čech complex. Its geometric realisation therefore corresponds to a graph with vertex set

$$V := \{v_k | k \in \{1, \dots, m\}\}$$

and edge set

$$E := \{e_k := (v_k v_{k+1}) | k \in \{1, \dots, m\}\},$$

which is a cycle graph. Every vertex v_k corresponds to an open set U_k , and every edge e_k corresponds to an intersection $U_k \cap U_{k+1}$.

Since there are no non-empty triple intersections, the Čech differential δ^1 vanishes, hence every 1-cochain is a 1-cocycle. A 1-cochain is determined by assigning signs $\sigma_{k,k+1} \in \mu_2$ to the edges of the cycle graph. A 0-cochain assigns elements of μ_2 to the vertices of the cycle graph, and its coboundary modifies the edge labels accordingly. Since the associated Čech nerve is a cycle graph, every cocycle is determined up to coboundary by its monodromy around the unique cycle,

$$\prod_{k=1}^m \sigma_{k,k+1} \in \mu_2.$$

This identifies

$$\check{H}^1(\mathcal{U}, \mu_2) \cong \mu_2 \cong \mathbb{Z}/2\mathbb{Z}.$$

Let $f \in \mathcal{O}_X^\times(X)$. As each U_k is contractible, there exists a local square root $g_k \in \mathcal{O}_X^\times(U_k)$ with

$$g_k^2 = f|_{U_k}.$$

On intersections $U_k \cap U_{k+1}$, the quotient

$$\sigma_{k,k+1} := g_{k+1}g_k^{-1}$$

satisfies $\sigma_{k,k+1} \in \mu_2(U_k \cap U_{k+1})$. Thus the local square roots define a Čech 1-cocycle $\sigma \in \check{Z}^1(\mathcal{U}, \mu_2)$. Changing the local square roots by signs changes σ by a coboundary

$$g_k \mapsto \varepsilon_k g_k.$$

Therefore the obstruction to gluing the g_k to a global square root is the cohomology class

$$[\sigma] \in \check{H}^1(\mathcal{U}, \mu_2).$$

Under the above identification, this class is trivial if and only if

$$\prod_{k=1}^m \sigma_{k,k+1} = 1.$$

To exhibit the non-trivial class, consider the sheaf of local square roots of the identity function z ,

$$T(U) := \{s \in \mathcal{O}(U) \mid s^2 = z|_U\}.$$

This is a μ_2 -torsor which represents a non-trivial cohomology class.

Equivalently, one could say that T is the pullback of the covering p along the identity morphism

$$z : X \rightarrow \mathbb{C}^\times$$

with

$$\begin{aligned} p : \mathbb{C}^\times &\rightarrow \mathbb{C}^\times \\ w &\mapsto w^2. \end{aligned}$$

Claim: The torsor T represents the non-trivial element of $\check{H}^1(\mathcal{U}, \mu_2)$.

Assume T is trivial. Then there exists a global section $s \in T(X)$ by Theorem 2.3.23, that is, a holomorphic map

$$\begin{aligned} s : \mathbb{C}^\times &\rightarrow \mathbb{C}^\times \\ s(z)^2 &= z \end{aligned}$$

for all $z \in \mathbb{C}^\times$ such that $p \circ s = \text{id}_X$. Applying the induced maps to fundamental groups, a section would induce

$$p_* \circ s_* = \text{id}_*,$$

however one computes

$$\begin{aligned} p_* : \pi_1(\mathbb{C}^\times) &\rightarrow \pi_1(\mathbb{C}^\times) \\ n &\mapsto 2n, \end{aligned}$$

which is multiplication by 2 with no right inverse:

The morphism

$$\begin{aligned} p : \mathbb{C}^\times &\rightarrow \mathbb{C}^\times \\ z &\mapsto z^2 \end{aligned}$$

is a two-sheeted covering (since every point $z \in \mathbb{C}^\times$ has two preimages, $\sqrt{z}$ and $-\sqrt{z}$). A generator of the fundamental group $\pi_1(\mathbb{C}^\times) \cong \mathbb{Z}$ is $\gamma(t) = e^{2\pi it}$. Applying p yields

$$p \circ \gamma(t) = (e^{2\pi it})^2 = e^{4\pi it}.$$

Therefore

$$p_*([\gamma]) = 2[\gamma].$$

Since $\pi_1(\mathbb{C}^\times) \cong \mathbb{Z}$ by Example A.4.49, the induced morphism $p_* : \mathbb{Z} \rightarrow \mathbb{Z}$ is $n \mapsto 2n$ which has no right inverse as a group homomorphism. Thus, no such section exists and T cannot be trivial.

Under the identification

$$\check{H}^1(\mathcal{U}, \mu_2) \cong \mu_2,$$

the torsor T represents the non-trivial class. Hence any cocycle representing T satisfies

$$\prod_{k=1}^m \sigma_{k,k+1} = -1.$$

Thus the associated cohomology class in $\check{H}^1(\mathcal{U}, \mu_2)$ is non-trivial. While the computations above show that not all functions have a global square root, one might ask if there exist functions that admit a global square root, and if so, what characterises them as such.

The obstruction admits a concrete analytic interpretation in terms of winding numbers. The Čech cohomology class records the parity of the winding number around the puncture for meromorphic functions. Moreover, this connects directly to the Laurent series:

Assume furthermore that $f \in \mathcal{O}_X^\times(X)$ extends meromorphically to a neighbourhood of 0. Let

$$f(z) = \sum_{n \in \mathbb{Z}} a_n z^n$$

be the Laurent expansion around 0, with

$$\kappa := \min\{n \mid a_n \neq 0\}.$$

Note that such a κ exists since f is meromorphic by assumption. Factoring z^κ out gives

$$f(z) = z^\kappa \cdot u(z)$$

where $u(z)$ is holomorphic and non-vanishing at 0. The integer κ is the order of f at 0; equivalently, it is the winding number of $f(\gamma)$ around 0 for a positively oriented small loop γ around the puncture.

If there exists a function g such that $g^2 = f$, then necessarily

$$g(z) = z^{\frac{\kappa}{2}} \cdot \sqrt{u(z)}.$$

Since $u(0) \neq 0$, one can define a holomorphic logarithm locally. Any possible obstruction lies purely in the monomial part z^κ . On $\mathbb{C}^\times$ $z^{\frac{\kappa}{2}}$ is single valued if and only if κ is even.

Thus the obstruction depends only on the induced representation

$$\pi_1(X) \cong \mathbb{Z} \rightarrow \mu_2,$$

which records the parity of the winding number. The Čech obstruction class corresponds exactly to the parity of the winding number, recovering the identification

$$\mathcal{H}^1(X, \mu_2) \cong \check{H}^1(X, \mu_2) \cong \operatorname{Hom}(\pi_1(X), \mu_2) \cong \mathbb{Z}/2\mathbb{Z}.$$

5.3 Obstructions on the Twice–Punctured Plane

Example 5.3.1. Consider the topological space $X := \mathbb{C} \setminus \{0, 1\}$, equipped with the Euclidean topology, and the long exact sequence induced by (5.0.1)

$$\dots \rightarrow \check{H}^0(X, \mathcal{O}_X^\times) \xrightarrow{(\cdot)^2} \check{H}^0(X, \mathcal{O}_X^\times) \xrightarrow{\check{\partial}} \check{H}^1(X, \mu_2) \rightarrow \check{H}^1(X, \mathcal{O}_X^\times) \rightarrow \dots$$

where $\check{\partial}$ is the connecting morphism from Construction 4.3.7. Since

$$\check{H}^0(X, \mathcal{O}_X^\times) = \Gamma(X, \mathcal{O}_X^\times) = \mathcal{O}_X^\times(X),$$

and

$$\check{H}^0(X, \mu_2) = \Gamma(X, \mu_2) = \mu_2(X),$$

the connecting morphism $\check{\partial}$ measures the obstruction to taking global square roots. Consider the following cover $\mathcal{U}$ with open sets

$$U_1 := (-\infty, 0) \times (-\infty, \infty)$$

$$U_2 := (0, 1) \times (-\infty, \infty)$$

$$U_3 := (1, \infty) \times (-\infty, \infty)$$

$$U_4 := (-\infty, \infty) \times (0, \infty)$$

$$U_5 := (-\infty, \infty) \times (-\infty, 0)$$

Then $U_i \cap U_j \neq \emptyset$ for $i \in \{1, 2, 3\}$ and $j \in \{4, 5\}$ and $U_m \cap U_n = \emptyset$ otherwise, and there are no triple overlaps, after passing to the normalised Čech nerve. Since X is paracompact by Theorem A.4.33 and all non-empty intersections are contractible,

and since μ_2 is a locally constant sheaf, Theorem 4.3.22 applies, that is, $\mathcal{U}$ is acyclic. Thus, Theorem 4.3.26 applies, that is,

$$\check{H}^1(\mathcal{U}, \mu_2) \cong \mathcal{H}^1(X, \mu_2).$$

The nerve $N(\mathcal{U})$ is a bipartite graph $K_{2,3}$ with vertices $V = \{v_1, v_2, v_3, v_4, v_5\}$ where v_k corresponds to U_k (with $k \in \{1, \dots, 5\}$) and edges $E = \{e_{ij} := v_i v_j\}$. As the first Betti number is

$$\beta_1(K_{2,3}) = |E| - |V| + 1 = 6 - 5 + 1 = 2,$$

the cycle space has dimension 2, and one may choose the independent cycles $l_1 := v_1 v_5 v_2 v_4 v_1$, and $l_2 := v_2 v_5 v_3 v_4 v_2$.

A Čech 0-cochain is a choice of signs $\varepsilon_1, \dots, \varepsilon_5$ with $\varepsilon_k \in \{\pm 1\}$. Let $f \in \mathcal{O}_X^\times(X)$. Since all sets U_i are contractible, there exist local square roots $g_i \in \mathcal{O}_X^\times(U_i)$ such that

$$g_i^2 = f|_{U_i}.$$

On overlaps, one may define Čech 1-cocycles with values in μ_2 as

$$\sigma_{ij} := g_i g_j^{-1}.$$

Since $\mu_2 = \{\pm 1\}$, one has $\sigma_{ij}^{-1} = \sigma_{ij}$. Every 1-cochain is a 1-cocycle, since there are no triple overlaps. Changing the local choices $g_k \mapsto \varepsilon_k g_k$ modifies the cocycle by a coboundary, making the obstruction to gluing local square roots the cohomology class

$$[\sigma] \in \check{H}^1(\mathcal{U}, \mu_2).$$

By Theorem 4.3.15

$$|\check{N}(\mathcal{U})| \simeq X.$$

Since $K_{2,3}$ has first Betti number 2,

$$H_{\text{sing}}^1(|\check{N}(\mathcal{U})|, \mu_2) \cong \mu_2^2.$$

One can define a μ_2 -torsor T over X as

$$T(U) = \{s \in \mathcal{O}_X(U) \mid s^2 = z(z-1)|_U\}$$

and show that T is a non-trivial μ_2 -torsor:

Assume that there exists a global section $s \in T(X)$, that is, a holomorphic map

$$\begin{aligned} s : \mathbb{C} \setminus \{0, 1\} &\rightarrow \mathbb{C}^\times \\ s(z)^2 &= z(z-1) \end{aligned}$$

for all $z \in \mathbb{C} \setminus \{0, 1\}$, and denote

$$\begin{aligned} f : X &\rightarrow \mathbb{C}^\times \\ z &\mapsto z(z-1) \end{aligned}$$

and the squaring morphism

$$\begin{aligned} q : \mathbb{C}^\times &\rightarrow \mathbb{C}^\times \\ w &\mapsto w^2. \end{aligned}$$

Then

$$(q \circ s)(z) = s(z)^2 = f(z),$$

and f would admit a lift through q . Furthermore, f induces a morphism

$$f_* : \pi_1(X) \rightarrow \pi_1(\mathbb{C}^\times) \simeq \mathbb{Z},$$

and

$$q_* \circ s_* = f_* \tag{5.3.2}$$

on the level of fundamental groups. Let γ_0 denote a loop around 0, and let γ_1 denote a loop around 1. Then

$$f_*([\gamma_l]) = 1 \in \mathbb{Z}$$

for $l \in \{0, 1\}$, hence $\text{im}(f_*) = \mathbb{Z}$. However, $\text{im}(q_*) = 2\mathbb{Z}$, which is a contradiction, since 5.3.2 would force

$$\mathbb{Z} = \text{im}(f_*) \subseteq \text{im}(q_*) = 2\mathbb{Z}.$$

Therefore, no such global section s exists, and T is a non-trivial μ_2 -torsor. Hence it represents a non-zero class in $\check{H}^1(X, \mu_2) \cong \mu_2^2$, corresponding to the two independent parity obstructions associated with loops around 0 and 1.

In contrast to Example 5.2.1, one has two obstructions. This raises the question

whether there exist functions that have a square root on a small neighbourhood around the punctures and cannot be continued globally. In terms of Čech cohomology, the identification

$$\check{H}^1(X, \mu_2) \cong \mathcal{H}^1(X, \mu_2) \cong H_{\text{sing}}^1(|\check{N}(\mathcal{U})|, \mu_2) \cong \mu_2^2 \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$$

gives the parity of the winding number around the respective puncture.

As before, assume that $f \in \mathcal{O}_X^\times(X)$ extends meromorphically to a neighbourhood of each puncture $x \in \{0, 1\}$, and let

$$f(z) = \sum_{n \in \mathbb{Z}} a_n (z - x)^n$$

be the Laurent expansion of a meromorphic function f around one of the punctures, that is, $x \in \{0, 1\}$, and let

$$\kappa := \min\{n \mid a_n \neq 0\}.$$

Factoring out $(z - x)^\kappa$ gives

$$f(z) = (z - x)^\kappa \cdot u(z)$$

where $u(z)$ is holomorphic and non-vanishing at x . The order κ records the local winding behaviour of f around x . Without loss of generality, let κ_1 be the winding number around 0, and κ_2 the winding number around 1, then

$$f(z) = z^{\kappa_1} \cdot (z - 1)^{\kappa_2} \cdot \tilde{u}(z).$$

Since f extends meromorphically across every puncture, and $\kappa_k = \text{ord}_{p_k}(f)$, the quotient

$$u(z) := f(z) \prod_{k=1}^m (z - p_k)^{-\kappa_k}$$

has removable singularities at all p_k . Hence u extends to a holomorphic function that is nowhere vanishing in a neighbourhood of every puncture. If there exists a function g such that $g^2 = f$, then

$$g(z) = z^{\frac{\kappa_1}{2}} \cdot (z - 1)^{\frac{\kappa_2}{2}} \cdot \sqrt{\tilde{u}(z)}.$$

One can define a holomorphic logarithm locally since $\tilde{u}(0), \tilde{u}(1) \neq 0$. Thus, a global square root exists if and only if κ_1 and κ_2 are even. If κ_1 (respectively κ_2) is odd, then no holomorphic square root exists on any punctured neighbourhood of 0 (respectively 1). Consequently, if both κ_1 and κ_2 are odd, then no global square root exists on X .

Assume that κ_1 is even and κ_2 is odd. Let $B_\varepsilon(0) \setminus \{0\}$ for $1 > \varepsilon > 0$ small enough be an open neighbourhood of 0, then $1 \notin B_\varepsilon(0) \setminus \{0\}$. Moreover, the factor $(z - 1)^{\kappa_2}$ is holomorphic and non-vanishing on $B_\varepsilon(0)$, and one can define a local square root of

$$f(z) = z^{\kappa_1} \cdot (z - 1)^{\kappa_2} \cdot \tilde{u}(z)$$

on $B_\varepsilon(0) \setminus \{0\}$, since $(z - 1)^{\kappa_2}$ is holomorphic and non-vanishing on $B_\varepsilon(0) \setminus \{0\}$.

5.4 Generalisation to n -th Roots on Finitely Punctured Complex Planes

The preceding examples suggest a common mechanism: the obstruction to extracting roots of holomorphic functions is governed by monodromy around punctures. This section aims to make this precise and unify the analytic, topological, and categorical perspectives. First of all, one might notice that

$$(\mathbb{C} \setminus \{0\}) \cap (\mathbb{C} \setminus \{1\}) = \mathbb{C} \setminus \{0, 1\}$$

and that $\mathbb{C} \setminus \{0\} \simeq \mathbb{C} \setminus \{1\}$.

One might ask how the results change for more punctures of the complex plane, or under a change of a morphism between the sheaves of groups.

Indeed, one can generalise the considerations as follows: Let $X := \mathbb{C} \setminus \{p_1, \dots, p_m\}$ where $p_1, \dots, p_m$ are pairwise distinct, equipped with the Euclidean topology. Topologically, X depends only on the number of m punctures and is homotopy equivalent to the wedge of m circles. However, the punctures cannot in general be moved to $0, \dots, m - 1$ without changing the holomorphic structure. Accordingly, the topological computation of $\mathcal{H}^1(X, \mu_n)$ will be performed independently of the precise positions of the punctures. For the special configuration $p_k = k - 1$, one may

visualise this computation using the following ladder cover.

Instead of

$$(\cdot)^2 : \mathcal{O}_X^\times \rightarrow \mathcal{O}_X^\times,$$

one may consider any morphism that gives a morphism of sheaves of abelian groups, such as

$$(\cdot)^n : \mathcal{O}_X^\times \rightarrow \mathcal{O}_X^\times$$

with $n \in \mathbb{N}_{\geq 2}$.

Consider the structure sheaf

$$\mathcal{O}_X : \text{Open}(X)^{\text{op}} \rightarrow \text{CRing}$$

where

$$\mathcal{O}_X^\times(U) = \{f : U \rightarrow \mathbb{C}^\times \text{ holomorphic}\}$$

is the sheaf of units, and

$$\mu_n := \ker((\cdot)^n : \mathcal{O}_X^\times \rightarrow \mathcal{O}_X^\times).$$

This yields a short exact sequence

$$1 \rightarrow \mu_n \rightarrow \mathcal{O}_X^\times \xrightarrow{(\cdot)^n} \mathcal{O}_X^\times \rightarrow 1$$

since $(\cdot)^n$ is locally surjective as a sheaf-morphism, as every nowhere-vanishing holomorphic function admits a local holomorphic logarithm, hence a local n -th root; 1 denotes the trivial group sheaf. This yields a long exact sequence

$$\dots \rightarrow \check{H}^0(X, \mathcal{O}_X^\times) \xrightarrow{(\cdot)^n} \check{H}^0(X, \mathcal{O}_X^\times) \xrightarrow{\check{\partial}} \check{H}^1(X, \mu_n) \rightarrow \check{H}^1(X, \mathcal{O}_X^\times) \rightarrow \dots$$

where $\check{\partial}$ is the connecting morphism as in Construction 4.3.7.

For the remainder of the explicit cover and the Mayer–Vietoris computations, assume $p_k = k - 1$ for $k = 1, \dots, m$. The general case can be recovered from the topological description and the winding–number criterion. Consider the following

ladder cover $\mathcal{U}$ (identifying $\mathbb{C} \simeq \mathbb{R}^2$), with the open sets

$$\begin{aligned} U_1 &:= (-\infty, 0) \times (-\infty, \infty) \\ U_k &:= (k-2, k-1) \times (-\infty, \infty) \\ U_{m+1} &:= (m-1, \infty) \times (-\infty, \infty) \\ U_{m+2} &:= (-\infty, \infty) \times (-\infty, 0) \\ U_{m+3} &:= (-\infty, \infty) \times (0, \infty) \end{aligned} \tag{5.4.1}$$

with $k \in \{2, \dots, m\}$. For $I := \{1, \dots, m+1\}$ and $J := \{m+2, m+3\}$, the intersections $U_{ij} := U_i \cap U_j \neq \emptyset$ for all $i \in I$ and $j \in J$, and empty otherwise. Furthermore, there are no triple overlaps.

Since X is locally contractible, paracompact by Theorem A.4.33, μ_n is locally constant, and the cover sets U_i are contractible, $\mathcal{U}$ is acyclic. Therefore, Theorem 4.3.26 applies, that is,

$$\check{H}^1(\mathcal{U}, \mu_n) \cong \mathcal{H}^1(X, \mu_n).$$

The associated abstract nerve $\check{N}(\mathcal{U})$ is a 1-dimensional simplicial complex. More precisely, it is the complete bipartite graph $K_{m+1,2}$ with vertex set $V = V_I \dot{\cup} V_J$.

Since $|V| = m+3$ and $|E| = 2(m+1)$,

$$\beta_1(K_{m+1,2}) = 2(m+1) - (m+3) + 1 = m.$$

Thus the cycle space has rank m . One possible basis is given by the cycles

$$\mathfrak{L}_i := (v_{m+3}v_iv_{m+2}v_{i+1}v_{m+3})$$

for $i \in I \setminus \{m+1\}$ from the consecutive vertical strips.

Since X deformation retracts onto a wedge of m circles, one has

$$\pi_1(X, x_0) \cong F_m$$

the free group on m generators. Since μ_n is a constant sheaf on X , the comparison

theorem and the homotopy equivalence $X \simeq \bigvee_{k=1}^m S^1$ give

$$\begin{aligned} \mathcal{H}^1(X, \mu_n) &\cong \operatorname{Hom}(H_1^{\text{sing}}(X, \mathbb{Z}), \mu_n) \\ &\cong \operatorname{Hom}(\mathbb{Z}^m, \mu_n) \\ &\cong \mu_n^m \\ &\cong (\mathbb{Z}/n\mathbb{Z})^m. \end{aligned}$$

Equivalently, because every homomorphism from F_m to the abelian group μ_n factors through the abelianisation $F_m^{\text{Ab}} \cong \mathbb{Z}^m$,

$$\mathcal{H}^1(X, \mu_n) \cong \operatorname{Hom}(\pi_1(X, x_0), \mu_n).$$

For each $k \in \{1, \dots, m\}$, let

$$X_k := \mathbb{C} \setminus \{p_k\}.$$

Then

$$X = \mathbb{C} \setminus \{p_1, \dots, p_m\} = \bigcap_{k=1}^m X_k.$$

Thus the m -punctured plane may be viewed set-theoretically as the intersection of m once-punctured planes.

For each X_k , consider the four-set cover $\mathcal{W}_k = \{W_l^{(k)} \mid l \in \{1, 2, 3, 4\}\}$ with

$$\begin{aligned} W_1^{(k)} &:= (-\infty, k-1) \times (-\infty, \infty) \\ W_2^{(k)} &:= (k-1, \infty) \times (-\infty, \infty) \\ W_3^{(k)} &:= (-\infty, \infty) \times (-\infty, 0) \\ W_4^{(k)} &:= (-\infty, \infty) \times (0, \infty). \end{aligned}$$

The nerve of $\mathcal{W}^{(k)}$ is a cycle graph, as in the once-punctured situation of Example 5.2.1. Hence $\mathcal{W}^{(k)}$ isolates the single cycle surrounding the puncture p_k and can be used to compute the corresponding obstruction.

Restricting each $\mathcal{W}^{(k)}$ to

$$X = \bigcap_{k=1}^m X_k$$

gives a cover of X . The ladder cover $\mathcal{U}$ is adapted simultaneously to these once-punctured models and separates the m individual cycles associated with the punc-

tures.

By applying Theorem 4.3.18 iteratively to suitable open sets of the rung decomposition $V_k \subseteq X$, one can decompose $\mathcal{H}^1(X, \mu_n)$ into contributions associated with the individual punctures. Each such contribution is subsequently identified with the cohomology of the corresponding once-punctured model X_k .

The only empty pairwise intersections are

$$W_1^{(k)} \cap W_2^{(k)}, \quad W_3^{(k)} \cap W_4^{(k)}$$

and there are no triple overlaps.

Since all $W_l^{(k)}$ and all intersections are locally contractible, and paracompact by Theorem A.4.33, every $\mathcal{W}^{(k)}$ is a good cover by Theorem 4.3.22, and hence

$$\check{H}^N(\mathcal{W}^{(k)}, \mu_n) \cong \mathcal{H}^N(X_k, \mu_n)$$

by Theorem 4.3.26. The corresponding Čech nerve $|\check{N}(\mathcal{W}^{(k)})|$ is the geometric realisation of a 1-dimensional simplicial complex, hence corresponds to the cycle graph $G^{(k)} = (V^{(k)}, E^{(k)})$ with

$$\begin{aligned} V^{(k)} &= \left\{ v_l^{(k)} \mid l \in \{1, 2, 3, 4\} \right\} \\ E^{(k)} &= \left\{ e_{l, l+1 \bmod 4} := (v_l, v_{l+1 \bmod 4}) \mid l \in \{1, 2, 3, 4\} \right\}. \end{aligned}$$

Consider a morphism of sheaves of abelian groups

$$(\cdot)^n : \mathcal{O}_{X_k}^\times \rightarrow \mathcal{O}_{X_k}^\times$$

where

$$\mathcal{O}_{X_k}^\times(U) = \{f : U \rightarrow \mathbb{C}^\times \text{ holomorphic}\}$$

is a sheaf of abelian groups under pointwise multiplication, and

$$\ker((\cdot)^n) =: \mu_n.$$

This gives rise to a short exact sequence

$$1 \rightarrow \mu_n \rightarrow \mathcal{O}_{X_k}^\times \xrightarrow{(\cdot)^n} \mathcal{O}_{X_k}^\times \rightarrow 1,$$

which in turn motivates a long exact sequence

$$\dots \rightarrow \mathcal{O}_{X_k}^\times \xrightarrow{(\cdot)^n} \mathcal{O}_{X_k}^\times \xrightarrow{\check{\delta}} \check{H}^1(X_k, \mu_n) \rightarrow \check{H}^1(X_k, \mathcal{O}_{X_k}^\times) \rightarrow \dots$$

with connecting morphism $\check{\delta}$ as in Construction 4.3.7.

Choose local n -th roots $g_l^{(k)} \in \mathcal{O}_{X_k}^\times(W_l^{(k)})$ such that $(g_l^{(k)})^n = f|_{W_l^{(k)}}$. A Čech 0-cochain is a choice of n -th roots $\varepsilon_1, \dots, \varepsilon_4$ with $\varepsilon_l \in \mu_n$.

On overlaps, one may define Čech 1-cocycles with values in μ_n as

$$\sigma_{l,l+1} := g_{l+1} \cdot g_l^{-1}.$$

Since there are no non-empty triple intersections after passing to the normalised Čech complex,

$$\check{C}^2(\mathcal{W}^{(k)}, \mu_n) = 0,$$

hence every 1-cochain is a 1-cocycle. A change of local choices

$$g_l \mapsto \varepsilon_l \cdot g_l$$

is a modification of a cocycle by a coboundary.

Since the cover $\mathcal{W}^{(k)}$ is good, the Nerve Theorem 4.3.15 implies that

$$\left| \check{N}(\mathcal{W}^{(k)}) \right| \simeq X_k.$$

As $W_l^{(k)} \cap W_{l+1}^{(k)} \neq \emptyset$ and as all other intersections are empty, the nerve is a cycle graph, hence homotopy-equivalent to S^1 . Together with the fact that μ_n is locally constant, and the monodromy representation

$$\mathcal{H}^1(S^1, \mu_n) \cong \text{Hom}(\pi_1(S^1), \mu_n) \cong \text{Hom}(\mathbb{Z}, \mu_n) \cong \mu_n,$$

this yields

$$\mathcal{H}^1(S^1, \mu_n) \cong \mathcal{H}^1(X_k, \mu_n) \cong \check{H}^1(\mathcal{W}^{(k)}, \mu_n) \cong \mu_n.$$

One can define a μ_n -torsor T over X_k as the sheaf of local n -th roots of the function

$$\begin{aligned} h : X_k &\rightarrow \mathbb{C}^\times \\ z &\mapsto z - (k - 1) \end{aligned}$$

as

$$T(U) = \{s \in \mathcal{O}_{X_k}(U) \mid s^n = h|_U\}.$$

By Theorem 2.3.23, one can show that T is a non-trivial μ_n -torsor by showing that there exists no global section.

Assume that T is trivial, and that there exists a global section s with

$$\begin{aligned} s : X_k &\rightarrow \mathbb{C}^\times \\ s(z)^n &= z - (k - 1) \end{aligned}$$

for all $z \in X_k$. However, a lift s of h along q , that is, $q \circ s = h$ with

$$\begin{aligned} q : \mathbb{C}^\times &\rightarrow \mathbb{C}^\times \\ w &\mapsto w^n \end{aligned}$$

would force the induced morphism on fundamental groups to satisfy $q_* \circ s_* = h_*$. The morphism h is a translation, hence induces an isomorphism

$$h_* : \pi_1(X_k) \xrightarrow{\sim} \pi_1(\mathbb{C}^\times).$$

On the other hand,

$$q_* : \pi_1(\mathbb{C}^\times) \rightarrow \pi_1(\mathbb{C}^\times)$$

is multiplication by n :

The morphism q is an n -sheeted covering since every point $z \in \mathbb{C}^\times$ has n preimages. In particular, if γ_k is a generator of $\pi_1(X_k)$, and if γ is a generator of $\pi_1(\mathbb{C}^\times)$, it holds that

$$h_*([\gamma_k]) = [\gamma].$$

A generator of the fundamental group $\pi_1(\mathbb{C}^\times) \cong \mathbb{Z}$ is $\gamma(t) = e^{2\pi it}$. Applying q yields

$$q \circ \gamma(t) = \left(e^{2\pi it}\right)^n = e^{2\pi int}.$$

Therefore,

$$q_*([\gamma]) = n[\gamma].$$

Since

$$\pi_1(X_k) \cong \pi_1(\mathbb{C}^\times) \cong \mathbb{Z}$$

by Example A.4.49, the induced morphism $q_* : \mathbb{Z} \rightarrow \mathbb{Z}$ is $N \mapsto nN$, which has no right inverse as a group homomorphism. Thus, no such section exists and T cannot be trivial.

Since

$$\mathcal{H}^1(X_k, \mu_n) \cong \mu_n \cong \mathbb{Z}/n\mathbb{Z},$$

there are n isomorphism classes of μ_n -torsors on X_k , consisting of the trivial class and $(n - 1)$ non-trivial classes.

Now consider a **rung decomposition** $\{V_i\}_{i \in \{0, \dots, m+1\}}$ of X with subsets

$$V_k := ((k - 2, k) \times (-\infty, \infty)) \setminus \{k - 1\} \subseteq X$$

$$V_0 := (-\infty, 0) \times (-\infty, \infty)$$

$$V_{m+1} := (m - 1, \infty) \times (-\infty, \infty)$$

for all $k \in \{1, \dots, m\}$. Define inductively

$$Y_0 := V_0 \quad Y_r := Y_{r-1} \cup V_r$$

for $r \in \{1, \dots, m + 1\}$. Since

$$Y_{r-1} \cap V_r = V_r \cap V_{r-1},$$

each new set V_r intersects only with V_{r-1} , ensuring that no compatibility conditions propagate across intersections. Applying Theorem 4.3.18 inductively on Y_r gives

$$\begin{aligned} 0 \rightarrow \mathcal{H}^0(Y_r, \mu_n) &\rightarrow \mathcal{H}^0(Y_{r-1}, \mu_n) \oplus \mathcal{H}^0(V_r, \mu_n) \rightarrow \mathcal{H}^0(Y_{r-1} \cap V_r, \mu_n) \rightarrow \dots \\ \dots \rightarrow \mathcal{H}^1(Y_r, \mu_n) &\rightarrow \mathcal{H}^1(Y_{r-1}, \mu_n) \oplus \mathcal{H}^1(V_r, \mu_n) \rightarrow \mathcal{H}^1(Y_{r-1} \cap V_r, \mu_n) \rightarrow \dots \end{aligned}$$

All intersections $Y_{r-1} \cap V_r = (r - 2, r - 1) \times (-\infty, \infty)$ are contractible, hence

$$\mathcal{H}^1(Y_{r-1} \cap V_r, \mu_n) = 0.$$

Moreover, since Y_{r-1} , V_r , and $Y_{r-1} \cap V_r$ are connected and μ_n is locally constant, one has

$$\mathcal{H}^0(Y_{r-1}, \mu_n) \cong \mathcal{H}^0(V_r, \mu_n) \cong \mathcal{H}^0(Y_{r-1} \cap V_r, \mu_n) \cong \mu_n,$$

and the restriction map

$$\mathcal{H}^0(Y_{r-1}, \mu_n) \oplus \mathcal{H}^0(V_r, \mu_n) \rightarrow \mathcal{H}^0(Y_{r-1} \cap V_r, \mu_n)$$

is surjective. Since $\mathcal{H}^1(Y_{r-1} \cap V_r, \mu_n) = 0$, exactness of the Mayer-Vietoris sequence yields a short exact sequence

$$0 \rightarrow \mathcal{H}^1(Y_r, \mu_n) \rightarrow \mathcal{H}^1(Y_{r-1}, \mu_n) \oplus \mathcal{H}^1(V_r, \mu_n) \rightarrow 0$$

and consequently,

$$\mathcal{H}^1(Y_r, \mu_n) \cong \mathcal{H}^1(Y_{r-1}, \mu_n) \oplus \mathcal{H}^1(V_r, \mu_n)$$

in each step, resulting in

$$\mathcal{H}^1(X, \mu_n) \cong \bigoplus_{r=1}^{m+1} \mathcal{H}^1(V_r, \mu_n) \cong \bigoplus_{r=1}^m \mathcal{H}^1(V_r, \mu_n).$$

Intuitively, every V_k isolates one puncture, contributing independently to $\mathcal{H}^1(X, \mu_n)$.

For every $k \in \{1, \dots, m\}$, translation by $-(k-1)$ identifies V_k with the punctured strip

$$(-1, 1) \times \mathbb{R} \setminus \{0\}.$$

Since the open strip is homeomorphic to $\mathbb{R}^2$, with the origin fixed under a suitable homeomorphism, one obtains $V_k \simeq \mathbb{C}^\times$. Likewise, $X_k \simeq \mathbb{C}^\times$ by translation.

Hence

$$\mathcal{H}^1(V_k, \mu_n) \cong \mathcal{H}^1(X_k, \mu_n) \cong \mathbb{Z}/n\mathbb{Z}.$$

Together with the preceding Mayer-Vietoris decomposition, this gives

$$\mathcal{H}^1(X, \mu_n) \cong \bigoplus_{k=1}^m \mathcal{H}^1(V_k, \mu_n) \cong (\mathbb{Z}/n\mathbb{Z})^m.$$

Thus each V_k , and hence each puncture p_k contributes one independent factor to

the obstruction group. The once-punctured space X_k provides the corresponding model for this factor, since

$$\mathcal{H}^1(V_k, \mu_n) \cong \mathcal{H}^1(X_k, \mu_n) \cong \mathbb{Z}/n\mathbb{Z}.$$

Consequently, the existence of a global n -th root is equivalent to the triviality of the associated class in $\check{H}^1(X, \mu_n)$, that is, to the vanishing of all its components in $\mathbb{Z}/n\mathbb{Z}$. Equivalently, each puncture contributes one independent obstruction, while the contractible overlaps in the Mayer–Vietoris decomposition introduce no additional $\mathcal{H}^1$ -contributions.

The cohomological obstruction is reflected analytically in the local behaviour of f near punctures: The induced morphism

$$\begin{aligned} q_* : \mathbb{Z} &\rightarrow \mathbb{Z} \\ N &\mapsto nN \end{aligned}$$

has no right inverse group homomorphism, but if the winding number is a multiple of n , the homotopy class lies in the image and a global n -th root exists, that is, the identification

$$\check{H}^1(X, \mu_n) \cong \mu_n^m \cong (\mathbb{Z}/n\mathbb{Z})^m \cong \text{Hom}(\pi_1(X), \mu_n)$$

records the winding number modulo n around each puncture. Here, $\text{Hom}(\pi_1(X), \mu_n)$ denotes the group of monodromy representations assigning to each loop the root of unity by analytically continuing the local n -th roots.

If f extends meromorphically to the punctures, let

$$f(z) = \sum_{N \in \mathbb{Z}} a_N^{(k)} (z - p_k)^N$$

be the Laurent expansion around p_k , where (k) is indexing the coefficients, with set

$$\kappa_k := \min\{N \in \mathbb{Z} \mid a_N^{(k)} \neq 0\} = \text{ord}_{p_k}(f).$$

If f does not extend meromorphically to p_k , κ_k is not well-defined. Since $\kappa_k = \text{ord}_{p_k}(f)$,

the quotient

$$u(z) := f(z) \prod_{k=1}^m (z - p_k)^{-\kappa_k}$$

extends holomorphically across every puncture and is nowhere vanishing. Factoring out the principal parts at each puncture yields a decomposition

$$f(z) = u(z) \cdot \prod_{k=1}^m (z - p_k)^{\kappa_k}$$

where $u(z)$ is holomorphic and nowhere vanishing on X . If g is a function such that $g^n = f$, then

$$g(z) = (u(z))^{\frac{1}{n}} \cdot \prod_{k=1}^m (z - p_k)^{\frac{\kappa_k}{n}}.$$

Since $u(z)$ is nowhere vanishing, one can define a holomorphic logarithm locally on simply connected subsets of X , hence a local n -th root of $u(z)$ exists. Therefore, all obstructions lie in the monomial part, which is single-valued if and only if

$$\kappa_k \equiv 0 \pmod{n}$$

for all $k \in \{1, \dots, m\}$. Thus, a global n -th root exists if and only if all winding numbers are divisible by n . In particular, on a punctured neighbourhood $B_\varepsilon(p_k) \setminus \{p_k\}$, a single-valued holomorphic n -th root exists if and only if $\kappa_k \equiv 0 \pmod{n}$. The preceding discussion can be summarised as follows:

Theorem 5.4.2. Let $X := \mathbb{C} \setminus \{p_1, \dots, p_m\}$, where the points $p_1, \dots, p_m$ are pairwise distinct, and let $n \geq 2$. For $f \in \mathcal{O}_X^\times(X)$, let

$$\check{\partial}(f) \in \check{H}^1(X, \mu_n)$$

be the connecting class associated to the short exact sequence

$$1 \rightarrow \mu_n \rightarrow \mathcal{O}_X^\times \xrightarrow{(\cdot)^n} \mathcal{O}_X^\times \rightarrow 1.$$

For each k , choose a positively oriented simple loop γ_k enclosing p_k and no other

puncture, and define

$$\kappa_k(f) := \text{wind}_0(f \circ \gamma_k) = \frac{1}{2\pi i} \int_{\gamma_k} \frac{f'(z)}{f(z)} dz.$$

Then the following statements are equivalent:

1. There exists $g \in \mathcal{O}_X^\times(X)$ such that $g^n = f$.
2. The obstruction class $\check{\partial}(f) \in \check{H}^1(X, \mu_n)$ is trivial.
3. One has

$$\kappa_k(f) \equiv 0 \pmod{n}$$

for every $k \in \{1, \dots, m\}$. Moreover,

$$\check{H}^1(X, \mu_n) \cong \mathcal{H}^1(X, \mu_n) \cong \text{Hom}(\pi_1(X), \mu_n) \cong (\mathbb{Z}/n\mathbb{Z})^m,$$

and under this identification,

$$\check{\partial}(f) = (\kappa_1(f) \bmod n, \dots, \kappa_m(f) \bmod n).$$

If f extends meromorphically across every puncture, then

$$\kappa_k(f) = \text{ord}_{p_k}(f),$$

and hence f admits a global holomorphic n -th root if and only if

$$\text{ord}_{p_k}(f) \equiv 0 \pmod{n}$$

for every k .

Remark 5.4.3. Note, that the preceding example, as well as the generalisation, depend not only on the underlying set, but also on the chosen topology.

If one chose, for example, the discrete topology, the topological definition of the winding number would immediately collapse, and every loop would have winding number 0. The analytic definition would lose its meaning as well, since contour integration fundamentally relies on the Euclidean topology of $\mathbb{C}$.

A Preliminaries

This chapter provides a brief overview of the necessary algebraic and topological foundations used throughout this thesis, and aims to fix the notation and conventions. Section [A.1](#) revisits several set-theoretic and graph-theoretic foundations. In Section [A.2](#), central concepts of complex analysis such as contour integrals, the Laurent series and the winding number are briefly revisited. Section [A.3](#) discusses briefly fundamental algebraic objects, and Section [A.4](#) revisits basic topological notions and concepts, as well as algebraic topological concepts such as simplicial complexes and fundamental groups.

A.1 A Small Selection of Set–theoretic and Graph–theoretic Foundations

This section collects a number of auxiliary notions from set theory and graph theory, that are used throughout the thesis, beginning with the Axiom of Choice, which underlies several existence results in category theory and homological algebra.

Axiom A.1.1 (Axiom of choice). Let $\mathcal{M}$ be a collection of disjoint non-empty sets. There exists a set M consisting of exactly one element from each element of $\mathcal{M}$. In particular, for each $m \in \mathcal{M}$ the intersection $m \cap M$ contains exactly one element (see [\[39, Ch. 1, §9, p. 59\]](#)).

One of the most important consequences of the Axiom of Choice is Zorn’s Lemma, which is used to establish the existence of maximal objects.

Lemma A.1.2 (Zorn). Any inductively ordered poset $(A, \leq)$ has a maximal element (see [\[40\]](#)).

Proof. Let

$$T := \{C \subseteq A : C \text{ is a chain in } A\}.$$

For each $C \in T$, define

$$\begin{aligned}\overline{U}_C &:= \{x \in A : x \text{ is an upper bound of } C\} \\ U_C &:= \{x \in A : x \text{ is a strict upper bound of } C\} = \overline{U}_C \setminus C.\end{aligned}$$

Then

$$\overline{U}_C = U_C \cup \{\max C\}$$

if the maximum element $\max C$ of C exists, and

$$\overline{U}_C = U_C$$

otherwise. Therefore, if $x \in \overline{U}_C$, $y \in A$, and $x < y$, then $y \in U_C$. If $\max C < y$ then $y \notin C$. Moreover, if $C_1, C_2 \in T$ and $U_{C_1} \not\subseteq U_{C_2}$, then $C_1 \cap \overline{U}_{C_2} = \emptyset$.

Indeed, if $x \in C_1 \cap \overline{U}_{C_2}$, then any $y \in U_{C_2}$ satisfies that $x < y$ by definition of U_{C_1} , and hence $y \in U_{C_1}$, which contradicts the assumption $U_{C_1} \not\subseteq U_{C_2}$. The Axiom of Choice yields a choice function f_0 for the family of non-empty subsets X of A , that is, $f_0(X) \in X$ for any such X . One defines a function

$$f : T \rightarrow A$$

satisfying that, for each $C \in T$,

$$f(C) := \begin{cases} f_0(U_C) \in U_C \subseteq A \setminus C & U_C \neq \emptyset \\ \max C \in C \cap \overline{U}_C & U_C = \emptyset. \end{cases}$$

If $U_C = \emptyset$, the assumption that $(A, \leq)$ is inductively ordered, implies that C has an upper bound $x \in A$ that is not strict, which satisfies $x \in C$ and must be the maximum element of C . By the construction of f , if $C_1, C_2 \in T$ and $U_{C_1} = U_{C_2} \neq \emptyset$, then $f(C_1) = f(C_2) \in U_{C_1} = U_{C_2}$.

Let C_0 denote the set of all $C \in T$ satisfying that for $S \subseteq C$ and $U_S \not\subseteq U_C$, $f(S) \in C$.

Let D denote the set of all $C \in C_0$ such that $C' \in C_0$ implies $C \subseteq C' \cup U_{C'}$, and let $C^* := \bigcup D$.

One proves that $C^* \in D$, by verifying the defining conditions of D as follows:

- Let $C' \in C_0$. Then each $C \in D$ satisfies $C \subseteq C' \cup U_{C'}$, and therefore $C^* \subseteq C' \cup U_{C'}$.
- For $C, C' \in D$ with $x \in C$ and $y \in C'$. Now $C \subseteq C' \cup U_{C'}$, therefore, either $x, y \in C'$ or $x \in U_{C'}$. Hence $y < x$, implying $C^* \in T$.
- Suppose that $S \subseteq C^*$ and

$$U_S \not\subseteq U_{C^*} = \bigcap_{C \in D} U_C.$$

Then $U_S \not\subseteq U_C$ for some $C \in D$. Now $S \cap U_C = \emptyset$, while

$$S \subseteq C^* \subseteq C \cup U_C.$$

Therefore $S \subseteq C$. Since $S \subseteq C$, one has $f(S) \in C \subseteq C^*$. Therefore $f(S) \in C^*$.

If $U_{C^*} = \emptyset$, then $f(C^*) = \max C^*$ is a maximal element of A , since otherwise, $C^* < y$ for some $y \in A$ and hence $y \in U_{C^*}$, which is a contradiction.

Let $u := f(C^*) \in U_{C^*}$ and $C^{**} := C^* \cup \{u\}$. Therefore $u = \max C^{**} \notin C^*$, and $C^{**} \not\subseteq C^*$, and $C^{**} \in T$ as $C^* \in T$. It is left to show that $C^{**} \in D$:

- Suppose that $S \subseteq C^{**}$ and $U_S \not\subseteq U_{C^{**}}$. Then $S \cap \overline{U_{C^{**}}} = \emptyset$, and hence $u \notin S$. This implies that $S \subseteq C^*$, and hence $U_{C^*} \subseteq U_S$.

If $U_S \subseteq U_{C^*}$, then $U_S = U_{C^*} \neq \emptyset$ and

$$f(S) = f(C^*) = u \in C^{**}.$$

If $U_S \not\subseteq U_{C^*}$, then $f(S) \in C^* \subseteq C^{**}$. Hence $f(S) \in C^{**}$.

- Let $C' \in C_0$. As $C^* \subseteq C' \cup U_{C'}$, it suffices to show that $u \in C' \cup U_{C'}$, or equivalently, $u \in C'$ if $u \notin U_{C'}$. Since $u \in U_{C^*}$, $U_{C^*} \not\subseteq U_{C'}$. Hence $C^* \cap U_{C'} = \emptyset$, while $C^* \subseteq C' \cup U_{C'}$ as above. Therefore $C^* \subseteq C'$, and $u = f(C^*) \in C'$.

However, $C^{**} \in D$ contradicts

$$C^{**} \not\subseteq C^* = \bigcup D.$$

Hence $U_{C^*} = \emptyset$, and therefore $\max C^*$ is a maximal element of A (see [40]). □

The following notions concern partially ordered sets and directed systems, which play a central role in the theory of filtered colimits.

Definition A.1.3 (Directed poset). Let $(I, \leq)$ be a poset, that is, a set with a partial order relation. It is called **directed** if for every non-empty finite subset $\{i_1, \dots, i_n\} \subseteq I$, there exists an element $k \in I$ such that $i_j \leq k$ for all $j \in \{1, \dots, n\}$ (see [8, Ch. 6, §5, Def. 6.5.15]).

Closely related to directed posets is the notion of cofinality, which allows one to replace an indexing system by a sufficiently large subcollection without changing the resulting colimit.

Definition A.1.4 (Cofinal). Let $(M, \leq)$ be a preordered set, that is, a set M equipped with a binary relation $\leq$ that is reflexive and transitive but not necessarily antisymmetric. A subset $S \subseteq M$ is **cofinal** in M if for all $m \in M$ there exists an $s \in S$ such that $m \leq s$ (see [51]).

Before turning to graph-theoretic notions, the definition of arbitrary Cartesian products of sets shall be recalled.

Definition A.1.5 (Cartesian product). Let J be an index set and let A be a set. A J -tuple of elements of A is formally a function

$$a : J \rightarrow A.$$

For $j \in J$, one denotes the value of a at j as a_j (instead of $a(j)$) and calls it the **j-th coordinate** of a . The function itself is often represented by the tuple $(a_j)_{j \in J}$.

For an indexed family of sets $(A_j)_{j \in J}$, the (infinite) **Cartesian product** is the set of all J -tuples $(a_j)_{j \in J}$ such that $a_j \in A_j$ for each $j \in J$ (see [39, Ch. 2, §19, p. 113]).

The remaining definitions introduce graph-theoretic terminology required for the discussion of diagrams and simplicial constructions, as well as providing the language for describing combinatorial structures such as the Čech nerve.

Definition A.1.6 (Graph). A (simple undirected) **graph** G is a pair of sets (V, E) , where V denotes the set of **vertices**, and E denotes the set of **edges**, where every edge is an unordered pair of distinct vertices of G (see [21, Ch. 1, §1, p. 1]).

A **subgraph** H of $G = (V, E)$ consists of a subset of vertices W , and a subset of edges F on the vertices W (see [21, Ch. 1, §2, p. 3]).

The notion of paths and cycles is fundamental for describing connectedness, and especially for the study of the Čech nerve in Chapter 5.

Definition A.1.7 (Path, cycle). A **path** in a graph from a vertex v to a vertex w is a sequence of vertices, starting with v and ending at w , such that any two consecutive vertices are distinct, and there exists an edge between any two consecutive vertices (see [21, Ch. 1, §2, p. 4]).

A **cycle** is a path starting and ending at a vertex v .

Paths naturally lead to the notion of connectedness.

Definition A.1.8 (Component). A graph is said to be **connected**, if there exists a path between any two vertices. A **component** of a graph is the maximal connected subgraph. The number of components of a graph G is denoted by $c(G)$, or simply c if it is clear what graph is considered (see [21, Ch. 1, §2, p. 4]).

An important class of graphs used in the study of planes with multiple punctures in Chapter 5, are bipartite graphs.

Definition A.1.9 (Bipartite Graph). A graph $G = (V, E)$ is bipartite, if there exist two disjoint subsets $A, B \subseteq V$ such that $A \dot{\cup} B = V$, and such that every edge $e \in E$ has one endpoint in A , and one in B . A complete bipartite graph is denoted as $K_{|A|,|B|}$ (see [21, Ch. 1, §3, p. 4])

For applications in category theory, it is convenient to allow multiple directed edges between vertices.

Definition A.1.10 (Directed multigraph). A **directed multigraph** is a pair (V, E) , where V is a set of vertices, and E is a set of directed edges, such that every edge e has a source vertex s and a target vertex t for $s \neq t$. One may write this as $e : s \rightarrow t$ or $e = (s, t)$ (see [8, Ch. 2, §2, Ex. 2.2.22]).

Remark A.1.11. 1. If between any *ordered* pair of vertices (s, t) there exists at most one edge $e : s \rightarrow t$, a graph is called **directed**.

2. A **loop** is an edge $e : v \rightarrow v$.

3. If one allows a graph to have multiple edges between two vertices, it is called a **multigraph**. Following [25], loops are not allowed in multigraphs. Other authors allow loops in multigraphs.

4. A graph that permits multiple edges between two vertices, as well as loops, is called a **pseudograph**.

For reference see [25, Ch. 2, p. 10].

Finally, the notion of Cartesian product of graphs shall be fixed, since the terminology varies between sources, and some limit constructions rely on this notion.

Definition A.1.12 (Cartesian product of graphs). Let $\Gamma := (V, E)$ and $\Lambda := (W, F)$ be directed graphs. The **Cartesian product** $\Gamma \otimes \Lambda$ is the directed graph

$$\Gamma \otimes \Lambda := (V \times W, V \times F \sqcup E \times W),$$

that is, there exists an edge $(v, w) \rightarrow (v', w')$ in $\Gamma \otimes \Lambda$ if and only if one of the following cases is true:

1. $v = v'$ and there exists an edge $w \rightarrow w'$ in Λ .

2. There exists an edge $v \rightarrow v'$ in Γ and $w = w'$ in Λ .

For reference, see [8, Ch. 6, §6, Def. 6.6.1].

The following example illustrates the concept of the Cartesian product of graphs concretely:

Example A.1.13. The Cartesian product of the two directed graphs

$$\Gamma := v_1 \longrightarrow v_2 \longrightarrow v_3 \quad \text{and} \quad \Lambda := \begin{array}{c} w_1 \\ \downarrow \\ w_2 \end{array}$$

is

$$\Gamma \otimes \Lambda = \begin{array}{ccccc} (v_1, w_1) & \longrightarrow & (v_2, w_1) & \longrightarrow & (v_3, w_1) \\ \downarrow & & \downarrow & & \downarrow \\ (v_1, w_2) & \longrightarrow & (v_2, w_2) & \longrightarrow & (v_3, w_2) \end{array}$$

(see [8, Ch. 6, §6, Def. 6.6.1]).

Remark A.1.14. The Cartesian product of graphs is not the product in the categorical sense. The categorical product of graphs is the tensor product, which is defined differently (see [8, Ch. 6, §6, Rem. 6.6.2]).

A.2 Analytic Foundations

One of the essential results of this thesis is the connection between Laurent expansions and the obstruction to extracting holomorphic roots. In this section, the notation for winding numbers and contour integrals is fixed, and basic analytic definitions such as holomorphy, isolated singularities, and Laurent series are recalled.

Definition A.2.1 (Holomorphy). Let $f : U \rightarrow \mathbb{C}$ be a function, with $U \subseteq \mathbb{C}$ open, and $z_0 \in U$. The **derivative** of f at a point $z_0 \in \mathbb{C}$ is defined as the limit

$$f'(z_0) = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}.$$

If the limit exists, f is said to be **complex differentiable at** z_0 .

A function is **holomorphic** on an open set U if it is complex differentiable at every point of U , and **holomorphic at a point** z_0 if it is holomorphic on some neighbourhood of z_0 (see [38, Ch. 10, §5, Def. 5.1]).

A basic analytic construction used repeatedly in the study of holomorphic roots is the holomorphic logarithm.

Example A.2.2 (Holomorphic logarithm). Let $f : X \rightarrow \mathbb{C}$ be a nowhere-vanishing holomorphic function. A **holomorphic logarithm** of f is a holomorphic function $g : X \rightarrow \mathbb{C}$ such that $e^g(z) = f(z)$ for all $z \in X$ (see [38, Ch. 10, §5, p. 211/212]).

A holomorphic logarithm need not exist globally. On simply connected domains however, it does exist, as Theorem A.4.52 shows.

Definition A.2.3 (Continuous function). A function $f : \mathbb{C} \rightarrow \mathbb{C}$ is continuous at z_0 if

$$\lim_{z \rightarrow z_0} f(z) = f(z_0)$$

exists and is well-defined.

The set of continuous functions $f : X \rightarrow Y$ where $X, Y \subseteq \mathbb{C}$ is denoted as $C^0(X, Y)$, or as $C^0(X)$ if $Y = \mathbb{C}$. The set of n -times continuous differentiable functions is denoted as $C^n(X, Y)$.

A function is **piecewise continuous** if it is not continuous in only finitely many points, and continuous everywhere else (see [10, Ch. 14, §1, Def. 14.1.1.3]).

for contour integration, continuity alone is not sufficient.

Definition A.2.4 (Contour). A **contour** in $\mathbb{C}$ is a piecewise C^1 -map

$$\gamma : [a, b] \rightarrow \mathbb{C}$$

(see [10, Ch. 8, §3, Def. 8.3.3.3]).

A standard example of contours used in this thesis are circle lines.

Example A.2.5. The positively oriented circle around $a \in \mathbb{C}$ with radius $r > 0$ is parametrised by

$$\gamma(t) = a + re^{it}, \quad t \in [0, 2\pi],$$

which is a smooth closed contour. The unit circle around 0 is parametrised by e^{it} , or the set-theoretic description S^1 . By definition, γ is a smooth closed contour.

This allows to consider especially circles around punctures in the complex plane, which are used in contour integrals and in Chapter 5.

Definition A.2.6 (Contour integral). Let $U \subseteq \mathbb{C}$ be an open subset, let $f : U \rightarrow \mathbb{C}$ be continuous, and let $\gamma : [a, b] \rightarrow U$ be a contour. The **contour integral of f along γ** is defined as

$$\int_{\gamma} f(z) dz := \int_a^b f(\gamma(t)) \dot{\gamma}(t) dt$$

where $\dot{\gamma}$ denotes the first derivative of γ . If γ is closed, one denotes the contour integral by $\oint$ (see [38, Ch. 10, §6, p. 222]).

Contour integrals not only define integration in the complex plane, but also provide numerical invariants describing the behaviour of closed curves. The most important example is the winding number.

Definition A.2.7 (Winding number). The **winding number of γ around z_0** of a closed, piecewise smooth *loop* $\gamma : [0, 1] \rightarrow \mathbb{C} \setminus \{z_0\}$ is

$$\text{wind}(\gamma, z_0) := \frac{1}{2\pi i} \oint_{\gamma} \frac{dz}{z - z_0}.$$

For a simple closed curve γ and a point z_0 in the bounded component of its complement, the orientation is called **positively oriented**, if $\text{wind}(\gamma, z_0) > 0$, and negatively oriented if $\text{wind}(\gamma, z_0) < 0$ (see [38, Ch. 10, §6, p. 233]).

Many results in complex analysis relate contour integrals around closed loops to the behaviour of holomorphic functions near singular points. This motivates the notion of isolated singularities.

Definition A.2.8 (Isolated singularity). Let f be holomorphic on $U \setminus \{a\}$ for some open neighbourhood $U \subseteq \mathbb{C}$ of a . Then a is called an **isolated singularity of f** if there exists some radius $r > 0$ such that f is holomorphic on $0 < |z - a| < r$ (see [10, Ch. 14, §3]).

Functions with isolated singularities can be analysed locally by means of Laurent series. Their coefficients are obtained by contour integrals and describe both the regular and singular behaviour of the function.

Definition A.2.9 (Laurent Series). Let $f : A \rightarrow \mathbb{C}$ be a complex function that is holomorphic on an **annulus**

$$A = \{z \in \mathbb{C} : r < |z - a| < R\}.$$

Let γ be the positively oriented circle of winding number 1 around a , lying in A .

Then f can be written as **Laurent series** around a :

$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z - a)^n$$

with coefficients

$$c_n = \frac{1}{2\pi i} \oint_{\gamma} \frac{f(z)}{(z - a)^{n+1}} dz.$$

Furthermore,

$$\sum_{n=-\infty}^{-1} c_n (z - a)^n$$

is called **principal part**, and

$$\sum_{n=0}^{\infty} c_n (z - a)^n$$

is called **regular** or **holomorphic part** (see [10, Ch. 14, §3]).

The principal part of the Laurent series describes singularities of the function f . Furthermore, the Laurent series can characterise whether the singularities can be removed.

Definition A.2.10 (Meromorphic). Let $U \subseteq \mathbb{C}$ be open. A function $f : U \rightarrow \mathbb{C} \cup \{\infty\}$ is called **meromorphic** if for every point $u \in U$,

- either f is holomorphic near u
- or u is an **isolated singularity** such that

$$|f(z)| \rightarrow \infty, \quad \text{for } z \rightarrow u$$

(see [10, Ch. 14, §3, Def. 14.3.5.2]).

If near u

$$f(z) = \frac{g(z)}{(z - u)^n}$$

with $g(u) \neq 0$, then u is called a **pole of order n** (see [10, Ch. 14, §3, Def. 14.3.5.4]).

Meromorphic functions are holomorphic away from isolated poles. In many situations one is interested in whether a holomorphic function defined on a smaller domain can be extended beyond its original domain of definition.

Definition A.2.11 (Holomorphic continuation). Let $U, V \subseteq \mathbb{C}$ be open subsets with $U \subseteq V$, and let $f : U \rightarrow \mathbb{C}$ be a holomorphic function. A holomorphic function $F : V \rightarrow \mathbb{C}$ is called a **holomorphic continuation** of f to V if

$$F|_U = f$$

(see [10, Ch. 14, §3, Def. 14.3.3]).

Holomorphic continuations allow unique extensions to a larger domain. An example is the Zeta function in number theory.

A.3 Algebraic Foundations

Throughout this thesis, several algebraic notions are used. This section aims to recall foundational notions such as groups, rings, and ideals, but also spectra.

Groups

Definition A.3.1 (Group). A **group** is a set G , equipped with a binary operation $\star$, such that the following properties are satisfied:

1. For $a, b, c \in G$, $\star$ is associative, that is,

$$(a \star b) \star c = a \star (b \star c) = a \star b \star c.$$

2. There exists a unique element $e \in G$, called the identity, such that

$$e \star a = a \star e = a$$

for all $a \in G$.

3. For each $a \in G$, there exists a unique element a^{-1} such that

$$a \star a^{-1} = a^{-1} \star a = e.$$

A group is called **abelian** if it is commutative (see [29, p. 7]).

A subset $H \subseteq G$ is called a **subgroup** of G if it satisfies all group axioms (see [29, Ch. 1, §1, p. 10]).

A **generator** of a group is a subset $S \subseteq G$ such that every element of G can be written as a finite combination with respect to $\star$ and their inverses. In that case, one writes $G = \langle S \rangle$ (see [17, Ch. 2, §4, p. 61]).

Example A.3.2. If one considers the set $\mathbb{Z}/n\mathbb{Z} := \{\overline{0}, \overline{1}, \dots, \overline{n-1}\}$ equipped with addition, is the **group of integers modulo n** .

In $\mathbb{C}^\times := \mathbb{C} \setminus \{0\}$, the **n -th roots of unity**

$$\mu_n := \{z \in \mathbb{C}^\times \mid z^n = 1\}.$$

equipped with multiplication are a group.

Groups arise throughout the thesis, for instance as fundamental groups, groups of roots of unity, automorphism groups, and kernels of morphisms of group-valued sheaves.

Definition A.3.3 (Group homomorphism). Let $(G, \star_G)$ and $(H, \star_H)$ be groups. A **group homomorphism** is a structure-preserving map

$$\phi : G \rightarrow H$$

such that

$$\phi(a \star_G b) = \phi(a) \star_H \phi(b)$$

for all $a, b \in G$ (see [17, Ch. 1, §6, p. 36]).

A **right inverse** of a group homomorphism $\phi : G \rightarrow H$ is a group morphism $\psi : H \rightarrow G$ such that $(\phi \circ \psi) = \text{id}_H$ (see [17, Ch. 0, §1, p.2]).

Some mathematical objects, such as automorphisms, form a group.

Definition A.3.4 (Automorphism group). An **automorphism** is an isomorphism where the target object and the source object agree, that is, an isomorphism $G \rightarrow G$. Together with the composition of maps $\circ$, automorphisms form a group under composition, called **automorphism group**, denoted by $\text{Aut}(G)$ (see [17, Ch. 4, §4, p. 133]).

Automorphism groups occur naturally, for example as groups of deck transformations and as targets of monodromy representations. Many mathematical objects carry a group action. Morphisms compatible with such actions are called equivariant.

Definition A.3.5 (Equivariant map). Let G be a group acting on sets X and Y , with actions described by

$$\rho_X : G \rightarrow \text{Aut}(X), \quad \rho_Y : G \rightarrow \text{Aut}(Y).$$

A map $f : X \rightarrow Y$ is called **G -equivariant** if

$$f(g \cdot x) = g \cdot f(x)$$

for every $g \in G$ and $x \in X$. Equivalently,

$$f \circ \rho_X(g) = \rho_Y(g) \circ f$$

for every $g \in G$ (see [2, Ch. 9, §9]).

Remark A.3.6. One can replace X by other objects, such as vector spaces, topological spaces, and sheaves, amongst others (see [2, Ch. 9, §9]).

Many invariants in algebraic topology, such as homology and cohomology groups, are abelian groups. Several notions concerning finitely generated abelian groups are therefore needed.

Definition A.3.7 (Torsion). Let A be an abelian group. An element a is called **torsion element** if there exists an integer

$$n \in \mathbb{Z} \setminus \{0\}$$

such that $na = 0$. The set of all torsion elements is called **torsion subgroup** of A , denoted by $\text{Tor}(A)$ (see [17, Ch. 5, §2, p. 158]).

Torsion elements measure the finite part of an abelian group. The remaining free part is described by its rank.

Definition A.3.8 (Rank). Let A be a finitely generated abelian group. By the structure theorem for finitely generated abelian groups, there exists an isomorphism

$$A \cong \mathbb{Z}^r \oplus \text{Tor}(A)$$

for a uniquely determined integer $r \geq 0$. The integer r is called the **rank** of A and is denoted by $\text{rank}(A) := r$ (see [17, Ch. 5, §2, p. 158]).

Free abelian groups form the building blocks of finitely generated abelian groups and appear naturally in chain complexes.

Definition A.3.9 (Free abelian group). A **free abelian group** on a set S is an abelian group $F(S)$ together with an injective map $\iota : S \hookrightarrow F(S)$ such that every element of $F(S)$ can be written uniquely as a finite sum

$$\sum_{s \in S} n_s \iota(s), \quad n_s \in \mathbb{Z}$$

with only finitely many $n_s \neq 0$ (see [17, Ch. 5, §2, p. 158]).

The following theorem shows that every finitely generated abelian group decomposes into a free part and into a torsion part.

Remark A.3.10 (Structure Theorem). Let A be a finitely generated abelian group. Then there exist integers

$$r \geq 0, \quad n_1, \dots, n_k \geq 2$$

such that

$$n_i \mid n_{i+1}$$

for all $1 \leq i < k$, and

$$A \cong \mathbb{Z}^r \oplus \mathbb{Z}/n_1\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/n_k\mathbb{Z}$$

(see [29, Ch. 1, §5, Theorem 5.17]).

Rings

Definition A.3.11 (Ring). A **ring** is a set R , equipped with two binary operations: addition $+$ and multiplication $\cdot$, such that the following axioms are satisfied:

1. $(R, +)$ is an abelian group (see Definition A.3.1) with an additive identity element which is usually denoted by 0.
2. The multiplication $\cdot$ is associative with a multiplicative identity element, usually denoted by 1.
3. The distributive laws hold, in essence, for all $a, b, c \in R$,

$$(a + b) \cdot c = a \cdot c + b \cdot c$$

and

$$a \cdot (b + c) = ab + ac.$$

A ring is **commutative** if the multiplication is commutative (see [2, Ch. 10, §1, Def. 1.3]).

Throughout this thesis, rings are assumed to have a multiplicative identity. A special subset of a ring are ideals, which are essential in ring theory and in objects such as spectra and local rings.

Definition A.3.12 (Ideal). Let R be a ring. A subset $\mathfrak{r}$ of R such that

1. $\mathfrak{r}$ is an additive subgroup of $(R, +)$
2. For every $r \in \mathfrak{r}$ and for every $x \in R$, it holds that $x \cdot r \in \mathfrak{r}$ and $r \cdot x \in \mathfrak{r}$

hold is called **ideal** (see [29, Ch. 3, §1, Def. 1.3]).

An ideal $\mathfrak{r}$ is called **maximal** if the following properties hold:

1. $\mathfrak{r} \neq R$
2. There exists no other ideal $\mathfrak{s}$ of R such that $\mathfrak{r} \subsetneq \mathfrak{s} \subsetneq R$

(see [29, Ch. 3, §3, Def. 3.2]).

A two-sided ideal $\mathfrak{r}$ is called **prime ideal** if $\mathfrak{r} \neq R$ and for all ideals $\mathfrak{a}, \mathfrak{b} \subseteq R$, it holds that if $\mathfrak{a}\mathfrak{b} \subseteq \mathfrak{r}$, it follows that $\mathfrak{a} \subseteq \mathfrak{r}$ or $\mathfrak{b} \subseteq \mathfrak{r}$ (see [17, Ch. 7, §4, p. 260]).

Ideals allow the construction of quotient rings and localisation.

Definition A.3.13 (Local ring). A commutative ring with a unique maximal ideal is called **local** (see [29, Ch. 3, Exercise 14]).

Local rings are essential to locally ringed spaces, which are a natural setting for schemes. Spectra are essential to schemes.

Definition A.3.14 (Spectrum). A **spectrum** of a commutative ring R is the set of all prime ideals of R , denoted by $\text{spec}(R)$ (see [17, Ch. 15, §5, p. 731]).

The spectrum of a ring carries a natural topology, which forms the foundation of algebraic geometry.

Definition A.3.15 (Zariski topology). Let R be a commutative ring. For an ideal $I \subseteq R$, define

$$V(I) := \{p \in \text{spec}(R) \mid I \subseteq p\}.$$

The **Zariski topology** on $\text{spec}(R)$ is the topology whose closed sets are precisely the subsets of the form $V(I)$ (see [17, Ch. 15, §2, p. 676]).

Modules

Definition A.3.16 (Module). Let R be a ring with identity. A **left R -module** is an abelian group $(M, +)$ together with a map

$$R \times M \rightarrow M, \quad (r, m) \mapsto rm,$$

called **scalar multiplication**, such that for all $r, s \in R$ and $m, n \in M$,

$$\begin{aligned}(r + s)m &= rm + sm, \\ r(m + n) &= rm + rn, \\ (rs)m &= r(sm), \\ 1_R m &= m.\end{aligned}$$

A **right R -module** is defined analogously with

$$M \times R \rightarrow M, \quad (m, r) \mapsto mr,$$

satisfying

$$\begin{aligned}m(r + s) &= mr + ms, \\ (m + n)r &= mr + nr, \\ (mr)s &= m(rs), \\ m1_R &= m\end{aligned}$$

(see [2, Ch. 12, §1]).

A.4 Topological Foundations

Topology is fundamental to the constructions developed in this thesis. This section recalls the basic notions of topological spaces, open and closed subsets, neighbourhoods, continuous maps, bases, and covers among other basic concepts and notions, before turning to homotopy theory.

Definition A.4.1 (Topological space). A **topological space** is a pair (X, τ) , where X is a set and $\tau \subseteq \mathcal{P}(X)$ is a set of subsets of X satisfying the following conditions:

1. $\emptyset$ and X are elements of τ .
2. If $T_i \in \tau$ for all $i \in I$, then the union $\bigcup_{i \in I} T_i$ is in τ for any index set I .

3. If $T_i \in \tau$ for all $i \in I$, then the finite intersection $\bigcap_{i \in I} T_i$ is in τ for every finite index set I .

The elements of τ are called **open sets** (see [39, Ch. 2, §12, p. 76]).

Remark A.4.2. In order to improve the reading flow, topological spaces are denoted by X instead of (X, τ) .

While a topology is specified by its open subsets, many topological properties are more naturally formulated in terms of their complements.

Definition A.4.3 (Closed set). Let X be a topological space. A set $U \subseteq X$ is **closed** if $X \setminus U$ is open (see [31, Ch. 1, §1]).

By passing to complements, the axioms for open subsets immediately yield the corresponding closure properties of closed subsets.

Theorem A.4.4. Let X be a topological space. The following conditions hold:

1. $\emptyset$ and X are closed.
2. Arbitrary intersections of closed sets are closed.
3. Finite unions of closed sets are closed

(see [39, Ch. 2, §17, Thm. 17.1]).

Proof. 1. $\emptyset$ and X are complements of open sets respectively (see [39, Ch. 2, §17, Thm. 17.1]).

The proof of statement 2 and 3 uses DeMorgan's laws. A more detailed proof can be found in [39, Ch. 2, §17, Thm. 17.1]. □

Open subsets describe the global structure of a topology. To study the topology locally around a point, one considers neighbourhoods.

Definition A.4.5 (Neighbourhood). Let X be a topological space and $x \in X$. A **neighbourhood** of x is a set U that contains an open set V containing x . The collection of all neighbourhoods of x , denoted by $\mathcal{U}_x$, is called the **neighbourhood system** at x (see [55, Ch. 2, §4, Def. 4.1]).

Remark A.4.6 (Open Neighbourhood). There are variations in the definition of a neighbourhood of a point. Sometimes it is required that the neighbourhood is open. In this thesis, a neighbourhood consisting of open sets is called **open neighbourhood**.

Neighbourhoods provide an equivalent local characterisation of open subsets.

Theorem A.4.7. A subset $V \subseteq X$ is open if and only if there exists an open neighbourhood $U_x \subseteq V$ for all $x \in V$ (see [31, Ch. 1, §1]).

Proof. Let (X, τ) be a topological space and $V \subseteq X$.

" $\Rightarrow$ ": Assume that V is open. For every $x \in V$, choose the open neighbourhood $U_x = V$, since V is open and $x \in V \subseteq V$, V is an open neighbourhood for all its points.

" $\Leftarrow$ ": Assume that for every $x \in V$, there exists an open set $U_x \in \tau$ such that $x \in U_x \subseteq V$. One can express V as the union of those open sets:

$$V = \bigcup_{x \in V} \{x\} \subseteq \bigcup_{x \in V} U_x \subseteq V.$$

Therefore $V = \bigcup_{x \in V} U_x$. Since the union of open sets is open, V is open (see [31, Ch. 1, §1]). □

Describing every open subset individually is often impractical. A topology can instead be determined by a smaller collection of distinguished open subsets.

Definition A.4.8 (Base). Let X be a set. A **base** for a topology on X is a collection $\mathcal{B}$ of subsets $(B_i)_{i \in I}$ of X , called **base elements**, such that

1. For each $x \in X$ there exists a base element with $x \in B_i$.
2. If $x \in B_i$ and $x \in B_j$, then there exists a base element $B_k \subseteq B_i \cap B_j$ such that $x \in B_k$.

The **topology** τ is **generated by** $\mathcal{B}$ as follows: A subset $U \subseteq X$ is open if for each $x \in U$ there exists a base element $B \in \mathcal{B}$ such that $x \in B$ and $B \subseteq U$. Every base element itself is element of τ (see [39, Ch. 2, §13, p. 78]).

The global notion of a base has a corresponding local version at each point.

Definition A.4.9 (Neighbourhood base). A **neighbourhood base** at x in a topological space X is a subcollection $\mathcal{B}_x$ of $\mathcal{U}_x$ such that for any $U \in \mathcal{U}_x$ there exists a $B \in \mathcal{B}_x$ such that $B \subseteq U$. That is,

$$\mathcal{U}_x = \{U \subseteq X \mid B \subseteq U \text{ for some } B \in \mathcal{B}_x\}$$

(see [55, Ch. 2, §4, Def. 4.3]).

This allows one to understand neighbourhood systems as cofinal systems. Let X be an arbitrary topological space and let $\mathcal{U}_x$ be a system of all neighbourhoods of $x \in X$. One can define an order relation $\leq$ via

$$U \leq V \iff V \subseteq U.$$

This gives rise to a neighbourhood base $\mathcal{B} \subseteq \mathcal{U}_x$ such that for any $V \in \mathcal{U}_x$ there exists $B \in \mathcal{B}$ such that $B \subseteq V$. Thus, $\mathcal{B}$ is a *cofinal* subset of $(\mathcal{U}_x, \leq)$.

Having established the local and global description of a topology, one can consider maps that preserve this structure.

Definition A.4.10 (Continuous function). Let X and Y be topological spaces. A function $f : X \rightarrow Y$ is **continuous** if and only if the preimage of any open subset $V \subseteq Y$ is open in X (see [39, Ch. 2, §18, p. 102]).

Continuous maps need not preserve all topological information. The maps that identify two spaces without changing their topological structure are the homeomorphisms.

Definition A.4.11 (Homeomorphism). A **homeomorphism** is a bijective, continuous map

$$f : X \rightarrow Y$$

between topological spaces X and Y , such that f^{-1} is continuous. One says that X and Y are **homeomorphic**, denoted by $X \simeq Y$ (see [39, Ch. 2, §18, p. 105]).

Different topologies may be imposed on the same underlying set. Their relative strength is measured by inclusion.

Definition A.4.12 (Coarseness). Let τ and τ' be two topologies on a set X .

1. If $\tau \subseteq \tau'$, then τ' is **finer** than τ , and τ is **coarser** than τ' .
2. If $\tau \subsetneq \tau'$, then τ' is **strictly finer** than τ , and τ is **strictly coarser** than τ' .

Two topologies are **comparable** if $\tau \subseteq \tau'$ or $\tau' \subseteq \tau$ (see [39, Ch. 2, §12, p. 77]).

The following example illustrates how a topology may be characterised by a universal continuity requirement.

Example A.4.13 (Disjoint union topology). Let $((X_i, \tau_i))_{i \in I}$ be a family of topological spaces and let $X := \coprod_{i \in I} X_i$ be the disjoint union of the underlying sets. The **canonical injection** is defined as follows: for each $i \in I$,

$$\begin{aligned} \iota_i : X_i &\rightarrow X \\ \iota_i(x) &= (x, i). \end{aligned} \tag{A.4.14}$$

The **disjoint union topology** on X is the finest topology on X , such that all canonical injections (A.4.14) are continuous (see [31, Ch. 1, §3], [8, Ch. 6, §5, Thm. 6.5.6]).

An important source of topologies is provided by distance functions.

Definition A.4.15 (Metric space). Let X be a non-empty set and let

$$d : X \times X \rightarrow [0, \infty)$$

be a function such that the following properties hold:

1. $d(x, y) \geq 0$
2. $d(x, y) = 0$ if and only if $x = y$
3. $d(x, y) = d(y, x)$
4. $d(x, z) \leq d(x, y) + d(y, z)$

Then d defines a **metric** on X , making (X, d) a **metric space** (see [10, Ch. 12, §2, Def. 12.2.1]).

Metric spaces carry some very useful properties, such as paracompactness (see Theorem A.4.33), which in turn allows to conclude what a "good" cover is. An essential metric is the euclidean metric:

Example A.4.16 (Euclidean Metric). The **euclidean metric** on $\mathbb{R}^n$ is defined as

$$d(x, y) := \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

(see [10, Ch. 12, §2, p. 677]).

Remark A.4.17. On $\mathbb{C}$, the euclidean metric can be defined as

$$d(z_1, z_2) := \sqrt{(\operatorname{Re}(z_1) - \operatorname{Re}(z_2))^2 + (\operatorname{Im}(z_1) - \operatorname{Im}(z_2))^2}$$

(see [10, Ch. 12, §2, p. 677]).

Every metric determines a topology through its open balls. Conversely, a topological space is called metrisable when its topology arises in this way.

Definition A.4.18 (Metrisable topological space). A topological space (X, τ) is **metrisable** if there exists a metric d that induces τ (see [31, Ch. 1, §2]).

Example A.4.19 (Euclidean topology). The topological space induced by the base of **open balls**

$$B_r(x) := \{y \in \mathbb{R}^n \mid \|y - x\| < r\}$$

with radius $r \in \mathbb{R}_{>0}$ and centre x on $\mathbb{R}^n$ with respect to the euclidean metric is called **euclidean topology** (see [39, Ch. 2, §20, pp. 122]).

Neighbourhoods also describe how a subset sits inside its ambient space. This leads to the notions of closure, interior, exterior, and boundary.

Definition A.4.20 (Closure). Let X be a metrisable topological space, and let $A \subseteq X$. The **closure** of A is the set

$$\overline{A} := \{x \in X \mid U \cap A \neq \emptyset \text{ for every neighbourhood } U \text{ of } x\}$$

(see [31, Ch. 1, §1]).

Remark A.4.21. There exists the notion of the *interior*, the *exterior*, and the *boundary* of a topological space. The interior of a set $U \subseteq X$ is defined as

$$\overset{\circ}{U} := \{x \in U \mid \exists V_x \subseteq U\}$$

where V_x is an open neighbourhood. The exterior is the interior of the complement $U^C := X \setminus U$, and the boundary of U is

$$\partial U := \{x \in X \mid V_x \cap (X \setminus U) \neq \emptyset \text{ and } V_x \cap U \neq \emptyset\}$$

for all open neighbourhoods V_x of x . The characterisation of open sets given in the preceding theorem is equivalent to the characterisation that a topological space is open if and only if it is equal to its interior (compare [39, Ch. 2 §17, p. 95]), [31, Ch. 1, §1]).

Separation properties express the extent to which distinct points or disjoint subsets can be distinguished by open neighbourhoods.

Definition A.4.22 (Hausdorff). A topological space X is called **Hausdorff** if for every pair of distinct points $x, y \in X$ there exist open sets $U, V \subseteq X$ such that

$$x \in U, \quad y \in V, \quad U \cap V = \emptyset$$

(see [31, Ch. 1, §7]).

Example A.4.23. The complex plane $\mathbb{C}$ is Hausdorff with the euclidean topology. For two points $x \neq y$, one can take balls with radius $\frac{d(x,y)}{3}$.

Hausdorffness separates distinct points. The stronger notion of normality requires the separation of disjoint closed subsets.

Definition A.4.24 (Normal topological space). A topological space is called **normal** if for every pair of disjoint closed sets $A, B \subseteq X$, there exist disjoint open sets $U, V \subseteq X$ such that $A \subseteq U$ and $B \subseteq V$ (see [39, Ch. 4, §31, p. 195]).

Normality not only separates closed subsets by open neighbourhoods, but also by continuous functions.

Lemma A.4.25 (Urysohn). Let X be a normal topological space, and let $A, B \subseteq X$ be closed such that $A \cap B = \emptyset$. Then there exists a continuous function

$$f : X \rightarrow [0, 1]$$

with $f(a) = 1$ for all $a \in A$, and $f(b) = 0$ for all $b \in B$ (see [31, Ch. 8, §1]).

Proof. Let A and B be closed disjoint subsets of X . Consider an increasing chain $\mathcal{U} = (A_0, \dots, A_r)$ of subsets of X such that

$$A = A_0 \subsetneq A_1 \subsetneq \dots \subsetneq A_r \subsetneq X \setminus B$$

with

$$\overline{A_{i-1}} \subseteq \overset{\circ}{A}_i.$$

Such intermediate sets can be constructed using normality: if $F \subseteq U$ with F closed and U open, then there exists an open set V such that

$$F \subseteq V \subseteq \overline{V} \subseteq U.$$

Indeed, F and $X \setminus U$ are disjoint closed subsets and can therefore be separated by disjoint open neighbourhoods.

Consider the uniform stair function

$$f(x) = \begin{cases} 1 & x \in A_0 \\ 1 - \frac{k}{r} & x \in A_k \setminus A_{k-1} \\ 0 & x \in A_r^C \end{cases}$$

with $A_{-1} = \emptyset$ and $A_{r+1} = X$. Note, that this stair function is generally not continuous. The idea is to refine the chain successively by inserting intermediate sets,

thereby decreasing the height of the jumps.

Using normality as above, one can insert between every two consecutive members A_{i-1} and A_i an intermediate set A'_i such that

$$\overline{A}_{i-1} \subseteq A'_i \subseteq \overline{A}'_i \subseteq A_i.$$

This yields a refinement

$$(A_0, A'_1, A_1, \dots, A'_r, A_r)$$

of the original chain.

Let $\mathcal{U}_0$ be a chain $(A, X \setminus B)$, and let $\mathcal{U}_{n+1}$ be a refinement of $\mathcal{U}_n$, and let f_n be the uniform stair function on $\mathcal{U}_n$. The sequence of functions $(f_n)_{n \geq 1}$ is pointwise monotone increasing, and bounded above by 1, and especially it is pointwise convergent. The limit function

$$f := \lim_{n \rightarrow \infty} f_n : X \rightarrow [0, 1]$$

satisfies the property that $f|_A \equiv 1$, and $f|_B \equiv 0$.

To show that f is continuous one can choose $\frac{1}{2^{n-1}} < \varepsilon$, as f_n varies on $\mathcal{U}_n$ at most $\frac{1}{2^n}$ (see [31, Ch. 8, §2]). □

Open covers play a central role in local-to-global constructions used throughout this thesis.

Definition A.4.26 (Cover). Let X be a topological space, and let I be an index set. A collection $\mathcal{U}$ of sets $\mathcal{U} := \{U_i \subseteq X, i \in I\}$ is called a **cover** of X if

$$\bigcup_{i \in I} U_i = X$$

holds. A cover is open if all its sets are open (see [39, Ch. 3, §26, p. 164]).

Remark A.4.27. Sometimes, a cover is referred to as *covering*. However, in this thesis the term covering is used for covering spaces.

A cover may be replaced by a finer cover whose members are better adapted to the local structure of the space.

Definition A.4.28 (Refinement). Let X be a topological space, and let $\mathcal{U} := \{U_i\}_{i \in I}$ be a cover of X . A **refinement** of the cover $\mathcal{U}$ is a cover $\mathcal{V} := \{V_j\}_{j \in J}$ such that for every V_j there exists an $k \in I$ such that $V_j \subseteq U_k$ (see [39, Ch. 6, §39, p. 245]).

Of particular importance are refinements that remain locally manageable, in the sense that only finitely many members meet a neighbourhood of any given point.

Definition A.4.29 (Locally finite cover). An open cover of a space X is **locally finite** if there exists a neighbourhood that intersects only finitely many sets in the cover for every $x \in X$ (see [39, Ch. 6, §39 p. 244]).

One way of controlling an open cover globally is to require that finitely many of its members already cover the entire space.

Definition A.4.30 (Compact). A topological space X is **compact** if every cover $\mathcal{U}$ has a finite subcover, that is, a subset of $\mathcal{U}$ covering X (see [31, Ch. 1, §8]).

Paracompactness provides a different form of control: rather than requiring a finite subcover, it requires the existence of a locally finite refinement.

Definition A.4.31 (Paracompact). A topological space X is **paracompact** if every open cover has a locally finite open refinement (see [39, Ch. 6, §41, p. 253]).

In some applications, paracompactness is required not only for the ambient space but also for its subspaces.

Definition A.4.32 (Hereditarily paracompact). A topological space X is **hereditarily paracompact** if every open subset $U \subseteq X$ is paracompact (see [50, p. 1]).

Metric spaces constitute the principal class of paracompact spaces used in this thesis.

Theorem A.4.33 (Stone). Metrisable spaces are paracompact (see [31, Ch. 8 §5]).

Proof. Let X be a metrisable space, let d be a metric inducing its topology, and let $\mathcal{U}$ be an open cover of X . One constructs a locally finite open refinement of $\mathcal{U}$.

Step 1: Construction of a countably locally open refinement.

Choose a well-ordering $<$ of $\mathcal{U}$. For every $U \in \mathcal{U}$ and $n \in \mathbb{N}$ define

$$S_n(U) := \{x \in X \mid B_{\frac{1}{n}}(x) \subseteq U\}$$

and

$$T_n(U) := S_n(U) \setminus \bigcup_{V < U} V.$$

Thus, $T_n(U)$ consists of those points whose $\frac{1}{n}$ -neighbourhood lies in U and which do not belong to any earlier member of the well-ordering.

For fixed n , the sets $T_n(U)$ are uniformly separated. Indeed, let $U, V \in \mathcal{U}$ be distinct, and assume without loss of generality that $U < V$. If $x \in T_n(U)$ and $y \in T_n(V)$, then $B_{\frac{1}{n}}(x) \subseteq U$, whereas $y \notin U$, since $U < V$. Consequently,

$$d(x, y) \geq \frac{1}{n}.$$

Define an open enlargement of $T_n(U)$ by

$$Q_n(U) := \bigcup_{x \in T_n(U)} B_{\frac{1}{3n}}(x).$$

Since $T_n(U) \subseteq S_n(U)$, one has $Q_n(U) \subseteq U$. Hence

$$\mathcal{Q}_n := \{Q_n(U) \mid U \in \mathcal{U}\}$$

is an open refinement of $\mathcal{U}$.

Moreover, $\mathcal{Q}_n$ is locally finite. In fact, a ball of radius $\frac{1}{6n}$ can meet at most one member of $\mathcal{Q}_n$. Otherwise, there would exist distinct $U, V \in \mathcal{U}$ and points $x_U \in T_n(U)$ and $x_V \in T_n(V)$ such that

$$d(x_U, x_V) < \frac{1}{n},$$

contradicting the separation established above.

The family

$$\mathcal{Q} := \bigcup_{n \in \mathbb{N}} \mathcal{Q}_n$$

covers X . Indeed, let $x \in X$, and let $U \in \mathcal{U}$ be the first member of the well-ordering containing x . Since U is open, there exists $n \in \mathbb{N}$ such that

$$B_{\frac{1}{n}}(x) \subseteq U.$$

Hence $x \in S_n(U)$. By the minimality of U , the point x lies in no member $V < U$, and therefore

$$x \in T_n(U) \subseteq Q_n(U).$$

Thus, $\mathcal{Q}$ is an open refinement of $\mathcal{U}$ which is a countable union of locally finite families.

Step 2: Construction of a locally finite refinement.

As above, let

$$\mathcal{Q} := \bigcup_{n \in \mathbb{N}} \mathcal{Q}_n$$

and set

$$V_n := \bigcup_{Q \in \mathcal{Q}_n} Q.$$

For every $Q \in \mathcal{Q}_n$, define

$$Y_n(Q) := Q \setminus \bigcup_{k < n} V_k,$$

and let

$$\mathcal{Y}_n := \{Y_n(Q) \mid Q \in \mathcal{Q}_n\}, \quad \mathcal{Y} := \bigcup_{n \in \mathbb{N}} \mathcal{Y}_n.$$

The family $\mathcal{Y}$ still refines $\mathcal{U}$ and covers X . To see the latter, let $x \in X$, and choose the smallest N such that $x \in Q$ for some $Q \in \mathcal{Q}_N$. Then $x \in Y_N(Q)$.

The family $\mathcal{Y}$ is locally finite. Fix $x \in X$, choose the smallest N for which $x \in Q_0$ for some $Q_0 \in \mathcal{Q}_N$, and, for each $n \in \{1, \dots, N\}$, choose a neighbourhood W_n of x meeting only finitely many members of $\mathcal{Q}_n$. Then

$$W := Q_0 \cap \bigcap_{n=1}^N W_n$$

is a neighbourhood of x . It meets only finitely many members of $\mathcal{Y}_n$ for $n \leq N$. For $n > N$, one has

$$W \subseteq Q_0 \subseteq V_N,$$

whereas every member of $\mathcal{Y}_n$ is disjoint from V_N . Therefore W meets no member of $\mathcal{Y}_n$ for $n > N$.

The sets in $\mathcal{Y}$ need not be open. Since every metrisable space is regular, consider the open cover

$$\mathcal{W} := \{W \subseteq X \mid W \text{ open and } \overline{W} \subseteq U \text{ for some } U \in \mathcal{U}\}.$$

By regularity, $\mathcal{W}$ covers X . Applying the preceding construction to $\mathcal{W}$, choose a locally finite family $\mathcal{Z}$ covering X and refining $\mathcal{W}$. Then

$$\mathcal{R} := \{\overline{Z} \mid Z \in \mathcal{Z}\}$$

is a locally finite closed cover refining $\mathcal{U}$.

It remains to enlarge members of $\mathcal{R}$ to open sets without destroying local finiteness. Let $\mathcal{P}$ be the collection of all open subsets of X that meet only finitely many members of $\mathcal{R}$. Since $\mathcal{R}$ is locally finite, $\mathcal{P}$ is an open cover of X . By the preceding construction, $\mathcal{P}$ has a locally finite, closed refinement $\mathcal{N}$ covering X .

For every $R \in \mathcal{R}$, set

$$\mathcal{N}(R) := \{N \in \mathcal{N} \mid N \subseteq X \setminus R\}$$

and define

$$Q(R) := X \setminus \bigcup_{N \in \mathcal{N}(R)} N.$$

Because $\mathcal{N}$ is a locally finite family of closed sets, the union of every subfamily of $\mathcal{N}$ is closed. Hence $Q(R)$ is open. Moreover, $R \subseteq Q(R)$.

The family $\{Q(R) \mid R \in \mathcal{R}\}$ is locally finite. Indeed, let $x \in X$, and choose a neighbourhood W of x meeting only finitely many members

$$N_1, \dots, N_r$$

of $\mathcal{N}$. Since $\mathcal{N}$ covers X , one has

$$W \subseteq N_1 \cup \cdots \cup N_r.$$

If W meets $Q(R)$, then some N_j meets $Q(R)$. This implies that N_j is not contained in $X \setminus R$, and hence

$$N_j \cap R \neq \emptyset.$$

Each N_j meets only finitely many members of $\mathcal{R}$, so W meets only finitely many sets $Q(R)$.

Finally, for each $R \in \mathcal{R}$, one can choose a member $U(R) \in \mathcal{U}$ such that $R \subseteq U(R)$, and define

$$\mathcal{D} := \{Q(R) \cap U(R) \mid R \in \mathcal{R}\}.$$

Each member of $\mathcal{D}$ is open and contained in a member of $\mathcal{U}$. Since

$$R \subseteq Q(R) \cap U(R),$$

and $\mathcal{R}$ covers X , the family $\mathcal{D}$ covers X . Moreover, $\mathcal{D}$ is locally finite because $\{Q(R) \mid R \in \mathcal{R}\}$ is locally finite.

Thus $\mathcal{D}$ is a locally finite open refinement of $\mathcal{U}$, and therefore X is paracompact (see [39, Ch. 6, §39]). $\square$

Example A.4.34. The complex plane $\mathbb{C}$ is hereditarily paracompact. By Theorem A.4.33, $\mathbb{C}$ is paracompact. Any open subset $U \subseteq \mathbb{C}$ is a metric space with respect to the restricted metric. Therefore every open subset U is paracompact.

Compact metric spaces admit an additional uniformity property for open covers, expressed by the existence of a Lebesgue number.

Lemma A.4.35 (Lebesgue lemma). Let (X, d) be a compact metric space and let $\mathcal{U} = \{U_i\}_{i \in I}$ be an open cover of X . Then there exists a $\lambda \in \mathbb{R}_{>0}$ such that every subset $A \subseteq X$ with diameter $\text{diam}(A) < \lambda$ is contained in some U_i . Such a λ is called **Lebesgue number** (see [9, Ch. 1, §9, Lemma 9.11]).

Proof. For every $x \in X$, choose an index $i(x) \in I$ such that $x \in U_{i(x)}$. Since $U_{i(x)}$

is open, there exists $\varepsilon_x > 0$ such that

$$B_{2\varepsilon_x}(x) \subseteq U_{i(x)}.$$

The balls $B_{\varepsilon_x}(x)$ form an open cover of X . By compactness, there are points $x_1, \dots, x_m \in X$ such that

$$X = \bigcup_{j=1}^m B_{\varepsilon_{x_j}}(x_j).$$

Set

$$\lambda := \min_{1 \leq j \leq m} \varepsilon_{x_j} > 0.$$

Let $A \subseteq X$ satisfy $\varnothing(A) < \lambda$. If $A = \emptyset$, there is nothing to prove. Assuming that $A \neq \emptyset$, one chooses $a \in A$. Since these balls cover X , there exists a j such that

$$a \in B_{\varepsilon_{x_j}}(x_j).$$

For every $y \in A$, one has

$$d(y, x_j) \leq d(y, a) + d(a, x_j) < \lambda + \varepsilon_{x_j} \leq 2\varepsilon_{x_j}.$$

Hence

$$A \subseteq B_{2\varepsilon_{x_j}}(x_j) \subseteq U_{i(x_j)}.$$

Therefore λ is a Lebesgue number of $\mathcal{U}$ (see [9, Ch. 1, §9, Lemma 9.11]). □

Paracompactness and separation properties allow an open cover to be encoded by continuous functions. This is achieved through partitions of unity subordinate to the cover.

Lemma A.4.36 (Existence of partitions of unity). Let X be a paracompact Hausdorff space and let $\{U_i\}_{i \in I}$ be an open cover of X . Then there exists a family of continuous functions $\{\varphi_i : X \rightarrow [0, 1]\}_{i \in I}$ such that

1. $\text{supp}(\varphi_i) \subseteq U_i$ for all $i \in I$
2. the family $\{\text{supp}(\varphi_i)\}_{i \in I}$ is locally finite
3. $\sum_{i \in I} \varphi_i(x) = 1$ for all $x \in X$,

where the sum is finite in a neighbourhood of each point (see [31, Ch. 8, §5]).

Proof. Since X is paracompact and Hausdorff, every open cover admits a locally finite shrinking. Hence there exist locally finite open families $\{V_i\}_{i \in I}$ and $\{W_i\}_{i \in I}$ such that

$$\overline{V_i} \subseteq W_i \subseteq \overline{W_i} \subseteq U_i$$

for every $i \in I$, and such that $\{V_i\}_{i \in I}$ covers X .

By normality and the Urysohn lemma, applied to the disjoint closed subsets $\overline{V_i}$ and $X \setminus W_i$, there exists for each $i \in I$ a continuous function

$$f_i : X \rightarrow [0, 1]$$

such that

$$f_i = 1 \text{ on } \overline{V_i}, \quad f_i = 0 \text{ on } X \setminus W_i.$$

Consequently,

$$\text{supp}(f_i) \subseteq \overline{W_i} \subseteq U_i.$$

The family $\{W_i\}_{i \in I}$ is locally finite and $\text{supp}(f_i) \subseteq \overline{W_i}$, the family $\{\text{supp}(f_i)\}_{i \in I}$ is locally finite. Hence the family of functions $\{f_i\}_{i \in I}$ is locally finite in the sense that every point has a neighbourhood on which all but finitely many f_i vanish.

Hence the sum

$$F(x) := \sum_{i \in I} f_i(x)$$

is well-defined and continuous, and strictly positive for all $x \in X$, as the sets V_i cover X . Define

$$\varphi_i(x) := \frac{f_i(x)}{F(x)}.$$

Then each φ_i is continuous, satisfies $\text{supp}(\varphi_i) \subseteq U_i$, and the family $\{\text{supp}(\varphi_i)\}$ remains locally finite. Moreover,

$$\sum_{i \in I} \varphi_i(x) = 1$$

for all $x \in X$. Thus $\{\varphi_i\}_{i \in I}$ is a partition of unity subordinate to $\{U_i\}$ (see [31, Ch. 8, §5]). □

The preceding results provide the point-set-topological tools needed for the study of covers and nerve constructions.

Connectedness and Homotopy

Besides separation properties, notions describing how a space behaves as a whole are equally important. The first of these is connectedness and its local counterpart.

Definition A.4.37 (Connected). A topological space X is **disconnected** if it is the union of two disjoint non-empty open sets. Otherwise, X is said to be **connected** (see [39, Ch. 3, §23, p. 148]).

A topological space X is **locally connected at** x if every neighbourhood of x contains a connected open neighbourhood of x , that is, if every point x has a neighbourhood base consisting of open connected sets.

A **locally connected space** is a space which is locally connected at every point $x \in X$ (see [39, Ch. 3, §25, p. 161]).

Connectedness alone does not describe how points are joined inside a space. For homotopy theory, the stronger notion of paths is required.

Definition A.4.38 (Path, loop). A **path** between two points x and y in a topological space X is a continuous map

$$\gamma : [0, 1] \rightarrow X$$

such that $\gamma(0) = x$ and $\gamma(1) = y$.

A **loop based at** x_0 in a topological space X is a path

$$l : [0, 1] \rightarrow X$$

such that $l(0) = l(1) = x_0$ (see [39, Ch. 3, §24, p. 155, Ch. 9, §52, p. 331]).

Using paths, one obtains stronger global and local connectedness properties.

Definition A.4.39 (Path-connected). A topological space X is **path-connected** if there exists a path in X between any two points in X .

A topological space is **locally path-connected at** x if every neighbourhood of x contains a path-connected open neighbourhood of x , that is, if the point x has a neighbourhood base consisting of path-connected open sets.

A **locally path-connected space** is a topological space that is locally path-connected at every point x (see [39, Ch. 3, §24, p. 155]).

Paths describe how points are connected. More generally, one is interested in continuous deformations of arbitrary continuous maps.

Definition A.4.40 (Homotopy). Let $f, g : X \rightarrow Y$ be continuous. A **homotopy** between f and g is a continuous map

$$H : X \times [0, 1] \rightarrow Y$$

such that

$$H(x, 0) = f(x)$$

and

$$H(x, 1) = g(x)$$

for all $x \in X$.

Furthermore, homotopy defines an equivalence relation $f \sim g$ for any two continuous maps that are homotopic. The corresponding equivalence classes are called **homotopy classes** (see [39, Ch. 9, §51, p. 323]).

Frequently, such deformations are required to leave a prescribed subspace fixed throughout the homotopy.

Definition A.4.41 (Homotopy relative to a subspace). Let $A \subseteq X$ and let $f, g : X \rightarrow Y$ be continuous maps such that $f|_A = g|_A$. A homotopy

$$H : X \times [0, 1] \rightarrow Y$$

from f to g is called a **homotopy relative to** A if

$$H(a, t) = f(a) = g(a)$$

for all $a \in A$ and all $t \in [0, 1]$. In this case, one writes $f \simeq_A g$ (see [31, Ch. 5, §1, p. 68/69]).

Homotopies also provide a notion of equivalence between topological spaces that preserve their essential topological structure.

Definition A.4.42 (Homotopy equivalent topological spaces). Let $f : X \rightarrow Y$ be a continuous map between topological spaces X and Y . The map f is a **homotopy equivalence** if there exists a continuous map $g : Y \rightarrow X$ such that

$$g \circ f \simeq \text{id}_X$$

and

$$f \circ g \simeq \text{id}_Y$$

are homotopic. The map g is the **homotopy inverse of f** . Two topological spaces are **homotopy equivalent** if there exists a homotopy equivalence between them (see [39, Ch. 9, §58, p. 363]).

An important special case is when a map can be continuously deformed to a constant map.

Definition A.4.43 (Null-homotopic). A continuous map $f : X \rightarrow Y$ is null-homotopic if it is homotopic to a constant map (see [39, Ch. 9, §51, p. 323]).

Applying this idea to the identity map leads to one of the strongest topological properties considered in this thesis.

Definition A.4.44 (Contractible). A topological space $X \neq \emptyset$ is **contractible** if it is homotopy equivalent to a single point, that is, if there exists a continuous map

$$H : X \times [0, 1] \rightarrow X$$

and a fixed point $p \in X$ such that

$$H(x, 0) = x$$

for all $x \in X$ and

$$H(x, 1) = p$$

for all $x \in X$.

A topological space X is **locally contractible at a point** x if for every neighbourhood U_x of x there exists a neighbourhood V_x of x such that the inclusion of $V_x \hookrightarrow U_x$ is null-homotopic in U_x .

A space is **locally contractible** if it is locally contractible at every point x (see [39, Ch. 9, §51, Ex. 3]).

Homotopy classes of loops carry additional algebraic structure, giving rise to one of the central invariants of algebraic topology.

Definition A.4.45 (Fundamental group). Let $\Omega(X, x_0)$ be the set of loops based at x_0 . On homotopy classes relative to the base point, define multiplication by concatenation $*$ as

$$(\gamma_1 * \gamma_2)(t) := \begin{cases} \gamma_1(2t) & \text{for } t \in [0, \frac{1}{2}] \\ \gamma_2(2t - 1) & \text{for } t \in (\frac{1}{2}, 1]. \end{cases}$$

The product of homotopy classes $[\gamma_1]$ and $[\gamma_2]$ is $[\gamma_1 * \gamma_2]$. This gives a group, called **fundamental group**, denoted as $\pi_1(X, x_0)$.

A based continuous map $f : (X, x_0) \rightarrow (Y, y_0)$ induces a group homomorphism

$$f_* : \pi_1(X, x_0) \rightarrow \pi_1(Y, y_0)$$

such that $f(x_0) = y_0$ for $x_0 \in X$ and $y_0 \in Y$, where for any loop γ based at x_0 ,

$$f_*([\gamma]) := [f \circ \gamma]$$

(see [39, Ch. 9, §52, p. 331]).

The fundamental group naturally motivates spaces whose loops are all homotopically trivial.

Definition A.4.46 (Simply connected). A topological space is **simply connected** if it is path-connected, and every loop is null-homotopic (see [39, Ch. 9, §52, p.

333]). A **locally simply connected** topological space X is a topological space that admits a base of simply connected sets.

The following elementary observation relates contractibility to the connectedness properties introduced above.

Lemma A.4.47. If a topological space X is contractible, it follows that it is also path-connected and simply connected (see [39, Ch. 9, §51, p. 329]).

Proof. Let X be a contractible space. Then there exists a homotopy

$$H : X \times [0, 1] \rightarrow X$$

such that $H(x, 0) = x$ and $H(x, 1) = x_0$ for some point x_0 , meaning that the identity map is homotopic to the constant map.

Furthermore, for any $x \in X$,

$$t \mapsto H(x, t)$$

is a path from x to x_0 . Hence X is path-connected.

Let $\gamma : S^1 \rightarrow X$ be an arbitrary loop. Composing with the contraction gives a continuous deformation of the loop to the constant loop at x_0 . Thus, every loop is null-homotopic and X is simply connected. $\square$

Besides global and local simple connectedness, a weaker local condition is needed in the study of covering spaces.

Definition A.4.48 (Semilocally simply connected). A topological space is called **semilocally simply connected** if for every $x \in X$ there exists an open neighbourhood U_x of x such that the induced map

$$\pi_1(U_x, x) \rightarrow \pi_1(X, x)$$

is trivial (see [39, Ch. 13, §82, p. 494/495]).

The fundamental group of the circle is the prototypical example and plays a central role in algebraic topology.

Example A.4.49 (Fundamental group of S^1). The fundamental group of the unit circle S^1 in $\mathbb{C}$ is

$$\pi_1(S^1) \cong \mathbb{Z}$$

and it is generated by $\gamma(t) = e^{2\pi it}$. A proof can be found in [A.4.49](#).

The **winding number** of a loop $\gamma : [0, 1] \rightarrow \mathbb{C} \setminus \{a\}$ around a point a is the integer corresponding to its homotopy class under the canonical isomorphism

$$\pi_1(\mathbb{C} \setminus \{a\}) \cong \mathbb{Z}$$

(see [\[39, Ch. 10, §65, p. 398\]](#)).

Consider the covering map

$$\begin{aligned} p : \mathbb{R} &\rightarrow S^1 \\ x &\mapsto \begin{pmatrix} \cos(2\pi x) \\ \sin(2\pi x) \end{pmatrix}. \end{aligned}$$

Let $e_0 = 0$ and let $b_0 = p(e_0)$, then the preimage

$$p^{-1}(b_0)$$

is the set of integers $\mathbb{Z}$. Note that for any covering map

$$p : E \rightarrow B$$

with $p(e_0) = b_0$, any path

$$f : [0, 1] \rightarrow B$$

beginning at b_0 has a unique lifting to a path $\tilde{f}$ in E beginning at e_0 (as can be seen in [\[39, Ch. 9, §54, Lemma 54.1\]](#)).

One can define the **lifting correspondence** for a covering map

$$p : E \rightarrow B$$

for $b_0 \in B$ and e_0 such that

$$p(e_0) = b_0.$$

Given an element $[f]$ of $\pi_1(B, b_0)$, let $\tilde{f}$ be the lifting of f to a path in E beginning at e_0 , and let

$$\phi([f]) := \tilde{f}(1).$$

Then ϕ is a well-defined set map

$$\phi : \pi_1(B, b_0) \rightarrow p^{-1}(b_0),$$

called **lifting correspondence** derived from the covering map p . Note that the correspondence depends on the choice of e_0 .

If E is simply connected, the lifting correspondence is bijective:

Let $[f]$ and $[g]$ be two elements of $\pi_1(B, b_0)$ such that

$$\phi([f]) = \phi([g]),$$

and let $\tilde{f}$ and $\tilde{g}$ be the liftings of f and of g , respectively, to paths in E beginning at e_0 , then

$$\tilde{f}(1) = \tilde{g}(1).$$

Since E is simply connected, there exists a path homotopy $\tilde{F}$ in E between $\tilde{f}$ and $\tilde{g}$. Then $p \circ \tilde{F}$ is a path homotopy in B between f and g . (Details as to why $\tilde{F}$ is a path homotopy can be found in [39, Ch. 9, §54, Lemma 54.2]).

To show surjectivity, let $e \in p^{-1}(b_0)$. Since E is simply connected, it is path-connected, so there exists a path

$$\tilde{f} : [0, 1] \rightarrow E$$

from e_0 to e . Then

$$f := p \circ \tilde{f}$$

is a loop in B based at b_0 , since

$$f(0) = p(e_0) = b_0, \quad \text{and} \quad f(1) = p(e) = b_0.$$

By uniqueness of path liftings, $\tilde{f}$ is the lift of f beginning at e_0 . Hence

$$\phi([f]) = \tilde{f}(1) = e.$$

Thus ϕ is surjective.

Since $\mathbb{R}$ is simply connected, the lifting correspondence

$$\phi : \pi_1(S^1, b_0) \rightarrow \mathbb{Z}$$

is bijective.

It is left to show that ϕ is a homomorphism.

Given $[f]$ and $[g]$ in $\pi_1(B, b_0)$, let $\tilde{f}$ and $\tilde{g}$ be their respective liftings to paths on $\mathbb{R}$ beginning at 0, and let

$$\tilde{f}(1) = n$$

and

$$\tilde{g}(1) = m,$$

then $\phi([f]) = n$ and $\phi([g]) = m$ by the definition of a lifting correspondence.

Let γ be the path

$$\gamma(t) = n + \tilde{g}(t)$$

on $\mathbb{R}$. Since $p(n + x) = p(x)$ for all $x \in \mathbb{R}$ by the periodicity of $\sin$ and $\cos$, the path γ is a lifting of g , beginning at n .

The product $\tilde{f} * \gamma$ is defined, and it is the lifting of $f * g$ beginning at 0 and ending at $\gamma(1) = n + m$. Then by definition of lifting correspondences,

$$\phi([f] * [g]) = n + m = \phi([f]) + \phi([g]).$$

(see [39, Ch. 9, §54], [39, Ch. 9, §53, Thm 53.1]).

Remark A.4.50. The topological definition of the winding number coincides with the analytic definition. The analytic winding number is invariant under homotopies in $\mathbb{C} \setminus \{a\}$, and additive under concatenation of loops. Consequently, it defines a group homomorphism

$$\pi_1(\mathbb{C} \setminus \{a\}) \rightarrow \mathbb{Z}.$$

The positively oriented loop

$$\gamma_a(t) = a + re^{2\pi it}$$

is mapped to 1. Hence this homomorphism agrees with the isomorphism

$$\pi_1(\mathbb{C} \setminus \{a\}) \rightarrow \mathbb{Z}$$

determined by sending the positively oriented generator to 1. Therefore, the analytic and the topological definitions of the winding number coincide.

Fundamental groups are closely related to covering spaces, which are used in the discussion of monodromy.

Definition A.4.51 (Covering). Let X and Y be topological spaces. A continuous surjective map

$$p : Y \rightarrow X$$

is called a **covering map** if every point $x \in X$ has an open neighbourhood $U \subseteq X$ such that

$$p^{-1}(U) = \bigcup_{a \in A} V_a$$

for a family of pairwise disjoint open subsets $V_a \subseteq Y$, and such that each restriction

$$p|_{V_a} : V_a \xrightarrow{\sim} U$$

as a homeomorphism.

The space Y is called **covering space of X** , and the open sets V_a are called **sheets** over U . The discrete set $p^{-1}(x)$ is called **fibre over x** .

If X is connected, the cardinality of $p^{-1}(x)$ is independent of x ; this cardinality is called the **degree** or **number of sheets** of the covering (see [39, Ch. 9, §53, p. 336]).

As an application of simple connectedness, the following classical result from complex analysis is used in the discussion of holomorphic logarithms and torsors.

Theorem A.4.52. Let $U \subseteq \mathbb{C}$ be a simply connected domain and let $f : U \rightarrow \mathbb{C}$ be nowhere vanishing and holomorphic. Then f admits a holomorphic logarithm (see [38, Ch. 10, §5, p. 211/212]).

Proof. Since f is nowhere vanishing, the quotient $\frac{f'}{f}$ is holomorphic on U . Since U is simply connected, every holomorphic function on U has a primitive holomorphic function by the fundamental theorem of complex analysis. Hence there exists $g : U \rightarrow \mathbb{C}$ such that $g' = \frac{f'}{f}$.

Claim: $\frac{e^g}{f}$ is constant.

Indeed,

$$\left(\frac{e^g}{f}\right)' = \frac{e^g \cdot g' \cdot f - e^g \cdot f'}{f^2} = \frac{e^g \cdot f}{f^2} \left(g' - \frac{f'}{f}\right) = 0.$$

Thus there exists a constant $c \in \mathbb{C}^\times$ such that

$$\frac{e^g}{f} = c.$$

Choose $\lambda \in \mathbb{C}$ with $e^\lambda = c^{-1}$ and set

$$h := g + \lambda.$$

Then

$$e^h = e^{g+\lambda} = e^g \cdot e^\lambda = c^{-1}e^g = f.$$

Hence h is a holomorphic logarithm of f . □

Homotopy Extension and Mapping Cylinders

A central concept in homotopy theory is that of continuously deforming a space onto one of its subspaces.

Definition A.4.53 (Deformation retract). A subspace A of X is a **deformation retract** of X if there exists a continuous map

$$H : X \times [0, 1] \rightarrow X$$

such that for all $x \in X$ and all $a \in A$

$$H(x, 0) = x$$

$$H(x, 1) \in A$$

$$H(a, t) = a \quad \forall t \in [0, 1]$$

(see [39, Ch. 9, §58, p. 361]).

To control homotopies on subspaces, the homotopy extension property is introduced.

Definition A.4.54 (Homotopy extension property). Let X be a topological space, and let $A \subseteq X$.

The pair (X, A) is said to satisfy the **homotopy extension property** if for every topological space Y , every continuous map

$$F : X \times \{0\} \rightarrow Y,$$

and every homotopy

$$H : A \times [0, 1] \rightarrow Y$$

satisfying $H(a, 0) = F(a, 0)$ for all $a \in A$, there exists a continuous map

$$\tilde{H} : X \times [0, 1] \rightarrow Y$$

such that $\tilde{H}(x, 0) = F(x, 0)$ for all $x \in X$, and $\tilde{H}(a, t) = H(a, t)$ for all $a \in A$ and all $t \in [0, 1]$ (see [26, Ch. 0, §1, p. 14]).

Remark A.4.55. A pair (X, A) has the homotopy extension property if A has a mapping cylinder neighbourhood in X , that is, there exists a closed neighbourhood $N \subseteq X$ of A , containing a subspace $B \subseteq N$ such that $N \setminus B$ is an open neighbourhood of A , together with a continuous map $f : B \rightarrow A$ and a homeomorphism

$$h : M(f) \rightarrow N$$

such that

$$h|_{A \cup B} = \text{id}_{A \cup B}.$$

In order to verify the homotopy extension property, notice that $[0, 1] \times [0, 1]$ retracts onto

$$([0, 1] \times \{0\}) \cup (\partial[0, 1] \times [0, 1]),$$

and hence

$$B \times [0, 1] \times [0, 1]$$

retracts onto

$$(B \times [0, 1] \times \{0\}) \cup (B \times \partial[0, 1] \times [0, 1]).$$

This retraction induces a retraction on $M(f) \times [0, 1]$ onto

$$M(f) \times \{0\} \cup (A \cup B) \times [0, 1].$$

Thus, $(M(f), A \cup B)$ has the homotopy extension property, and so does the homeomorphic pair $(N, A \cup B)$.

Given a map $X \rightarrow Y$ and a homotopy of its restriction to A , one can take the constant homotopy on $X \setminus (N \setminus B)$ and then extend it over N by applying the homotopy extension property for $(N, A \cup B)$ to the given homotopy on A and the constant homotopy on B (see [26, Ch. 0, §1, Ex. 0.15]).

The following proposition illustrates the strength of the homotopy extension property by relating it to homotopy equivalences and deformation retracts.

Proposition A.4.56. Suppose (X, A) and (Y, A) satisfy the homotopy extension property, and $f : X \rightarrow Y$ is a homotopy equivalence with

$$f|_A = \text{id}_A.$$

Then f is a homotopy equivalence relative A . Moreover, if (X, A) satisfies the homotopy extension property, and if the inclusion $A \hookrightarrow X$ is a homotopy equivalence, then A is a deformation retract of X (see [26, Ch. 0, §1, Prop. 0.19, Corollary 0.20]).

Proof. Let $g : Y \rightarrow X$ be a homotopy inverse of f . Then there exist homotopies

$$g \circ f \simeq \text{id}_X$$

and

$$f \circ g \simeq \text{id}_Y.$$

1. One constructs a homotopy from g to a map $G(-, 1)$ with $G(-, 1)|_A = \text{id}_A$.

Let

$$H : X \times [0, 1] \rightarrow X$$

be a homotopy from $g \circ f$ to id_X . Since $f|_A = \text{id}_A$, the restriction $H|_{A \times [0,1]}$ is a homotopy from $g|_A$ to id_A . Since (Y, A) has the homotopy extension property by assumption, this homotopy can be extended to a homotopy

$$G : Y \times [0, 1] \rightarrow X$$

with

$$G(y, 0) = g(y).$$

Then

$$G(-, 1)|_A = \text{id}_A.$$

2. One shows that

$$G(-, 1) \circ f \simeq_A \text{id}_X.$$

The ordinary homotopy from $G(-, 1) \circ f$ to id_X is obtained by first following the homotopy G backwards from $G(-, 1)$ to g , and then following H from $g \circ f$ to id_X . More precisely, define $K : X \times [0, 1] \rightarrow X$ by

$$K(x, t) := \begin{cases} G(f(x), 1 - 2t), & 0 \leq t \leq \frac{1}{2} \\ H(x, 2t - 1) & \frac{1}{2} \leq t \leq 1. \end{cases}$$

Then K is a homotopy from $G(-, 1) \circ f$ to id_X .

Since $f|_A = \text{id}_A$ and $G|_{A \times [0,1]} = H|_{A \times [0,1]}$, one has

$$K(-, t)|_A = K(-, 1 - t)|_A$$

for every $t \in [0, 1]$. Moreover,

$$K(-, 0)|_A = K(-, 1)|_A = \text{id}_A.$$

Thus the path of maps $K(-, t)|_A$ traverses the same homotopy forwards and backwards.

Define a homotopy of this homotopy on A by

$$L : A \times [0, 1] \times [0, 1] \rightarrow X$$

$$L(a, t, u) := \begin{cases} K(a, (1-u)t), & 0 \leq t \leq \frac{1}{2} \\ K(a, 1 - (1-u)(1-t)), & \frac{1}{2} < t \leq 1. \end{cases}$$

This is well-defined and continuous at $t = \frac{1}{2}$, since

$$K\left(-, \frac{1-u}{2}\right)|_A = K\left(-, \frac{1+u}{2}\right)|_A.$$

For $u = 0$ one recovers the original homotopy,

$$L(a, t, 0) = K(a, t),$$

while for $u = 1$ one has $L(a, t, 1) = a$. Furthermore,

$$L(a, 0, u) = L(a, 1, u) = a$$

for all $a \in A$ and $u \in [0, 1]$. Since (X, A) has the homotopy extension property, so does $(X \times [0, 1], A \times [0, 1])$.

Therefore the homotopy L of $K|_{A \times [0, 1]}$ extends to a homotopy

$$\tilde{L} : X \times [0, 1] \times [0, 1] \rightarrow X$$

satisfying

$$\tilde{L}(x, t, 0) = K(x, t)$$

and

$$\tilde{L}(a, t, u) = L(a, t, u)$$

for all $a \in A$.

Traversing the left, top, and right edges of the (t, u) -square yields a homotopy from $G(-, 1) \circ f$ to id_X which is constant on A . Hence

$$G(-, 1) \circ f \simeq_A \text{id}_X.$$

3. It remains to show that

$$f \circ G(-, 1) \simeq_A \text{id}_Y .$$

Hence $G(-, 1)$ is a homotopy inverse of f relative to A . Therefore f is a homotopy equivalence relative to A .

$$f \circ G(-, 1) \simeq f \circ g \simeq \text{id}_Y .$$

Apply steps 1 and 2 again, with the roles of f and $G(-, 1)$ interchanged. This produces a map

$$f_1 : X \rightarrow Y$$

such that

$$f_1|_A = \text{id}_A$$

and

$$f_1 \circ G(-, 1) \simeq_A \text{id}_Y .$$

Since $G(-, 1) \circ f \simeq_A \text{id}_X$, one obtains

$$\begin{aligned} f_1 &\simeq_A f_1 \circ (G(-, 1) \circ f) \\ &= (f_1 \circ G(-, 1)) \circ f \\ &\simeq_A f . \end{aligned}$$

Therefore

$$f \circ G(-, 1) \simeq_A f_1 \circ G(-, 1) \simeq_A \text{id}_Y .$$

Hence $G(-, 1)$ is a homotopy of f relative to A . Thus f is a homotopy equivalence relative to A .

For the second statement, suppose that the inclusion

$$\iota_A : A \hookrightarrow X$$

is a homotopy equivalence. By the first statement, ι is a homotopy equivalence relative to A . Hence it admits a homotopy inverse

$$r : X \rightarrow A$$

such that

$$r \circ \iota = \text{id}_A$$

and

$$\iota \circ r \simeq_A \text{id}_X.$$

Thus the latter homotopy is a deformation retraction of X onto A . Therefore A is a deformation retract of X (compare [26, Ch. 0, §1, Prop. 0.19, Corollary 0.20]). $\square$

Since mapping cylinders provide a natural setting for the homotopy extension property, their definition is recalled next.

Definition A.4.57 (Mapping cylinder). Let $f : X \rightarrow Y$ be a continuous map between topological spaces X and Y . The **mapping cylinder** of f is the quotient space

$$M(f) := (X \times [0, 1] \dot{\cup} Y) / \sim,$$

where the equivalence relation is given by

$$(x, 1) \sim f(x)$$

for all $x \in X$ (see [26, p. 2]).

Remark A.4.58. The mapping cylinder contains a copy of X and a copy of Y , namely

$$\iota_X : X \times \{0\} \rightarrow M(f), \quad (x, 0) \mapsto [(x, 0)]$$

where ι_X denotes the inclusion, and

$$\iota_Y : Y \rightarrow M(f), \quad y \mapsto [y].$$

Moreover, $M(f)$ deformation retracts onto Y (see [26, p. 2]).

Complexes of spaces

In the proof of the Nerve Theorem, it is convenient to assign a topological space to every simplex of a simplicial complex in a functorial way. This leads to the notion of a complex of spaces.

Definition A.4.59 (Complex of spaces over a simplicial complex). Let K be a simplicial complex. A **complex of spaces over K** consists of a topological space X_σ for every simplex $\sigma \in K$, together with continuous maps

$$\iota_{\sigma,\tau}^X : X_\sigma \rightarrow X_\tau$$

for every face $\tau \subseteq \sigma$, such that

$$\iota_{\tau,\rho}^X \circ \iota_{\sigma,\tau}^X = \iota_{\sigma,\rho}^X$$

whenever $\rho \subseteq \tau \subseteq \sigma$, and

$$\iota_{\sigma,\sigma}^X = \text{id}_{X_\sigma}.$$

Equivalently, a complex of spaces is a contravariant functor from the face poset of K to Top .

Its **realisation** is

$$|\mathfrak{X}| := \left(\coprod_{\sigma \in K} X_\sigma \times \Delta^\sigma \right) / \sim,$$

where the equivalence relation is generated by

$$(x, j_{\tau,\sigma}(t)) \sim (\iota_{\sigma,\tau}^X(x), t)$$

for every face $\tau \subseteq \sigma$, every $x \in X_\sigma$, and every $t \in \Delta^\tau$, where

$$j_{\tau,\sigma} : \Delta^\tau \hookrightarrow \Delta^\sigma$$

is the corresponding face inclusion.

A morphism of complexes of spaces

$$f : \mathfrak{X} \rightarrow \mathfrak{Y}$$

is a family of continuous maps

$$f_\sigma : X_\sigma \rightarrow Y_\sigma$$

such that for every face $\tau \subseteq \sigma$ the diagram

$$\begin{array}{ccc} X_\sigma & \xrightarrow{\iota_{\sigma,\tau}^X} & X_\tau \\ f_\sigma \downarrow & & \downarrow f_\tau \\ Y_\sigma & \xrightarrow{\iota_{\sigma,\tau}^Y} & Y_\tau \end{array}$$

commutes.

Such a morphism induces a continuous map

$$|f| : |\mathfrak{X}| \rightarrow |\mathfrak{Y}|$$

defined by

$$|f|([x, t]) = [f_\sigma(x), t]$$

(see [26, Ch. 4, §G, pp. 457 ff.]).

In particular, complexes of spaces naturally arise from open covers.

Remark A.4.60. For $X_\sigma = U_\sigma$, where U_σ denotes the corresponding intersection of sets of a cover $\mathcal{U}$ of a topological space X , the realisation

$$|\mathfrak{X}| = \left(\coprod_{\sigma} U_\sigma \times \Delta^\sigma \right) / \sim$$

is exactly the thickened Čech space $\check{\mathbb{X}}$.

Simplices and simplicial complexes

Many constructions throughout this thesis are based on simplices and simplicial complexes. Their basic definitions are therefore recalled next.

Definition A.4.61 (Standard simplex). For $n \in \mathbb{N}$, the **standard n -simplex** is

$$\Delta^n := \left\{ (t_0, \dots, t_n) \in \mathbb{R}^{n+1} \mid t_i \geq 0, \sum_{i=0}^n t_i = 1 \right\}.$$

The numbers $t_0, \dots, t_n$ are called the **barycentric coordinates** of points in Δ^n . The distinguished point

$$b_n := \left(\frac{1}{n+1}, \dots, \frac{1}{n+1} \right)$$

is called the **barycentre**.

A **face** of Δ^n is obtained by setting at least one barycentric coordinate to zero. The **i -th face map**

$$\varphi_i : \Delta^{n-1} \rightarrow \Delta^n, \quad (t_0, \dots, t_{n-1}) \mapsto (t_0, \dots, t_{i-1}, 0, t_i, \dots, t_{n-1})$$

is obtained by inserting 0 at position i .

The **i -th degeneracy map**

$$\sigma_i : \Delta^{n+1} \rightarrow \Delta^n, \quad (t_0, \dots, t_{n+1}) \mapsto (t_0, \dots, t_{i-1}, t_i + t_{i+1}, t_{i+2}, \dots, t_n)$$

is obtained by repeating the i -th vertex.

The **boundary** of Δ^n is

$$\partial\Delta^n := \{(t_0, \dots, t_n) \in \Delta^n \mid t_i = 0 \text{ for at least one } i\}.$$

Equivalently, $\partial\Delta^n$ is the union of all proper faces of Δ^n (see [31, Ch. 7, pp. 100 ff.]).

Remark A.4.62. For a mapping cylinder, it holds that $M(f)_\sigma := M(f_\sigma)$. The only thing that has to be verified is that face maps are preserved. Suppose $\tau \subseteq \sigma$. Since the induced diagram

$$\begin{array}{ccc} X_\sigma & \longrightarrow & X_\tau \\ \downarrow & & \downarrow \\ Y_\sigma & \longrightarrow & Y_\tau \end{array}$$

commutes, there is an induced map $M(f_\sigma) \rightarrow M(f_\tau)$ given by $[(x, t)] \mapsto [(\iota_{\sigma, \tau}(x), t)]$ on the cylinder part and $y \mapsto \iota_{\sigma, \tau}(y)$ on the Y -part. Therefore the spaces $M(f_\sigma)$ again form a complex of spaces over the same simplicial complex K .

Often only the incidence relations between simplices are relevant. This leads to the notion of an abstract simplicial complex.

Definition A.4.63 (Abstract simplices and abstract simplicial complexes). Let V be a set. An **abstract simplicial complex** on V is a collection $\mathcal{K}$ of non-empty finite subsets of V such that whenever $\sigma \in \mathcal{K}$ and $\emptyset \neq \tau \subseteq \sigma$, one has $\tau \in \mathcal{K}$.

The elements $\sigma \in \mathcal{K}$ are called **simplices**, and the elements of a simplex are called its **vertices**. A simplex $\tau \subseteq \sigma$ is called a **face** of σ . A maximal proper face is called a **facet**.

The **dimension** of a simplex σ is

$$\dim(\sigma) := |\sigma| - 1.$$

The dimension of $\mathcal{K}$ is

$$\dim(\mathcal{K}) := \sup_{\sigma \in \mathcal{K}} \dim(\sigma).$$

The n -**skeleton** $\mathcal{K}^{(n)}$ is the subcomplex consisting of all simplices of dimension at most n (see [31, Ch. 7, pp. 100 ff.]).

To relate simplicial objects to topological spaces, one considers continuous maps from standard simplices.

Definition A.4.64 (Singular simplices). Let Δ^n be a standard simplex, and let X be a topological space. A **singular simplex** is a continuous map $\sigma : \Delta^n \rightarrow X$ (see [31, Ch. 7, pp. 100 ff.]).

For arguments involving covers, it is useful to restrict attention to simplices contained in individual elements of the cover.

Definition A.4.65 ($\mathcal{U}$ -small simplices and chains). Let X be a topological space, and let $\mathcal{U} = \{U_i\}_{i \in I}$ be an open cover of X . A singular simplex $\sigma : \Delta^n \rightarrow X$ is **$\mathcal{U}$ -small** if there exists an $i \in I$ such that $\sigma(\Delta^n) \subseteq U_i$.

A singular chain $c = \sum_k a_k \sigma_k$ is $\mathcal{U}$ -small if every simplex σ_k in the sum is $\mathcal{U}$ -small (see [26, Ch. 3, pp. 102 ff.]).

A fundamental tool for refining singular simplices is barycentric subdivision.

Definition A.4.66 (Barycentric subdivision). Let Δ^n be the standard n -simplex. For every non-empty face $F \subseteq \Delta^n$, let b_F denote its barycentre. The **barycentric subdivision** of Δ^n is a simplicial complex whose vertices are the barycentres of the non-empty faces of Δ^n , and the simplices are the convex hulls of $(b_{F_0}, \dots, b_{F_k})$, where $F_0 \subsetneq F_1 \subsetneq \dots \subsetneq F_k$ is a strictly increasing chain of non-empty faces of Δ^n (see [26, Ch. 3, pp. 102 ff.]).

B Algebraic Categories

This chapter provides an introduction to algebraic categories. Algebraic categories come equipped with an underlying set–functor, which allows many constructions to be studied by comparison with $\mathbf{Set}$. This makes it possible to transfer suitable $\mathbf{Set}$ –theoretic constructions to algebraic categories such as $\mathbf{Ab}$.

Definition B.0.1 (Signatures and algebraic types). A **signature** is a set Ω together with an arity map $\Omega \rightarrow \mathbb{N}_0$. The elements $F \in \Omega$ are called **operation symbols** and their image $n \in \mathbb{N}$ is called **arity**. One denotes a logical operator of arity n by $F^{[n]}$.

An **term** in the variables $X_1, \dots, X_m$ with respect to the signature Ω is a formal term, arising from the following rules:

1. X_i is a term in the variables $X_1, \dots, X_m$ for $1 \leq i \leq m$.
2. If $T_1, \dots, T_n$ are terms in the variables $X_1, \dots, X_m$ and if $F^{[n]} \in \Omega$, then $F(T_1, \dots, T_n)$ is a term in the variables $X_1, \dots, X_m$.
3. A term can be generated only by these rules.

An **equation** is an ordered pair of two terms in finitely many variables.

A **type** of an algebraic structure is a tuple $\tau = (\Omega, \mathcal{E})$, where Ω is a signature and $\mathcal{E}$ is a set of equations.

A signature specifies the formal operations available in an algebraic theory, while the equations prescribe the identities that these operations must satisfy. One can interpret this syntactic data on an underlying set.

Definition B.0.2 (Structures of a type). Let $\tau = (\Omega, \mathcal{E})$ be a type. A **structure** A of type Ω consists of a set $X = \mathfrak{F}(A)$, the so-called **underlying set** and an

association that maps any symbol $F^{[n]} \in \Omega$ to a n -ary operation

$$|F|_A : X^n \rightarrow X.$$

For any structure A of type Ω and $x \in X^m$, one can assign an element $|T|_A(x) \in X$ to any term T in the variables $X_1, \dots, X_m$, by defining recursively

$$\begin{aligned} |X_i|_A(x) &= x_i \\ |F(T_1, \dots, T_n)|_A(x) &:= |F|_A(|T_1|_A(x), \dots, |T_n|_A(x)). \end{aligned}$$

A structure A of type Ω **satisfies the equation (S,T)** in the variables $X_1, \dots, X_m$, if

$$|S|_A(x) = |T|_A(x)$$

for all $x \in X^m$.

A **structure of type τ** is a structure A of type Ω , that satisfies all equations in $\mathcal{E}$ (see [8, p. 84]).

In particular, a nullary operation corresponds to the choice of a distinguished element of X .

Example B.0.3. For abelian groups, the type $\tau = (\Omega, \mathcal{E})$ is as follows: The signature Ω consists of

- one binary operation $+^{[2]}$,
- one unary operation $-^{[1]}$,
- one nullary operation (constant) $0^{[0]}$.

A structure A therefore consists of a set X together with

$$+ : X \times X \rightarrow X, \quad - : X \rightarrow X,$$

and $0 \in X$.

The equations $\mathcal{E}$ express commutativity, that is, associativity, identity, the equations expressing associativity, the identity element, inverses, and commutativity.

Thus, addition has arity two, inversion has arity one, and the distinguished zero element is represented by a nullary operation.

Having specified the objects determined by an algebraic type, one next defines the mappings that preserve their algebraic structure.

Definition B.0.4 (Homomorphisms of structures). Let A, B be structures of type $\tau = (\Omega, \mathcal{E})$ with underlying sets $\mathfrak{F}(A)$ and $\mathfrak{F}(B)$. A **homomorphism** $f : A \rightarrow B$ is a mapping $f : \mathfrak{F}(A) \rightarrow \mathfrak{F}(B)$ that is compatible with all operations, that is, for all $F^{[n]} \in \Omega$ the diagram

$$\begin{array}{ccc} \mathfrak{F}(A)^n & \xrightarrow{f^n} & \mathfrak{F}(B)^n \\ \downarrow |F|_A & & \downarrow |F|_B \\ \mathfrak{F}(A) & \xrightarrow{f} & \mathfrak{F}(B) \end{array}$$

commutes. Thus, for all $(x_1, \dots, x_n) \in \mathfrak{F}(A)^n$, and with

$$f^n(x_1, \dots, x_n) := (f(x_1), \dots, f(x_n)),$$

$$f(|F|_A(x_1, \dots, x_n)) = |F|_B(f(x_1), \dots, f(x_n))$$

and $\mathfrak{F}(f) : \mathfrak{F}(A) \rightarrow \mathfrak{F}(B)$ is the underlying mapping between sets of the homomorphism f (see [8, p. 84]).

For a nullary operation symbol, the compatibility condition states that the corresponding distinguished element is preserved.

Example B.0.5. If $\Omega = \{+, -, 0\}$, then the commuting diagrams become

- Addition: $f(x + y) = f(x) + f(y)$
- Inverse: $f(-x) = -f(x)$,
- Zero: $f(0) = 0$,

which are precisely the axioms for a homomorphism of abelian groups.

Structures of a fixed type and their homomorphisms are closed under identity mappings and composition, and therefore form a category.

Definition B.0.6 (Category of structures). In the category of structures of type τ , denoted by $S(\tau)$, the objects are structures of type τ , and the morphisms are homomorphisms. An **algebraic category** of finite arity is a category $\mathcal{C}$ together with an equivalence $\mathcal{C} \xrightarrow{\sim} S(\tau)$, such that the diagram of functors

$$\begin{array}{ccc} \mathcal{C} & \xrightarrow{\sim} & S(\tau) \\ & \searrow & \swarrow \\ & \text{Set} & \end{array}$$

commutes (see [8, p. 85]).

Example B.0.7. Consider the type $\tau = (\{+, -, 0\}, \mathcal{E})$, where $\mathcal{E}$ consists of the abelian group axioms. Then the objects of $S(\tau)$ are abelian groups, and the morphisms are group homomorphisms.

The preceding equivalence equips every algebraic category with an underlying set functor. The corresponding inverse construction assigns to a set the algebraic structure generated freely by its elements.

Definition B.0.8 (Forgetful functor, free structures). Let $\tau = (\Omega, \mathcal{E})$ be a type. The **forgetful functor**

$$\mathfrak{F} : S(\tau) \rightarrow \text{Set}$$

assigns to each structure its underlying set, and to each homomorphism its underlying function. A **free structure** on a set X is a structure $\mathfrak{U}(X)$ of type τ , together with a function

$$\eta_X : X \rightarrow \mathfrak{F}(\mathfrak{U}(X))$$

such that for every structure A and every function $f : X \rightarrow \mathfrak{F}(A)$, there exists a unique homomorphism

$$\phi : \mathfrak{U}(X) \rightarrow A$$

with $\mathfrak{F}(\phi) \circ \eta_X = f$ (see [61]).

Remark B.0.9 (Free-forgetful adjunction). The forgetful functor $\mathfrak{F} : \mathcal{C} \rightarrow \text{Set}$ admits an adjoint functor $\mathfrak{U} : \text{Set} \rightarrow \mathcal{C}$, called **free object functor**. The adjunction $\mathfrak{U} \dashv \mathfrak{F}$ is called **free–forgetful adjunction** (see [61], [60], [59]).

Example B.0.10. If one considers $\mathbf{Ab}$, the forgetful functor

$$\mathfrak{F} : \mathbf{Ab} \rightarrow \mathbf{Set}$$

assigns to every abelian group its underlying set. Its left-adjoint

$$\mathfrak{U} : \mathbf{Set} \rightarrow \mathbf{Ab}$$

maps a set X to the free abelian group generated by X . Explicitly,

$$\mathfrak{U}(X) = \bigoplus_{x \in X} \mathbb{Z},$$

whose elements are finite formal integer linear combinations

$$\sum_{x \in X} n_x [x], \quad n_x \in \mathbb{Z},$$

and $n_x = 0$ for all but finitely many x . The unit

$$\eta_X : X \rightarrow \mathfrak{F}(\mathfrak{U}(X))$$

sends each element $x \in X$ to the corresponding basis element. Given an abelian group A and any function

$$f : X \rightarrow \mathfrak{F}(A),$$

there exists a unique group homomorphism

$$\phi : \mathfrak{U}(X) \rightarrow A$$

extending f .

Every adjunction determines a monad. In the case of a free-forgetful adjunction, the resulting monad records the operations of forming formal terms and substituting terms into one another.

Definition B.0.11 (Monad). A **monad** on a category $\mathcal{C}$ is a triple (T, η, μ) consisting of

- an *endofunctor* $T : \mathcal{C} \rightarrow \mathcal{C}$,

- a natural transformation $\eta : \text{id}_{\mathcal{C}} \rightarrow T$, which is called **unit**,
- a natural transformation $\mu : T \circ T \rightarrow T$, which is called the **multiplication**,

such that the following diagrams commute:

$$\begin{array}{ccc}
 T^3 & \xrightarrow{T\mu} & T^2 \\
 \downarrow \mu T & & \downarrow \mu \\
 T^2 & \xrightarrow{\mu} & T
 \end{array}
 \quad
 \begin{array}{ccccc}
 T & \xrightarrow{T\eta} & T^2 & \xleftarrow{\eta T} & T \\
 & \searrow \text{id}_T & \downarrow \mu & \swarrow \text{id}_T & \\
 & & T & &
 \end{array}$$

(see [4, Ch. 3, §1, p. 82]).

A monad T on Set is called **finitary** if its underlying functor preserves filtered colimits.

Example B.0.12. The free-forgetful adjunction

$$\mathfrak{U} : \text{Set} \rightleftarrows \text{Ab} : \mathfrak{F}$$

induces a monad on Set . Its underlying endofunctor is $T = \mathfrak{F} \circ \mathfrak{U}$, which assigns to a set its underlying free abelian group. The unit

$$\eta : \text{id}_{\text{Set}} \rightarrow T$$

maps every element to the corresponding generator, while the multiplication

$$\mu : T^2 \rightarrow T$$

flattens a formal linear combination of formal linear combinations into a single formal linear combination.

The example of abelian groups is representative of the general situation: a finitary algebraic theory can be recovered from the monad induced by its free-forgetful adjunction.

Proposition B.0.13 (Algebraic categories are monadic). Let $\mathcal{C}$ be an algebraic category of finite arity, and let

$$\mathfrak{F} : \mathcal{C} \rightarrow \text{Set}$$

be its underlying-set functor. Then $\mathfrak{F}$ admits a left-adjoint. Moreover, the induced monad

$$T := \mathfrak{F} \circ \mathfrak{U} : \mathbf{Set} \rightarrow \mathbf{Set}$$

is finitary, and $\mathcal{C}$ is equivalent over $\mathbf{Set}$ to the *Eilenberg–Moore category* $\mathbf{Set}^T$ of T -algebras (see [6, Ch. 4, §2, Prop. 4.6.2]).

Proof. Since $\mathcal{C}$ is an algebraic category of finite arity, there exists an algebraic type $\tau = (\Omega, \mathcal{E})$ whose operation symbols have finite arity and an equivalence

$$E : \mathcal{C} \xrightarrow{\sim} S(\tau)$$

commuting with the respective underlying-set functors. It therefore suffices to prove the assertion for $S(\tau)$.

Let X be a set. Denote by $\mathbf{Term}_\Omega(X)$ the set of all formal Ω -terms in variables indexed by the elements of X . Every operation symbol $G^{[n]} \in \Omega$ induces an operation

$$G : \mathbf{Term}_\Omega(X)^n \rightarrow \mathbf{Term}_\Omega(X), \quad (t_1, \dots, t_n) \mapsto G(t_1, \dots, t_n).$$

Let $\sim_{\mathcal{E}}$ be the smallest congruence on $\mathbf{Term}_\Omega(X)$ containing all instances of the equations in $\mathcal{E}$. Define

$$\mathfrak{U}(X) := \mathbf{Term}_\Omega(X) / \sim_{\mathcal{E}}.$$

Since $\sim_{\mathcal{E}}$ is a congruence, the operations on the term algebra descend to the quotient, and $\mathfrak{U}(X)$ is a structure of type τ .

The map

$$\eta_X : X \rightarrow \mathfrak{F}(\mathfrak{U}(X)), \quad x \mapsto [x],$$

assigns to every element its corresponding variable term. Let $A \in S(\tau)$ and let

$$f : X \rightarrow \mathfrak{F}(A)$$

be a function. Every term t can be evaluated in A after substituting $f(x)$ for the variable x . Recursive evaluation defines the unique homomorphism of Ω -structures

$$\tilde{f} : \mathbf{Term}_\Omega(X) \rightarrow A.$$

Since A satisfies all equations in $\mathcal{E}$, the map $\tilde{f}$ is constant on the equivalence classes of $\sim_{\mathcal{E}}$. It therefore induces a homomorphism

$$\bar{f} : \mathfrak{U}(X) \rightarrow A$$

satisfying

$$\mathfrak{F}(\bar{f}) \circ \eta_X = f.$$

Since every element of $\mathfrak{U}(X)$ is represented by a term in the generators $[x]$, this homomorphism is unique. Hence there is a natural bijection

$$\mathrm{Hom}_{S(\tau)}(\mathfrak{U}(X), A) \cong \mathrm{Hom}_{\mathrm{Set}}(X, \mathfrak{F}(A)).$$

Thus

$$\mathfrak{U} : \mathrm{Set} \rightarrow S(\tau)$$

is left-adjoint to $\mathfrak{F}$.

Let

$$T := \mathfrak{F} \circ \mathfrak{U}$$

be the monad induced by this adjunction. Its unit is the map

$$\eta_X : X \rightarrow T(X)$$

above, and its multiplication

$$\mu_X : T(T(X)) \rightarrow T(X)$$

is given by substitution of terms into terms.

For a structure $A \in S(\tau)$, define

$$\alpha_A : T(\mathfrak{F}(A)) \rightarrow \mathfrak{F}(A), \quad [t] \mapsto |t|_A$$

by evaluating every formal term in A . Evaluation of a variable gives

$$\alpha_A \circ \eta_{\mathfrak{F}(A)} = \mathrm{id}_{\mathfrak{F}(A)},$$

while compatibility of evaluation with substitution gives

$$\alpha_A \circ T(\alpha_A) = \alpha_A \circ \mu_{\mathfrak{F}(A)}.$$

Hence $(\mathfrak{F}(A), \alpha_A)$ is a T -algebra. Conversely, let (X, α) be a T -algebra. For every operation symbol $G^{[n]} \in \Omega$, define an n -ary operation on X by

$$|G|_\alpha(x_1, \dots, x_n) := \alpha([G(x_1, \dots, x_n)]).$$

Since terms identified by the congruence generated by $\mathcal{E}$ represent the same element of $T(X)$, these operations satisfy all equations in $\mathcal{E}$. Thus they define a structure of type τ on X .

The unit and associativity axioms of the T -algebra imply, respectively, that variables are interpreted as themselves and that the interpretation of a composite term agrees with recursive evaluation. Consequently, the two constructions above are mutually inverse on objects.

A function

$$h : X \rightarrow Y$$

is a morphism of T -algebras precisely when

$$h \circ \alpha = \beta \circ T(h).$$

Applied to the terms corresponding to the basic operation symbols, this condition is equivalent to

$$h(|G|_\alpha(x_1, \dots, x_n)) = |G|_\beta(h(x_1), \dots, h(x_n)).$$

Hence the morphisms of T -algebras are exactly the homomorphisms of structures. Therefore

$$S(\tau) \simeq \text{Set}^T.$$

Finally, because all operation symbols have finite arity, every term involves only finitely many variables. Consequently, the underlying functor T preserves filtered colimits, so the monad T is finitary.

Transporting the free functor, the monad, and the equivalence along

$$\mathcal{C} \simeq S(\tau)$$

proves the assertion for $\mathcal{C}$ (see [4, Thm. 3.14 p. 100, Cor. 4.3 p. 105, 3.4 p. 107]). □

Example B.0.14. Consider the free-forgetful adjunction

$$\mathfrak{U} : \mathbf{Set} \rightleftarrows \mathbf{Ab} : \mathfrak{F},$$

where $\mathfrak{U}$ assigns to a set its free abelian group and $\mathfrak{F}$ is the forgetful functor. The induced monad is

$$T = \mathfrak{F} \circ \mathfrak{U}.$$

For a set X , the set $T(X)$ consists of all finite formal integer linear combinations of elements of X . If A is an abelian group, then the underlying set $\mathfrak{F}(A)$ carries a natural T -algebra structure

$$\alpha : T(\mathfrak{F}(A)) \rightarrow \mathfrak{F}(A),$$

obtained by evaluating formal sums using the addition in A . Conversely, every T -algebra arises in this way. Hence

$$\mathbf{Ab} \simeq \mathbf{Set}^T.$$

The preceding example illustrates the monadic description in the familiar case of abelian groups. One can isolate the finiteness property of the associated monad needed below. Its proof makes the role of the finitary assumption explicit: every element of a free algebra depends on finitely many generators.

Proposition B.0.15. Let $\tau = (\Omega, \mathcal{E})$ be an algebraic type whose operation symbols all have finite arity, and let

$$T : \mathbf{Set} \rightarrow \mathbf{Set}$$

be the monad induced by the free–forgetful adjunction. Then T is finitary; equivalently, T preserves filtered colimits (see [4]).

Proof. Claim: The elements of $T(X)$ have finite support.

Fix a set X . An element of $T(X)$ is represented by a τ -term in variables from X , modulo the congruence generated by $\mathcal{E}$. Since all operation symbols have finite arity, every term is a finite term tree, hence it contains only finitely many occurrences of variables. Therefore, for every $t \in T(X)$ there exists a finite subset $S \subseteq X$ such that t can be represented by a term only involving variables from S . Thus, t has finite support.

Equations do not change this: applying an equation replaces a subterm by another subterm, both involving finitely many variables.

Let $(X_i)_{i \in I}$ be a filtered diagram in $\mathbf{Set}$, with colimit object X and morphisms

$$\lambda_i : X_i \rightarrow X.$$

Claim: For every finite subset $S \subseteq X$, there exists some $j \in I$ such that $S \subseteq \text{im}(\lambda_j)$.

Each element of S comes from some stage $X_{i(s)}$; since I is filtered and S is finite, there exists a single stage j dominating all $i(s)$, hence all elements of S come from X_j .

Claim: The canonical morphism $\text{colim } T(X_i) \rightarrow T(\text{colim } X_i)$ is bijective.

Consider the canonical map

$$\Phi : \text{colim}_{i \in I} T(X_i) \rightarrow T(\text{colim}_{i \in I} X_i) = T(X).$$

Take $t \in T(X)$. As it was already shown, t has finite support $S \subseteq X$, and S is contained in the image of some $\lambda_j : X_j \rightarrow X$. Choose representatives of the variables in S inside X_j ; then the term representing t already defines an element $t_j \in T(X_j)$ whose image in $T(X)$ is t . Hence Φ is surjective.

Suppose that two elements $t_1 \in T(X_i)$ and $t_2 \in T(X_j)$ have the same image in $T(X)$. Choose term representatives for t_1 and t_2 . Their equality in $T(X)$ means that their images as terms in variables from X are equivalent modulo the congruence generated by $\mathcal{E}$. Such an equivalence is witnessed by a finite sequence of applications of equations from $\mathcal{E}$, substitutions, and compatibility with the operations. Hence only finitely many elements of X occur in this derivation. Each of these elements is represented at some stage of the filtered diagram, and all equal-

ities between representatives that occur in the derivation hold at some sufficiently large stage. By filteredness, there exists an object $k \in I$, receiving morphisms from i and j , at which the entire finite derivation is represented. Consequently, the images of t_1 and t_2 in $T(X_k)$ are equal. Hence t_1 and t_2 represent the same element of $\operatorname{colim}_{i \in I} T(X_i)$, so Φ is injective.

This shows that T preserves filtered colimits, so the associated monad is finitary (see [4]). □

The finite-support property also controls the behaviour of filtered colimits in the algebraic category itself. Combined with the usual componentwise constructions of limits and the existence of free algebras and quotients, it yields the following fundamental properties of algebraic categories of finite arity.

Corollary B.0.16. Let $\mathcal{C}$ be an algebraic category of finite arity. Then

1. $\mathcal{C}$ is complete and cocomplete
2. limits are created by the forgetful functor $\mathfrak{F} : \mathcal{C} \rightarrow \mathbf{Set}$
3. filtered colimits commute with finite limits in $\mathcal{C}$

(see [4]).

Proof. **Claim 1:** The category $\mathcal{C}$ is complete, and limits are created by $\mathfrak{F}$.

Let $\mathbf{D} : J \rightarrow \mathcal{C}$ be a diagram, and let L be the limit in $\mathbf{Set}$ with projections

$$\operatorname{pr}_j : L \rightarrow \mathfrak{F}(\mathbf{D}(j)).$$

Define a τ -structure on the set L by defining each basic operation $F^{[n]} \in \Omega$ on L and consider the map

$$\langle |F|_{\mathbf{D}(j)} \circ (\operatorname{pr}_j)^n \rangle_{j \in J} : L^n \rightarrow \prod_{j \in J} \mathfrak{F}(\mathbf{D}(j)).$$

Since $L = \lim \mathfrak{F}(\mathbf{D}(j))$, giving morphisms $L^n \rightarrow \mathfrak{F}(\mathbf{D}(j))$ compatible with the diagram is equivalent to giving a unique map $L^n \rightarrow L$. The morphisms are homomorphisms, hence commute with $|F|$. Thus, there exists a unique function

$$|F|_L : L^n \rightarrow L$$

such that

$$\mathrm{pr}_j \circ |F|_L = |F|_{\mathbf{D}(j)} \circ (\mathrm{pr}_j)^n$$

for all j . If done for every $F \in \Omega$, then L becomes a τ -structure, and the pr_j become homomorphisms by constructions.

Since each pr_j is a homomorphism and the equations from $\mathcal{E}$ hold in every $\mathbf{D}(j)$, they also hold after composing with each pr_j ; since the pr_j are jointly monomorphic in $\mathbf{Set}$, the equations hold in L .

Finally, the universal property in $\mathcal{C}$ is forced by the universal property in $\mathbf{Set}$ and the uniqueness of the induced operations. Hence L is the limit in $\mathcal{C}$, created by $\mathfrak{F}$.

Claim 2: The category $\mathcal{C}$ is cocomplete.

It is sufficient that all coproducts and coequalisers exist.

Claim 2.A: All coproducts exist.

Given a family $(A_i)_{i \in I}$ in $\mathcal{C}$, take the disjoint union of underlying sets

$$X := \coprod_{i \in I} \mathfrak{F}(A_i).$$

From the free τ -algebra $\mathfrak{U}(X)$ on X . Each inclusion $\mathfrak{F}(A_i) \rightarrow X$ gives a map

$$\mathfrak{F}(A_i) \rightarrow \mathfrak{F}(\mathfrak{U}(X)).$$

Impose the relations saying that each A_i sits inside a subalgebra: in essence, quotient $\mathfrak{U}(X)$ by the smallest congruence forcing the induced map $A_i \rightarrow \mathfrak{U}(X)$ to be a homomorphism. The resulting algebra, with the induced maps

$$A_i \rightarrow \coprod_i A_i$$

satisfies the coproduct universal property.

Claim 2.B: All coequalisers exist.

Let $\sim$ be the smallest τ -congruence on the underlying set $\mathfrak{F}(B)$ such that

$$f(a) \sim g(a)$$

for all $a \in \mathfrak{F}(A)$. Let

$$q : \mathfrak{F}(B) \rightarrow \mathfrak{F}(B) / \sim$$

be the quotient map. Because $\sim$ is a congruence, the operations on B descend to well-defined operations on $B/\sim$, making

$$q : B \rightarrow B/\sim.$$

Then

$$q \circ f = q \circ g$$

by construction. If $h : B \rightarrow C$ is an arbitrary homomorphism with

$$h \circ f = h \circ g,$$

then h identifies all pairs $f(a), g(a)$, hence identifies the congruence they generate, so h factors uniquely through q , making q the coequaliser morphism.

Since $\mathcal{C}$ has all coproducts and coequalisers, it is cocomplete.

Claim 3: Filtered colimits are created by the forgetful functor $\mathfrak{F}$.

Let

$$A := \operatorname{colim}_I \mathfrak{F}(\mathbf{D}(i))$$

in $\mathbf{Set}$. For each basic operation in $F^{[n]}$, define $|F|_A$ on A^n :

Represent each input by some stage and use the filteredness to move them to a common stage k , apply $|F|_{\mathbf{D}(k)}$, and pass it to the colimit. Finite arity is crucial for finitely many inputs to unify at one stage.

Define

$$|F|_A(a_1, \dots, a_n) := \lambda_j \left(|F|_{\mathbf{D}(j)} \left(\mathfrak{F}(\mathbf{D}(\alpha_1))(x_1), \dots, \mathfrak{F}(\mathbf{D}(\alpha_n))(x_n) \right) \right) \in A$$

which is independent of the choices of representatives (i_k, x_k) and of the object j . Fix the representatives and choose another object $j' \in I$ with morphisms

$$\alpha'_k : i_k \rightarrow j'.$$

By the filteredness of I , there exist $m \in I$ and morphisms

$$\beta : j \rightarrow m, \quad \beta' : j' \rightarrow m.$$

Using that $\mathbf{D}(\beta)$ and $\mathbf{D}(\beta')$ are homomorphisms, one computes in $\mathfrak{F}(\mathbf{D}(m))$

$$\mathfrak{F}(\mathbf{D}(\beta))(|F|_{\mathbf{D}(j)}(\dots)) = |F|_{\mathbf{D}(m)}\left(\mathfrak{F}(\mathbf{D}(\beta \circ \alpha_1))(x_1), \dots, \mathfrak{F}(\mathbf{D}(\beta \circ \alpha_n))(x_n)\right)$$

and similarly for j' . Applying λ_m and using

$$\lambda_m \circ \mathfrak{F}(\mathbf{D}(\beta)) = \lambda_j$$

and

$$\lambda_m \circ \mathfrak{F}(\mathbf{D}(\beta')) = \lambda_{j'}$$

both choices yield the same element of A . Hence the value is independent of the choice j .

Fix the object and change the representatives. For each k let $x'_k \in \mathfrak{F}(\mathbf{D}(i'_k))$ be another representative of a_k , that is, $\lambda_{i'_k}(x'_k) = a_k$. By the description of colimits in $\mathbf{Set}$, there exist $t_k \in I$ and morphisms $i_k \rightarrow t_k$ and $i'_k \rightarrow t_k$ such that the images of x_k and x'_k in $\mathfrak{F}(\mathbf{D}(t_k))$ coincide. Since I is filtered and n is finite, there exists $t \in I$ receiving morphisms from all t_k . Transporting all representatives to $\mathfrak{F}(\mathbf{D}(t))$, the two n -tuples become equal; hence applying $|F|_{\mathbf{D}(t)}$ gives the same result, and thus the value in A is independent of the representatives. Therefore the operation $|F|_A$ is well-defined.

For $i \in I$ and $y_1, \dots, y_n \in \mathfrak{F}(\mathbf{D}(i))$, choosing $i = j$ and $\alpha_k = \text{id}_i$ yields

$$|F|_A(\lambda_i(y_1), \dots, \lambda_i(y_n)) = \lambda_i(|F|_{\mathbf{D}(i)}(y_1, \dots, y_n)),$$

so each λ_i preserves F , hence is a homomorphism.

Let $(S, T) \in \mathcal{E}$ be an equation in m variables and $(a_1, \dots, a_m) \in A^m$. Choose $j \in I$ and representatives $x_1, \dots, x_m \in \mathfrak{F}(\mathbf{D}(j))$ mapping to $a_1, \dots, a_m$, which is possible since I is filtered and m is finite. Since $\mathbf{D}(j)$ satisfies (S, T) and λ_j is a homomorphism, one obtains

$$\begin{aligned} |S|_A(a_1, \dots, a_m) &= \lambda_j(|S|_{\mathbf{D}(j)}(x)) \\ &= \lambda_j(|T|_{\mathbf{D}(j)}(x)) \\ &= |T|_A(a_1, \dots, a_m). \end{aligned}$$

Thus A is a structure of type τ .

Let $B \in \text{Ob}(\mathcal{C})$ and let $(f_i : D(i) \rightarrow B)_{i \in I}$ be a cocone in $\mathcal{C}$. Applying $\mathfrak{F}$, one obtains a cocone of sets, so there is a unique factorisation

$$f : A \rightarrow \mathfrak{F}(B)$$

with

$$f \circ \lambda_i = \mathfrak{F}(f_i)$$

for all i . To see that f is a homomorphism, take $F^{[n]} \in \Omega$ and $(a_1, \dots, a_n) \in A^n$. Choose $j \in I$ and representatives $z_k \in \mathfrak{F}(D(j))$ with $a_k = \lambda_j(z_k)$. Then

$$\begin{aligned} f(|F|_A(a_1, \dots, a_n)) &= f(\lambda_j(|F|_{\mathbf{D}(j)}(z_1, \dots, z_n))) \\ &= \mathfrak{F}(f_j)(|F|_{\mathbf{D}(j)}(z_1, \dots, z_n)) \\ &= |F|_B(\mathfrak{F}(f_j)(z_1), \dots, \mathfrak{F}(f_j)(z_n)) \\ &= |F|_B(f(\lambda_j(z_1)), \dots, f(\lambda_j(z_n))) \\ &= |F|_B(f(a_1), \dots, f(a_n)). \end{aligned}$$

Hence f is a homomorphism and therefore a unique morphism in $\mathcal{C}$ with

$$f \circ \lambda_i = f_i.$$

Thus (A, λ_i) is the colimit of $\mathcal{D}$ in $\mathcal{C}$.

If A carried another τ -structure for which all λ_i are homomorphisms, then for each $F^{[n]}$ and each $(a_1, \dots, a_n) \in A^n$ the value of $|F|_A(a_1, \dots, a_n)$ would be forced by the identity

$$|F|_A(\lambda_i(y_1), \dots, \lambda_i(y_n)) = \lambda_i(|F|_{\mathbf{D}(i)}(y_1, \dots, y_n)),$$

and the fact that every a_k has a representative $\lambda_i(y_k)$. Hence the induced structure is unique, and $\mathfrak{F}$ creates filtered colimits. Thus $\mathfrak{F}$ creates finite limits and filtered colimits.

Claim 4: Filtered colimits commute with finite limits in $\mathcal{C}$.

Filtered colimits and finite limits commute in Set (Theorem 1.5.32). Therefore, for

any filtered diagram of finite-limit diagrams in $\mathcal{C}$,

$$\operatorname{colim}_i \left(\lim_j \mathbf{D}(i, j) \right) \rightarrow \lim_j \left(\operatorname{colim}_i \mathbf{D}(i, j) \right)$$

identifies it with the corresponding comparison map in $\mathbf{Set}$, which is an isomorphism because filtered colimits commute with finite limits in $\mathbf{Set}$. Since the underlying-set functor reflects isomorphisms, the comparison morphism is an isomorphism in $\mathcal{C}$ (see [4]). □

C An Intermediate Embedding Theorem

This chapter aims to prove an intermediate version for cocomplete categories of the Mitchell embedding theorem. More precisely, it is shown that every small full exact subcategory of a cocomplete abelian category admitting a projective generator has a full exact embedding into the category of modules. The general Mitchell embedding theorem will then be stated as a consequence of the fact that every small abelian category admits a full exact embedding into a cocomplete abelian category with a projective generator.

A word of caution: Here, an **embedding** is defined as a fully faithful functor.

Definition C.0.1 (Projective object). Let $\mathcal{C}$ be an abelian category. An object G is **projective** if the functor

$$\mathrm{Hom}(G, -) : \mathcal{C} \rightarrow \mathbf{Ab}$$

is exact (see [30, p. 8]).

Hence G is projective, if and only if, for every morphism

$$p : G \rightarrow B$$

and any epimorphism

$$e : A \rightarrow B,$$

there exists a lift

$$\tilde{p} : G \rightarrow A$$

of p , such that

$$e \circ \tilde{p} = p$$

(see [30, p. 8]).

The proof relies on the fact that projective objects are preserved by coproducts.

Proposition C.0.2. If $\{P_j\}$ is a family of projectives in an abelian category, then the coproduct $\coprod_j P_j$ is projective, if it exists (see [30, Prop. 3.6])

Proof. A map

$$p : \coprod_j P_j \rightarrow B$$

is given by individual maps

$$p_j : P_j \rightarrow B.$$

If

$$e : A \rightarrow B$$

is an epimorphism, there are componentwise lifts

$$\tilde{p}_j : P_j \rightarrow A,$$

that is,

$$e \circ \tilde{p}_j = p_j.$$

These collect into a map

$$\tilde{p} : \coprod_j P_j \rightarrow A,$$

which lifts p , since these maps agree on each P_j :

$$(e \circ \tilde{p}) \circ \iota_j = e \circ (\tilde{p} \circ \iota_j) = e \circ (\tilde{p}_j) = p_j = p \circ \iota_j,$$

where ι_j is the j -th inclusion into the sum. □

To obtain sufficiently many epimorphisms from a single projective object, one uses the standard characterisation of generators in a cocomplete abelian category.

Proposition C.0.3. In a cocomplete abelian category $\mathcal{C}$, G is a generator if and only if for every $A \in \text{Ob}(\mathcal{C})$ the map

$$\coprod_{\text{Hom}(G,A)} G \rightarrow A$$

is epimorphic (see [30, Prop. 3.10.]).

Proof. Let G be a generator. Suppose that there is some $A \in \text{Ob}(\mathcal{C})$ with

$$A \rightarrow B := \text{coker} \left(\coprod_{\text{Hom}(G,A)} G \rightarrow A \right) \neq 0.$$

Then there is a map $G \rightarrow A$ with

$$g \rightarrow A \rightarrow B \neq 0,$$

but this contradicts that

$$\coprod_{\text{Hom}(G,A)} G \rightarrow A \rightarrow B = 0.$$

Conversely, suppose that

$$\coprod_{\text{Hom}(G,A)} G \rightarrow A$$

is an epimorphism, then its cokernel is zero by Lemma 1.6.22. Suppose for a contradiction that there is some $A \rightarrow B \neq 0$ such that every $G \rightarrow A$ has

$$G \rightarrow A \rightarrow B = 0.$$

Then

$$\coprod_{\text{Hom}(G,A)} G \rightarrow A \rightarrow B = 0,$$

so $A \rightarrow B$ factors through

$$\text{coker} \left(\coprod_{\text{Hom}(G,A)} G \rightarrow A \right) = 0.$$

Thus $A \rightarrow B = 0$ (see [30, Prop. 3.10.]). □

With the preceding propositions installed, one can now prove the following intermediate embedding theorem:

Theorem C.0.4 (Intermediate Embedding Theorem). A cocomplete abelian category with a projective generator is fully abelian, that is, for every full small exact

subcategory $\mathcal{D}$ of $\mathcal{C}$ there exists a ring R and a full exact embedding of $\mathcal{D}$ into $R - \text{Mod}$ (see [30, Theorem 4.1]).

Proof. Let $\mathcal{D}$ be a small full exact subcategory of a cocomplete category $\mathcal{C}$. Let G be a projective generator for $\mathcal{C}$.

One modifies G :

For each $A \in \text{Ob}(\mathcal{D})$, one considers the epimorphism

$$\coprod_{\text{Hom}(G,A)} G \rightarrow A$$

from Proposition C.0.3. Let

$$I := \dot{\bigcup}_{A \in \text{Ob}(\mathcal{D})} \text{Hom}_{\mathcal{C}}(G, A),$$

and define

$$P := \coprod_I G.$$

By Proposition C.0.2, P is a projective generator, with the additional property, that for each $A \in \text{Ob}(\mathcal{D})$, there is an epimorphism $P \rightarrow A$.

Let R be the ring of endomorphisms on P . Here, multiplication in $R = \text{End}_{\mathcal{C}}(P)$ is written in the order $rs := s \circ r$.

For every $A \in \text{Ob}(\mathcal{C})$, the abelian group $\text{Hom}_{\mathcal{C}}(P, A)$ has a canonical $R - \text{Mod}$ structure: given

$$x : P \rightarrow A$$

in $\text{Hom}_{\mathcal{C}}(P, A)$, and

$$r : P \rightarrow P$$

in R , define

$$r \cdot x = x \circ r \in \text{Hom}_{\mathcal{C}}(P, A).$$

For every map $y \in \text{Hom}_{\mathcal{C}}(A, B)$, the induced morphism

$$\text{Hom}_{\mathcal{C}}(P, A) \xrightarrow{y \circ -} \text{Hom}_{\mathcal{C}}(P, B)$$

is R -linear:

$$\begin{aligned}
 (y \circ -)(r \cdot x) &= y \circ (r \cdot x) \\
 &= y \circ (x \circ r) \\
 &= (y \circ x) \circ r \\
 &= r \cdot (y \circ x) \\
 &= r((y \cdot -)x).
 \end{aligned}$$

Thus, one obtains a functor

$$F = \text{Hom}(P, -) : \mathcal{C} \rightarrow R\text{-Mod},$$

which is an exact embedding: the functor

$$\text{Hom}(P, -) : \mathcal{C} \rightarrow \text{Ab}$$

is an exact embedding by definition of P as a projective generator, and by the definition of generators, it is faithful. But $R\text{-Mod}$ is an exact subcategory of Ab . The forgetful inclusion functor $R\text{-Mod} \rightarrow \text{Ab}$ has a left-adjoint and a right-adjoint, so it preserves finite limits and colimits, so it is exact.

The restriction $F|_{\mathcal{D}}$ is therefore an exact embedding. It remains to show that it is full.

Let $A, B \in \text{Ob}(\mathcal{D})$, and $\tilde{y} \in \text{Hom}_{R\text{-Mod}}(F(A), F(B))$. Since $A, B \in \text{Ob}(\mathcal{D})$, there are exact sequences

$$0 \rightarrow K \rightarrow P \rightarrow A \rightarrow 0$$

and

$$P \rightarrow B \rightarrow 0$$

in $\mathcal{C}$ coming from the epimorphisms $P \rightarrow A$ and $P \rightarrow B$, with

$$K \rightarrow P = \ker(P \rightarrow A).$$

Since $F(P) = R$, taking F gives the following commutative diagram in $R - \text{Mod}$:

$$\begin{array}{ccccccc} 0 & \longrightarrow & F(K) & \longrightarrow & R & \longrightarrow & F(A) \longrightarrow 0 \\ & & & & \downarrow f & & \downarrow \tilde{y} \\ & & & & R & \longrightarrow & F(B) \longrightarrow 0 \end{array}$$

where f is a lift of

$$R \rightarrow F(A) \xrightarrow{\tilde{y}} F(B),$$

where R is projective, and $R \rightarrow F(B)$ is an epimorphism.

Since R is a ring, one has $\text{End}(R) \cong R^{\text{op}}$, that is, any endomorphism on R is given by multiplication on the right by some R -element.

Hence for all $s \in R$ with $r : P \rightarrow P$ in R ,

$$f(s) = s \cdot r = s \circ r.$$

In the diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & K & \longrightarrow & P & \longrightarrow & A \longrightarrow 0 \\ & & & & \downarrow r & & \\ & & & & P & \longrightarrow & B \longrightarrow 0 \end{array}$$

in $\mathcal{C}$, one has

$$K \rightarrow P \xrightarrow{r} P \rightarrow B = 0,$$

as

$$F(K) \rightarrow R \xrightarrow{f} R \rightarrow F(B) = 0$$

and F is an embedding. Hence

$$P \xrightarrow{r} P \rightarrow B$$

factors through the cokernel $P \rightarrow A$: there exists a morphism $y : A \rightarrow B$ such that

$$\begin{array}{ccc} P & \longrightarrow & A \\ \downarrow r & & \downarrow y \\ P & \longrightarrow & B \end{array}$$

commutes. Hence

$$\begin{array}{ccc} R & \longrightarrow & F(A) \\ \downarrow f & & \downarrow F(y) \\ R & \longrightarrow & F(B) \end{array}$$

commutes. Therefore

$$R \rightarrow F(A) \xrightarrow{F(y)} F(B) = R \xrightarrow{f} R \rightarrow F(B) = R \rightarrow F(A) \xrightarrow{\tilde{y}} F(B).$$

Since $R \rightarrow F(A)$ is epimorphic, $F(y) = \tilde{y}$. □

One can show that this holds not only for cocomplete abelian categories, but for all abelian categories.

Remark C.0.5. The preceding theorem is the projective-generator case of the Freyd–Mitchell embedding theorem. To obtain the general result, one shows that every small abelian category admits a full exact embedding into a cocomplete abelian category possessing a projective generator. Applying the preceding theorem to its essential image then yields a full exact embedding into a category of modules.

The proof relies on essential extensions, injective envelopes, the Yoneda-embedding, torsions, and pure subobjects, among other things.

D Elaborations on Sheaves, Sheafification, and Sieves

This chapter collects several auxiliary results on Grothendieck topologies, sieves, sheafification, and sheaves with values in Grothendieck categories that are used throughout this thesis.

D.1 Elaborations on Sieves and Topological Spaces

The ordinary topology of a topological space X induces a Grothendieck topology on the category $\text{Open}(X)$ of open subsets of X . In this setting, covering sieves can be characterised entirely in terms of ordinary open covers. The following lemmas make the correspondence between covering families and covering sieves explicit.

Lemma D.1.1. In the usual Grothendieck topology on $\text{Open}(X)$ induced by the topology of X , a sieve S on an open set $U \in \text{Ob}(\text{Open}(X))$ is a covering sieve of U if and only if

$$\bigcup_{V \in S} V = U$$

(see [32, Ch. 16, §1, pp. 390 ff.]).

Proof. $\Rightarrow$: Suppose S is a cover. By definition there exists a covering family $\{U_i\}_{i \in I}$ with $U_i \subseteq U$ for all $i \in I$ and

$$\bigcup_i U_i = U$$

such that $\langle U_i \rangle \subseteq S$. In particular, each $U_i \in \langle U_i \rangle$, hence $U_i \in S$. Therefore

$$U = \bigcup_i U_i \subseteq \bigcup_{V \in S} V \subseteq U,$$

so $\bigcup S = U$.

$\Leftarrow$: Suppose $\bigcup S = U$. Consider the family of inclusions $\{V \rightarrow U\}_{V \in S}$. Since its union is U , it is a covering family, and since S is a downward closed sieve, the sieve generated by the family $\{V \rightarrow U\}_{V \in S}$ is precisely S itself. Hence S is a covering sieve (see [32, Ch. 16, §1, pp. 390 ff.]). $\square$

The previous lemma characterises covering sieves. Conversely, every ordinary open cover determines a canonical covering sieve, as shown below.

Lemma D.1.2. Let $\mathcal{U} := \{U_i\}_{i \in I}$ be a topological cover of U . Then there exists an associated covering sieve

$$S_{\{U_i\}} := \{V \subseteq U \mid \exists i \in I : V \subseteq U_i\}$$

(see [32, Ch. 16, §1, pp. 390 ff.]).

Proof. Let $U \in \text{Ob}(\text{Open}(X))$ and let $\{U_i \rightarrow U\}_{i \in I}$ be a covering family. Define the associated sieve $S_{\{U_i\}}$ on U by

$$S_{\{U_i\}} := \{V \subseteq U \text{ open} \mid \exists i \in I : V \subseteq U_i\}.$$

$S_{\{U_i\}}$ is the smallest sieve on U that contains all arrows $U_i \rightarrow U$, often denoted as $S_{\{U_i\}} = \langle U_i \rightarrow U \rangle_{i \in I}$.

Claim: $S_{\{U_i\}}$ is a sieve.

Take $V \in S_{\{U_i\}}$. Then $V \subseteq U_i$ for some i . If $W \subseteq V$, that is, $W \rightarrow V \rightarrow U$, then $W \subseteq U_i$, hence $W \in S_{\{U_i\}}$. Thus $S_{\{U_i\}}$ is downward closed and therefore a sieve.

Claim: $S_{\{U_i\}}$ is a cover.

Compute

$$\bigcup_{V \in S_{\{U_i\}}} V.$$

For every $i \in I$, one has $U_i \in S_{\mathcal{U}}$ since $U_i \subseteq U_i$. Therefore

$$\bigcup_{i \in I} U_i \subseteq \bigcup_{V \in S_{\mathcal{U}}} V.$$

Hence, each U_i lies in the sieve. Since every $V \in S_{\{U_i\}}$ is contained in some U_i ,

$$\bigcup_{V \in S_{\{U_i\}}} V \subseteq \bigcup_i U_i.$$

Thus

$$\bigcup_{V \in S_{\{U_i\}}} V = \bigcup_i U_i = U.$$

By Lemma D.1.1, $S_{\{U_i\}}$ is a covering sieve (see [32, Ch. 16, §1, pp. 390 ff.]). $\square$

D.2 Sheafification

In general, a presheaf is not automatically a sheaf. The process of turning a presheaf into a sheaf, called sheafification, and is often non-trivial.

Definition D.2.1 (Sheafification). Let $F : \text{Open}(X)^{\text{op}} \rightarrow \mathcal{C}$ be a presheaf. A **sheafification** of F is a sheaf F^{Sh} together with a morphism

$$\text{Sh} : F \rightarrow F^{\text{Sh}}$$

such that for every sheaf $\mathcal{G}$, every morphism $F \rightarrow \mathcal{G}$ factors uniquely through Sh , that is, there exists a unique morphism $F^{\text{Sh}} \rightarrow \mathcal{G}$ making the diagram

$$\begin{array}{ccc} F & \xrightarrow{\quad} & \mathcal{G} \\ \text{Sh} \searrow & & \nearrow \\ & F^{\text{Sh}} & \end{array}$$

commute.

Equivalently, sheafification is left-adjoint to the inclusion

$$\text{Sh}(X, \mathcal{C}) \hookrightarrow \text{PSh}(X, \mathcal{C}).$$

For sheaves on a topological space, the canonical morphism Sh induces an isomorphism on every stalk (see [8, Ch. 9, §5, Ex. 9.5.12]).

The definition does not assert either the existence or the exactness of the sheafification. For presheaves on a topological space and with values in an algebraic category, the following construction used in Lemma , guarantees the existence of sheafification.

Construction D.2.2 (Sheafification). Let $F \in \text{Ob}(\text{PSh}(X, \mathcal{C}))$, where $\mathcal{C}$ denotes an algebraic category. For each $x \in X$, let

$$F_x := \text{colim}_{x \in U} F(U)$$

be the stalks of F at x . Define

$$E_F := \coprod_{x \in X} F_x$$

and let

$$\text{pr} : E_F \rightarrow X$$

be the projection sending an element of F_x to x . For an open subset $U \subseteq X$ and a section $s \in F(U)$, define

$$\tilde{s} : U \rightarrow E_F, \quad x \mapsto s_x.$$

Give E_F the topology generated by subsets

$$\tilde{s}(U) \subseteq E_F$$

for all open $U \subseteq X$ and all $s \in F(U)$. Then $\text{pr} : E_F \rightarrow X$ is a local homeomorphism. The sheafification of F is the sheaf

$$F^{\text{Sh}}(U) := \{\sigma : U \rightarrow E_F \mid \text{pr} \circ \sigma = \text{id}_U, \sigma \text{ is continuous}\}.$$

The canonical morphism $F \rightarrow F^{\text{Sh}}$ is

$$\eta_U : F(U) \rightarrow F^{\text{Sh}}(U), \quad s \mapsto \tilde{s}$$

(see [33, Ch. 2, §6]).

The étalé-space construction above is specific to presheaves on topological spaces. On an arbitrary small site, set-valued sheafification may instead be constructed by the double-plus construction.

Theorem D.2.3 (Exactness of sheafification). [2.2.242.2.24](#) Let $(\mathcal{C}, J)$ be a small site. Then for the inclusion

$$\iota : \mathrm{Sh}((\mathcal{C}, J), \mathrm{Set}) \hookrightarrow \mathrm{PSh}((\mathcal{C}, J), \mathrm{Set})$$

there exists a left-adjoint, namely the sheafification functor

$$s^\# : \mathrm{PSh}((\mathcal{C}, J), \mathrm{Set}) \rightarrow \mathrm{Sh}((\mathcal{C}, J), \mathrm{Set})$$

(see [52, Ch. 7, §10], [32, Ch. 17, §4]).

Proof. Let $F : \mathcal{C}^{\mathrm{op}} \rightarrow \mathrm{Set}$ be a presheaf. For an object $U \in \mathrm{Ob}(\mathcal{C})$ and a covering sieve $R \in J(U)$, one considers the limit in Set of the diagram determined by the restrictions of F to the arrows belonging to R , that is,

$$\lim_{(V \rightarrow U) \in R} F(V).$$

This limit consists of the compatible families of sections of F indexed by the sieve R .

Define the **plus construction** by

$$F^+(U) := \mathrm{colim}_{R \in J(U)} \lim_{(V \rightarrow U) \in R} F(V),$$

where the colimit is taken over covering sieves ordered by refinement. Since $\mathcal{C}$ is small, the relevant diagrams are small, and in Set , all required limits and colimits exist.

The usual restriction morphisms make F^+ into a presheaf, and there is a natural morphism

$$\eta_F : F \rightarrow F^+.$$

Claim: F^+ is separated.

Let $U \in \text{Ob}(\mathcal{C})$, and let $R \in J(U)$ be a covering sieve. One has to show that the canonical map

$$F^+(U) \rightarrow \lim_{(V \rightarrow U) \in R} F^+(V)$$

is injective.

Let $a, b \in F^+(U)$, and suppose that their restrictions along every arrow

$$f : V \rightarrow U$$

belonging to R agree in $F^+(V)$.

One can choose covering sieves $R_a, R_b \in J(U)$ and matching families

$$s = (s_g)_{g \in R_a} \in \lim_{(V \rightarrow U) \in R_a} F(V), \quad t = (t_g)_{g \in R_b} \in \lim_{(V \rightarrow U) \in R_b} F(V)$$

representing a and b , respectively. Replacing R_a and R_b by their intersection, which is again a covering sieve, and restricting both matching families, one may assume that they are defined on the same covering sieve

$$R_0 := R_a \cap R_b.$$

Fix an arrow $f : V \rightarrow U$ belonging to R . The restrictions $a|_V$ and $b|_V$ are represented by s and t to the pullback sieve

$$f^*R_0 := \{h : W \rightarrow V \mid f \circ h \in R_0\}.$$

Since $a|_V = b|_V$ in $F^+(V)$, there exists a covering sieve

$$Q_f \in J(V), \quad Q_f \subseteq f^*R_0,$$

such that the two restricted matching families agree on Q_f . Thus, for every arrow $h : W \rightarrow V$ in Q_f ,

$$s_{f \circ h} = t_{f \circ h}.$$

Define Q to be the sieve on U generated by all composites $f \circ h$, where

$$f : V \rightarrow U \in R, \quad h : W \rightarrow V \in Q_f.$$

For every $f : V \rightarrow U$ in R , the pullback sieve f^*Q contains Q_f and is therefore a cover. Since R is a cover, the local character axiom of the Grothendieck topology implies that Q is a covering sieve on U .

Moreover, each Q_f is contained in f^*R_0 , so every generating composite $f \circ h$ belongs to R_0 . Hence $Q \subseteq R_0$. The matching families s and t agree on the generators of Q . Since they are matching families, they also agree on every arrow of the sieve generated by those arrows. Consequently,

$$s|_Q = t|_Q.$$

Thus the representatives s and t agree after restriction to the common covering refinement Q . They therefore determine the same element of

$$F^+(U) = \operatorname{colim}_{R' \in J(U)} \lim_{(V \rightarrow U) \in R'} F(V).$$

Hence $a = b$, proving that F^+ is separated.

Claim: If G is a separated presheaf, then G^+ is a sheaf.

Let $U \in \operatorname{Ob}(\mathcal{C})$, let $R \in J(U)$ be a covering sieve, and let

$$(a_f)_{f \in R} \in \lim_{(f:V \rightarrow U) \in R} G^+(V)$$

be a matching family. For every morphism $f : V \rightarrow U$ in R , one can choose a covering sieve $R_f \in J(V)$ and a matching family

$$(s_{f,h})_{h \in R_f} \in \lim_{(h:W \rightarrow V) \in R_f} G(W)$$

representing a_f .

Let Q be the sieve on U generated by the composites

$$f \circ h, \quad f \in R, \quad h \in R_f.$$

For each $f \in R$, the pullback f^*Q contains R_f and is therefore covering. Since R is covering, the local character axiom 2.2.3 implies that Q is a covering sieve on U .

The sections $s_{f,h}$ determine a matching family of G on Q . Suppose that a morphism $k : Z \rightarrow U$ admits two factorisations

$$k = f \circ h = \tilde{f} \circ \tilde{h},$$

where $f, \tilde{f} \in R$, $h \in R_f$, and $\tilde{h} \in R_{\tilde{f}}$. Since $(a_f)_{f \in R}$ is a matching family in G^+ , the corresponding restrictions of a_f and $a_{\tilde{f}}$ agree in $G^+(Z)$. Hence there exists a covering sieve $S \in J(Z)$ on which the restrictions of $s_{f,h}$ and $s_{\tilde{f},\tilde{h}}$ agree. Since G is separated, the two sections are already equal in $G(Z)$. The same argument, after restriction, proves compatibility with all morphisms into Z . Consequently, the sections $s_{f,h}$ form a matching family on Q .

This matching family defines an element

$$a \in G^+(U).$$

By construction, the restriction of a along each $f \in R$ is a_f . Thus every matching family in G^+ admits an amalgamation. Since G^+ is separated by the preceding claim, this amalgamation is unique. Therefore G^+ is a sheaf.

Hence, applying the plus-construction once more to F^+ , yields a sheaf F^{++} .

Define

$$s^\# F := F^{++}.$$

Let $\mathcal{G}$ be a sheaf and let

$$\varphi : F \rightarrow \iota(\mathcal{G})$$

be a morphism of presheaves. For each $U \in \text{Ob}(\mathcal{C})$, define

$$\varphi_U^+ : F^+(U) \rightarrow \mathcal{G}(U)$$

as follows:

If an element of $F^+(U)$ is represented by a matching family $(s_f)_{f \in R}$ on a covering sieve $R \in J(U)$, let

$$\varphi_U^+([(s_f)_{f \in R}])$$

be the unique amalgamation of the matching family

$$(\varphi_V(sf))_{(f:V \rightarrow U) \in R}$$

in $\mathcal{G}$. If two matching families represent the same element of $F^+(U)$, then they agree after restriction to a common covering refinement. Their amalgamation in $\mathcal{G}(U)$ are therefore equal, since $\mathcal{G}$ is separated. Hence φ_U^+ is well-defined.

The uniqueness of amalgamation also implies that the morphisms φ_U^+ commute with restrictions. Thus they define a morphism of presheaves

$$\varphi^+ : F^+ \rightarrow \mathcal{G}.$$

Moreover,

$$\varphi^+ \circ \eta_F = \varphi.$$

Applying the same construction to $\varphi^+ : F^+ \rightarrow \mathcal{G}$ yields a morphism

$$\tilde{\varphi} : F^{++} \rightarrow \mathcal{G}$$

such that

$$\tilde{\varphi} \circ \eta_{F^+} = \varphi^+.$$

Consequently, if

$$\eta_F^\# : F \rightarrow F^{++}$$

denotes the composite

$$F \xrightarrow{\eta_F} F^+ \xrightarrow{\eta_{F^+}} F^{++},$$

then

$$\tilde{\varphi} \circ \eta_F^\# = \varphi.$$

It remains to prove uniqueness. Let

$$\psi : F^{++} \rightarrow \mathcal{G}$$

be another morphism satisfying

$$\psi \circ \eta_F^\# = \varphi.$$

Every element of $F^+(U)$ is locally represented by sections coming from F . Since $\mathcal{G}$ is separated, this implies that

$$\psi \circ \eta_{F^+} = \varphi^+.$$

Similarly, every element of $F^{++}(U)$ is locally represented by sections coming from $F^+(U)$, and hence separatedness of $\mathcal{G}$ implies

$$\psi = \tilde{\varphi}.$$

Therefore composition with $\eta_F^\#$ induces a bijection

$$\mathrm{Hom}_{\mathrm{Sh}((\mathcal{C}, J), \mathrm{Set})}(F^{++}, \mathcal{G}) \cong \mathrm{Hom}_{\mathrm{PSh}((\mathcal{C}, J), \mathrm{Set})}(F, \iota(\mathcal{G})),$$

natural in F and $\mathcal{G}$. Thus $s^\#$ is left-adjoint to ι . □

The preceding double-plus construction establishes the existence of sheafification for set-valued presheaves on a small site. Its proof relies essentially on the elementwise description of matching families and therefore does not immediately extend to presheaves with values in an arbitrary abelian category. For the abelian-valued setting required later in this thesis, one instead uses the construction of Kashiwara–Schapira. Under the hypotheses stated below, the associated sheaf functor exists, is left-adjoint to the inclusion of sheaves into presheaves, and is exact.

Definition D.2.4 (Projector). Let $\mathcal{C}$ be a category. A **projector** on $\mathcal{C}$ is a pair (P, ε) , where $P : \mathcal{C} \rightarrow \mathcal{C}$ is an endofunctor, and $\varepsilon : \mathrm{id}_{\mathcal{C}} \rightarrow P$ is a natural transformation such that, for every object $A \in \mathrm{Ob}(\mathcal{C})$, both morphisms

$$P(\varepsilon_A), \quad \varepsilon_{P(A)} : P(A) \rightarrow P^2(A)$$

are isomorphisms (see [32, Ch. 2, Ex. 2.9, p. 66]).

The projector used in the construction of sheafification is not defined directly on the original site, but on a larger presheaf category, where covering data can be encoded by single morphisms. This motivates the definition of enlarged sites.

Definition D.2.5 (Enlarged category of a site). Let X be a small site with underlying category $\mathcal{C}_X$. Define

$$\widehat{\mathcal{C}}_X := \text{PSh}(\mathcal{C}_X, \text{Set}) = [\mathcal{C}_X^{\text{op}}, \text{Set}].$$

The original category then embeds into this larger category through the Yoneda embedding

$$h_X : \mathcal{C}_X \rightarrow \widehat{\mathcal{C}}_X, \quad U \mapsto h_U := \text{Hom}_{\mathcal{C}_X}(-, U)$$

(see [32, Ch. 17, §1]).

The point is, that $\widehat{\mathcal{C}}_X$ contains not only the original objects U , but also objects obtained from them using arbitrary small colimits. For example, a covering family $(U_i \rightarrow U)_{i \in I}$ can be represented by the morphism

$$\coprod_{i \in I} h_{U_i} \rightarrow h_U.$$

This allows covering families to be treated as single morphisms. Local epimorphisms and local isomorphisms are defined inside $\widehat{\mathcal{C}}_X$.

Definition D.2.6 (Local Isomorphism). Let $(\mathcal{C}, J)$ be a small site and let $f : A \rightarrow B$ be a morphism in

$$\widehat{\mathcal{C}} := [\mathcal{C}^{\text{op}}, \text{Set}].$$

The morphism f is called a **local epimorphism** if every section of B admits local lifts along a covering, that is, for every object $U \in \text{Ob}(\mathcal{C})$ and every $t \in B(U)$, there exists a covering sieve $R \in J(U)$ such that, for every morphism

$$u : V \rightarrow U$$

belonging to R , the restricted section $t|_V$ lies in the image of

$$f_V : A(V) \rightarrow B(V).$$

Moreover, u is called a **local monomorphism** if the diagonal morphism

$$A \rightarrow A \times_B A$$

is a local epimorphism.

If f is both, a local monomorphism and a local epimorphism, it is called a **local isomorphism** (see [32, Ch. 16, §2, Def. 16.2.1]).

The class of local isomorphisms is denoted by $\mathcal{LI}$. For an object $A \in \text{Ob}(\widehat{\mathcal{C}})$, let $\mathcal{LI}_A$ denote the category whose objects are local isomorphisms

$$u : B \rightarrow A,$$

and whose morphisms are given by

$$\text{Hom}_{\mathcal{LI}_A}((B \xrightarrow{u} A), (C \xrightarrow{v} A)) = \{w : B \rightarrow C, u = v \circ w\}$$

(see [32, Ch. 16, §2, Notation 16.2.5]).

To apply this enlarged setting to presheaves originally defined on the site X , one needs a canonical way of extending them from representable objects to arbitrary objects of $\widehat{\mathcal{C}}_X$. This is provided by the following extension functor.

Definition D.2.7 (The extension functor). Consider an $\mathcal{A}$ -valued presheaf

$$F : \mathcal{C}_X^{\text{op}} \rightarrow \mathcal{A}.$$

Its extension

$$h_X^\dagger F : \widehat{\mathcal{C}}_X^{\text{op}} \rightarrow \mathcal{A}$$

is given by

$$(h_X^\dagger F)(A) \cong \lim_{(h_U \rightarrow A) \in \text{Ob}((\mathcal{C}_X)_A)} F(U).$$

Here, the category $(\mathcal{C}_X)_A$ is the category whose objects are pairs (U, u) , where $U \in \text{Ob}(\mathcal{C}_X)$, and

$$u : h_U \rightarrow A,$$

whose morphisms are evident commuting triangles (see [32, Ch. 17, §1, p. 407]).

Informally, $h_X^\dagger F(A)$ is determined by looking at all representable objects mapping into A and taking the compatible limit over their values under F . For a representable $A = h_U$, the Yoneda lemma gives

$$(h_X^\dagger F)(h_U) \cong F(U).$$

Thus $h_X^\dagger F$ really extends F .

Definition D.2.8 (Connected category). A category $\mathcal{C}$ is called **connected** if it is non-empty and for every pair of objects $A, B \in \text{Ob}(\mathcal{C})$ there exists a finite sequence of objects

$$A = C_0, C_1, \dots, C_n = B,$$

such that, for every $i = 0, \dots, n-1$, at least one of $\text{Hom}_{\mathcal{C}}(C_i, C_{i+1})$ or $\text{Hom}_{\mathcal{C}}(C_{i+1}, C_i)$ is non-empty (see [32, Ch. 1, §2, p. 13]).

The construction below involves colimits indexed by categories of local isomorphisms. To replace such indexing categories by smaller ones without changing the resulting colimits, one uses cofinal functors.

Definition D.2.9 (Cofinal functor). A functor

$$\varphi : \mathcal{J} \rightarrow \mathcal{I}$$

is called **cofinal** if, for every object $i \in \text{Ob}(\mathcal{I})$, the comma category $(i \downarrow \varphi)$ is non-empty and connected (see [32, Ch. 2, §5, Def. 2.5.1]).

Cofinality allows one to compute a colimit over a category by passing to a suitable subcategory. This motivates the following smallness condition on possibly large indexing categories.

Definition D.2.10 (Cofinally small category). A category $\mathcal{I}$ is called **cofinally small** if there exists a small category $\mathcal{J}$ and a cofinal functor

$$\varphi : \mathcal{J} \rightarrow \mathcal{I}$$

(see [32, Ch. 2, §5, Def. 2.5.5]).

One can amplify this notion to the categories of local isomorphisms introduced above. Although these categories need not themselves be small, their opposites admit small cofinal indexing categories; this is precisely what ensures that colimits appearing in the construction of the projector $(-)^b$ are legitimate.

Lemma D.2.11. Let $\mathcal{C}$ be a small category endowed with a Grothendieck topology, let

$$\widehat{\mathcal{C}} := [\mathcal{C}^{\text{op}}, \text{Set}],$$

and let $A \in \text{Ob}(\widehat{\mathcal{C}})$. Then $(\mathcal{LI}_A)^{\text{op}}$ is cofinally small (see [32, Ch. 2, §5, Def. 2.5.5, Corollary 2.5.6]).

Proof. The category $\mathcal{LI}_A$ admits finite limits, and is therefore cofiltered. Hence $(\mathcal{LI}_A)^{\text{op}}$ is filtered.

Set

$$I := \{(U, s) \mid U \in \mathcal{C}, s \in A(U)\}.$$

Since $\mathcal{C}$ is small, I is a small set. For $i = (U, s) \in I$, write $U_i := U$. By the Yoneda lemma, the elements $s \in A(U)$ induce a canonical epimorphism

$$\coprod_{i \in I} h_{U_i} \rightarrow A.$$

For $J \subseteq I$, set

$$C_J := \coprod_{j \in J} h_{U_j}.$$

Let Σ be the collection of diagrams

$$C_J \twoheadrightarrow S \rightarrow A$$

in which the first morphism is an epimorphism, and the second morphism is a local isomorphism. Since I is small and every C_J has only a small set of quotients up to isomorphism, Σ is small.

The objects $S \rightarrow A$ occurring in Σ form a cofinal family in $(\mathcal{LI}_A)^{\text{op}}$. Let

$$B \rightarrow A$$

be a local isomorphism and let

$$B_1 := \text{im}(B \rightarrow A).$$

Define

$$J := \{(U, s) \in I \mid s \in B_1(U)\}.$$

Then the canonical morphism

$$C_J \rightarrow B_1$$

is an epimorphism. Since $B(U) \rightarrow B_1(U)$ is surjective for every $U \in \mathcal{C}$, this morphism lifts to a morphism

$$C_J \rightarrow B.$$

Set

$$S := \text{im}(C_J \rightarrow B).$$

One obtains from the factorisation

$$C_J \twoheadrightarrow S \rightarrow B \rightarrow A.$$

Since $B \rightarrow A$ and $B_1 \rightarrow A$ are local isomorphisms, $B \rightarrow B_1$ is a local isomorphism. As $B \rightarrow B_1$ is an epimorphism, $C_J \rightarrow B$ is a local epimorphism. Hence $S \rightarrow B$ is a local epimorphism. Since it is also a monomorphism, it is a local isomorphism. Consequently, $S \rightarrow A$ is a local isomorphism as well.

Thus $C_J \twoheadrightarrow S \rightarrow A$ belongs to Σ , and $S \rightarrow B$ defines a morphism

$$(S \rightarrow A) \rightarrow (B \rightarrow A)$$

in $\mathcal{LI}_A$, or equivalently, a morphism

$$(B \rightarrow A) \rightarrow (S \rightarrow A)$$

in $(\mathcal{LI}_A)^{\text{op}}$.

Therefore every object of $(\mathcal{LI}_A)^{\text{op}}$ admits a morphism to an object in a small family. Since $(\mathcal{LI}_A)^{\text{op}}$ admits a morphism to an object in a small family. Since $(\mathcal{LI}_A)^{\text{op}}$ is filtered, it is cofinally small (see [32, Ch. 2, §5, Def. 2.5.5, Corollary 2.5.6]). $\square$

Cofinally small means that colimits over $\mathcal{I}$ can be computed over $\mathcal{J}$. The category itself may be large, but for the purpose of computing colimits, a small part of it

suffices. This guarantees that in the following generalisation,

$$F^b(A) = \operatorname{colim}_{(B \rightarrow A) \in \operatorname{Ob}(\mathcal{LT}_A^{\operatorname{op}})} F(B)$$

is a legitimate small colimit.

One can generalise Theorem 2.2.24 as follows:

Let X be a site, and let $\mathcal{A}$ be a category satisfying the following conditions:

1. *$\mathcal{A}$ admits small limits and small colimits;*
2. *small filtered colimits are exact;*
3. *$\mathcal{A}$ satisfies the IPC-property, that is, for every small family of small filtered categories $(I_s)_{s \in S}$, and every family of functors $\alpha_s : I_s \rightarrow \mathcal{A}$, the canonical morphism*

$$\operatorname{colim}_{(i_s)_{s \in S} \in \prod_{s \in S} I_s} \prod_{s \in S} \alpha_s(i_s) \rightarrow \prod_{s \in S} \operatorname{colim}_{i_s \in I_s} \alpha_s(i_s)$$

is an isomorphism.

Then the inclusion

$$\iota : \operatorname{Sh}(X, \mathcal{A}) \rightarrow \operatorname{PSh}(X, \mathcal{A})$$

admits a left-adjoint

$$a : \operatorname{PSh}(X, \mathcal{A}) \rightarrow \operatorname{Sh}(X, \mathcal{A}).$$

*For $F \in \operatorname{Ob}(\operatorname{PSh}(X, \mathcal{A}))$, the sheaf $F^a := a(F)$ is called the **sheaf associated with F** , or the **sheafification of F** . If, in addition, $\mathcal{A}$ is abelian, then the functor a is exact (see [32, Ch. 17, §4]).*

Proof. The proof is divided into several steps.

Step 1: The categories of local isomorphisms.

For $A \in \operatorname{Ob}(\widehat{\mathcal{C}})_X$, let $\mathcal{LT}_A$ denote the category whose objects are local isomorphisms

$$p : B \rightarrow A,$$

and whose morphisms from $p : B \rightarrow A$ to $p' : B' \rightarrow A$ are morphisms $v : B \rightarrow B'$ satisfying $p = p' \circ v$.

The category $\mathcal{LI}_A$ is cofiltered, and its opposite $\mathcal{LI}_A^{\text{op}}$ is filtered and cofinally small. Consequently, the colimits used below are well-defined.

Step 2: Construction of the functor F^b .

Let $F : \widehat{X} \rightarrow \mathcal{A}$ be a presheaf. For $A \in \text{Ob}(\widehat{\mathcal{C}})$, define

$$F^b(A) := \text{colim}_{(B \rightarrow A) \in \text{Ob}(\mathcal{LI}_A)} F(B). \quad (\text{D.2.12})$$

Since F is contravariant, it defines a functor

$$\alpha_A : (\mathcal{LI}_A)^{\text{op}} \rightarrow \mathcal{A}, \quad (B \rightarrow A) \rightarrow F(B),$$

and the expression in (D.2.12) is the colimit of α_A .

One now defines the restriction morphism of F^b .

Let

$$u : A' \rightarrow A$$

be a morphism in $\widehat{\mathcal{C}}_X$. The pullback along u defines a functor

$$\lambda_u : \mathcal{LI}_A \rightarrow \mathcal{LI}_{A'}, \quad (B \rightarrow A) \rightarrow (B \times_A A' \rightarrow A').$$

For every local isomorphism $B \rightarrow A$, the projection

$$B \times_A A' \rightarrow B$$

induces a morphism

$$F(B) \rightarrow F(B \times_A A').$$

Passing to colimits gives

$$F^b(u) : F^b(A) \rightarrow F^b(A').$$

Explicitly, this morphism is the composite

$$\text{colim}_{(B \rightarrow A) \in \mathcal{LI}_A} F(B) \rightarrow \text{colim}_{(B \rightarrow A) \in \mathcal{LI}_A} F(B \times_A A') \rightarrow \text{colim}_{(B' \rightarrow A') \in \mathcal{LI}_{A'}} F(B'),$$

where the second morphism is induced by λ_u .

The compatibility of pullbacks with identities and compositions shows that

$$A \rightarrow F^b(A)$$

defines a presheaf on $\widehat{X}$. Hence one obtains a functor

$$(-)^b : \text{PSh}(\widehat{X}, \mathcal{A}) \rightarrow \text{PSh}(\widehat{X}, \mathcal{A}).$$

Step 3: The natural morphism $F \rightarrow F^b$.

For every A , the identity id_A is a local isomorphism and hence an object of $\mathcal{LI}_A$. Therefore, the canonical morphism from the corresponding term into the colimit gives a morphism

$$\varepsilon_{b,F}(A) : F(A) \rightarrow F^b(A).$$

These morphisms are natural in A and in F . Thus there is a natural transformation

$$\varepsilon_b : \text{id}_{\text{PSh}(\widehat{X}, \mathcal{A})} \rightarrow (-)^b. \quad (\text{D.2.13})$$

Step 4: Invariance under local isomorphisms.

Suppose that

$$u : A' \rightarrow A$$

is itself a local isomorphism. Besides the pullback functor λ_u , there is a functor

$$\mu_u : \mathcal{LI}_{A'} \rightarrow \mathcal{LI}_A, \quad (B \rightarrow A') \rightarrow (B \rightarrow A' \xrightarrow{u} A).$$

The pair (μ_u, λ_u) is an adjoint pair. Hence the induced morphism

$$F^b(A) \rightarrow F^b(A')$$

is an isomorphism. Thus

$$F^b(A) \xrightarrow{\sim} F^b(A') \quad (\text{D.2.14})$$

for every local isomorphism $A' \rightarrow A$.

Step 5: The projector property of $(-)^b$.

It holds that

$$\varepsilon_{b,F^b} = (\varepsilon_{b,F})^b : F^b \rightarrow F^{bb},$$

and that this is an isomorphism.

Indeed, for $A \in \text{Ob}(\widehat{\mathcal{C}}_X)$, one has

$$F^{bb}(A) = \text{colim}_{(B \rightarrow A) \in \mathcal{LI}_A} F^b(B).$$

By (D.2.14), every morphism

$$F^b(A) \rightarrow F^b(B)$$

induced by local isomorphism $B \rightarrow A$ is an isomorphism. Hence the canonical morphism

$$F^b(A) \rightarrow \text{colim}_{(B \rightarrow A) \in \mathcal{LI}_A} F^b(B) = F^{bb}(A)$$

is an isomorphism. Consequently, the pair $((-)^b, \varepsilon_b)$ is a projector on $\text{PSh}(\widehat{X}, \mathcal{A})$.

Step 6: Preservation of small limits.

Recall that $\text{PSh}(X, \mathcal{A})$ is identified with the full subcategory of $\text{PSh}(\widehat{X}, \mathcal{A})$ consisting of presheaves which commute with small limits.

Let $F \in \text{Ob}(\text{PSh}(\widehat{X}, \mathcal{A}))$ commute with small limits. One shows that F^b also commutes with small limits. Since every small limit can be constructed from small products and fibre products, it is enough to show that F^b commutes with small products and fibre products.

First, let $(A_i)_{i \in I}$ be a small family of objects. There is a canonical morphism

$$F^b\left(\prod_{i \in I} A_i\right) \rightarrow \prod_{i \in I} F^b(A_i).$$

The categories $\mathcal{LI}_{A_i}$ are filtered. Their product

$$K := \prod_{i \in I} \mathcal{LI}_{A_i}$$

is cofiltered, while K^{op} is filtered and cofinally small, and the functor

$$K \rightarrow \mathcal{LI}_{\prod_i A_i}, \quad (B_i \rightarrow A_i)_{i \in I} \rightarrow \left(\prod_i B_i \rightarrow \prod_i A_i \right)$$

is cofinal.

It follows that

$$\begin{aligned} F^b\left(\prod_{i \in I} A_i\right) &\cong \text{colim}_{(B_i \rightarrow A_i)_{i \in I}} F\left(\prod_{i \in I} B_i\right) \\ &\cong \text{colim}_{(B_i \rightarrow A_i)_{i \in I}} \prod_{i \in I} F(B_i) \\ &\cong \prod_{i \in I} \text{colim}_{(B_i \rightarrow A_i) \in \mathcal{LI}_{A_i}} F(B_i) \\ &= \prod_{i \in I} F^b(A_i). \end{aligned}$$

Here, the second isomorphism uses the fact that F commutes with fibre products, while the third uses the exactness of small filtered colimits in $\mathcal{A}$.

Similarly, suppose that

$$B \xrightarrow{u} A \xleftarrow{v} C$$

is a diagram. Using the cofinal category of compatible local isomorphisms over A , B , and C , one obtains

$$\begin{aligned} F^b(B \times_A C) &\cong \text{colim} F(B' \times_{A'} C') \\ &\cong \text{colim} (F(B') \times_{F(A')} F(C')) \\ &\cong F^b(B) \times_{F^b(A)} F^b(C). \end{aligned}$$

The second isomorphism uses the fact that F commutes with fibre products, and the third uses the exactness of filtered colimits.

Hence F^b commutes with small limits whenever F does.

Step 7: The construction of F^a .

Now let $F \in \text{Ob}(\text{PSh}(X, \mathcal{A}))$, and let

$$h_X^\dagger : \text{PSh}(X, \mathcal{A}) \rightarrow \text{PSh}(\widehat{X}, \mathcal{A})$$

denote the canonical extension functor and let

$$(h_X)_* : \text{PSh}(\widehat{X}, \mathcal{A}) \rightarrow \text{PSh}(X, \mathcal{A})$$

denote restrictions to X .

By step 6, $(h_X^\dagger F)^b$ commutes with small limits. Hence, by the characterisation of the essential image of the Yoneda extension, there exists a presheaf F^a on X such that

$$(h_X^\dagger F)^b \cong h_X^\dagger(F^a).$$

Equivalently, one may define

$$F^a := (h_X)_*((h_X^\dagger F)^b). \quad (\text{D.2.15})$$

For $U \in \text{Ob}(\mathcal{C}_X)$, this construction gives the natural isomorphism

$$F^a(U) \cong \text{colim}_{(A \rightarrow U) \in (\mathcal{L}\mathcal{I}_U)^{\text{op}}} F(A). \quad (\text{D.2.16})$$

The natural morphism (D.2.13) induces a natural morphism

$$\varepsilon_F : F \rightarrow F^a. \quad (\text{D.2.17})$$

Step 8: The presheaf F^a is a sheaf.

By the characterisation of sheaves in terms of local isomorphisms, a presheaf G^{pre} on X is a sheaf if and only if its canonical extension $h_X^\dagger G$ to $\widehat{\mathcal{C}}_X$ sends every local isomorphism to an isomorphism.

By step 4, $(h_X^\dagger F)^b$ sends every local isomorphism to an isomorphism. Therefore

$$F^a \in \text{Sh}(X, \mathcal{A}).$$

Moreover, if F is already a sheaf, then $h_X^\dagger F$ already sends local isomorphisms to isomorphisms. Consequently,

$$\varepsilon_F : F \rightarrow F^a$$

is an isomorphism.

Step 9: The universal property.

Let $G \in \text{Ob}(\text{Sh}(X, \mathcal{A}))$. One shows that composition with ε_F induces a natural isomorphism

$$\text{Hom}_{\text{Sh}(X, \mathcal{A})}(F^a, G) \xrightarrow{\sim} \text{Hom}_{\text{PSh}(X, \mathcal{A})}(F, G). \quad (\text{D.2.18})$$

Let $f : F \rightarrow G$ be a morphism of presheaves. Applying the sheafification gives

$$f^a : F^a \rightarrow G^a.$$

Since G is a sheaf, the natural morphism

$$\varepsilon_G : G \rightarrow G^a$$

is an isomorphism. Hence one obtains a morphism

$$\tilde{f} := \varepsilon_G^{-1} \circ f^a : F^a \rightarrow G.$$

Naturality of ε gives

$$\tilde{f} \circ \varepsilon_F = f.$$

Conversely, if $g : F^a \rightarrow G$ satisfies

$$g \circ \varepsilon_F = f,$$

then using the projector identity

$$\varepsilon_{F^a} = (\varepsilon_F)^a$$

and the fact that ε_{F^a} is an isomorphism, one obtains

$$g = \varepsilon_G^{-1} \circ f^a.$$

Thus $\tilde{f}$ is unique, proving (D.2.18). Therefore $a \dashv \iota$.

Step 10: Exactness.

Assume now that $\mathcal{A}$ is abelian. The goal is to show that

$$(-)^a : \text{PSh}(X, \mathcal{A}) \rightarrow \text{Sh}(X, \mathcal{A})$$

is left-exact.

For every $A \in \text{Ob}(\widehat{\mathcal{C}}_X)$, the functor

$$\text{PSh}(X, \mathcal{A}) \rightarrow \mathcal{A}, \quad F \mapsto F(A) := (h_X^\dagger F)(A)$$

is left-exact. Indeed,

$$F(A) \cong \lim_{h_U \rightarrow A} F(U)$$

and evaluation at every $U \in \text{Ob}(\mathcal{C}_X)$ is exact, while limits are left-exact.

Let

$$0 \rightarrow F' \rightarrow F \rightarrow F'' \rightarrow 0 \tag{D.2.19}$$

be a short exact sequence in $\text{PSh}(X, \mathcal{A})$. Exactness in $\text{PSh}(X, \mathcal{A})$ is computed objectwise. Hence, for every $A \in \text{Ob}(\widehat{\mathcal{C}}_X)$, the sequence

$$0 \rightarrow F'(A) \rightarrow F(A) \rightarrow F''(A) \rightarrow 0$$

is exact.

For $U \in \text{Ob}(\mathcal{C}_X)$, one has

$$F^a(U) \cong \text{colim}_{(A \rightarrow h_U) \in \mathcal{L}\mathcal{I}_{h_U}} F(A),$$

and similarly for F' and F'' . Since $(\mathcal{L}\mathcal{I}_{h_U})^{\text{op}}$ is filtered and filtered colimits are exact in $\mathcal{A}$, applying the filtered colimit to (D.2.19) gives an exact sequence

$$0 \rightarrow F'^a(U) \rightarrow F^a(U) \rightarrow F''^a(U) \rightarrow 0.$$

Thus, $(-)^a$ is left-exact.

By step 9, the functor $(-)^a$ is left-adjoint to the inclusion

$$\iota : \text{Sh}(X, \mathcal{A}) \hookrightarrow \text{PSh}(X, \mathcal{A}).$$

Therefore it preserves colimits, in particular cokernels, and is right-exact. Consequently, $(-)^a$ is exact (see [32, Ch. 17, §4]). $\square$

D.3 Sheaves over Grothendieck Categories

The proof that sheaf categories over Grothendieck categories are again Grothendieck uses several general results from localisation theory.

Definition D.3.1 (Exact localisation). Let $\mathcal{C}, \mathcal{D}$ be categories. A **reflective localisation** of $\mathcal{C}$ consists of a functor $F : \mathcal{C} \rightarrow \mathcal{D}$ such that there exists a fully faithful functor $\iota : \mathcal{D} \rightarrow \mathcal{C}$ with $F \dashv \iota$.

Equivalently, the counit of the adjunction

$$\varepsilon : F \circ \iota \rightarrow \text{id}_{\mathcal{D}}$$

is a natural isomorphism. Then $\mathcal{D}$ is called a **reflective localisation** of $\mathcal{C}$, and F is called **localisation functor**.

If $\mathcal{C}$ and $\mathcal{D}$ are abelian categories, and if F is exact, the localisation is called **exact localisation** (see [52, Ch. 7, §25]).

Example D.3.2. The sheafification is an exact localisation. Let $\mathcal{C}$ be a category, let

$$\iota : \text{Sh}(X, \mathcal{C}) \hookrightarrow \text{PSh}(X, \mathcal{C})$$

be the inclusion of sheaves into presheaves, and let

$$s : \text{PSh}(X, \mathcal{C}) \rightarrow \text{Sh}(X, \mathcal{C})$$

be the sheafification functor, which is left-adjoint to the inclusion. The inclusion functor is fully faithful since Sh is a full subcategory of PSh , that is, $\text{Ob}(\text{Sh}) \subseteq \text{Ob}(\text{PSh})$ and $\text{Mor}(\text{Sh}) \subseteq \text{Mor}(\text{PSh})$ (see [52, Ch. 7, §25, Lemma 7.25.2]).

An exact localisation annihilates a distinguished class of objects. Exactness forces this class to be stable under subobjects, quotients, and extensions, leading naturally to the notion of a Serre subcategory.

Definition D.3.3 (Serre Subcategory). Let $\mathcal{C}$ be an abelian category. A full subcategory $\mathcal{D} \subseteq \mathcal{C}$ is called a **Serre subcategory** if for every short exact sequence

$$0 \rightarrow A_1 \rightarrow A_2 \rightarrow A_3 \rightarrow 0$$

one has $A_2 \in \text{Ob}(\mathcal{D})$ if and only if $A_1, A_3 \in \text{Ob}(\mathcal{D})$ (see [52, Ch. 12, §10, Def. 12.10.1]).

Remark D.3.4. Equivalently,

1. every subobject of objects in the subcategory is again an object in the subcategory,
2. every quotient object of objects in the subcategory is again an object in the subcategory,
3. every extension of two objects in the subcategory is again an object in the subcategory

(see [52, Ch. 12, §10]).

If $F : \mathcal{C} \rightarrow \mathcal{D}$ is an exact localisation, then

$$\ker(F) = \{A \in \text{Ob}(\mathcal{C}) \mid F(A) \cong 0\}$$

is automatically a Serre subcategory. Indeed, the exactness of F implies that for any short exact sequence

$$0 \rightarrow A_1 \rightarrow A_2 \rightarrow A_3 \rightarrow 0,$$

one has

$$F(A_2) = 0 \iff F(A_1) = F(A_3) = 0.$$

Thus, the kernel is closed under subobjects, quotients, and extensions.

A Serre subcategory may be regarded as a class of objects that one wishes to make trivial. The corresponding universal construction is the Serre quotient category.

Definition D.3.5 (Serre quotient category). Let $\mathcal{C}$ be an abelian category, and let $\mathcal{E} \subseteq \mathcal{C}$ be a Serre subcategory. A **Serre quotient**, also called **quotient category** of

$\mathcal{C}$ by $\mathcal{E}$ is an abelian category $\mathcal{C}/\mathcal{E}$ together with an exact functor

$$q : \mathcal{C} \rightarrow \mathcal{C}/\mathcal{E}$$

such that

1. $q(A) \cong 0$ for every object $A \in \text{Ob}(\mathcal{E})$, and
2. for every abelian category $\mathcal{D}$ and every exact functor $F : \mathcal{C} \rightarrow \mathcal{D}$ satisfying $F(A) \cong 0$ for all $A \in \text{Ob}(\mathcal{E})$, there exists an exact functor $\bar{F} : \mathcal{C}/\mathcal{E} \rightarrow \mathcal{D}$ and a natural isomorphism $F \cong \bar{F} \circ q$, where $\bar{F}$ is unique up to natural isomorphism (see [54, Ch. 10, §3, Ex. 10.3.2, Ex. 10.4.4]).

In terms of diagrams, this means that

$$\begin{array}{ccc} \mathcal{C} & \xrightarrow{F} & \mathcal{D} \\ \downarrow q & \searrow \bar{F} & \\ \mathcal{C}/\mathcal{E} & & \end{array}$$

commutes up to a unique natural isomorphism.

Definition D.3.6 (Localising Serre subcategory). Let $\mathcal{C}$ be an abelian category. A Serre subcategory $\mathcal{E} \subseteq \mathcal{C}$ is called **localising** if the quotient functor

$$q : \mathcal{C} \rightarrow \mathcal{C}/\mathcal{E}$$

admits a fully faithful right-adjoint (see [18]).

To identify a quotient category with the target of a localisation, one must show that the induced functor is an equivalence.

Definition D.3.7 (Essentially surjective). A functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is called **essentially surjective** if for every object $A \in \text{Ob}(\mathcal{D})$ there exists an object $B \in \text{Ob}(\mathcal{C})$ and an isomorphism $F(B) \xrightarrow{\sim} A$ (see [52, Ch. 4, §2, Def. 4.2.9]).

Essential surjectivity alone does not determine an equivalence, since one must also control morphisms. The appropriate notion is an equivalence of categories, defined by the existence of a quasi-inverse.

Definition D.3.8 (Equivalence of categories). Let $\mathcal{C}$ and $\mathcal{D}$ be categories. A functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is called an **equivalence of categories** if there exists a functor $G : \mathcal{D} \rightarrow \mathcal{C}$ together with natural isomorphisms

$$\eta : \text{id}_{\mathcal{C}} \rightarrow G \circ F, \quad \varepsilon : F \circ G \rightarrow \text{id}_{\mathcal{D}}.$$

The functor G is called a **quasi-inverse** of F (see [52, Ch. 4, §2, Def. 4.2.17]).

Although equivalences are defined using quasi-inverses, they are usually verified through the more concrete properties of fullness, faithfulness, and essential surjectivity.

Theorem D.3.9 (Criterion for equivalences of categories). Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a functor. Then the following are equivalent:

1. F is an equivalence of categories.
2. F is full, faithful, and essentially surjective

(see [52, Ch. 4, §2, Lemma 4.2.19]).

Proof. " $\Rightarrow$ ": Assume first that F is an equivalence with quasi-inverse $G : \mathcal{D} \rightarrow \mathcal{C}$. Since $F \circ G \cong \text{id}_{\mathcal{D}}$, every object $A \in \text{Ob}(\mathcal{D})$ is isomorphic to $F(G(A))$. Hence F is essentially surjective.

To prove fullness, let

$$\alpha : F(A) \rightarrow F(B)$$

be a morphism in $\mathcal{D}$. Consider the composite

$$A \xrightarrow{\eta_A} (G \circ F)(A) \xrightarrow{G(\alpha)} (G \circ F)(B) \xrightarrow{\eta_B^{-1}} B.$$

Applying F and using the naturality of η together with

$$F \circ G \cong \text{id}_{\mathcal{D}},$$

shows that this morphism is mapped to α . Hence F is full.

For faithfulness, suppose that $f, g : A \rightarrow B$ satisfy $F(f) = F(g)$. Applying G yields

$$(G \circ F)(f) = (G \circ F)(g).$$

Conjugating by the natural isomorphisms

$$\eta_A : A \xrightarrow{\sim} (G \circ F)(A), \quad \eta_B : B \xrightarrow{\sim} (G \circ F)(B)$$

gives $f = g$. Thus F is faithful.

" $\Leftarrow$ ": Assume that F is full, faithful, and essentially surjective. For every object $A \in \text{Ob}(\mathcal{D})$, choose an object $G(A) \in \text{Ob}(\mathcal{C})$ together with an isomorphism

$$\varepsilon_A : F(G(A)) \xrightarrow{\sim} A.$$

Let $h : A \rightarrow B$ be a morphism. Since F is full, there exists a unique morphism

$$G(h) : G(A) \rightarrow G(B)$$

satisfying

$$F(G(h)) = \varepsilon_B^{-1} \circ h \circ \varepsilon_A.$$

The uniqueness follows from faithfulness, and one can check directly, that G preserves identities and composition. Thus G is a functor.

The chosen isomorphisms ε_A form a natural isomorphism

$$F \circ G \rightarrow \text{id}_{\mathcal{D}}.$$

Since F is fully faithful, for every object $A \in \text{Ob}(\mathcal{C})$ there exists a unique isomorphism

$$\eta_A : A \rightarrow (G \circ F)(A)$$

whose image under F is

$$\varepsilon_{F(A)}^{-1}.$$

These assemble into a natural isomorphism

$$\text{id}_{\mathcal{C}} \rightarrow G \circ F.$$

Hence F is an equivalence of categories (see [52, Ch. 4, §2, Lemma 4.2.19]). $\square$

Remark D.3.10. The choice of an object $G(A) \in \text{Ob}(\mathcal{C})$ is a mild form of the Axiom of Choice, that is, a choice of one representative from each isomorphism class in

the essential image.

The preceding notions now allow the target of an exact localisation to be described intrinsically. Gabriel's theorem identifies it with the Serre quotient by the kernel of the localisation functor.

Theorem D.3.11 (Gabriel). Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be an exact localisation of abelian categories. Then $\ker(F)$ is a localising Serre subcategory of $\mathcal{C}$, and the induced functor

$$\mathcal{C}/\ker(F) \rightarrow \mathcal{D}$$

is an equivalence of categories (see [41, pp. 5 ff.]).

Proof. Define

$$\mathcal{E} := \ker(F) = \{A \in \text{Ob}(\mathcal{C}) \mid F(A) = 0\}.$$

Note that $\mathcal{E}$ is a Serre subcategory. Let

$$q : \mathcal{C} \rightarrow \mathcal{C}/\mathcal{E}$$

be the quotient functor. Since F eliminates every object of $\mathcal{E}$, the universal property of the quotient gives a unique exact functor

$$\bar{F} : \mathcal{C}/\mathcal{E} \rightarrow \mathcal{D}$$

such that

$$F = \bar{F} \circ q.$$

This yields a diagram

$$\begin{array}{ccc} \mathcal{C} & \xrightarrow{F} & \mathcal{D} \\ \downarrow q & \nearrow \bar{F} & \\ \mathcal{C}/\mathcal{E} & & \end{array}$$

Claim: $\bar{F} : \mathcal{C}/\mathcal{E} \rightarrow \mathcal{D}$ is an equivalence.

Because $\iota : \mathcal{D} \rightarrow \mathcal{C}$ is right-adjoint to F , and since ι is fully faithful, the counit

$$(F \circ \iota)(A) \rightarrow A$$

is an isomorphism for every $A \in \text{Ob}(\mathcal{D})$. Therefore every object $A \in \text{Ob}(\mathcal{D})$ is isomorphic to $F(\iota(A))$. But since

$$F(\iota(A)) = \bar{F}(q(\iota(A))),$$

$\bar{F}$ is essentially surjective.

The quotient category $\mathcal{C}/\mathcal{E}$ is constructed precisely so that morphisms whose kernel and cokernel lie in $\mathcal{E}$ become isomorphisms. Consider the unit

$$\eta_A : A \rightarrow (\iota \circ F)(A).$$

Since the counit is an isomorphism, applying F yields

$$F(\eta_A) : F(A) \rightarrow F(\iota \circ F(A))$$

is an isomorphism. By the exactness of F , this implies that

$$F(\ker(\eta_A)) = 0, \quad F(\text{coker}(\eta_A)) = 0.$$

Thus $\ker(\eta_A), \text{coker}(\eta_A) \in \mathcal{E}$. Therefore $q(\eta_A)$ is an isomorphism in $\mathcal{C}/\mathcal{E}$.

Hence every object of the quotient category is isomorphic to an object of the form $q(\iota(A))$ for some $A \in \text{Ob}(\mathcal{D})$. It therefore suffices to consider morphisms between objects of this form.

Recall that the Serre quotient can be obtained by localising $\mathcal{C}$ at the class of morphisms whose kernels and cokernels belong to $\mathcal{E}$. Thus a morphism

$$q(\iota(A)) \rightarrow q(\iota(B))$$

can be represented by a proof

$$\iota(A) \xleftarrow{s} X \xrightarrow{g} \iota(B),$$

where $\ker(s), \text{coker}(s) \in \mathcal{E}$. Since $\mathcal{E} = \ker(F)$ and F is exact, $F(s)$ is an isomorphism. Using the counit isomorphisms

$$F(\iota(A)) \cong A, \quad F(\iota(B)) \cong B,$$

define

$$h := F(g) \circ F(s)^{-1} : A \rightarrow B.$$

Then

$$F(g) = h \circ F(s).$$

By the adjunction $F \dashv \iota$, there is a natural bijection

$$\mathrm{Hom}_{\mathcal{C}}(X, \iota(B)) \cong \mathrm{Hom}_{\mathcal{D}}(F(X), B).$$

Under this bijection, the morphisms g and $\iota(h) \circ s$ correspond to $F(g)$ and $h \circ F(s)$, respectively. Hence

$$g = \iota(h) \circ s.$$

Since $q(s)$ is an isomorphism in $\mathcal{C}/\mathcal{E}$, the morphism represented by the *roof* is

$$\begin{aligned} q(g) \circ q(s)^{-1} &= q(\iota(h)) \circ q(s) \circ q(s)^{-1} \\ &= q(\iota(h)). \end{aligned}$$

Thus every morphism

$$q(\iota(A)) \rightarrow q(\iota(B))$$

is induced by a morphism $h : A \rightarrow B$.

Moreover, if

$$q(\iota(h)) = q(\iota(k)),$$

then applying $\overline{F}$ and using the counit isomorphism

$$F \circ \iota \cong \mathrm{id}_{\mathcal{D}}$$

gives $h = k$. Consequently,

$$\mathrm{Hom}_{\mathcal{C}/\mathcal{E}}\left(q(\iota(A)), q(\iota(B))\right) \cong \mathrm{Hom}_{\mathcal{D}}(A, B).$$

Therefore $\overline{F}$ is fully faithful.

Therefore

$$\overline{F} : \mathcal{C} / \ker(F) \rightarrow \mathcal{D}$$

is an equivalence, so

$$\mathcal{D} \simeq \mathcal{C} / \ker(F)$$

(see [41, pp. 5 ff.]). □

Gabriel's theorem identifies an exact localisation with a quotient by a localising Serre subcategory. It remains to verify that the defining properties of a Grothendieck category—abelianness, cocompleteness, exactness of filtered colimits, and the existence of a generator—pass to the localised category.

Theorem D.3.12. Let $\mathcal{C}$ be a Grothendieck category, and let

$$F : \mathcal{C} \rightarrow \mathcal{D}$$

be an exact localisation. Then $\mathcal{D}$ is again a Grothendieck category (compare [41, pp. 5 ff.], [52, Ch. 19, §10]).

Proof. 1. **Claim:** $\mathcal{D}$ is abelian.

Since F is exact and has a fully faithful right-adjoint, the kernel $\ker(F)$ is a localising Serre subcategory, and the quotient $\mathcal{C} / \ker(F)$ is abelian. Moreover, $\mathcal{D} \simeq \mathcal{C} / \ker(F)$.

2. **Claim:** All small colimits exist.

Since $F \dashv \iota$, F is left-adjoint and therefore preserves all colimits by Theorem 1.7.6. Consider a diagram $\mathbf{D} : I \rightarrow \mathcal{D}$. Then

$$\operatorname{colim}_{\mathcal{D}} \mathbf{D} \cong F(\operatorname{colim}_{\mathcal{C}} \iota \circ \mathbf{D}).$$

Thus every small colimit exists.

3. **Claim:** Filtered colimits are exact.

Let $0 \rightarrow A_j \rightarrow B_j \rightarrow C_j \rightarrow 0$ be a filtered diagram of a short exact sequence in $\mathcal{D}$. Since ι is a right-adjoint, it is left-exact, and hence

$$0 \rightarrow \iota(A_j) \rightarrow \iota(B_j) \rightarrow \iota(C_j)$$

is exact in $\mathcal{C}$. Let

$$Q_j := \text{coker}(\iota(B_j) \rightarrow \iota(C_j)).$$

Then

$$0 \rightarrow \iota(A_j) \rightarrow \iota(B_j) \rightarrow \iota(C_j) \rightarrow Q_j \rightarrow 0$$

is exact.

Applying the exact functor F and using $F \circ \iota \cong \text{id}_{\mathcal{D}}$ shows that

$$F(Q_j) = 0.$$

Since $\mathcal{C}$ is Grothendieck, filtered colimits are exact, so

$$0 \rightarrow \text{colim}_j \iota(A_j) \rightarrow \text{colim}_j \iota(B_j) \rightarrow \text{colim}_j \iota(C_j) \rightarrow \text{colim}_j Q_j \rightarrow 0$$

is exact. Applying F , using exactness and preservation of colimits, gives

$$0 \rightarrow \text{colim}_j A_j \rightarrow \text{colim}_j B_j \rightarrow \text{colim}_j C_j \rightarrow F(\text{colim}_j Q_j) \rightarrow 0.$$

Finally,

$$F(\text{colim}_j Q_j) \cong \text{colim}_j F(Q_j) = 0.$$

Hence filtered colimits are exact in $\mathcal{D}$.

4. Claim: $\mathcal{D}$ has a generator.

Let G be a generator of $\mathcal{C}$. Since F preserves coproducts, one can test $F(G)$ to be a generator.

Suppose $f, g : A \rightarrow B$ are distinct morphisms in $\mathcal{D}$. Applying a fully faithful functor ι , it follows that $\iota(f) \neq \iota(g)$ inside $\mathcal{C}$. Because G is a generator, there exists

$$u : G \rightarrow \iota(A)$$

such that

$$\iota(f) \circ u \neq \iota(g) \circ u.$$

Using the adjunction,

$$\text{Hom}_{\mathcal{C}}(G, \iota(A)) \cong \text{Hom}_{\mathcal{D}}(F(G), A),$$

which corresponds to

$$\bar{u} : F(G) \rightarrow A.$$

The faithfulness of ι implies

$$f \circ \bar{u} \neq g \circ \bar{u}.$$

Thus $F(G)$ is a generator.

Therefore $\mathcal{D}$ is abelian, has all small colimits, has exact filtered colimits, and a generator. Hence $\mathcal{D}$ is a Grothendieck category (compare [41, pp. 5 ff.], [52, Ch. 19, §10]). $\square$

Now one can prove Theorem 2.2.26 Let $\mathcal{D}$ be a Grothendieck category, and let $(\mathcal{C}, J)$ be a small site. Then the category $\text{Sh}((\mathcal{C}, J), \mathcal{D})$ is again a Grothendieck abelian category (compare [52, Ch. 6, §18, Lemma 6.18.2, Ch. 6, §29]).

Proof. Claim: The category of presheaves $\text{PSh}(\mathcal{C}, \mathcal{D})$ is Grothendieck.

Limits and colimits in $\text{PSh}(\mathcal{C}, \mathcal{D})$ are computed objectwise. Since $\mathcal{D}$ is abelian, so is the presheaf category. Filtered colimits are computed objectwise and are therefore exact because filtered colimits are exact in $\mathcal{D}$. Finally, since $\mathcal{C}$ is small and $\mathcal{D}$ possesses a generator, the presheaf category also admits a generator. Hence the presheaf category is Grothendieck.

The sheafification functor is exact and left-adjoint to the inclusion

$$\text{Sh}((\mathcal{C}, J), \mathcal{D}) \hookrightarrow \text{PSh}(\mathcal{C}, \mathcal{D}).$$

Thus $\text{Sh}((\mathcal{C}, J), \mathcal{D})$ is an exact reflective localisation of a Grothendieck category. Exact localisations of Grothendieck categories are again Grothendieck. Hence $\text{Sh}((\mathcal{C}, J), \mathcal{D})$ is Grothendieck. $\square$

The preceding proof uses the fact that sheafification is an exact reflective localisation. The following definitions recall the localisation-theoretic notions required for this argument.

D.3.1 The Proof of Theorem 3.3.7

The existence of enough injective objects in a Grothendieck category is proved by a transfinite enlargement construction. Starting from an object M , one repeatedly adjoins extensions of all morphisms from subobjects of a fixed generator. After sufficiently many stages, a smallness argument ensures that every morphism from such a subobject already occurs at an earlier stage and therefore extends over the generator. Baer's criterion then implies that the resulting object is injective.

The proof of Theorem 3.3.7 requires several auxiliary facts connecting continuous transfinite chains, cofinality, factorisation through sufficiently early stages, and the behaviour of monomorphisms under pushouts.

Definition D.3.13 (Continuous chain). A chain $(B_\alpha)_{\alpha < \lambda}$ is called **continuous** if

$$B_\mu = \operatorname{colim}_{\alpha < \mu} B_\alpha$$

for every limit ordinal $\mu < \lambda$ (see [52, Ch. 19, §11]).

Whether a morphism into the colimit of a continuous chain factors through an earlier stage depends on the length of the chain relative to the size of its domain. The relevant measure of the length of an ordinal is its cofinality.

Definition D.3.14 (Cofinality). The **cofinality** $\operatorname{cf}(\lambda)$ of an ordinal λ is the smallest cardinality of a subset $S \subseteq \lambda$ such that every $\beta < \lambda$ is bounded above by some element of S ; equivalently, it is the least cardinality of an unbounded (cofinal) subset of λ (see [52, Ch. 19, §11]).

If the cofinality of the indexing ordinal is larger than the number of subobjects of a given object, an increasing chain of inverse images cannot acquire genuinely new subobjects at every stage. This yields the following factorisation result.

Lemma D.3.15 (Factorisation/ Smallness Lemma). Let $\mathcal{C}$ be a Grothendieck category and let $(B_\beta)_{\beta < \lambda}$ be a continuous increasing chain of subobjects with colimit

$$B = \operatorname{colim}_{\beta < \lambda} B_\beta.$$

Let $M \in \operatorname{Ob}(\mathcal{C})$. Assume that

1. the cardinality of the set of subobjects of M is at most $|M|$,
2. $\text{cf}(\lambda) > |M|$.

Then every morphism $f : M \rightarrow B$ factors through one of the objects B_β (see [52, Ch. 19, §11]).

Proof. For every ordinal $\beta < \lambda$, define

$$M_\beta := f^{-1}(B_\beta) \subseteq M.$$

Since $B_\beta \subseteq B_{\beta'}$ for $\beta \leq \beta'$, the subobjects M_β form an increasing chain. Because filtered colimits are exact in a Grothendieck category,

$$B = \text{colim}_{\beta < \lambda} B_\beta.$$

Hence inverse images commute with this filtered colimit:

$$\text{colim}_{\beta < \lambda} M_\beta = f^{-1}(B) = M.$$

By assumption, M possesses at most $|M|$ distinct subobjects. Consequently, among the family $(M_\beta)_{\beta < \lambda}$ there occur at most $|M|$ distinct objects. Choose one ordinal β_i for each distinct subobject. Since there are at most $|M|$ of these ordinals, and $\text{cf}(\lambda) > |M|$, their supremum $\bar{\beta} = \sup_i \beta_i$ still satisfies $\bar{\beta} < \lambda$. For every $\beta < \lambda$, the subobject M_β is equal to M_{β_i} for some chosen representative $\beta_i \leq \bar{\beta}$. Since the chain is increasing, it follows that

$$M_{\bar{\beta}} = \text{colim}_{\beta < \lambda} M_\beta = M.$$

Hence $f(M) \subseteq B_{\bar{\beta}}$, which means precisely that $f : M \rightarrow B$ factors as

$$M \rightarrow B_{\bar{\beta}} \rightarrow B$$

(see [52, Ch. 19, §11]). □

The preceding lemma controls morphisms into sufficiently long transfinite colimits. To recognise when the object obtained at the end of the construction is injective, it is enough to test extensions from subobjects of a fixed generator.

Lemma D.3.16 (Baer's criterion for Grothendieck categories). Let $\mathcal{C}$ be a Grothendieck category with generator G . An object $I \in \text{Ob}(\mathcal{C})$ is injective if and only if for every subobject $A \subseteq G$, every morphism $A \rightarrow I$ extends to a morphism $G \rightarrow I$ (see [52, Ch. 19, §11, Lemma 19.11.6]).

Proof. " $\Rightarrow$ ": If I is injective, then every morphism from a subobject $A \subseteq G$ extends over G .

" $\Leftarrow$ ": Assume the extension property for all $A \subseteq G$. Let $B \subseteq C$ be a monomorphism and let $\varphi : B \rightarrow I$ be a morphism. One wants to extend φ to C . Consider the set S of pairs (B', φ') where $B \subseteq B' \subseteq C$ and $\varphi' : B' \rightarrow I$ extends φ .

Order S by extension. By Lemma A.1.2, S has a maximal element (B', φ') . Indeed, if (B_t, φ_t) is a chain, let

$$B_\infty := \text{colim}_t B_t.$$

Since filtered colimits are exact and each $B_t \rightarrow C$ is monomorphic, the induced morphism

$$B_\infty \rightarrow C$$

is monomorphic. Thus B_∞ may be regarded as a subobject of C . The compatible morphisms φ_t induce a morphism

$$\varphi_\infty : B_\infty \rightarrow I,$$

so $(B_\infty, \varphi_\infty)$ is an upper bound.

Claim: $B' = C$.

Suppose not. Indeed, if every morphism $G \rightarrow C$ factored through B' , then $\text{Hom}(G, C/B') = 0$. Since G is a generator, this would imply $C/B' = 0$, contradicting $B' \subsetneq C$.

Furthermore, there exists a morphism

$$\psi : G \rightarrow C$$

which does not factor through B' . Define

$$N := B' \cap \text{im}(\psi), \quad M := \psi^{-1}(N) \subseteq G.$$

Then the composite

$$M \rightarrow N \rightarrow B' \xrightarrow{\varphi'} I$$

extends to a morphism

$$\chi : G \rightarrow I$$

by assumption. Since $\ker(\psi) \subseteq M$, and χ agrees with the above composite on M , one gets

$$\chi|_{\ker(\psi)} = 0.$$

Therefore χ factors through $\text{im}(\psi)$:

$$G \twoheadrightarrow \text{im}(\psi) \xrightarrow{\psi''} I.$$

Now ψ'' and φ' agree on $N = B' \cap \text{im}(\psi)$. Hence they glue to a morphism

$$B' + \text{im}(\psi) \rightarrow I.$$

But $B' \subsetneq B' + \text{im}(\psi)$, because ψ does not factor through B' . This contradicts the maximality of (B', φ') . Therefore $B' = C$, so φ extends to C . Hence I is injective. (see [52, Ch. 19, §11, Lemma 19.11.6]). $\square$

The enlargement construction is based on pushouts. The following lemma ensures that each enlargement preserves the embedding of the original object.

Lemma D.3.17. In an abelian category, pushouts preserve monomorphisms (see [57, Prop. 3.3]).

Proof. Let $\mathcal{C}$ be an abelian category, let $\iota : A \rightarrow B$ be monomorphic, and let $f : A \rightarrow C$ be any morphism. One forms the pushout

$$\begin{array}{ccc} A & \xrightarrow{\iota_1} & B \\ \downarrow f & & \downarrow g \\ C & \xrightarrow{\iota_2} & P. \end{array}$$

In an abelian category, the pushout P can be realised as the cokernel of

$$h := \operatorname{coker} \begin{pmatrix} \iota_1 \\ -f \end{pmatrix} : A \rightarrow B \oplus C.$$

Thus there is an exact sequence

$$A \xrightarrow{h} B \oplus C \xrightarrow{q} P \rightarrow 0,$$

and the induced morphism $\iota_2 : C \rightarrow P$ is given by

$$\iota_2 = q \circ j_C,$$

where $j_C : C \rightarrow B \oplus C$ is the canonical inclusion.

Let $k : K \rightarrow C$ be a morphism such that

$$\iota_2 \circ k = 0.$$

Then

$$q \circ j_C \circ k = 0.$$

Since $\ker(q) = \operatorname{im}(h)$, the morphism $j_C \circ k$ factors through $\operatorname{im}(h)$. Moreover, h is a monomorphism: if

$$a : T \rightarrow A$$

satisfies

$$h \circ a = 0,$$

then

$$\iota \circ a = 0,$$

and hence $a = 0$, since ι is monomorphic. Thus $\operatorname{im}(h) \cong A$, and there exists a morphism

$$\alpha : K \rightarrow A$$

such that

$$j_C \circ k = h \circ \alpha.$$

Applying $\operatorname{pr}_1 : B \oplus C \rightarrow B$ gives $0 = \iota \circ \alpha$. Since ι is monomorphic, $\alpha = 0$, and

consequently

$$j_C \circ k = 0.$$

Since j_C is monomorphic, $k = 0$. Therefore ι_2 is a monomorphism (see [57, Prop. 3.3]). $\square$

The auxiliary results can now be assembled. The pushout construction produces an increasing transfinite chain of enlargements, the smallness lemma ensures that morphisms from subobjects of the generator factor through an earlier stage, and Baer's criterion then proves that a sufficiently late stage is injective.

Theorem D.3.18. Let $\mathcal{C}$ be a Grothendieck category. Then every object of $\mathcal{C}$ admits a monomorphism into an injective object (see [52, Ch. 19, §11, Thm. 19.11.7]).

Proof. Let G be a generator of $\mathcal{C}$. For an object $A \in \text{Ob}(\mathcal{C})$, one constructs an *enlargement* $\mathbf{E}(M)$.

For every subobject $N \subseteq G$ and every morphism

$$\varphi : N \rightarrow M,$$

one forms the coproduct of all such data and defines $\mathbf{E}(M)$ by the pushout

$$\begin{array}{ccc} \bigoplus_{N \subseteq G} \bigoplus_{\varphi \in \text{Hom}(N, M)} N & \longrightarrow & M \\ \downarrow & & \downarrow \\ \bigoplus_{N \subseteq G} \bigoplus_{\varphi \in \text{Hom}(N, M)} G & \longrightarrow & \mathbf{E}(M). \end{array}$$

Since coproducts are exact in a Grothendieck category, the coproduct of monomorphisms

$$N \hookrightarrow G$$

is again a monomorphism. Therefore Lemma D.3.17 shows that the canonical morphism

$$M \rightarrow \mathbf{E}(M)$$

is a monomorphism.

One iterates this construction *transfinitely*. Define

$$\mathbf{E}_0(M) := M, \quad \mathbf{E}_{\alpha+1}(M) := \mathbf{E}(\mathbf{E}_\alpha(M)),$$

and, for a limit ordinal Λ ,

$$\mathbf{E}_\Lambda(M) := \operatorname{colim}_{\alpha < \Lambda} \mathbf{E}_\alpha(M).$$

Because $\mathcal{C}$ satisfies *AB5*, filtered colimits are exact. Hence the transition morphisms

$$\mathbf{E}_\alpha(M) \rightarrow \mathbf{E}_\beta(M)$$

are monomorphisms for all $\alpha \leq \beta$.

Choose a limit ordinal λ such that $\operatorname{cf}(\lambda) > |G|$. For example, one may take the initial ordinal corresponding to the successor cardinal of $|G|$.

Claim: $I(M) := \mathbf{E}_\lambda(M)$ is injective.

Let $N \subseteq G$ be a subobject, and let

$$\varphi : N \rightarrow I(M) = \mathbf{E}_\lambda(M)$$

be a morphism. Since $N \subseteq G$, every subobject of N is also a subobject of G . Hence

$$|N| \leq |G| = \kappa.$$

Since $\operatorname{cf}(\lambda) > \kappa$, the smallness lemma [D.3.15](#) implies that φ factors through some earlier stage $\mathbf{E}_\alpha(M)$ with $\alpha < \lambda$, that is,

$$N \rightarrow \mathbf{E}_\alpha(M) \rightarrow \mathbf{E}_\lambda(M).$$

By construction of

$$\mathbf{E}_{\alpha+1}(M) = \mathbf{E}(\mathbf{E}_\alpha(M)),$$

every morphism $N \rightarrow \mathbf{E}_\alpha(M)$ extends to a morphism $U \rightarrow \mathbf{E}_{\alpha+1}(M)$. Composing with the canonical map

$$\mathbf{E}_{\alpha+1}(M) \rightarrow \mathbf{E}_\lambda(M)$$

gives an extension

$$G \rightarrow \mathbf{E}_\lambda(M).$$

Therefore every morphism $N \rightarrow I(M)$ extends to U . By the Baer criterion [D.3.16](#), $I(M)$ is injective.

The canonical morphism

$$M = \mathbf{E}_0(M) \rightarrow \mathbf{E}_\lambda(M) = I(M)$$

is a monomorphism. Hence every object embeds into an injective object. Therefore $\mathcal{C}$ has enough injectives (see [\[52, Ch. 19, §11, Thm. 19.11.7\]](#)). □

E The Comparison of Sheaf Cohomology and Singular Cohomology

The passage from local representatives to globally compatible data requires more than equality of germs: one must refine the underlying cover so that the representatives agree as actual sections on overlaps. The following lemma provides precisely this refinement step.

Lemma E.0.1 (Refinement of germwise compatible sections). Let U be a paracompact Hausdorff space, let F be a presheaf on U , and let $\{U_i\}_{i \in I}$ be an open cover of U with sections $s_i \in F(U_i)$. Assume that for every $x \in U_i \cap U_j$ the germs agree, that is,

$$(s_i)_x = (s_j)_x.$$

Then there exists an open refinement $\{V_\lambda\}_{\lambda \in \Lambda}$ of $\{U_i\}_{i \in I}$ and indices $i(\lambda) \in I$ with $V_\lambda \subseteq U_{i(\lambda)}$ such that the sections

$$s_{i(\lambda)}|_{V_\lambda}$$

agree on pairwise intersections (see [44, Ch. 1, §1, Lemma 1.6, Lemma 1.13]).

Proof. **Claim 1:** Let $\{Z_j\}_{j \in J}$ be a locally finite open cover of U , and let $\{W_j\}_{j \in J}$ be a shrinking, that is, $\overline{W_j} \subseteq Z_j$ for every $j \in J$. Then for every $x \in U$ there exists an open neighbourhood N_x of x such that, with

$$\Lambda_x := \{j \in J \mid N_x \cap W_j \neq \emptyset\},$$

the following properties hold:

1. Λ_x is finite,
2. for every $j \in \Lambda_x$, $x \in \overline{W_j}$, and $N_x \subseteq W_j$,
3. if $N_x \cap N_y \neq \emptyset$, then there exists $j \in J$ such that $N_x \cup N_y \subseteq Z_j$.

Indeed, since the family $\{W_j\}_{j \in J}$ is locally finite, every point $x \in U$ has an open neighbourhood M_x of x such that the set

$$\Lambda_x^0 := \{j \in J \mid M_x \cap W_j \neq \emptyset\}$$

is finite. Define

$$M'_x := M_x \cap \bigcap_{\substack{j \in \Lambda_x^0 \\ x \notin \overline{W_j}}} (U \setminus \overline{W_j}).$$

This is an open neighbourhood of x , since the intersection is finite. Moreover, if

$$M'_x \cap W_j \neq \emptyset,$$

then $j \in \Lambda_x^0$, and necessarily

$$x \in \overline{W_j}.$$

Set

$$\Lambda'_x := \{j \in J \mid M'_x \cap W_j \neq \emptyset\}.$$

The set Λ'_x is finite. Define

$$N_x := M'_x \cap \bigcap_{j \in \Lambda'_x} Z_j.$$

Since $x \in \overline{W_j} \subseteq Z_j$ for every $j \in \Lambda'_x$, the set N_x is an open neighbourhood of x .

Now let

$$\Lambda_x := \{j \in J \mid N_x \cap W_j \neq \emptyset\}.$$

Then $\Lambda_x \subseteq \Lambda'_x$, so Λ_x is finite. Moreover, for every $j \in \Lambda_x$, one has $x \in \overline{W_j}$ and $N_x \subseteq W_j$. This proves the first two assertions.

Suppose now that

$$N_x \cap N_y \neq \emptyset,$$

and choose

$$z \in N_x \cap N_y.$$

Since $\{W_j\}_{j \in J}$ is a cover of U , there exists $j \in J$ such that $z \in W_j$. Hence

$$j \in \Lambda_x \cap \Lambda_y.$$

By the second assertion, $N_x \subseteq Z_j$ and $N_y \subseteq Z_j$. Therefore

$$N_x \cup N_y \subseteq Z_j,$$

which proves the claim (compare [44, Ch. 1, §1, Lemma 1.13]).

Since U is paracompact Hausdorff, the open cover $\{U_i\}_{i \in I}$ admits a locally finite open refinement $\{Z_j\}_{j \in J}$ together with a shrinking $\{W_j\}_{j \in J}$. For each $j \in J$, choose an index $\kappa(j) \in I$ such that

$$Z_j \subseteq U_{\kappa(j)}.$$

Apply claim 1 to the families $\{Z_j\}_{j \in J}$ and $\{W_j\}_{j \in J}$.

For every $x \in U$ there exists an open neighbourhood V_x such that V_x intersects only finitely many members of the shrinking, and whenever $V_x \cap V_y \neq \emptyset$, there exists an index i with

$$V_x \cup V_y \subseteq U_i.$$

Assume that $V_x \cap V_y \neq \emptyset$. By construction there exists an index k with

$$V_x \cup V_y \subseteq U_k.$$

For every $x \in U$, choose $j(x) \in J$ such that $x \in W_{j(x)}$, and set

$$i(x) := \kappa(j(x)).$$

Since $j(x) \in \Lambda_x$, one has $x \in U_{i(x)} \cap U_{\kappa(j)}$. Hence the germs of $s_{i(x)}$ and s_k agree at x . Therefore there exists an open neighbourhood

$$O_{x,j} \subseteq N_x$$

of x such that

$$s_{i(x)}|_{O_{x,j}} = s_{\kappa(j)}|_{O_{x,j}}.$$

Since Λ_x is finite, the intersection

$$V_x := N_x \cap \bigcap_{j \in \Lambda_x} O_{x,j}$$

is an open neighbourhood of x .

Claim 2: Let $W_1, W_2 \subseteq U$ be open, let $s \in F(W_1)$ and $t \in F(W_2)$, and define

$$E(s, t) := \{x \in V \cap W \mid s_x = t_x\}.$$

Then $E(s, t)$ is open. More precisely, for every $x \in E(s, t)$ there exists an open neighbourhood $N_x \subseteq W_1 \cap W_2$ of x such that

$$s|_{N_x} = t|_{N_x}.$$

Indeed, if $x \in E(s, t)$, then $s_x = t_x$. By the definition of equality of germs, there exists an open neighbourhood $N_x \subseteq W_1 \cap W_2$ of x such that

$$s|_{N_x} = t|_{N_x}.$$

For every $y \in N_x$, this equality implies $s_y = t_y$. Hence $N_x \subseteq E(s, t)$, and therefore $E(s, t)$ is open.

Thus by claim 2, the set of points where these germs agree is open, and therefore, after shrinking V_x if necessary, one obtains

$$s_{i(x)}|_{V_x} = s_k|_{V_x}.$$

Similarly, after shrinking V_y if necessary,

$$s_{i(y)}|_{V_y} = s_k|_{V_y}.$$

Consequently, $s_{i(x)} = s_k = s_{i(y)}$ on $V_x \cap V_y$. The resulting family of shrunk neighbourhoods forms an open refinement of the original cover satisfying the desired compatibility. □

The preceding refinement lemma allows germwise compatible local representatives

to be replaced by genuinely compatible sections. Combined with the gluing property of a presheaf, this yields a useful criterion ensuring that every section of the sheafification is represented by a global section of the original presheaf.

Proposition E.0.2 (Standard paracompactness criterion for sheafification). Let X be hereditarily paracompact, and let F be a presheaf. Assume F satisfies the **gluing property**:

for every open cover $\mathcal{U} = \{U_i\}_{i \in I}$ of U and every family $s_i \in F(U_i)$ satisfying

$$s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$$

for all $i, j \in I$, there exists a section $s \in F(U)$ such that

$$s|_{U_i} = s_i.$$

Then the natural morphism

$$F(U) \rightarrow F^{\text{Sh}}(U)$$

is surjective for every open subset $U \subseteq X$ (see [50, Prop. 0.2]).

Proof. Let $\sigma \in F^{\text{Sh}}(U)$. By the construction of sheafification D.2.2, there is an open cover $\{U_i\}_{i \in I}$ of U and sections $s_i \in F(U_i)$ representing σ locally.

The sections s_i need not agree on $U_i \cap U_j$. However, since they represent the same section σ of the sheafification, their germs agree at every point of $U_i \cap U_j$.

Since X is hereditarily paracompact, the open subset $U \subseteq X$ is paracompact. Hence Lemma E.0.1 applies. Therefore there exists an open refinement

$$\{V_\lambda\}_{\lambda \in \Lambda}$$

of $\{U_i\}_{i \in I}$ and indices $i(\lambda) \in I$ with

$$V_\lambda \subseteq U_{i(\lambda)}$$

such that the sections

$$t_\lambda := s_{i(\lambda)}|_{V_\lambda}$$

agree on pairwise intersections, that is,

$$t_\lambda|_{V_\lambda \cap V_\mu} = t_\mu|_{V_\lambda \cap V_\mu}$$

for all $\lambda, \mu \in \Lambda$.

Thus the family $(t_j)_{j \in J}$ is compatible on pairwise intersections. By the gluability assumption on F , there exists a section $t \in F(U)$ such that

$$t|_{V_j} = t_j$$

for all $j \in J$. The image of t in $F^{\text{Sh}}(U)$ agrees with σ on the cover $\{V_j\}_{j \in J}$. Since F^{Sh} is a sheaf, the two sections are equal. Hence σ lies in the image of

$$F(U) \rightarrow F^{\text{Sh}}(U).$$

Therefore this map is surjective (see [50, Prop. 0.2]). □

This condition is strictly weaker than being a sheaf, as only the gluing condition is required, and not uniqueness. Furthermore, suppose that $s \in F^{\text{Sh}}(U)$. By the Definition of sheafification, s is represented locally by $s_i \in F(U_i)$. Since the s_i agree as germs, they agree after shrinking. Paracompactness lets one refine the cover so that these local representatives can actually be glued inside the presheaf. Without paracompactness this may fail.

The criterion above applies particularly well to singular cochains. Their presheaf admits extensions of compatible local cochains, and its restriction morphisms are surjective. This makes it possible to prove that the associated sheafified singular cochain sheaves are flabby.

Lemma E.0.3 (Flabbiness of sheafified singular cochains). Let X be a hereditarily paracompact topological space, let $G \in \text{Ob}(\text{Ab})$, and let

$$S^n(U, G) := \text{Hom}_{\text{Ab}}(C_n(U), G)$$

be the presheaf of singular n -cochains. Let

$$\mathcal{S}^n := (S^n(-, G))^{\text{Sh}}$$

denote its sheafification. Then $\mathcal{S}^n$ is flabby (compare [50, Lemma 0.1]).

Proof. Let $V \subseteq U$ be open. One shows that the restriction morphism

$$\mathcal{S}^n(U) \rightarrow \mathcal{S}^n(V)$$

is surjective.

Since X is hereditarily paracompact, the open subsets U and V are paracompact. For the presheaf $\mathcal{S}^n(-, G)$, the natural morphism

$$\mathcal{S}^n(W, G) \rightarrow \mathcal{S}^n(W)$$

is surjective for every open subset $W \subseteq X$. Indeed the presheaf $\mathcal{S}^n(-, G)$ satisfies the usual gluability condition:

Let $\{U_i\}_{i \in I}$ be a cover of U , and let

$$\varphi_i \in \mathcal{S}^n(U_i, G)$$

be a compatible family. Thus

$$\varphi_i|_{U_i \cap U_j} = \varphi_j|_{U_i \cap U_j}$$

for all $i, j \in I$. One defines a homomorphism

$$\varphi : C_n(U) \rightarrow G$$

on the free generators of $C_n(U)$, that is, on singular simplices $\sigma : \Delta^n \rightarrow U$. If $\sigma(\Delta^n) \subseteq U_i$ for some i , set

$$\varphi(\sigma) := \varphi_i(\sigma).$$

This is well-defined:

If also $\sigma(\Delta^n) \subseteq U_j$, then

$$\sigma(\Delta^n) \subseteq U_i \cap U_j,$$

and compatibility gives

$$\varphi_i(\sigma) = \varphi_j(\sigma).$$

If $\sigma(\Delta^n)$ is not contained in any member of the cover, define

$$\varphi(\sigma) = 0.$$

Since $C_n(U)$ is the free abelian group generated by singular n -simplices in U , this assignment extends uniquely to a homomorphism

$$\varphi \in \text{Hom}_{\text{Ab}}(C_n(U), G) = S^n(U, G).$$

By construction,

$$\varphi|_{U_i} = \varphi_i$$

for every $i \in I$. Hence $S^n(-, G)$ satisfies the gluing property.

Hence the standard paracompactness criterion for sheafification Proposition [E.0.2](#) applies.

Now let $s \in S^n(V)$. By the previous argumentation, there exists

$$\varphi \in S^n(V, G)$$

whose image in $S^n(V)$ is s . The restriction map

$$S^n(U, G) \rightarrow S^n(V, G)$$

is surjective: a cochain on V can be extended to a cochain on U by choosing arbitrary values on singular simplices whose image is not contained in V . One can choose an extension $\Phi \in S^n(U, G)$ of φ :

One recalls that

$$S^n(U, G) = \text{Hom}(C_n(U), G),$$

and

$$C_n(V) \subseteq C_n(U)$$

is a subgroup, as every simplex in V is also a simplex in U . Thus there exists an inclusion

$$\iota : C_n(V) \hookrightarrow C_n(U).$$

Since $C_n(U)$ is the free abelian group on singular simplices and $C_n(V)$ is generated

by a subset of those generators, one can write

$$C_n(U) = C_n(V) \oplus M$$

for some free subgroup M . One can define

$$\Phi : C_n(U) \rightarrow G$$

by $\Phi = \varphi$ on $C_n(V)$, and choose

$$\Phi(m) = 0$$

for every basis element outside V . Then

$$\Phi|_{C_n(V)} = \varphi.$$

Its image in $\mathcal{S}^n(U)$ restricts to s . Hence

$$\mathcal{S}^n(U) \rightarrow \mathcal{S}^n(V)$$

is surjective, and therefore $\mathcal{S}^n$ is flabby (compare [50, Lemma 0.1]). □

To compare the global sections of the sheafified singular cochain complex with ordinary singular cochains, one must control singular simplices relative to an arbitrary open cover. The required control is obtained from barycentric subdivision and its natural chain homotopy to the identity.

Lemma E.0.4. Let $S : C_\bullet(X) \rightarrow C_\bullet(X)$ denote barycentric subdivision. Then S is a chain map and there exists a natural chain homotopy

$$T : C_\bullet(X) \rightarrow C_{\bullet+1}(X)$$

such that

$$\partial \circ T + T \circ \partial = S - \text{id}.$$

Moreover, if a singular simplex $\sigma : \Delta^n \rightarrow X$ has image contained in an open set $U \subseteq X$, then every singular simplex occurring in $S(\sigma)$ and in $T(\sigma)$ also has image contained in U (compare [50, pp. 8 ff.]).

Proof. Start with the standard simplex Δ^n . Let $b_n \in \Delta^n$ denote its barycentre. If

$$\alpha : \Delta^k \rightarrow \Delta^n$$

is an affine singular simplex, define

$$b_n * \alpha : \Delta^{k+1} \rightarrow \Delta^n$$

to be the affine simplex whose vertices are $b_n, \alpha(e_0), \dots, \alpha(e_k)$. Extending linearly gives a cone operator

$$B_n : C_k(\Delta^n) \rightarrow C_{k+1}(\Delta^n)$$

on affine chains in Δ^n .

For an affine simplex $[v_0, \dots, v_k]$, one has

$$\partial[b_n, v_0, \dots, v_k] = [v_0, \dots, v_k] - \sum_{i=0}^k (-1)^i [b_n, v_0, \dots, \widehat{v_i}, \dots, v_k].$$

Thus for every $k \geq 0$

$$\partial \circ B_n + B_n \circ \partial = \text{id} - \eta \circ \varepsilon,$$

where

$$\varepsilon : C_0(\Delta^n) \rightarrow \mathbb{Z}$$

is the augmentation and

$$\eta : \mathbb{Z} \rightarrow C_0(\Delta^n)$$

sends 1 to the constant simplex at b_n . In particular, if c is a cycle of augmentation zero, then

$$\partial \circ B_n(c) = c.$$

Define a barycentric subdivision inductively:

For $n = 0$, set

$$S(\text{id}_{\Delta^0}) = \text{id}_{\Delta^0}.$$

For $n > 0$, assume that S has already been defined in dimension $< n$. Let

$$\iota_n : \Delta^n \rightarrow \Delta^n$$

be the identity simplex. Define

$$S(\iota_n) := B_n\left(S(\partial(\iota_n))\right).$$

Since $\partial^2 = 0$,

$$\partial(\partial(\iota_n)) = 0.$$

Because S is already defined in dimensions $< n$ and satisfies

$$\partial \circ S = S \circ \partial$$

there, it follows that

$$\partial \circ S(\partial(\iota_n)) = S(\partial^2(\iota_n)) = 0.$$

Hence $\partial(\iota_n)$ is a cycle. It has augmentation zero. Hence

$$\partial \circ S(\iota_n) = \partial \circ B_n\left(S(\partial(\iota_n))\right) = S(\partial(\iota_n)).$$

Thus

$$\partial \circ S(\iota_n) = S(\partial(\iota_n)).$$

Let

$$\sigma_{\#} : C_{\bullet}(\Delta^n) \rightarrow C_{\bullet}(X)$$

denote the chain map induced by σ . For a singular simplex $\sigma : \Delta^n \rightarrow X$, define

$$S(\sigma) := \sigma_{\#}(S(\iota_n)).$$

Extending linearly gives a homomorphism

$$S : C_n(X) \rightarrow C_n(X).$$

The identity just proved implies

$$(\partial \circ S)(\sigma) = \partial\left(\sigma_{\#}(S(\iota_n))\right) = \sigma_{\#}\left(\partial(S(\iota_n))\right) = \sigma_{\#}\left(S(\partial(\iota_n))\right) = S(\partial(\sigma)).$$

Therefore $\partial \circ S = S \circ \partial$, so S is a chain map.

It remains to construct a chain homotopy. Set $T_0 = 0$. Suppose T has been defined in dimensions $< n$ and satisfies

$$\partial \circ T + T \circ \partial = S - \text{id}.$$

For the identity simplex $\iota_n : \Delta^n \rightarrow \Delta^n$, define

$$c_n := S(\iota_n) - \iota_n - T(\partial(\iota_n)).$$

Then

$$\begin{aligned} \partial(c_n) &= (\partial \circ S)(\iota_n) - \partial \iota_n - (\partial \circ T)(\partial(\iota_n)) \\ &= S(\partial(\iota_n)) - \partial(\iota_n) - \left(S(\partial(\iota_n)) - \partial(\iota_n) - (T \circ \partial^2)(\iota_n) \right) \\ &= 0. \end{aligned}$$

Since $n > 0$ and $\partial c_n = 0$, the cone identity gives

$$\partial \circ B_n(c_n) = c_n.$$

Define

$$T(\iota_n) := B_n(c_n).$$

For a singular simplex $\sigma : \Delta^n \rightarrow X$, define

$$T(\sigma) := \sigma_{\#}(T(\iota_n)),$$

and extend linearly. Then

$$\begin{aligned} (\partial \circ T)(\sigma) &= \sigma_{\#}((\partial \circ T)(\iota_n)) \\ &= \sigma_{\#}(c_n) \\ &= S(\sigma) - \sigma - T(\partial(\sigma)). \end{aligned}$$

Thus

$$\partial \circ T(\sigma) + T(\partial(\sigma)) = S(\sigma) - \sigma.$$

Since $C_n(X)$ is the free abelian group generated by singular n -simplices, the iden-

tity extends to all chains by linearity. This proves

$$\partial \circ T + T \circ \partial = S - \text{id}.$$

Finally, every simplex occurring in $S(\sigma)$ or $T(\sigma)$ is of the form $\sigma \circ a$ for some affine map $a : \Delta^m \rightarrow \Delta^n$. Hence its image is contained in $\sigma(\Delta^n)$. Therefore, if $\sigma(\Delta^n) \subseteq U$, then every simplex occurring in $S(\sigma)$ and $T(\sigma)$ also has image contained in U (compare [50, pp. 8 ff.]). $\square$

Repeated barycentric subdivision eventually decomposes every singular simplex into simplices contained in members of a prescribed open cover. The preceding chain homotopy then allows this local subdivision process to be organised into a chain homotopy equivalence.

Theorem E.0.5. Let $\mathcal{U} = \{U_i\}_{i \in I}$ be an open cover of X , and let $C_n^{\mathcal{U}}(X)$ be a subgroup of $C_n(X)$ generated by those singular simplices $\sigma : \Delta^n \rightarrow X$ whose image is contained in some U_i . Then the inclusion

$$\iota : C_{\bullet}^{\mathcal{U}}(X) \hookrightarrow C_{\bullet}(X)$$

induces an isomorphism in cohomology, that is,

$$I : H_{\text{sing}}^n(X, A) \xrightarrow{\cong} H^n(\text{Hom}_{\text{Ab}}(C_{\bullet}^{\mathcal{U}}(X), A))$$

for all $n \geq 0$ (see [50]).

Proof. Let

$$S : C_{\bullet}(X) \rightarrow C_{\bullet}(X)$$

be a barycentric subdivision, and let

$$T : C_{\bullet}(X) \rightarrow C_{\bullet+1}(X)$$

be the chain homotopy from Lemma E.0.4 so that

$$\partial \circ T + T \circ \partial = S - \text{id}.$$

Let $\sigma : \Delta^n \rightarrow X$ be a singular simplex. The sets

$$\sigma^{-1}(U_i), \quad i \in I$$

form an open cover of the compact metric space Δ^n . Hence, by the Lebesgue number (see Lemma A.4.35), there exists $\lambda_\sigma > 0$ such that every subset of Δ^n of diameter $< \lambda_\sigma$ is contained in some $\sigma^{-1}(U_i)$.

Since iterated barycentric subdivision makes the diameters of all affine subsimplices tend to zero, there exists $m_\sigma \geq 0$ such that every simplex occurring in $S^{m_\sigma} \sigma$ has image contained in some U_i . Thus

$$S^{m_\sigma} \sigma \in C_n^{\mathcal{U}}(X).$$

Choose the integers m_σ inductively over the dimension of σ requiring additionally that

$$m_\sigma \geq m_\tau$$

for every face τ of σ .

Define

$$D(\sigma) := \sum_{k=0}^{m_\sigma-1} T(S^k(\sigma)),$$

and extend linearly to a homomorphism

$$D : C_n(X) \rightarrow C_{n+1}(X).$$

Now define

$$\rho(\sigma) := \sigma + \partial \circ D(\sigma) + D(\partial(\sigma)),$$

and extend linearly to

$$\rho : C_n(X) \rightarrow C_n(X).$$

Then

$$\rho - \text{id} = \partial \circ D + D \circ \partial.$$

Hence

$$\partial \rho = \rho \partial,$$

so ρ is a chain map.

Claim: ρ takes values in $C_{\bullet}^{\mathcal{U}}(X)$.

Indeed, using $\partial \circ T = S - \text{id} - T \circ \partial$, one can compute

$$\begin{aligned} \rho(\sigma) &= \sigma + \partial \left(\sum_{k=0}^{m_{\sigma}-1} T(S^k \sigma) \right) + D(\partial \sigma) \\ &= \sigma + \sum_{k=0}^{m_{\sigma}-1} (S^{k+1} \sigma - S^k \sigma - T(S^k \partial \sigma)) + D(\partial \sigma) \\ &= S^{m_{\sigma}} \sigma - \sum_{k=0}^{m_{\sigma}-1} T(S^k \partial \sigma) + D(\partial \sigma). \end{aligned}$$

Write $\partial \sigma = \sum_j \varepsilon_j \tau_j$ as the alternating sum of the faces of σ . Then

$$D(\partial \sigma) = \sum_j \varepsilon_j \sum_{k=0}^{m_{\tau_j}-1} T(S^k \tau_j),$$

whereas

$$\sum_{k=0}^{m_{\sigma}-1} T(S^k \partial \sigma) = \sum_j \varepsilon_j \sum_{k=0}^{m_{\sigma}-1} T(S^k \tau_j).$$

Since $m_{\sigma} \geq m_{\tau_j}$ for every face τ_j , it follows that

$$\rho(\sigma) = S^{m_{\sigma}} \sigma - \sum_j \varepsilon_j \sum_{k=m_{\tau_j}}^{m_{\sigma}-1} T(S^k \tau_j).$$

By construction, $S^{m_{\sigma}} \sigma$ is $\mathcal{U}$ -small. Moreover, for $k \geq m_{\tau_j}$, the chain $S^k \tau_j$ is $\mathcal{U}$ -small. By Lemma E.0.4, the operator T preserves $\mathcal{U}$ -smallness. Therefore every term $T(S^k \tau_j)$ is $\mathcal{U}$ -small. Hence $\rho(\sigma) \in C_n^{\mathcal{U}}(X)$. Thus ρ defines a chain map

$$\rho : C_{\bullet}(X) \rightarrow C_{\bullet}^{\mathcal{U}}(X).$$

Let

$$\iota : C_{\bullet}^{\mathcal{U}}(X) \hookrightarrow C_{\bullet}(X)$$

be the inclusion. Since

$$\iota \rho - \text{id}_{C_{\bullet}(X)} = \partial D + D \partial,$$

the composite $\iota \rho$ is chain homotopic to $\text{id}_{C_{\bullet}(X)}$.

It remains to compare $\rho\iota$ with $\text{id}_{C^\mathcal{U}_\bullet(X)}$.

Let σ be a $\mathcal{U}$ -small simplex. Then every simplex occurring in $S^k\sigma$ is again $\mathcal{U}$ -small by Lemma E.0.4. Moreover, the same Lemma implies that every simplex occurring in $T(S^k\sigma)$ is $\mathcal{U}$ -small. Hence

$$D(\sigma) = \sum_{k=0}^{m_\sigma-1} T(S^k\sigma)$$

belongs to $C^\mathcal{U}_{n+1}(X)$ whenever $\sigma \in C^\mathcal{U}_n(X)$. Thus D restricts to a homomorphism

$$D : C^\mathcal{U}_n(X) \rightarrow C^\mathcal{U}_{n+1}(X).$$

Restricting the identity

$$\rho - \text{id} = \partial \circ D + D \circ \partial$$

to $C^\mathcal{U}_\bullet(X)$ gives

$$\rho\iota - \text{id}_{C^\mathcal{U}_\bullet(X)} = \partial D + D\partial.$$

Therefore $\rho\iota$ is chain homotopic to $\text{id}_{C^\mathcal{U}_\bullet(X)}$.

Consequently, ι is a chain homotopy equivalence. Applying $\text{Hom}_{\text{Ab}}(-, A)$ gives a cochain homotopy equivalence

$$\text{Hom}_{\text{Ab}}(C_\bullet(X), A) \simeq \text{Hom}_{\text{Ab}}(C^\mathcal{U}_\bullet(X), A).$$

Hence the induced map on cohomology is an isomorphism:

$$H_{\text{sing}}^n(X, A) \cong H^n(\text{Hom}_{\text{Ab}}(C^\mathcal{U}_\bullet(X), A))$$

for all $n \geq 0$ (see [50]). □

The Small Chain Theorem compares ordinary singular cochains with cochains defined on chains subordinate to a fixed cover. Passing over all open covers now relates the original singular cochain complex to the global sections of its sheafification.

Lemma E.0.6. Let X be a paracompact topological space, let $G \in \text{Ob}(\text{Ab})$ and let

$\mathcal{S}^n$ be the sheafification of the presheaf

$$U \mapsto \operatorname{Hom}_{\mathbf{Ab}}(C_n(U), G).$$

Then the natural morphism of cochain complexes

$$S^\bullet(X, G) \rightarrow \Gamma(X, \mathcal{S}^\bullet)$$

is a quasi-isomorphism (see [50]).

Proof. A global section of $\mathcal{S}^n$ is represented locally by singular cochains. Hence every global section is determined by compatible cochains on some open cover $\mathcal{U}$ of X .

This identifies $\Gamma(X, \mathcal{S}^\bullet)$ with the directed colimit over open covers

$$\varinjlim_{\mathcal{U}} \operatorname{Hom}_{\mathbf{Ab}}(C_\bullet^\mathcal{U}(X), G),$$

where $C_\bullet^\mathcal{U}(X)$ denotes the complex of $\mathcal{U}$ -small chains. For every open cover $\mathcal{U}$, Theorem E.0.5 implies that the inclusion

$$C_\bullet^\mathcal{U}(X) \hookrightarrow C_\bullet(X)$$

is a chain homotopy equivalence. Therefore

$$\operatorname{Hom}_{\mathbf{Ab}}(C_\bullet(X), G) \rightarrow \operatorname{Hom}_{\mathbf{Ab}}(C_\bullet^\mathcal{U}(X), G)$$

is a quasi-isomorphism. Passing to the filtered colimit over all open covers preserves cohomology, since filtered colimits are exact in $\mathbf{Ab}$. Hence the natural morphism

$$S^\bullet(X, G) \rightarrow \Gamma(X, \mathcal{S}^\bullet)$$

is a quasi-isomorphism. □

The singular cochain sheaves form an acyclic resolution of the constant sheaf, while the previous lemma identifies the cohomology of their global sections with singular cohomology.

The preceding results provide the three ingredients required for the comparison Theorem 4.2.17: the sheafified singular cochain complex is an exact resolution of the constant sheaf, its terms are flabby and hence acyclic, and its complex of global sections computes ordinary singular cohomology. These facts can now be assembled to identify sheaf cohomology with singular cohomology.

Let X be a hereditarily paracompact and locally contractible topological space, and let $G^{\text{cnst}} \in \text{Ob}(\text{Sh}(X, \text{Ab}))$ be a constant sheaf associated to an abelian group G . Then the sheaf cohomology of G^{cnst} agrees with singular cohomology, that is,

$$\mathcal{H}^n(X, G^{\text{cnst}}) \cong H_{\text{sing}}^n(X, G)$$

for all $n \geq 0$ (see [50]).

Proof. The idea of the proof is to construct an acyclic resolution

$$0 \rightarrow G^{\text{cnst}} \rightarrow \mathcal{S}^0 \rightarrow \mathcal{S}^1 \rightarrow \dots$$

such that

1. the complex of global sections $\Gamma(X, \mathcal{S}^\bullet)$ computes singular cohomology,
2. the sequence is exact,
3. each sheaf $\mathcal{S}^n$ is acyclic.

Then the sheaf cohomology of G^{cnst} is computed by the global sections of this resolution, and the cohomology of those global sections coincides with singular cohomology.

For an open subset $U \subseteq X$, define

$$S^n(U, G) := \text{Hom}_{\text{Ab}}(C_n(U), G),$$

where $C_n(U)$ denotes the group of singular n -chains on U . If $V \subseteq U$, the inclusion $V \hookrightarrow U$ induces a morphism

$$C_n(V) \rightarrow C_n(U).$$

Precomposition therefore gives a restriction morphism

$$S^n(U, G) \rightarrow S^n(V, G).$$

Thus

$$U \mapsto S^n(U, G)$$

defines a presheaf on X . Let $\mathcal{S}^n := (S^n(U, G))^{\text{Sh}}$ denote its sheafification.

The singular coboundary induces morphisms of presheaves

$$\delta : S^n(-, G) \rightarrow S^{n+1}(-, G),$$

defined by

$$(\delta\varphi)(c) := \varphi(\partial c)$$

for $\varphi \in S^n(U, G)$ and $c \in C_{n+1}(U)$.

These morphisms are compatible with restrictions. Indeed, if $V \subseteq U$, $\varphi \in S^n(U, G)$, and $c \in C_{n+1}(V)$, then

$$((\delta\varphi)|_V)(c) = (\delta\varphi)(c) = \varphi(\partial c)$$

and

$$\delta(\varphi|_V)(c) = (\varphi|_V)(\partial c) = \varphi(\partial c).$$

Hence

$$(\delta\varphi)|_V = \delta(\varphi|_V).$$

After sheafification, this yields a morphism of sheaves

$$\delta^{\text{Sh}} : \mathcal{S}^n \rightarrow \mathcal{S}^{n+1},$$

making $\mathcal{S}^\bullet$ a complex of sheaves.

A section of the constant sheaf G^{cst} over an open subset $U \subseteq X$ determines a singular 0-cochain by evaluation on singular 0-simplices. This yields a morphism of sheaves

$$G^{\text{cst}} \rightarrow \mathcal{S}^0.$$

Hence one obtains a sequence of sheaves

$$0 \rightarrow G^{\text{cnst}} \rightarrow \mathcal{S}^0 \rightarrow \mathcal{S}^1 \rightarrow \dots$$

The exactness is verified on stalks. Fix a point $x \in X$. Since X is locally contractible, arbitrarily small contractible neighbourhoods of x form a neighbourhood basis. If U is contractible, then singular cohomology satisfies

$$H_{\text{sing}}^0(U, G) \cong G$$

and

$$H_{\text{sing}}^q(U, G) = 0$$

for all $q > 0$ by Theorem 4.1.11. Equivalently, the augmented singular cochain complex

$$0 \rightarrow G \rightarrow S^0(U, G) \rightarrow S^1(U, G) \rightarrow \dots$$

is exact.

Passing to stalks gives the complex

$$0 \rightarrow G \rightarrow \mathcal{S}_x^0 \rightarrow \mathcal{S}_x^1 \rightarrow \dots$$

Since stalks are filtered colimits over neighbourhoods of x , and filtered colimits preserve exactness in $\mathbf{Ab}$, the stalk complex is exact. Hence,

$$0 \rightarrow G^{\text{cnst}} \rightarrow \mathcal{S}^0 \rightarrow \mathcal{S}^1 \rightarrow \dots$$

is an exact sequence of sheaves.

Each sheaf $\mathcal{S}^n$ is flabby. The presheaf $\mathcal{S}^n(-, G)$ has surjective restriction maps: if $V \subseteq U$, then every cochain on V extends to a cochain on U by choosing arbitrary values on singular simplices not contained in V . Therefore, by Lemma E.0.3, its sheafification $\mathcal{S}^n$ is flabby.

By Theorem 4.2.16, flabby sheaves are acyclic. Hence

$$H^q(X, \mathcal{S}^n) = 0$$

for all $q > 0$. Thus

$$0 \rightarrow G^{\text{cst}} \rightarrow \mathcal{S}^0 \rightarrow \mathcal{S}^1 \rightarrow \dots$$

is an acyclic resolution of G^{cst} .

Since the resolution is acyclic, the sheaf cohomology is computed by the cohomology of the global section complex by Theorem 4.2.8

$$\mathcal{H}^n(X, G^{\text{cst}}) \cong H^n(\Gamma(X, \mathcal{S}^\bullet)).$$

There is a natural morphism of cochain complexes

$$S^n(X, G) \rightarrow \Gamma(X, \mathcal{S}^n)$$

from the singular cochain complex to the global sections of the sheafified singular cochain complex. By Lemma E.0.6, this morphism is a quasi-isomorphism. Therefore

$$H^n(\Gamma(X, \mathcal{S}^\bullet)) \cong H_{\text{sing}}^n(X, G).$$

Combining these isomorphisms gives

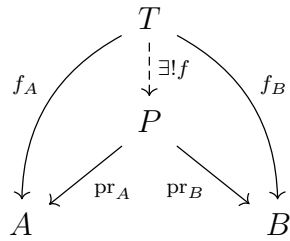
$$\mathcal{H}^n(X, G^{\text{cst}}) \cong H^n(\Gamma(X, \mathcal{S}^\bullet)) \cong H_{\text{sing}}^n(X, G)$$

(see [50]). □

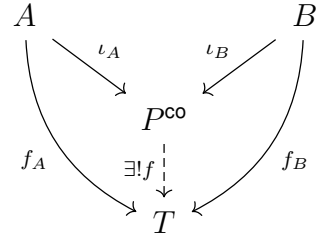
The hereditary paracompactness assumption enters the argument above throughout the refinement and flabbiness results. One can develop a different construction using nested cochains, which avoids this dependence on paracompactness and yields comparison results under weaker hypotheses. In this approach, neighbourhood data are assigned not merely to individual simplices, but compatibly to chains of faces, providing the additional local control required for sheafification (see [50]).

Overview of the Universal Properties

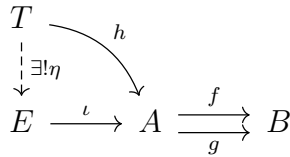
Binary product



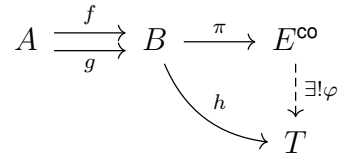
Binary coproduct



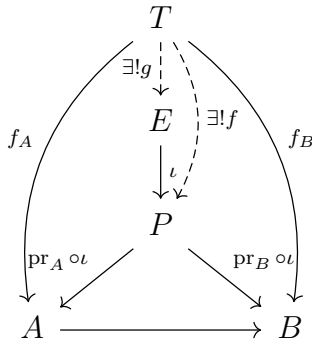
Equaliser



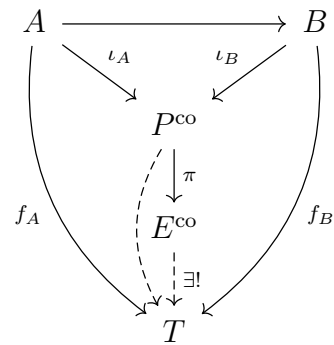
Coequaliser



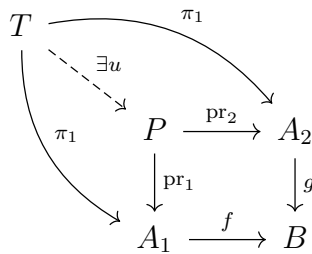
Limit



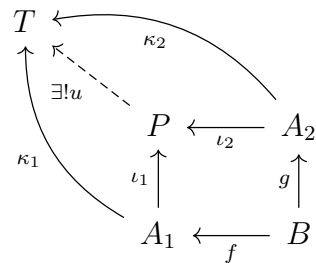
Colimit



Fibre Product (Pullback)



Fibre Coproduct (Pushout)



Binary Biproduct

$$\begin{array}{ccccc} & \text{pr}_A & & \text{pr}_B & \\ & \curvearrowright & & \curvearrowright & \\ A & & A \oplus B & & B \\ & \curvearrowleft & & \curvearrowleft & \\ & \iota_A & & \iota_B & \end{array}$$

Overview of the Categories

Category	Objects	Morphisms
$[\mathcal{C}, \mathcal{D}]$	Functors $\mathcal{C} \rightarrow \mathcal{D}$	Natural transformations
$\Pi_1(X)$	points in a topological space X	homotopy classes of paths
Ab	Abelian groups	Group homomorphisms
Add	Contravariant additive functors	Natural transformations
Ch	Chains	Chain morphisms
Coch	Cochains	Cochain morphisms
$\text{Cov}(X)$	Open covers of a topological space X	Refinements
CRing	Commutative rings with 1	Ring homomorphisms
Δ	$[n]$	order-preserving maps
Grp	Groups	Group homomorphisms
LRS	Locally ringed spaces	Sheaf morphisms
$\text{Open}(X)$	Open sets in the topological space X	Inclusion maps
PSh	Presheaves	Natural transformations
Sch	Schemes	Morphisms of locally ringed spaces
Set	Sets	Maps
Sh	Sheaves	Sheaf morphisms
Top	Topological spaces	Continuous maps
Top_*	Punctured topological spaces	Continuous maps

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Name: Jeannine L. Döhrer

Matrikelnummer: 1074486

Erklärung

Ich erkläre, dass ich die Arbeit selbständig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel verwendet habe.

Ulm, den ..



Jeannine L. Döhrer